

Client Advisor Competency Standards (CACS) Study Guide

Paper 2

Wealth Management Industry and Product Knowledge

2

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The Institute of Banking & Finance

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Preface

In April 2011, the private banking industry launched a Code of Conduct for the private banking industry in Singapore (PB Code). The PB Code, which takes effect from 1 September 2011, has been developed to foster professional standards, enhance transparency to clients and confidence in the private banking industry in Singapore.

It sets out standards of good practice on competency and market conduct expected of financial institutions (including their staff) operating in Singapore which are providing financial services to High Net Worth Individuals.

Such financial institutions are termed “Covered Entities” within the PB Code and their client-facing staff who provide financial advisory service(s) to High Net Worth Individuals are termed “Covered Persons”. Please refer to www.abs.org.sg for a copy of the PB Code.

Client Advisor Competency Standards (CACS)

Under the PB Code, Covered Persons are expected to pass a common competency assessment called the Client Advisor Competency Standards (CACS) before they can provide any financial advice. The CACS is intended to broaden and deepen the capabilities of these professionals in the provision of comprehensive private banking services.

This assessment will be introduced from 1 September 2011. The CACS will assess the knowledge dominant competencies of Covered Persons in the areas of market conduct and product knowledge. The Institute of Banking and Finance (IBF) has been appointed as the administrator of the CACS.

The CACS is organized as a two-paper assessment: Paper 1 will cover concepts on Wealth Management Legislations, Regulations and Industry Codes of Practice; and Paper 2 will cover concepts on Wealth Management Industry and Product Knowledge. Candidates will have to pass both papers in order to be deemed to have passed the CACS.

The CACS assessment will be conducted at IBF’s Assessment Centre. The Assessment will comprise of multiple-choice questions, and will include both questions testing on concepts, as well as computations. Candidates are expected to have prepared themselves adequately for the Assessment through the use of the Study Guide as well as other relevant reference materials.

Please refer to the IBF’s website at www.ibf.org.sg for the rules and regulations regarding the CACS assessment.

Paper 2: Wealth Management Industry and Product Knowledge

Objective of Study Guide

The objective of this Study Guide is to provide candidates with a good understanding of the following topics, in preparing for the CACS Paper 2 assessment:

- Introduction on investment concepts of risk and returns;
- The portfolio management process;
- Investment products such as fixed income and equity securities, options, futures, foreign exchange and swap contracts, structured products, unit trust and fund products, alternative investment products and services; and
- The use of credit and leverage.

All references made to “Covered Person” or “Covered Entity” within this Study Guide is to be taken in the context of the PB Code. The standards in the Code are intended to provide broad guidance and are not meant to be exhaustive or to replace any legislations, regulations, or guidelines issued by the relevant authorities in Singapore.

While this Study Guide highlights obligations imposed on Covered Persons and Covered Entities, it should be read in conjunction with the corresponding sections of the PB Code and all applicable legislation, regulations and guidelines, for a holistic understanding of the desired professional standards of a Covered Person and Covered Entity.

Organisation of the Study Guide

This Study Guide consists of 8 chapters, starting with an overview of basic investment concepts and the concepts of portfolio construction. Subsequent chapters provide elaboration on investment products such as fixed income and equity securities, options, futures, foreign exchange and swap contracts, structured products, unit trust and fund products, alternative investment products and services. The final chapter provides an insight into the use of credit and leverage.

Each chapter starts off with a list of learning objectives, followed by a chapter introduction which provides an overview of the chapter. Tips / key points are provided at various points within each chapter to highlight important learning points. Examples and case studies are also used extensively in the study guide to enhance candidate’s understanding and application of issues discussed.

The appendix at the end of each chapter provides further discussion on some of the theories or computational concepts addressed in the chapter and should be read. A formulae sheet summarizing all the formulae found within this study guide will be provided to candidates during the CACS assessment.

A summary of each chapter is provided below:-

- **Chapter1: Investment Concepts – Risk and Return** – In this chapter, you will learn about the various measures of risk and return, the factors that determine an investors' required rate of return and the risk-return profile of various investments. It also provides an introduction to financial statement analysis and commonly used financial ratios, as well as discusses the measures of economic performance and the key economic indicators, and the impact of fiscal and monetary policies.
- **Chapter 2: Portfolio Management** – This chapter discusses how the portfolio of a client is constructed, monitored and managed, and the roles and obligations of a client advisor in the portfolio management process, including risk management. Central to portfolio management is Modern Portfolio Theory (MPT) which focuses on diversification of risks and the Capital Asset Pricing Model (CAPM).
- **Chapter 3: Fixed Income and Equity Securities** – The objective of this chapter is to introduce two conventional asset classes – Fixed Income (which includes money market instrument and bonds) and Equities. These two asset groups represent traditional vehicles for investors to earn a return above bank deposit rates.
- **Chapter 4: Options, Futures, Foreign Exchange and Swap Contracts** – This chapter will discuss the different types of derivatives available to investors, their key features, benefits, risks, and the differences between exchange traded and over-the-counter traded derivatives.
- **Chapter 5: Structured Products** – This chapter provides an insight on the components, key features and generic risks of structured products. It will introduce common strategies used in putting together structured products.
- **Chapter 6: Unit Trusts and Fund Products** – This chapter focuses on Collective Investment Schemes, which include unit trusts, investment trusts and mutual funds. It will also highlight the advantages and disadvantages of investing in unit trusts and provide an overview of some of the criteria used to evaluate unit trusts.
- **Chapter 7: Alternative Investment Products and Services** – In this chapter, the more commonly used alternative investment assets will be discussed. These include Real Estate, Hedge Funds, Private Equity, Commodities and Passion investments.
- **Chapter 8: Using Credit and Leverage** – This chapter provides an overview of the bank lending environment and how lending activities in the private banking sector fit into the overall landscape. It also discusses the credit and lending needs of certain private banking clients and how private banks can meet these needs.

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Chapter 1:

Investment Concepts – Risk and Return

Objective

The candidate should be able to:

- ✓ Understand the measures of risk and return
- ✓ Appreciate the determinants of rates of return
- ✓ Evaluate quality of future earnings and financial ratios
- ✓ Implication of economic factors, policies and its inter-relationship on capital markets and client's investments

1.1 Introduction

In this chapter, we learn about the various measures of risk and return, the factors that determine an investors' required rate of return and the risk-return profile of various investments. It also provides an introduction to financial statement analysis, in particular, the income statement of a company and commonly used financial ratios. This chapter further discusses the measures of economic performance and the key economic indicators, and the impact of fiscal and monetary policies.

1.2 Measuring Return

It is important to differentiate between returns and yields. The **return** on investment computes the historical investment performance. It includes both the capital appreciation as well as the absolute dollars (or termed dollar-returns) or as a percentage of the initial investment.

The **yield** from an investment is an annualized rate of return computed from the regular income as a percentage of the initial investment. It is a forward-looking measure of income but does not include capital gains. It is based on the assumption that past income will continue to be received at the same amount.

Actual returns are ex-post computations as opposed to expected returns which are computed ex-ante. Actual returns computations measure the past performance

whereas the expected return is an indication of projected returns. It is also necessary to state the period over which returns are measured. That is, whether it is measured over a month, a quarter or on an annual basis.

1.2.1 Calculating Simple Returns

Return measurements are relatively simple if there are no cash flows during the measurement period. That is, there are no withdrawals or increased investments contributed during the “investment period”. In this situation, the rate of return on the investment is given by:

$$\text{Return on the investment, } (R_i) = \frac{V_E - V_B + D}{V_B}$$

Where,

R_i = Return on the investment

V_E = End value of the investment

V_B = Beginning value of the investment

D = Distributions (i.e. dividends or interest paid)

Example 1.1

If an investment has a market value of \$30 million at the beginning of a quarter and a market value of \$35 million at the end of the quarter, and there are no distributions during the quarter, the return of the investment during the quarter is:

$$\begin{aligned} R_i &= \frac{\$35\text{m} - \$30\text{m} + 0}{\$30\text{m}} \\ &= 16.67\% \text{ for the quarter} \end{aligned}$$

1.2.2 Deposits or Withdrawals of Funds during the Measurement Period

Measuring the return of an investment becomes more complicated when the client adds or withdraws funds during the measurement period. When that happens, it is necessary to compute a *Dollar-Weighted Rate of Return* (DWR) or *Time-Weighted Rate of Return* (TWR).

Dollar-Weighted Rate of Return (DWR)

The DWR is equivalent to the internal rate of return (IRR). It is the discount rate which equates the initial cash outlay with cash flows which occur during the investment period.

Example 1.2

Consider an investment with a market value of \$50m at the beginning of a quarter. Suppose the client invests a further \$5m in the middle of the quarter, and subsequently the market value of the investment at the end of the quarter is \$63m. What is the return of the investment during the quarter?

$$\begin{aligned} \$50\text{m} &= \frac{-\$5\text{m}}{(1+r)} + \frac{\$63\text{m}}{(1+r)^2} \\ r &= 7.36\% \text{ per half quarter} \end{aligned}$$

Since this r is the return for half a quarter, the quarterly return would be compounded to obtain $(1+0.0736)^2 - 1$ or 15.268% per quarter.

Time-Weighted Rate of Return (TWR)

The DWR method is affected by the size and timing of the cash flows by a client over which an investment manager has no control. Therefore, this method of evaluating the performance of the investment is sometimes regarded as inappropriate.

For evaluating the performance of the investment manager, the Time-Weighted Rate of Return is more appropriate as it measures the performance of the investment independent of the addition or withdrawal of cash by the client. The TWR requires information about the value of the investment before each cash flow occurs.

Example 1.3

Investment value at the beginning of quarter = \$50m

Investment value just before deposit = \$49m

Deposit in the middle of the quarter = \$5m

Investment value immediately after deposit = \$54m

Investment value at the end of the quarter = \$63m

Return of the Investment for the first half of the quarter is:

$$\begin{aligned} R_1 &= \frac{\$49\text{m} - \$50\text{m}}{\$50\text{m}} \\ &= -2.0\% \text{ for half a quarter} \end{aligned}$$

Return of the Investment for the second half of the quarter is:

$$\begin{aligned} R_2 &= \frac{\$63\text{m} - \$54\text{m}}{\$54\text{m}} \\ &= 16.67\% \text{ for half a quarter} \end{aligned}$$

The quarterly return is computed by linking the two rates of return as follows:

$$1 + R_i = [(1 - 0.02)(1 + 0.1667)]$$

$$R_i = 14.3 \% \text{ per quarter}$$

1.2.3 Annual Returns

To restate quarterly returns into annual returns, the formula is:

$$\text{Annual return, } 1 + R = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)]$$

$$R = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)] - 1$$

Where,

r_1, r_2, r_3, r_4 = quarterly returns

Actual Return for Clients vs. Returns Reported by the Fund

For Example 1.1 above, the quarterly return was 16.67%. If the same return is received in the next three quarters, the annual return can be computed as:

$$R = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)] - 1$$

$$= [(1 + 0.1667)(1 + 0.1667)(1 + 0.1667)(1 + 0.1667)] - 1$$

$$= 85.28\% \text{ p.a.}$$

The fund's return is typically reported at quarterly intervals. Thus, there is a need to compound it to report an annual return. However, the reported returns may be a gross return before subtracting expenses. When expenses are deducted, it is a **return net of fees**.

1.2.4 Expected Return

When returns are uncertain, there is a need to compute the "expected return" based on the probability of each given scenario occurring and the returns obtained. The expected rate of return is a point estimate of the investor's most likely return on investment over the possible scenarios.

$$\text{Expected Return, } E(R_i) = \sum_{j=1}^N (\text{Probability of Return}) (\text{Possible Return})$$

$$= \sum_{j=1}^N (P_i) (R_i)$$

Where,

P_j is the probability of event j associated with return R_j for security, i

Assuming there are two economic scenarios as listed below, the calculation of the Expected Return on security i is as follows:

Example		
Economic Scenario	Probability	Rate of Return
Strong Economy	75%	+0.20
Weak Economy	25%	-0.10

$$E(R_i) = [(0.75)(0.20)] + [(0.25)(-0.10)]$$

$$= 0.125, \text{ or } 12.5\%$$

1.2.5 Components of Return

Generally, the rate of return can be calculated as the sum of three main components. The following formula illustrates this:

$$\text{Required Return} = r + i + \sigma$$

Where,

r = Real interest rate

i = Inflation rate

σ = Risk premium

The real interest rate is the return that investor will expect without any risk and with no inflation. The inflation rate is the general level of increase in the prices of goods and services. An increase in the inflation rate results in a loss of purchasing power. The risk premium is the minimum rate of return that an investor will demand to compensate him for investing in a risky asset.

1.3 Measuring Risk

Risk is the uncertainty associated with an investment. The difference between risk and uncertainty is that uncertainty cannot be measured. Risk can be measured in various ways. One measure of risk is the **dispersion** of returns around the arithmetic mean, which is defined by the statistical measures of variance (σ^2) and its square root form, the standard deviation (σ). Variance and standard deviation are absolute measures of risk.

Generally, risk is dependent on a number of factors which include business, financial, liquidity, exchange rate and country risks. In the context of a portfolio of securities, one needs to measure portfolio risk. Portfolio risk depends on the correlation between the returns of the individual securities in the portfolio. This will be discussed in Chapter 2. Appendix 1.1 provides formulas on risk measures.

1.4 Determinants of Rates of Return

$$\begin{aligned}\text{Required Return} &= r + i + \sigma \\ &= \text{Nominal Interest Rate} + \text{Risk Premium, } \sigma\end{aligned}$$

Where,

$$\text{Nominal Interest Rate} = r + i$$

The Capital Asset Pricing Model (CAPM) is used to determine the theoretically required rate of return on an asset, R_i . CAPM assumes there is a nominal “risk-free interest rate” termed RFR and a risk premium which is approximated by a volatility factor called beta B_i and a market risk premium RM. The equation for CAPM is shown in the equation below.

$$E(R_i) = \text{RFR} + B_i (\text{RM} - \text{RFR})$$

Where,

RFR = Expected Return on government securities

B_i = Beta or Relative risk of the Stock i to the Market

RM = Expected Return on the Stock Market Index

Example

Suppose the client wishes to invest in a stock that has Beta of 2, and the expected market return is 12% per annum. A short term government security (which is risk free) has a return of 2%. Applying these variables to the CAPM formula, the expected return is:

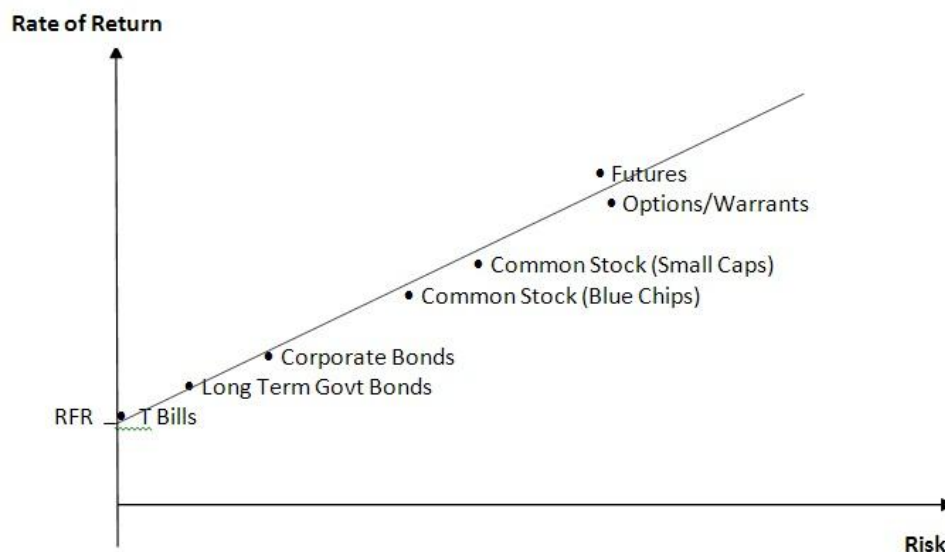
$$\begin{aligned}E(R_i) &= \text{RFR} + B_i (\text{RM} - \text{RFR}) \\ &= 2\% + 2 \times (12\% - 2\%) \\ &= 22\%\end{aligned}$$

Further elaboration will be discussed in Chapter 2.

1.5 Risk and Return Profile of Various Investments

The graph below shows the relative risk and return profiles of bonds, stocks and derivatives in general.

Exhibit 1.1 Risk and Return Profile of Various Investments



The various asset classes listed in Exhibit 1.1 will be discussed in the various chapters in this Study Guide.

1.6 Understanding Financial Statements

The key to valuation of financial securities is to understand the financial health of a firm. A quick assessment of a firm's current financial status can be gleaned from its financial statements. Although financial statements are historical documents, they provide useful indications of current and potential problems regarding the future growth of a company, including its earnings and assets.

The financial statements of a company needed for financial analysis are:

- Consolidated Statement of Comprehensive Income
- Balance Sheets
- Statements of Changes in Equity
- Consolidated Cash Flow Statements

1.6.1 Earnings Quality

The purpose of investigating a firm's quality of earnings is to gain insights into the variability of its earnings and to facilitate its valuation. When assessing the quality of future earnings, the factors to consider would include:

- The current macro-economic environment;

- Social and political factors that could impact the industry or the company;
- The growth potential of the industry;
- The competitive structure of the industry the company is in;
- The cyclical nature of business and revenues;
- The experience, competency and track record of management;
- Diversity of revenue sources;
- Cost structure and dependency on key commodities or sources;
- Product/service innovation; and
- Brand value.

Exhibit 1.2 lists some indicators regarding potential problems associated with earnings quality. Yet, there may be other warning signs of potential negative earnings surprises. These include:

- Weak accounting disclosures such as limited segment reporting, failure to disclose changes in accounting policy, limited discussion on negative factors or bad news;
- Related party transactions which are substantial and numerous;
- High percentage of compensation tied to short term profitability;
- High turnover in management staff and/or directors;
- Frequent changes or disputes with supporting professionals such as auditors, bankers and lawyers;
- Violation of securities law and late filings;
- Significant non-audit services performed by external auditors; and
- Declining market share and / or profit margins.

Exhibit 1.2 Indications of Potential Problems with Earnings Quality

Category	Observation / Examples
Revenues (overstated, or aggressive revenue recognition)	▪ Fictitious revenue/customers; ▪ Repeated recognition of the same goods/services produced/sold; ▪ Booking of prepayments as current revenue; ▪ Booking of the full revenue when only a deposit is provided; ▪ Booking revenues prior to customer acceptance; ▪ Classifying non-operating gains as operating income; and ▪ Aggressive recognition of consignment sales with recourse.
Expenses (understated, or liberal treatment of expense recognition)	▪ Wrongly capitalizing expenses; ▪ Lengthening the period of depreciation; ▪ Low level of provision for bad debts or other potential liabilities; and ▪ Usage of other accounting assumptions or treatments to recognize lower expenses.
Balance Sheet	▪ Use of off-balance sheet financing and other transactions especially to hide liabilities.
Operating Cash Flow	▪ Operating earnings are not synchronized with operating cash flow; ▪ Increase in capital expenditure in previous years not yielding increase in operating cash flow in future years; and ▪ Treating bank overdraft increase as operating cash flow.

1.6.2 Using Financial Ratios

An analysis of a company's financial ratios will provide insights and an understanding of its financial health and may unearth hidden problems. A single financial ratio by itself may not be very useful. Comparing the same ratios over a number of years or with its competitors and an industry average would be very helpful.

Past studies have revealed that there are five main groups of financial ratios which will substantially characterize a company. These are:

- **Solvency Ratios** (also known as Liquidity Ratios) – This category of ratios indicates the ability of the company to meet its current liabilities. Examples: Current and Quick ratios;
- **Activity Ratios** (also known as Turnover Ratios) – This category of ratios provides a view on the activity level of assets in generating sales. Examples: Total Asset Turnover, Inventory Turnover and Receivables Turnover ratios;
- **Leverage Ratios** (also known as Debt ratios) – This category of ratios focuses on the company's use of debt and its ability to service the debt interest and repayments. Examples: Debt to Equity and Interest Coverage ratios;
- **Efficiency Ratios** (also known as Profitability Ratios) – This category of ratios shows the profits as a percentage of sales, equity capital or total capital. Examples: Net Profit Margin, Return on Assets and Return on Equity ratios; and
- **Growth Ratios** – This category of ratios includes the growth of sales and earnings as well as the sustainable growth rate of earnings, which is used to analyze the growth potential of companies.

The details and formulas for the ratios in the 5 categories listed above can be found in Appendix 1.2.

1.6.3 Limitations of Financial Ratios

When using financial ratios, the user has to recognize its limitations. Below are some points to ponder when applying financial ratios:

- Accounting treatment of the subject companies should be comparable;
- The subject companies should be comparable and homogenous;
- The ratios should be within a reasonable range from the industry's figures;
- Window dressing may mask the true financial status of a firm; and
- Seasonal factors may cause ratios to fluctuate within the same year.

1.7 Understanding the Economic Environment

The health of an economy and the direction it is heading towards will have major impacts on whether prices of financial assets, like stocks and bonds, will go up or down. Because the economy and asset prices tend to be closely associated, it is important to understand various aspects of the economy. This is best done by appreciating some of the tools used by economists to monitor and measure economic performance.

Economic analysis is a key part of the investment valuation process. It forms the first stage in a “top-down” approach termed fundamental analysis.

1.7.1 Circular Flow Model of the Economy, GDP and GNP

It is common to measure the economic performance of a country by computing its Gross Domestic Product (**GDP**). This is defined as the value of all final goods and services produced within a country during a given period, usually a calendar year.

A good way to understand an economy is to see its linkages and how each part is connected. Wages, rent, interest, and profits comprise the economy’s aggregate income. Consumers and businesses **spend** the **income** while the government provides transfer payments such as welfare and unemployment benefits. Therefore, spending (expenditure) must equal to income in an economy.

The main point is that what is spent (the value of total expenditure) ends up as income (for other individuals in the economy). This is the **circular flow model** of the economy which depicts the flow of money in the economy. Thus,

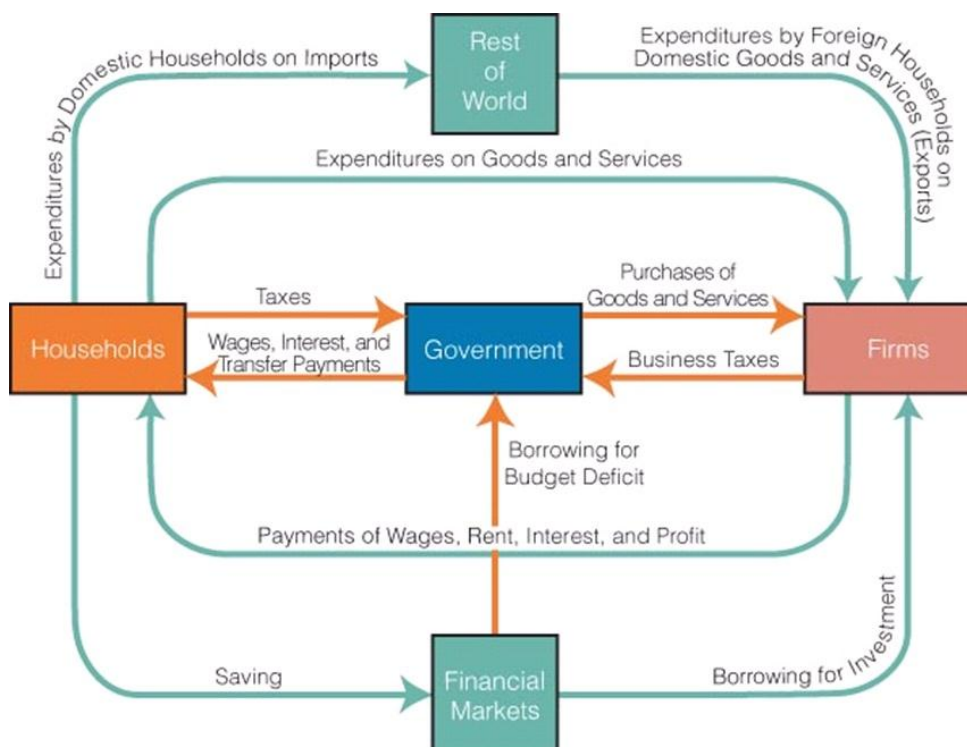
$$\begin{aligned}\text{Aggregate Expenditure, (AE)} &= \text{Aggregate Income (wages, rent, interest, etc)} \\ \text{AE} &= C + I + G + (X - M)\end{aligned}$$

Where,

- C = Consumption
- I = Investment
- G = Government Purchases
- X = Exports
- M = Imports
- X-M = Net exports

For the economy as a whole, expenditures and income are the same. This is illustrated in Exhibit 1.3 below.

Exhibit 1.3 Incomes and Expenditures



Source: Robert L. Sexton, *Essentials of Economics*, Thomson South Western.

GDP measures the Aggregate Income in the domestic economy. If the income earned by domestic citizens (that is, Singaporeans) working abroad is included while the income earned by non-Singaporeans working in Singapore is excluded, a new measure termed the Gross National Product (**GNP**) is available.

Since inflation is a reality, a dollar's purchasing power today is very different from its purchasing power ten years ago. Because of this, it will be meaningless to compare the GDP figure of today and the GDP figure recorded ten years ago unless adjustments are made. This is done by deflating the nominal numbers by a price index. That is,

$$\text{Real GDP} = (\text{Nominal GDP}) \times 100 / (\text{Price Index})$$

If Singapore's nominal GDP in 2010 is S\$8.1 billion, and its price index in 2010 relative to 2005 is 110.9, then

$$\text{Real GDP} = (8.1) \times 100 / (110.9) = \text{S\$7.3 billion (in 2005 constant dollars)}$$

1.7.2 Inflation

Inflation is the continuing rise in overall price levels over time. The opposite situation is deflation, where prices fall. Both inflation and deflation impact the economy. For financial markets, it may lead to a change in asset prices.

Inflation lowers the purchasing power of individuals with fixed-dollar incomes, like wage earners. Likewise, inflation can hurt creditors. If a lender does not correctly anticipate the higher rate of inflation, the borrower will be paying back with dollars that have less purchasing power.

The uncertainty that inflation creates can also discourage investment and economic growth. Moreover, inflation can raise a nation's price level relative to price levels in other countries. And this could affect that nation's export competitiveness. But there are also winners in an inflationary scenario. For instance, as mentioned, debtors would pay back dollars that are worth less in purchasing power than the sum borrowed.

To measure the changing purchasing power of the dollar, price indices are constructed. A price index attempts to measure the prices paid for a specific basket of goods and services over time. For instance, the Consumer Price Index (**CPI**) measures the trend in the prices of a fixed basket of goods and services that a typical consumer would purchase over the period of measurement.

1.7.3 Unemployment

When discussing employment of the labour force, economists compute the **unemployment rate**. The **labour force** includes the number of people over the age of sixteen who are available for employment.

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed}}{\text{Size of the Labour Force}}$$

There are four types of unemployment: frictional unemployment, structural unemployment, cyclical unemployment, and natural rate of unemployment (See Appendix 1.3 for definitions).

When all the resources of an economy, namely, labour, land, and capital, are fully employed, the economy is said to be producing at its **potential output**. It means that the economy is providing employment for all who are willing and able to work. It also means that capital and land are fully employed.

1.7.4 Balance of Payments

The **balance of payments** is a record of all of the international financial transactions of a nation over a year. It provides information about a nation's global trade position. Basically, the balance of payments has three main elements:

- The current account;
- The capital account; and
- A statistical discrepancy ("error term") if any.

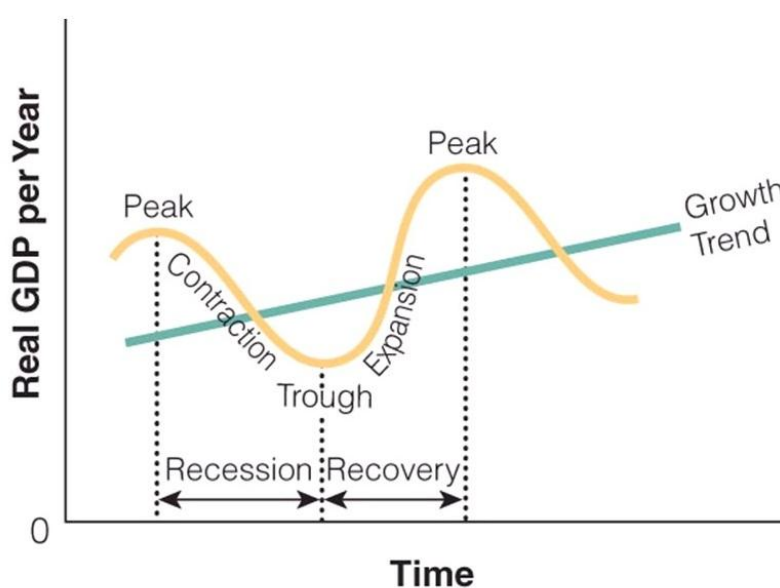
(Reference to Appendix 1.3 for definition of Balance of Payments components)

1.7.5 Business Cycles

Economies tend to experience cycles or termed **business cycles**. This refers to the short-term variations in economic activity. A business cycle can be divided into four phases: expansion, peak, contraction, and trough, as depicted in the Exhibit 1.4 below.

Business cycles may stretch over a few years. There may also be seasonal periods within the same year. For instance, department store sales are seasonal, higher during Christmas and somewhat lower during Easter. Some industries also experience seasonal fluctuations in output.

Exhibit 1.3 Incomes and Expenditures



Source: Robert L. Sexton, *Essentials of Economics*, Thomson South Western.

1.7.6 Economic Indicators

Many economists analyse published data to forecast economic growth. They look for **leading indicators**, **coincident indicators** and **lagging indicators**. A common leading indication of growth is the number of new housing starts in a country. Another common leading indicator is the stock market index.

1.7.7 Fiscal Policies

Fiscal policy is the use of government expenditure, taxes, and transfer payments to alter the output level (GDP) and the price level of an economy. For instance, if the government wishes to stimulate the economy by increasing aggregate demand (AD), it will increase government purchases of goods and services, increase transfer payments, lower taxes, or use some combination of these approaches. Any of these options will reduce budget surplus (which may also lead to a budget deficit) and result in an **expansionary fiscal policy**.

If the government wishes to dampen an economic boom by reducing AD, it will implement a contractionary **fiscal policy**. This will lead to a budget surplus, or reduction of the budget deficit. A **balanced budget** occurs when government expenditure equals income.

1.7.8 Monetary Policies

Monetary policy uses monetary means to stimulate or contract the economy. The usual ways are to adjust interest rates and control the money supply. Money supply, in turn, is managed by purchasing (or selling) government bonds, increasing (or lowering) the reserve requirements of commercial banks, and fixing the discount rate for the bills of commercial banks and financial institutions.

1.7.9 Sources of Economic Growth and Recessions

Economic growth is important to financial markets because growth will generally lead to higher asset prices. Generally, most economists agree that the availability and productivity of the major factors of production (labour, land, capital, and technology) are important. Other factors leading to growth include adequate savings, research and development, property rights, free trade, entrepreneurship and appropriate government policies.

Recession, on the other hand, is the opposite of economic growth. Technically, a recession occurs when there is “a decline in general economic activity over two successive calendar quarters”. A recession is usually accompanied by a higher unemployment rate in the labour force.

1.8 Chapter Summary

This chapter has covered many aspects of investment concepts, focusing on risk and return. Aspects covering the understanding of financial statements and the economic environment were also discussed. The salient points are:

- Returns include income and capital gains while yield is a measure of income and not capital gains;
- Return on an investment over time should take into account any deposits or withdrawals of funds during the measurement period. Dollar Weighted Return and Time Weighted Return are two methods used to calculate the return on investment;
- Variance and standard deviation of returns are common measures of absolute risk;
- The expected return on an investment depends on the real interest rate, inflation rate and the risk premium. Investors who take on higher risks expect to be compensated by higher returns. For example, derivative securities will have higher rates of return than government securities;
- The financial status, quality and sustainability of earnings are important considerations when valuing a company. An analyst or investor would use financial statements, and other reports to evaluate a company;
- The 5 major categories of financial ratios: solvency, activity, leverage, efficiency and growth, provide a good understanding of the financial health of a company. A comparison of financial ratios across firms is only meaningful if the companies belong to the same industry and adopt similar accounting policies;
- The health of the economy underpins the financial performance of industries and companies. Common measures of economic activity are Gross Domestic Product (GDP) and Gross National Product (GNP);
- GDP can be measured by computing Aggregate Expenditure (AE) or Aggregate Income. To conduct a meaningful comparison of GDP at different points in time, a price index is used to deflate nominal GDP to same base year;
- Important attributes of an economy include inflation, interest rate and unemployment;
- Inflation creates uncertainty and discourages investment, leading to potential currency depreciation and affects a nation's export competitiveness;
- The four phases of a business cycle are expansion, peak, contraction and trough;

- Economists employ economic indicators to anticipate direction of economic growth. The indicators could be leading, coincident or lagging the underlying economy;
- Fiscal policy is the use of government purchases, taxes, and transfer payments to manage the GDP and price levels of an economy; and
- Monetary policy is the use of government securities (purchasing or selling by the government), reserve requirements on banks and bank discount rate to influence money supply, interest rates and liquidity.

Appendices

Appendix 1.1: Formula for Risk Measures

Variance, (σ^2)	$= \sum_{i=1}^n (\text{Probability}) [\text{Possible Return} - E(R)]^2$
σ^2	$= \sum_{i=1}^n (P_i) [R_i - E(R_i)]^2$
Standard Deviation	$= \sqrt{\sigma^2} = \sigma$

Coefficient of Variation, (CV)	$= \frac{\text{Standard Deviation of Returns}}{\text{Expected Rate of Return}}$
	$= \frac{\sigma}{E(R)}$

Beta, (B_i)	$= \frac{\text{Cov}_{I,m}}{\text{Cov}_{m,m}}$
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Appendix 1.2: Financial Ratios

Solvency (Liquidity) Ratios

i) Current Ratio

This is the most commonly-used solvency ratio. A ratio of less than 1 means the company does not have sufficient current assets to meet current liabilities within the same operating cycle (usually 1 year). Solvency ratio indicates the company's ability to meet short term financial obligations.

Current Ratio	$= \frac{\text{Current Asset}}{\text{Current Liability}}$
---------------	---

ii) Quick Ratio

Because inventories and some other assets are not liquid and might not be converted to meet current liabilities in time, inventory is subtracted from Total Current Assets to compute the Quick Ratio.

$$\text{Quick Ratio} = \frac{\text{Current Asset} - \text{Inventory}}{\text{Current Liability}}$$

Activity (Turnover) Ratios

i) Total Asset Turnover

This ratio indicates the effective use of all assets. A capital intensive business will likely have a smaller ratio compared to the service-based business. It is also affected by leasing arrangements.

$$\text{Total Asset Turnover} = \frac{\text{Net Sales}}{\text{Average Total Net Assets}}$$

ii) Inventory Turnover

This ratio reflects the utilization of inventory. A ratio that is too low means that the company's capital is tied up in excessive inventory and there may be obsolete inventory. A high ratio is good, but if it is too high, it means that there is inadequate stock to support the current sales volume.

$$\text{Inventory Turnover} = \frac{\text{Cost of Sales}}{\text{Average Inventory}}$$

iii) Receivables Turnover and Annual Collective Period

This ratio measures the turnover of receivables. It reflects the credit policy of the company. The faster the receivables are turning over, the sooner the company is able to free up receivables into cash.

$$\text{Receivables Turnover} = \frac{\text{Net Annual Sales}}{\text{Average Receivables}}$$

Given the Receivables Turnover figure, the receivables collection period can be computed. The higher the turnover, the shorter the receivables collection period measured in days.

$$\text{Average Receivables Collection Period} = \frac{365}{\text{Annual Turnover}}$$

Leverage (Debt) Ratios

Financial risk is an additional uncertainty on returns faced by equity-holders due to the company's use of debt. It reflects the risk of default.

i) Debt / Equity Ratio

This ratio relates long-term debt to total equity. Total equity includes all forms of equity, including preference shares.

$$\text{Debt / Equity Ratio} = \frac{\text{Long-Term Debt}}{\text{Total Equity}}$$

ii) Debt / Total Capital Ratio

This ratio indicates the proportion of total debt to total capital (which consists of both debt and equity).

$$\text{Debt / Capital Ratio} = \frac{\text{Total Debt}}{\text{Total Capital}}$$

iii) Interest Coverage

This ratio indicates the amount of earnings available to meet the required interest payments.

$$\text{Interest Coverage} = \frac{\text{Net Income} + \text{Interest Expense} + \text{Taxes}}{\text{Interest Expense}}$$

iv) Total Fixed Charge Coverage

This ratio computes the coverage ratio when a company has lease obligations.

$$\text{Fixed Charge Coverage} = \frac{\text{EBIT} + \text{Lease Payments}}{\text{Interest Expense} + \text{Lease Payments} + \text{Preferred Dividend} / (1 - \text{Tax Rate})}$$

Efficiency (Profitability) Ratios

i) Gross Profit Margin

This ratio indicates the basic cost structure of a company by relating its gross profit to sales.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Net Sales}}$$

ii) Operating Profit Margin

This ratio shows the operating efficiency of a company by relating its operating profit to sales.

$$\text{Operating Profit Margin} = \frac{\text{Operating Profit}}{\text{Net Sales}}$$

iii) Net Profit Margin

This ratio computes the net profit after tax divided by sales.

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

iv) Return on Capital

This ratio relates the net income received (before interest payments) to the total capital employed.

$$\text{Return of Capital} = \frac{\text{Net Income} + \text{Interest Expense}}{\text{Total Capital}}$$

v) Return on Equity (ROE)

This ratio measures the returns available to equity-holders. It is the returns earned by providers of equity capital (both common and preferred shareholders).

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Total Equity}}$$

Growth Ratios

The sustainable growth rate of the company, g^* , measures the rate of growth that is achievable by a company if it does not want to raise new equity capital nor change its debt-equity profile.

$$g^* = \text{RR} \times \text{ROE}$$

Where,

RR = Retention rate of earnings
 = $1 - \text{Dividends} / \text{Net Income}$
 ROE = Return on Equity

Appendix 1.3

How GDP Is Calculated

There are a few ways to calculate the GDP of a country. One way of doing this is to calculate the GDP of an economy using the **expenditure approach**. This is calculated by adding how much market participants spend on final goods and services over a specific period of time. For convenience and for analytical purposes, economists usually group spending into four categories. They include:

Consumption, C ;
 Investment, I ;
 Government Purchases, G ;
 Net exports, which equal exports (X) minus imports (M), or $X - M$.

Essentially, $GDP = C + I + G + (X - M)$

Example

Given that Singapore's 2010 consumption expenditure is S\$2.4 billion, its investment expenditure is S\$2.3 billion, its government expenditure is S\$3.0 billion, its exports is S\$4.2 billion, and its imports is S\$3.8 billion, then its GDP is:

$$\begin{aligned} GDP &= \text{S\$2.4 billion} + \text{S\$2.3 billion} + \text{S\$3.0 billion} + (\text{S\$4.2 billion} - \text{S\$3.8 billion}) \\ &= \text{S\$8.1 billion} \end{aligned}$$

- **Consumption** – Refers to the purchase of consumer goods and services by households. This does not include purchases by businesses or government.
- **Investment** – Refers to the expenditure on capital goods such as power plants, equipment or machinery tools to produce goods and services. It also refers to investment in inventories. This definition of investment differs from the popular use of the same term referring to financial securities, such as bonds and equities.
- **Government Expenditure** – Refers to government spending, including salaries of civil servants and payments to private firms and individuals to provide various goods and services.
- **Net Exports** – Refers to goods and services exported less those imported.

Types of Unemployment

- **Frictional Unemployment** – Is the unemployment caused by changing market conditions and represented by individuals with transferable skills who change jobs, including search time. There is downtime when people are looking for suitable jobs and firms are looking for suitable workers.
- **Structural Unemployment** – Is caused by structural changes in the economy that eliminate some jobs and create other jobs. It may occur when workers lack the new skills for jobs that are available or have particular skills that are no longer in demand.

Frictional and structural unemployment are simply unavoidable in any economy. To a large extent, both result from imperfections in the labour market. Adding the frictional unemployment rate and structural unemployment rate gives the **natural rate of unemployment**. **Full employment** exists when the economy is operating at its natural unemployment rate.

Some joblessness may result from cyclical fluctuations in the economy called **cyclical unemployment**. For example, unemployment rate is higher during an economic downturn and lower when the economy peaks.

Balance of Payment Components

- **Current Account** – Includes all payments related to the purchase and sale of goods and services. Components of the account include exports, imports, and net unilateral transfers abroad.
- **Capital account** – Includes all payments related to the purchase and sale of assets and to borrowing and lending activities. Components include the outflow of capital away from a country and the inflow of foreign capital. A surplus in the capital account means there are more funds flowing into the country.
- **Statistical Discrepancy** – Credits and debits in the balance of payments must be equal. However, errors and omissions may occur. Hence, a statistical discrepancy is included as a balancing item.

Appendix 1.4

Basic Mathematics of Finance

Many decisions in Finance rely on the concept, “time-value of money”. This is because everyone prefers to receive \$100 today instead of the same sum of money next year. Money received today is worth more than money received later. There is an opportunity cost when the receipt of money is deferred. In other words, people have a time preference for money.

Computing Future Values

If the interest rate, r , is 10% per annum, \$100 is worth \$110 if invested and received the following year. Similarly, the \$100 grows to \$121 in Year 2 and \$133.10 in Year 3.

\$100 invested annually @ 10%				
$r = 10\%$				
Time Line	$t = 0$	$t = 1$	$t = 2$	$t = 3$
Future Value	\$100	\$110	\$121	\$133.10
	$100(1+r)^0$	$100(1+r)^1$	$100(1+r)^2$	$100(1+r)^3$
Future Value, (FV) = $100 (1+r)^t$				

Example

Suppose, in Year 0, you invest \$1,000 for 10 years at 10% per year. What is the future value in Year 10?

$$\text{Future Value in Year 10} = 1,000 (1 + 10\%)^{10} = \$2,593.74$$

Computing Present Values

If the opportunity cost of money is 10%, \$1000 received in Year 3 is worth only \$751.31 today. The further in future, the less is the worth of that sum in today's dollars. A rate of interest, r , is used to "discount" the worth of a sum of money (to be received in future) in today's dollars.

\$1,000 received in Year 3 discounted @ 10% per annum				
$r = 10\%$				
Time Line	$t = 0$	$t = 1$	$t = 2$	$t = 3$
Present Value	\$751.31			\$1,000
	$1,000/(1+r)^3$			1,000
Present Value, (PV) = $\frac{1,000}{(1+r)^t}$				

Example

Suppose you will receive \$1million in Year 20 and the “discount rate” is 10%, what is the present value today?

$$\begin{aligned}\text{Present Value in Year 0} &= \frac{1,000,000}{(1 + 10\%)^{10}} \\ &= \$148,643.63\end{aligned}$$

Chapter 2:

Portfolio Management

Objective

The candidate should be able to:

- ✓ Understand how portfolios are constructed and managed
- ✓ Understand how to calculate risk and return of a portfolio
- ✓ Understand the concepts of diversification
- ✓ Understand the Capital Asset Pricing Model
- ✓ Understand how to monitor and manage the risks of a portfolio
- ✓ Understand how to review a portfolio with the client

2.1 Introduction

This chapter discusses how the portfolio of a client is constructed, monitored and managed. It also discusses about the roles and obligations of a Covered Person in the portfolio management process, including risk management. Central to portfolio management is the literature on the Modern Portfolio Theory (MPT) which focuses on diversification of risks and the Capital Asset Pricing Model (**CAPM**).

2.2 Process of Portfolio Management

Portfolio management comprises a set of activities as illustrated in Exhibit 2.1. The main iterative steps are:

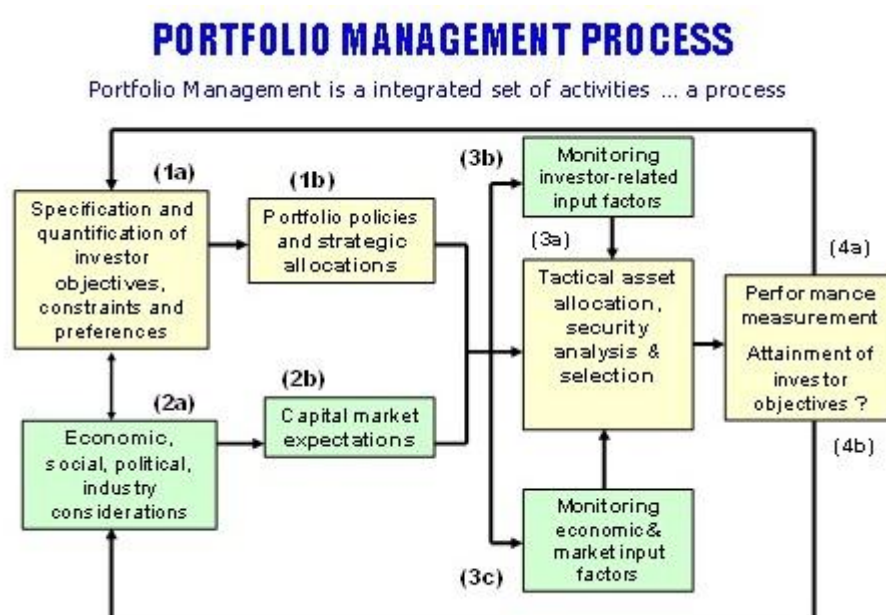
- Understanding the client's objectives, constraints and preferences;
- Understanding the macro-economic environment and capital market conditions;
- Constructing a policy portfolio through strategic allocation of assets;
- Selecting stocks through security analysis;
- Monitoring the environment and making tactical allocations; and
- Portfolio performance review

A Covered Person should first understand the objectives, constraints and preferences of his client. The client's needs have to be considered against the

economic, social, political environment and industry situation. Based on all available information, through **strategic allocation**, an appropriate **policy portfolio** will be constructed for the client. In constructing the portfolio, **security analysis** is carried out.

However, market conditions change and the Covered Person may need to make adjustments to the asset weights in the portfolio through **tactical allocation**. In subsequent periods, a **portfolio review** is carried out and this may result in looping back to construct a refreshed portfolio reflecting new market conditions.

Exhibit 2.1 Portfolio Management Process



Source: Adapted from "Managing Investment Portfolios", John Maginn, Donald Tuttle, Dennis Pinto and Jerald McLeavey. John Wiley, 2007, p6.

- **Strategic Asset Allocation** – Is a long-term strategy to meet the investment objectives of a client. This means that for the next 3 to 5 years, unless there is a significant change in the market conditions or the client's situation, the portfolio should be managed broadly in line with the strategic asset allocation.
- **Tactical Asset Allocation** – Refers to active management of the portfolio, with deviations from the portfolio's strategic asset allocation in order to take advantage of opportunities in the financial market. For example, after a natural disaster, the stock market in a country may trend downwards.

If the Portfolio Manager feels that selling in the market has been overdone and therefore equities are undervalued, he may temporarily increase the portfolio's exposure to equities beyond what was stated in the strategic asset allocation. The

Portfolio Manager will bring the asset allocation of the portfolio back to its long-term allocation once the situation normalizes (when stocks rebound).

Studies have shown that as much as 90% or more of a portfolio's return comes from asset allocation and less than 10% from activities like security selection. Furthermore, it was found that asset allocation has a much greater impact on reducing total exposure to risk than picking an individual security in any single asset category. Therefore, asset allocation is an important aspect of portfolio management.

The portfolio management process is iterative. Having constructed an initial portfolio, a Covered Person will continuously monitor the portfolio on an on-going basis, taking into account changes in market conditions as well as the client's personal objectives and tolerance for risks.

2.3 Portfolio Construction

2.3.1 Considerations for Portfolio Composition

A Covered Person has to assist his client to construct a portfolio of assets based on a number of important considerations discussed below.

Goals of a Client

A client may be investing with specific goals in mind. He may want to draw out a certain sum of money when the time comes for his children to go to college. He may also aim to retire at a certain age at which cash inflow from his job will stop and he needs to start drawing on his investments to meet his expenses. All these goals must be taken into consideration by the Covered Person when constructing the portfolio.

Risk Ability and Risk Tolerance

The risk profile of a client has significant influence on how a portfolio will be structured and managed. The client's return objectives and tolerance for risk have to be evaluated carefully. This is because higher returns come about when an investor is willing to take on higher risks.

There is a distinction between **risk ability** and **risk tolerance**. Risk ability is a client's ability to take risks based on various factors. A young person who has a long working life ahead of him is able to take more investment risks because he has many income-generating years ahead of him. A person who is close to retirement and needs to draw on his investment portfolio for livelihood is less able to take risks.

Yet the young person who has high risk ability may have low risk tolerance due to his personality. On the other hand, a person close to retirement, who has a low risk profile, may like to take higher risks. A conservative approach will note the lower of

the risk ability and risk profile of a client in constructing a portfolio. In the case of the two individuals above, this means that a lower-risk strategy will be adopted for both the younger person as well as the retiree.

Level of Wealth and Amount to be Invested

Another factor that affects a client's ability to take risks is the level of wealth and proportion of total wealth that will be invested in a particular portfolio. An extremely wealthy High Net Worth Individual (HNWI) may be able and may want to invest in high risks assets in his portfolio. This is also the case if he has other assets already held in low risk securities.

Liquidity Requirements

A client may have liquidity requirements that must be taken into consideration. The Covered Person will need to ensure that there is sufficient liquidity when the client has spending needs. He may need to make payments for a new house in the near future. To meet liquidity needs, funds should be invested in high quality, short maturity debt instruments such as Treasury Bills and Money Market Funds which can be easily liquidated. Similarly, only high quality stocks with market liquidity should be purchased.

Investment Universe

A client may sometimes prefer to invest in traditional asset classes, such as equities and bonds and may not want to invest in alternative investments such as private equity or commodities. Yet, other clients may wish to invest in real estate, art and gold. It is important to ascertain the preferred investment universe of the client as early as possible.

Client's Investment Experience

It is important to consider the past investment experience of a client. A Covered Person must take additional steps in explaining the nature of complex investments, e.g. derivatives, to his client to ensure that the client understands fully the risks and returns.

Tax Considerations

The Covered Person has to consider the tax position of his client. A client paying high taxes should be recommended tax-exempt securities if other taxable securities are not able to generate higher after-tax returns.

Client's Unique Preferences

The Covered Person should understand if a client has any preferences or restrictions with regards to investments in any types of securities or asset classes. Some clients may not wish to invest in casinos (and other gaming related equities). Others may be risk averse and do not wish to invest in emerging markets.

Time Horizon

Time horizon is defined as the length of time over which available funds will be invested. This is often linked to the goals of the client or related to the stage of his life cycle. Generally, a younger client would have a longer horizon for investment compared to a retiree.

The investment horizon of the client affects the type of investments that will be included in the portfolio. An investor who has a longer horizon could consider more volatile investments. Likewise, an investor who has a shorter time horizon would prefer less volatile portfolios consisting of more bonds than equities.

Life Cycle Approach of Investing

The Financial Planning Life Cycle approach is a common method used as a first-pass in assessing a client's risk and return preferences.

At the *early-career stage*, the client generally focuses on wealth accumulation for immediate needs such as buying a home. He may also be interested in longer-term investments to save up for retirement or the education of his children. Because the client is young, he would be able to take on more risky investments with higher returns. A portfolio for this person tends to have higher exposure to equities rather than fixed income securities.

When the same client enters the *mid-career stage*, he would have a higher level of income and relatively less liquidity needs. While he may still be keen to accumulate wealth, wealth preservation may begin to be more important.

At the *late-career stage*, as the client prepares for retirement, the emphasis shifts towards wealth preservation and gifting. Investments in his portfolio may switch to lower risk assets. Assets with larger dividends or interest payments would be preferred. The portfolio would typically have a higher weightage of fixed income securities than equities.

While the life cycle approach is useful in positioning the risk and return profile of a client, it is important to note that each client's risk tolerance is unique. Other factors to consider are a client's net worth, future financial commitments, experience with investments and emotional temperament.

2.3.2 The Investment Policy Statement

Having understood the client's profile, a Covered Person should come up with an Investment Strategy. This can be documented in an "Investment Policy Statement". This is basically a plan to guide the Covered Person as he executes the investment plan. In the plan, there would be benchmarks for evaluating portfolio performance. Benchmarks are typically used in discretionary accounts but not always used with advisory accounts.

For example, if it has been decided that 60% of a client's money should be placed in equities and the client wants to have his equity exposure in the Asia Pacific region outside Japan, a benchmark such as the MSCI AC Asia Pacific ex Japan Index could be used. If he wants to have a global bond portfolio consisting only of government issued bonds, a suitable index could be the Citigroup World Government Bond Index.



Tips / Key Points

Know your client well so that you are able to customize a portfolio that will meet the client's needs and preferences.

2.3.3 Constructing the Portfolio

Once the asset allocation has been determined for a client, the Covered Person constructs a portfolio for the client by breaking down the allocations into specific asset classes, e.g. equities, fixed income and cash instruments. The Covered Person's main objective is to maximise the return for the client in accordance with the investment objectives laid out by the client.

If a client does not want too much deviation from the chosen benchmark, the Covered Person may construct his portfolio with reference indices in mind. For example, if the benchmark of the fund is the MSCI AC Asia Pacific Index, the Covered Person may start with an analysis of the stocks that are in this benchmark and their weights in the benchmark. If the client does not mind his portfolio returns having large deviations from the benchmark, the Covered Person has more freedom to select securities.

2.4 Portfolio Theory

2.4.1 Portfolio Returns

The return of a portfolio is a weighted average of the returns of the individual securities in the portfolio.

$$R_p = \sum_{i=1}^n w_i r_i$$

Where,

R_p = Actual or expected return on the portfolio

w_i = Portfolio weight for the i^{th} security

r_i = Actual or expected return on the i^{th} security

n = Number of different securities in the portfolio

$\sum w_i = 1.0$

2.4.2 Portfolio Risks

The risk of a portfolio of securities is the volatility of its returns. The more volatile the returns of a portfolio are, the greater the risks are. The degree of volatility is measured by its standard deviation.

Portfolio Diversification and Diversifiable Risks

An investor is advised to own a portfolio consisting of a large number of securities to minimize the total portfolio risks. Diversification is the financial equivalent of the old saying, "Don't put all your eggs in one basket."

Every security has two forms of risks:

- **Systematic Risks** – This is the risk that comes from changes in the general economic environment; and
- **Unsystematic Risks** – This is the idiosyncratic risk which is specific to that security.

Studies have shown that as the number of securities in a portfolio increases, the risk of the portfolio decreases. This is because unsystematic risk can be diversified away if the returns of these securities are not correlated. The risk that remains even after diversifying into many stocks is called *market risk* or *systematic risk*, risk that is attributed to market-wide factors.

Systematic risk results from factors outside the firm's control. It is caused by general factors such as changes in the economic, political and sociological environments, which affect the prices of all marketable securities.

Interest rate risk and inflation risk are examples of other systematic risks that cannot be eliminated by diversification. Since systematic risk cannot be diversified away, there is little the investor can do to protect himself against such risks (other than to avoid investing entirely).

Correlation between Securities

The extent to which portfolio risks can be reduced through diversification depends on the degree to which the securities in the portfolio are correlated.

Correlation coefficient measures the extent to which the returns of any two securities are related. If the correlation coefficient is +1.0, it means that the two securities are perfectly positively correlated and the returns move in the same direction by the same magnitude. In other words, the two securities have identical return patterns.

If the correlation coefficient is -1.0, it means that the returns of the two securities are perfectly negatively correlated and the returns move in the opposite direction by the same magnitude. If the correlation coefficient is zero, the returns of the two securities are unrelated.

In the real world, the correlation of securities will lie between zero and 1.0. If the correlation is high, i.e. the correlation coefficient is close to +1.0, there is less diversification benefit than if the correlation is low, i.e. the correlation coefficient is close to zero. The following relationships exist between security price correlation and portfolio risk reduction:

- Combining two securities that are perfectly positively correlated to each other will not reduce portfolio risk. The risk of the resulting portfolio is simply a weighted average of the risks of the individual securities.
- Combining two securities that have zero correlation to each other has a great impact on reducing the risk of the portfolio. Yet portfolio risk cannot be totally eliminated.
- Combining two securities with perfect negative correlation with each other can eliminate risk altogether. This is the principle employed in hedging strategies. But if one forgoes the risks, the returns are also reduced.

The less correlated securities are to each other, the greater the potential for risk reduction. This explains why investors choose to combine securities of different countries in their portfolio. This is because movement in prices in different countries are less correlated than prices in the same country are to one another.

The stocks within Singapore are highly correlated to each other (correlation coefficient of about 0.95) even though the stocks are from different sectors. An investor from Singapore who wants to reduce his portfolio risk can do so by diversifying into stocks outside Singapore, e.g. U.S. stock market (which has correlation coefficient of about 0.85) or the Japanese stock market (which has correlation coefficient of about 0.6).



Tips / Key Points

Have a portfolio of stocks that are not well correlated to each other in order to effectively reduce the risk of the portfolio.

Total Risk of Portfolio

Although the expected return of a portfolio is a weighted average of the expected returns of the securities in that portfolio, total portfolio risk (as measured by the standard deviation or variance) is not a simple weighted average of the risk of the individual stocks in the portfolio.

For a two-asset case, the formula for calculating portfolio risk is as follows:

$$\text{Portfolio risk } (\sigma_p) = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

Where,

w_1 and w_2 = Portfolio weights given to security 1 and security 2 respectively

σ_1 and σ_2 = Standard deviation of security 1 and security 2 respectively

$\rho_{1,2}$ = Correlation coefficient of security 1 and 2

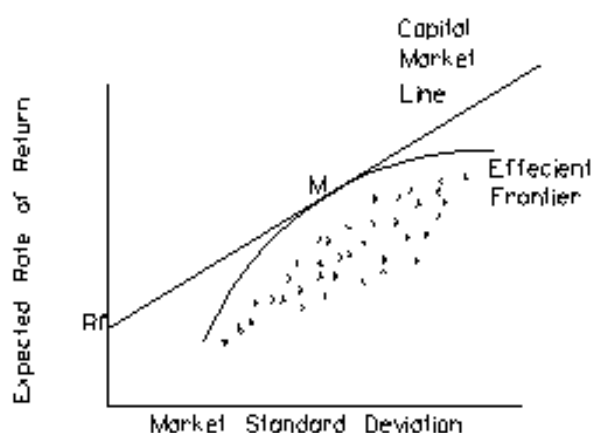
From this formula, it can be seen that the more uncorrelated the two stocks are to each other, i.e. the lower $\rho_{1,2}$ is, the lower will be the risk of the portfolio. Please refer to the Case Study (after the Chapter Summary) on how an investor can reduce the risks of his investments by diversification by adding another stock to his portfolio that is not perfectly positively correlated to his first stock.

2.4.3 The Capital Market Line (CML)

It is possible to compute the return and risk of every security or asset. If these returns and risks are plotted in a graph, they would be “dots” as illustrated in Exhibit 2.2. If an investor prefers assets with higher returns for the same level of risk, and lower risk for assets with the same return, we derive a set of portfolios on a curve, called the *efficient frontier*.

Portfolios along this line are “efficient” because you cannot find another portfolio that gives higher return for the same risk or lower risk for the same return.

Exhibit 2.2 Capital Market Line (CML)



Portfolios that fall below the efficient frontier are inefficient portfolios because it is possible to buy a more efficient portfolio by choosing one that lies on the efficient frontier. Portfolios which lie above the efficient frontier are not achievable as the efficient frontier already gives you the best possible combination of risk and return.

When there is a “capital market” (where there is the opportunity to borrow and lend at the risk free rate, denoted as R_f), a Capital Market Line (CML) can be drawn as shown in Exhibit 2.2. There is also a portfolio, M, which lies on the CML. Portfolio M represents the optimal set of assets on the CML.

The CML is a line which shows an increase in returns with an increase in the risk of an asset (measured by the standard deviation of returns). The equation of the CML is:

$$E(R_p) = R_f + \frac{E(R_M) - R_f}{\sigma_M} \sigma_p$$

Where,

$E(R_p)$ = Expected return on any efficient portfolio on the CML

R_f = Risk-free rate of return

$E(R_M)$ = Expected return on market portfolio M

σ_M = Standard deviation of the returns of market portfolio

σ_p = Standard deviation of the efficient portfolio being considered

Please see Appendix 2.2 for an example on how the Capital Market Line (CML) formula is applied to calculate the expected return of a portfolio.

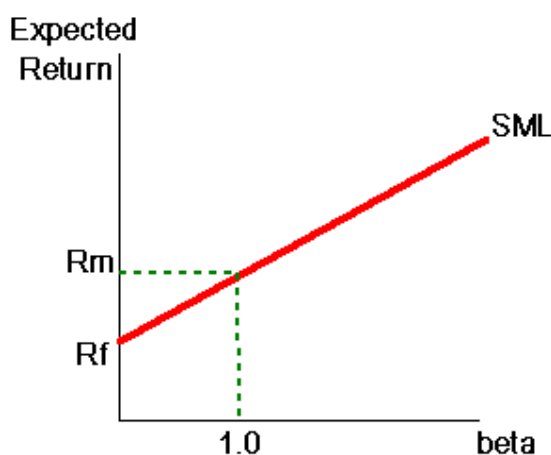
2.4.4 The Capital Asset Pricing Model (CAPM)

Further theoretical developments by William Sharpe, John Lintner and Jan Mossin led to a conceptual framework called the Capital Asset Pricing Model (CAPM). The essence of the CAPM is illustrated in Exhibit 2.3 which shows the Security Market Line (SML). The SML posits that the expected return on an asset is determined by its systematic risk (as measured by beta).

The beta of an asset is its risk factor relative to the beta of portfolio M, which has a value of 1.0. If a security has a beta greater than one, it has a higher volatility factor relative to Portfolio M (also termed the Market Portfolio). For example, a security with beta of 1.5 means that, on average, the returns of this security is 1.5 times more volatile than the market. The reverse is true meaning that if the beta of a security is less than one, it is less risky than the market portfolio. The risk-free asset has a beta of zero.

The Security Market Line (SML) depicts the relationship between the required rate of return of individual assets and their betas. The higher the beta of a security, the greater the risk it brings to the portfolio and the greater the returns investors will require from this security.

Exhibit 2.3 Security Market Line (SML)



The equation for the Security Market Line (SML) is:

$$E(R_i) = R_f + B_i[E(R_m) - R_f]$$

Where,

$E(R_p)$ = Expected return on security or portfolio

R_f = Risk-free rate of return

$E(R_m)$ = Expected return of overall market

B_i = Beta of security or portfolio

Please refer to Appendix 2.3 for an example on how to calculate the expected return of a security based on the Security Market Line (SML) formula.



Tips / Key Points

The Capital Asset Pricing Model (CAPM) teaches that you can reduce the risks of your portfolio by having a portion in risk-free assets. You can increase the risk/return profile of your portfolio by utilizing credit.

2.5 Portfolio Monitoring

The portfolio management process requires regular monitoring. Financial markets and the client's situation will change over time. These changes may require a Covered Person to make adjustments to the client's portfolio.

The frequency of monitoring would depend on assets in the client's portfolio and market conditions. If the portfolio consists mainly of low volatility securities such as government bonds, the frequency of monitoring would be less than a portfolio of high volatility stocks. When the financial markets are stable, less monitoring is needed than during turbulent conditions.

After each portfolio review, it may be necessary to rebalance the weights of the assets in the client's portfolio. Rebalancing a portfolio means that reducing some securities and increasing others to achieve a new portfolio mix which will fulfil the client's mandate and the agreed portfolio strategy.



Tips / Key Points

Monitor the client's portfolio frequently and rebalance if necessary to stay in line with the investment strategy.

2.6 Portfolio Risk Management

The Covered Person aims to maximize the return of the portfolio while keeping within risk levels that are acceptable to the client. Having determined the performance benchmarks for the client's portfolio, a Covered Person will set limits as to how far he will deviate from this benchmark.

For example, if the asset allocation of the portfolio is 60% bonds and 40% equities, the client may state that he does not want to deviate more than plus or minus 5%

from this allocation. If this is the case, the minimum equity exposure of the portfolio would be 35% and maximum 45%.

Another method to measure how well the portfolio is following a determined benchmark is to measure its tracking error. The tracking error is the standard deviation of the *differences* between the portfolio's return and the benchmark index returns. A portfolio that stays close to a particular benchmark will have a tracking error which is close to zero.

A client may have an understanding with a Covered Person on the maximum level of risk exposure and set targets in *absolute* terms. For example, the portfolio should not have an absolute loss of more than 20% in a year. Targets may also be stated *relative* to a benchmark. For example, the client may not want his portfolio returns to underperform the set benchmark by more than 3% per year.

How Adding More Stocks to a Portfolio can Reduce Portfolio Risk

Peter has just started investment and only owns one stock – Stock A, which has an expected return of 18% and a standard deviation of 20%. As a Covered Person, you advised Peter that by adding another stock to his portfolio that is not positively correlated to Stock A, he will be able to reduce his portfolio risk.

You further explained that the less correlated the new stock is to Stock A, the more potential the reduction in risks. Peter is excited about the possibility of being able to reduce the risk of his portfolio. He tells you that he is considering adding one of three new stocks to his portfolio with 60% weighting – Stock B, Stock C or Stock D.

All three stocks have an expected return of 9% and a standard deviation of 10%. However, Stock B has a correlation of 0.8 to Stock A; Stock C has a correlation of 1.0 to Stock A; while Stock D is uncorrelated to Stock A, i.e. zero correlation. He asked you to show him through calculation, how his portfolio risk can be reduced if he chose to add the stock with the lowest correlation to Stock A.

Consider the following portfolio which consists of two securities, Stock A and Stock B.

Stock	Expected Return	Standard Deviation	Correlation to Stock A
A	18%	20%	
B	9%	10%	0.8
C	9%	10%	1.0
D	9%	10%	0.0

The correlation coefficient between Stock A and Stock B is 0.8. The expected return on the portfolio consisting of Stock A and Stock B:

$$\begin{aligned}
 &= (40\%)(18\%) + (60\%)(9\%) \\
 &= 7.2\% + 5.4\%
 \end{aligned}$$

$$= 12.6\%$$

Portfolio risk (A & B):

$$\begin{aligned}
 &= \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2} \\
 &= \sqrt{0.4^2 \times 0.2^2 + 0.6^2 \times 0.1^2 + 2(0.4)(0.6)(0.8)(0.2)(0.1)} \\
 &= \sqrt{0.0064 + 0.0036 + 0.00768} \\
 &= \sqrt{0.01768} \\
 &= 0.1330 \text{ or } 13.30\%
 \end{aligned}$$

In the above example, $\rho = 0.8$, resulting in a portfolio risk of 13.30% as measured by the standard deviation, σ . If the same calculation is done for Stock C and Stock D, it would show the following portfolio risks:

For a portfolio of A with Stock C, $\rho = 1$, therefore $\sigma_p = 14\%$.

For a portfolio of A with Stock D, $\rho = 0$, therefore $\sigma_p = 10\%$.

With these calculations, a Covered Person is able to show that the less correlated stocks are to each other, the greater the potential to reduce portfolio risk. In this case, the risk of the portfolio decreases steadily from 14% to 10% as the correlation coefficient ρ declines from +1.0 to 0.

2.6.1 Concentration Risks

Quite often, a client's portfolio may be too focused on a few assets and this result in a portfolio with "concentration risks". If a portfolio contains too few stocks, the portfolio return may be very volatile as any adverse performance of the same few stocks would have a significant impact on the client's portfolio performance.

2.6.2 Currency Risks

An investor who invests globally needs to monitor currency movements. The Covered Person should assist his client to be aware of currency exposure. The client may set a limit as to how much foreign currency exposure is desirable.

2.6.3 Counterparty Risks

When there are trades which involve another party, there will be counterparty risk. For example, using derivatives will result in counterparty risks as the contracts depend on the performance of the other party. If these parties are unable to fulfil their obligations, the trade will fail.

There should be risk control mechanisms in place to manage counterparty risks. The Covered Person may limit this risk by ensuring that no more than a certain percentage, say 10% of the portfolio, would be exposed to counterparty risk from a single financial organization.

2.6.4 Disclosure of Risks

It is important for a Covered Person to be transparent and reveal all pertinent information to his client with regard to the risk of the portfolio. Special attention needs to be given to the use of financial derivatives that may increase the risks of the portfolio.



Tips / Key Points

Understand the client's risk profile and manage the portfolio within acceptable risk levels.

2.7 Portfolio Review

2.7.1 A Holistic Performance Review

A Covered Person has the responsibility to explain to his client the portfolio performance. When meeting a client for the purpose of a portfolio review, the factors that should be covered include how his portfolio as a whole has performed, the main investments that have contributed to this performance and the economic and market conditions during the period of review.

As the portfolio may have multi-currency exposure, a Covered Person should also explain how currency movements have affected the return of his portfolio when translated to the home currency. Presentations should avoid technical jargons and pitch at the appropriate level for a client to be engaged.

A "Portfolio Review" is commonly conducted once a year – the frequency of which will depend on the mandate which the Covered Entity has signed with the client. The most current market valuation of client's investments should be used for the review. It may be useful to show the changes between reviews and decisions made during that intervening period.

Discussions with clients and important details at each review should be documented for future reference. For the more important agreements such as significant changes to investment strategy or the implementation of a very different strategic asset allocation at the Covered Entity's discretion, it would be the role and responsibility of the Covered Person to obtain client's documentary endorsement to pre-empt potential disputes.

A Covered Person's approach to "Portfolio Review" with clients should first consider how an investment account or investment portfolio stands in context of the overall client relationship. If there are multiple investment portfolios under the same client relationship, consider the diverse purposes and inter-relationship between client's different investment portfolios. The following are some scenarios to think about:

- Covered Entities and their Covered Persons should act with professionalism and ensure that their activities are guided by appropriate ethical values, prudence and integrity. A client relationship may have multiple investment portfolios with the same Covered Entity, involving different investment strategies and investment preferences. Such a scenario could potentially translate into the client's need for a consolidated review of all assets and liabilities under the relationship. For example, there could be a personal savings and investment account held in joint name with a family member (e.g. spouse or child) for cash flow reasons. There could be a Charitable Trust, Foundation or an Education Trust set up for the benefit of family members as part of client's estate planning purposes;
- The multiple investment portfolios of a client relationship could be linked to the extension of collateralized credit facilities based on assets cross-pledged between client's different investment portfolios. Collateral shortfalls may trigger the liquidation of certain investment assets into cash or less risky assets with higher lending value, which can produce a positive or negative impact on the original investment strategy and performance of individual investment portfolios of the client; and
- Perhaps a client relationship has only one investment portfolio but the types of investment assets in the portfolio are linked to religious, tax-related or some other special preferences of the client. This needs to be accounted for in the review beyond the typical analysis of currency risk and appreciation, interpreting portfolio return and risk in relation to economic or market conditions, explanation of tactical allocations, securities selection, proposed investment time horizon of assets, hedges, leveraged positions, etc.

A holistic performance review may result in remedial actions to be taken in a timely manner by the Covered Person in consultation with the client.



Tips / Key Points

Plan your client's "Portfolio Review" based on the agreed frequency and prepare for it by referring to your notes on the intended purposes of and meaningful links between multiple investment portfolios belonging to the same client relationship, notes on the client's investment preferences, decisions, sophistication and communication style. It is also important to establish any remedies and deliverables agreed upon during the review.

2.7.2 Disclosure of Relevant Fees and Costs

It is important that a Covered Person discloses all the relevant fees and costs to the client.

If the Covered Entity manages the assets of the client on a discretionary basis, it will charge a fee for all the assets under management. There may also be performance fees associated with the portfolio that should be disclosed to the client. If the client's wealth is managed on an advisory basis, the Covered Entity will usually charge an agent fee that is linked to transactions, i.e. when securities are purchased or sold.

A Covered Person should also explain to the client that there are third party charges involved in certain transactions. For example, when he purchases a unit trust, there may be upfront loading fees that he has to pay as well as incur management fees on those funds. These fees are covered in detail in Chapter 6: Unit Trusts and Fund Products.

There are also service fees that the Covered Entity charges for such things as:

- Opening of accounts
- Monthly maintenance of accounts
- Money transfer
- Currency conversion
- Overdraft facilities
- Custodian services

Some of the fees that may be incurred by the client are explained below.

Upfront Loading Fee

This is the upfront fee that the investor pays to participate in the fund. If he is investing \$100,000 in the fund and the upfront loading fee is 5%, only \$95,000 goes into the fund and is invested. The remaining \$5,000 is used to pay the administrative expenses of setting up the fund.

Performance Fee

The client may also be required to pay a performance fee to the manager of the portfolio. The quantum of the performance fee and conditions under which the performance fee is paid will be spelled out clearly in the mandate. It is usually paid out when the portfolio outperforms the target it sets for itself. For example, the performance fee could be 15% of the portfolio's excess return over the benchmark.

Management Fee

This is to pay for professionals to invest the money in the portfolio and is charged directly to the portfolio's NAV. It may also include some administrative charges to do the work of providing information about the portfolio's holdings and performance.



Tips / Key Points

A Covered Person needs to disclose to the client all the relevant fees and costs.

2.7.3 Performance Measures

Measuring Portfolio Performance

A discretionary portfolio is usually managed relative to a benchmark so that the investor is able to compare the portfolio's return versus the benchmark and see how the portfolio has performed. If the portfolio's return is higher than the benchmark return, on a purely return basis, the portfolio has done well and vice versa.

A benchmark for the portfolio is determined when the strategic asset allocation for a client has been decided. For example, the client may want 10% of his portfolio in cash, 30% in world bonds and 60% in equities. For the cash portion, he would like to put in S\$ deposits, for the bond portion in global government bonds and for the equities portion stocks of the Asia Pacific markets other than Japan.

With this information, a benchmark can be generated for the client consisting of 10% weighting in 3-month S\$ interbank bid rate, 30% weighting in returns from Citigroup World Government Bond Index and 60% weighting in returns from the MSCI AC Asia Pacific ex Japan Index.

Portfolio Performance and Risks

Besides referencing a portfolio's returns against a benchmark, there is also a need to account for the portfolio risks. A portfolio may have done well because of taking excessive risks. The following are some of the risk measures for a portfolio:

- **Standard Deviation or Variance** – Standard deviation and variance are statistical measures of a portfolio's *absolute* risk. They measure the volatility of a portfolio's returns from its average return. The higher the variance, the higher the portfolio's volatility. Please refer to the Appendix 2.4 on how to calculate the standard deviation or variance of a portfolio. When doing portfolio review, a Covered Person should consider the volatility of the portfolio returns and see how this compares to the volatility of the benchmark.

- **Beta (β)** – Another way to look at risk is beta, which measures the volatility of a portfolio's return *relative* to the benchmark index or a "market index". The higher the beta, the more volatile a portfolio's return is compared to the benchmark. E.g. for a portfolio with a beta of 1.5, if the benchmark goes up by 10% during a period, it will on average go up by 15% and vice versa. The benchmark, by definition, has a beta of one. A Covered Person should look at the beta of the client's portfolio to see whether the portfolio is assuming higher market risks.

If the risk measures of a portfolio are known, e.g. standard deviation and beta, the portfolio's risk-adjusted performance can then be calculated to determine if the portfolio has done well on a risk-adjusted basis. These risk-adjusted measures include Sharpe Ratio, Treynor Ratio, etc., which are discussed below.

- **Sharpe Ratio** – This is a measure of a portfolio's annualized return in excess of the risk-free rate of interest and adjusted for the portfolio's **total risk**, as defined by its standard deviation.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where,

R_p = Return on the portfolio

R_f = Risk-free rate of return

σ_p = Standard deviation of the portfolio

The higher a portfolio's Sharpe Ratio, the better its return relative to the amount of total risk taken. Please refer to the Appendix 2.5 on how to calculate the Sharpe Ratio.

- **Treynor Ratio** – The Treynor Ratio relates the excess return of a portfolio to the portfolio's beta and is calculated by the formula:

$$\text{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

The higher a portfolio's Treynor Ratio, the better its return relative to the amount of systematic risk taken. Please refer to the Appendix 2.6 on how to calculate the Treynor Ratio.



Tips / Key Points

A Covered Person should compare the client's portfolio returns with the proper benchmarks as well as on a risk-adjusted basis.

2.7.4 Performance Attribution

A Covered Person has a duty to review the client's portfolio and explain outperformance or underperformance. It is also useful to perform an *attribution of performance*. Performance attribution seeks to decompose the total performance of a portfolio into specific components to determine the reasons why a portfolio did better or worse than the benchmark or the target return.

Performance attribution analysis often begins with an analysis of the asset allocation choices and progressively focuses on the finer details of investment decision-making, such as the choice of countries or sectors for investment. Common ways of performance attribution include:

- Analysing the contributions of each asset class to the final performance numbers;
- Analysing the sectors of good and bad performances within each asset class; and
- Analysing the component contributions of the Covered Person.

In order to carry out portfolio attribution to explain a Covered Person's contribution to a client's performance, the first step is to construct a benchmark (or bogey) portfolio to compare with. This bogey portfolio measures the returns if a passive strategy were adopted, i.e. if the portfolio does not deviate in terms of asset allocation or security selection. See an illustration in Example 2.1.

Example 2.1

Consider the two portfolios below, one being the client's portfolio and the other being the benchmark portfolio.

Client's Portfolio

Asset Class	Weight	Annual Return	Contribution to Portfolio Performance
Equities	70%	7.50%	5.25%
Bond	10%	2.00%	0.20%
Cash	20%	0.50%	0.10%
Portfolio Return			5.55%

Benchmark Portfolio

Asset Class	Weight	Annual Return	Weighted Return
Equities	60%	6.00%	3.60%
Bond	30%	1.50%	0.45%
Cash	10%	0.50%	0.05%
Benchmark Return			4.10%

The client's portfolio outperformed the benchmark portfolio by $5.55\% - 4.10\% = 1.45\%$. The next step is to determine what accounted for this 1.45% outperformance.

A. Asset Allocation Decision

The effect of the Covered Person's asset allocation choice can be computed by comparing the asset weighting of the client's portfolio and the benchmark portfolio and relating this to the benchmark return of the asset class.

Asset Class	Actual Weight	Benchmark Weight	Over/ (Under) Weight	Benchmark Return	Attribution of Performance
Equities	70%	60%	10%	6.00%	0.60%
Bonds	10%	30%	-20%	1.50%	-0.30%
Cash	20%	10%	10%	0.50%	0.05%
					0.35%

The above table shows that asset allocation decisions contributed 0.35% of the portfolio's excess return of 1.45% over the benchmark portfolio. Within the asset allocation decision, the major factor that contributed to outperformance is the higher weighting allocated to equities, which gave a higher return as compared to the other asset classes.

B. Security Selection Decision

Asset Class	Actual Return	Benchmark Return	Excess Return	Actual Weight	Attribution of Performance
Equities	7.50%	6.00%	1.50%	70%	1.05%
Bonds	2.00%	1.50%	0.50%	10%	0.05%
Cash	0.50%	0.50%	0.00%	20%	0.00%
					1.10%

The above table shows that superior security selection within the equity and bond asset classes contributed 1.10% out of the 1.45% excess return of the portfolio vs. the benchmark portfolio. The major contribution came from superior security selection within the equity asset class.

A Covered Person should consider whether a client may want to continue with an overweight in equities or reduce its weight and lock in some gains. He should also consider how he can add more value to the client in terms of security selection in the bond portion of the portfolio. Perhaps there has not been enough active management of the bond portion causing it to perform only in line with the benchmark.

2.8 Roles and Obligations of Covered Persons

A Covered Person has an important role to play in every step of the portfolio management process. He is responsible for helping the client formulate investment objectives and constructing the portfolio. He also monitors the portfolio in order to help the client manage portfolio risks and to generate returns that are in line with the client's investment objectives.

A Covered Person also reviews the portfolio with the client to explain the portfolio's performance over the review period. The frequency and extent of review depends on the mandate the Covered Entity has with the client.

There are two types of mandates that a Covered Entity have with a client. If the portfolio is managed on a discretionary basis, a Covered Person has the support of an internal Fund Manager who will assist to manage the client's portfolio in line with the investment strategy.

While the actual management of the portfolio may be handled by the internal Fund Manager, a Covered Person has the obligation to ensure that his client's investment strategy would be carried out and that the investment objectives would be met. If the mandate is one that is non-discretionary, a Covered Person will recommend to the client suitable investments for the client' to make informed decisions.

2.9 Chapter Summary

This chapter has covered many aspects of Portfolio Management. The salient points are:

- Portfolio management comprises a set of activities:
 - Understanding the client's objectives, constraints and preferences;
 - Understanding the macro-economic environment and capital market conditions;
 - Constructing a policy portfolio through strategic allocation of assets;
 - Selecting stocks through security analysis;
 - Monitoring the environment and making tactical allocations; and
 - Portfolio performance review.
- Tactical asset allocation refers to active management of the portfolio to deviate from the portfolio's strategic asset allocation in order to take advantage of market pricing anomalies.
- There are a number of important considerations that a Covered Person must be mindful of when constructing a portfolio.
- The Life Cycle Approach is a common starting point to ascertain a client's likely risk and return profile.

- An Investment Policy Statement guides the Covered Person as he executes the investment plan. This plan would also include reference benchmarks for evaluating portfolio performance.
- The return of a portfolio is a weighted average of the returns of the individual securities in the portfolio.
- The risk of a portfolio of securities is the volatility of its returns.
- A diversified portfolio seeks to lower the volatility of returns of the portfolio.
- The extent to which portfolio risks can be reduced through diversification depends on the degree to which the securities in the portfolio are correlated.
- The Capital Asset Pricing Model (CAPM) is used to determine a theoretically required rate of return of an asset.
- A Covered Person aims to maximize the return of the portfolio while keeping within risk levels that are acceptable to the client.
- A Covered Person has the responsibility to meet the client to explain the performance of his portfolio and the main factors that contribute to this performance. The frequency and extent of reviews depend on the mandate a Covered Entity has with the client.
- There are two types of mandates that a Covered Person receives from his client, namely discretionary or non-discretionary.

Appendices

Appendix 2.1 – How to Calculate Portfolio Returns

A portfolio consists of two stocks. Stock A has a return of 10% during the year and has a 40% weighting in the portfolio. Stock B has a return of 15% during the year and has a 60% weighting in the portfolio.

$$\begin{aligned}\text{Return of the portfolio} &= (10\% \times 40\%) + (15\% \times 60\%) \\ &= 13\%\end{aligned}$$

Appendix 2.2 – How to Calculate the Expected Return of a Portfolio Based on the Capital Market Line (using a total risk approach)

An investor owns a diversified portfolio of stocks. Because of the well-diversified nature of the portfolio, it is an “efficient” portfolio that is situated on the Capital Market Line (CML) postulated by the Capital Asset Pricing Model (CAPM).

Given that the risk-free rate is 4%, expected return on the market portfolio $E(R_m)$ is 10%, standard deviation of the returns of the market portfolio (σ_m) is 20%, and standard deviation of the returns of the client’s portfolio (σ_p) is 18%. Calculate the expected return of this portfolio.

$$\begin{aligned}\text{Expected return of the portfolio} &= 4\% + \left(\frac{10\% - 4\%}{20\%} \times 18\% \right) \\ &= 9.4\%\end{aligned}$$

Appendix 2.3 – How to Calculate the Expected Return of a Security Based on the Security Market Line (using the Capital Asset Pricing Model)

Assume that the beta for Stock A is 1.20. Given a risk-free rate of 5% and an expected market return of 18%, the required rate of return for Stock A is:

$$\begin{aligned}E(R) &= 0.05 + 1.20 (0.18 - 0.05) \\ &= 0.206 \text{ or } 20.6\%\end{aligned}$$

Appendix 2.4 – How to Calculate the Standard Deviation or Variance of a Portfolio

Suppose there are two portfolios with the following monthly returns:

Portfolio A: 5%, 4%, 5%, 6%

Portfolio B: 5%, -10%, 20%, 5%

Although both portfolios have the same average monthly return of 5%, Portfolio B is riskier because the returns fluctuate greatly versus its average return. The variance in this example is calculated as follows:

$$\begin{aligned}\text{Portfolio A's variance } (\sigma^2) &= \frac{(5-5)^2 + (4-5)^2 + (5-5)^2 + (6-5)^2}{4} \\ &= 0.5\end{aligned}$$

$$\begin{aligned}\text{Portfolio B's variance } (\sigma^2) &= \frac{(5-5)^2 + (-10-5)^2 + (20-5)^2 + (5-5)^2}{4} \\ &= 112.5\end{aligned}$$

The variance is an exaggerated number because all the differences are squared in the calculation. Therefore, to arrive at a true number to describe the fluctuation, you take the square root of the variance to arrive at the standard deviation.

$$\text{Portfolio A's standard deviation } (\sigma) = \sqrt{0.5} = 0.7$$

$$\text{Portfolio B's standard deviation } (\sigma) = \sqrt{112.5} = 10.6$$

As Portfolio A has a standard deviation of 0.7, it means that while the portfolio has an average return of 5%, the returns tend to fluctuate by 0.7% around this average.

As Portfolio B has a standard deviation of 10.6, it means that while the portfolio has an average return of 5%, the returns tend to fluctuate by 10.6% around this average.

Appendix 2.5 – How to Calculate the Sharpe Ratio

A portfolio has an annual volatility of 15% for the past one year. The risk-free rate is 2.5% p.a. and the 1-year return on the portfolio is 10%. What is the portfolio's Sharpe Ratio?

$$\begin{aligned}\text{Sharpe Ratio} &= \frac{10\% - 2.5\%}{15\%} \\ &= 0.5\end{aligned}$$

Appendix 2.6 – How to Calculate the Treynor Ratio

A portfolio has a beta of 1.5 for the past one year. The risk-free rate is 2.5% p.a. and the 1-year return on the portfolio is 10%.

$$\begin{aligned}\text{Treynor Ratio} &= \frac{10\% - 2.5\%}{1.5} \\ &= 0.05\end{aligned}$$

Chapter 3:

Fixed Income and Equity Securities

Objective

The candidate should be able to:

- ✓ Identify the characteristics of each asset class - money market instruments, bonds and stocks
- ✓ Evaluate the return and risk characteristics inherent in money market instruments, bonds and stocks
- ✓ Understand the motivation behind investment in each asset class
- ✓ Understand the applicability of each asset class to an investor, as well as its applicability in different economic environments
- ✓ Understand the variables that enter into the determination of price of these asset classes
- ✓ Interpret and apply valuation indicators contained in analyst reports and in the financial press
- ✓ Appreciate the differences between fundamental and technical analysis
- ✓ Determine the appropriateness of an investment for any particular investor
- ✓ Understand the various ways in which an investor can gain an exposure to each asset category

3.1 Introduction

The objective of this chapter is to introduce two conventional asset classes – Fixed Income (which includes money market instrument and bonds) and Equities. These two asset groups represent traditional vehicles for investors to earn a return above bank deposit rates.

They have different risk/return characteristics and should be included in an investor's portfolio in accordance with his risk tolerance and return requirements as

dictated by his individual needs and circumstances. In addition, capital market expectations must be taken into consideration in coming up with the final portfolio.

An investor can achieve a direct exposure to these asset classes via any of the following routes:

- **Primary Market** – The market for the issuance of new securities which includes Initial Public Offerings (IPOs); and
- **Secondary Market** – The resale market, where securities already issued are bought and sold by investors on an organized exchange or over-the-counter.

By tenure, these asset classes may also be classified as follows:

- **Money Market** – The market for short term finance, comprising securities with a maximum tenure of one year; and
- **Capital Market** – The market for long-term finance, comprising securities beyond one year maturity. These include bonds and equities.

3.2 Money Market Securities

Money market securities offer a regular and fixed income to investors. The objective of investing in a money market security is to provide both liquidity as well as a slightly higher yield than traditional bank deposits (with the exception of Treasury bills).

The money market is the market for *short-term* finance. It is the market where securities up to one year maturity are traded. The following instruments are traded in the money market:

- Treasury bills
- Certificates of Deposit
- Repurchase agreements
- Commercial paper
- Banker's acceptances

As far as the issuing entities are concerned, the purpose of issuing money market instruments is to meet short-term financing needs. Issuing entities could range from governments and banks to large corporations.

Known for their liquidity, these short-term instruments have low risk but minimum denominations are normally large. While Singapore Government Securities (SGS) Treasury bills sell in \$1,000 denominations, most of them sell in large denominations of at least \$250,000. Hence, they are normally not purchased by a retail investor.

The easiest way for retail investors to access a diversified portfolio of money market instruments is through money market funds. Managed by professional fund

managers, money market funds provide diversification across a range of money market instruments with just one investment and are available from denominations of \$1,000.



Tips / Key Points

Money market instruments are highly liquid, short term securities with maturity of less than and up to a year. They are suitable for investors who wish to earn slightly higher returns than regular bank deposits and yet want liquidity. These have slightly higher risk than bank deposits (with the exception of Treasury bills) and are available in a variety of forms, ranging from direct investment in money market instruments to investment in money market funds.

3.2.1 Key Features of Money Market Securities

Par value

Money Market securities generally have par value or face value of S\$1,000.

Price quote

Prices for money market instruments such as Treasury bills are quoted on a bank discount basis. This means they are quoted at a discount from face value and the investor earns the difference between the price paid and the face value. The yield is calculated as follows:

$$Y = \frac{\text{Face Value} - \text{Purchase Price}}{\text{Face Value}} \times \frac{360}{\text{No of Days to Maturity}}$$

$$= \frac{D}{F} \times \frac{360}{t}$$

Where,

Y = Annualised yield on a bank discount basis (in decimal form) or discount basis yield

D = Dollar discount

F = Face value

t = Number of days remaining to maturity

Example

A Treasury bill has 180 days to maturity and is priced at \$980 with a face value of \$1,000. What is the discount basis yield?

$$D = \$1,000 - \$980 = \$20$$

$$Y = \frac{\$20}{\$1,000} \times \frac{360}{180} = 4\%$$

This 180-day Treasury bill would be quoted on a yield basis, rather than a price basis, as 4% p.a.¹

3.2.2 Types of Money Market Securities

Treasury Bills

Treasury bills are issued by sovereign government and considered to be the safest of all money market instruments. SGS are traded in S\$1,000 denominations and on a rate of discount basis. The Singapore Government also issues T-bills of 3-month and 1-year maturities.

Certificates of Deposit (CDs)

A CD is a time deposit with a bank, and the bank pays interest and principal to the depositor only at the end of the fixed term. CDs are issued by banks and may be Domestic CDs, Eurodollar CDs (U.S. dollar denominated CDs issued by banks outside of U.S.) and Yankee CDs (U.S. dollar denominated CDs issued by foreign banks with offices in U.S.). Negotiable CDs, which are higher denomination CDs, typically greater than US\$100,000, are marketable and can be sold to another investor before its maturity date.

Repurchase Agreements (Repos)

Repos are loans collateralized by securities. Under a repo agreement, the dealer (borrower) sells government securities to an investor with an agreement to buy back the securities at a higher price on a later date. With overnight repos, the repurchase will be made the next day, while a term repo may be 30 days or more. These loans are considered very safe because they are backed by government securities.

¹ For US Treasury bills, the year has 360 days. The day count convention for SGS Treasury bills is 365 days, and yields are computed based on actual/365 day basis.

Commercial Paper (CP)

CPs are unsecured short-term debt (often less than one or two months) issued by large well-known corporations. They are often issued in multiples of \$100,000.

Banker's Acceptances (BA)

A banker's acceptance (BA), used commonly in international trade, is a time draft written by a borrower and guaranteed by the bank on which the draft is drawn. It is not unlike a post-dated cheque given to a bank to pay a certain sum of money at a future date. When the bank accepts the order, it assumes responsibility for payment to the holder of the acceptance, usually an exporter. The borrower is an importer who writes the draft in payment for the goods.

BAs are traded in the secondary market at a discount from face value. They are considered very safe because traders can substitute the bank's credit standing for their own. For a borrower, a BA is a short-term fixed rate loan enabling him to obtain low cost financing and to defer payment until the BA matures. For an exporter, a BA enables him to receive immediate payment for the goods he sells.

3.2.3 Returns and Risks

Because money market securities have short tenures, money market instruments tend to have low yields. These securities are favoured when there are concerns over health of the economy and in times of turmoil. They are suitable for meeting the liquidity needs of investors. They are low risk and are normally issued by good quality credits.

The risk of investing in money market instruments is that, due to their low yields, the returns may not keep pace with inflation and the investor may lose purchasing power. Another risk is interest rate risk. Interest rates and price of money market instruments are inversely related. If interest rates increase, their prices fall and conversely, if interest rates fall, their prices rise.

On a portfolio basis, a large commitment to this asset category is not recommended except for the most conservative investors.



Tips / Key Points

Quotes for money market instruments are given in terms of yields rather than prices. They are typically sold in very large denominations of at least \$250,000 except for SGS Treasury bills which sell in \$1,000 denominations. Individual investors can gain an exposure to a diversified basket of money market instruments by investing in money market funds or money market ETFs.

3.3 Bond Market Securities

A bond is a loan from the investor to an issuer - who could be a company, the government or supranational borrower - for a certain time period. Bond issuers could be companies - domestic or foreign - government, government agencies or supranational borrowers. As of 2009, the size of the global bond market, at about USD82 trillion, is almost two times the size of the global stock market, at about USD40-50 trillion.

For a typical plain vanilla bond, the investor earns a regular income in the form of coupons and receives the principal at maturity. The amount of coupon to be paid is stated at the outset when the bonds are issued and is expressed as a percentage of par value. Bonds appeal to investors who seek a regular income, and yet wish to preserve their capital. This is because the typical bond pays out regular interest, either semi-annually or annually, till maturity.

Each bond issue is bounded by a contractual agreement between the issuer and the investor called a *bond indenture*. The basic terms of the bond issue such as the principal, coupon rate, maturity date and rights and duties of investors; total amount of bonds issued; description of collateral if applicable; sinking fund provisions; call provisions; and details of protective covenants are all included in the bond indenture. Fulfilment of the conditions in the bond indenture is overseen by a trustee, who also ensures the timely payment of principal and interest.

3.3.1 Issuers of debt

The major issuers of debt in Singapore are:

- **Singapore Government** – The objective of issuing SGS bills and bonds is to foster the growth of an active secondary bond market and provide a benchmark for the pricing of corporate issues.
- **Domestic Companies** – Domestic SGD bonds are issued by publicly-listed, private companies and Statutory Boards of the Singapore Government. These Boards are autonomous organizations and their debts are considered among the most liquid of the Singapore corporate bond issues.
- **Non-Domestic Companies** – Non-domestic SGD corporate bond issuers mainly comprise supranational agencies, foreign banks, and foreign corporations from the United States, Europe and Asia. These include Citigroup Inc., Merrill Lynch & Co. Inc., and the Asian Development Bank.

In Singapore, SGS bonds, domestic bonds and non-domestic bonds are traded over-the-counter in the secondary market.

3.3.2 Key Features of Bonds

Bonds are differentiated from equity securities. Firstly, bond holders are the creditors of a company and are not owners. Secondly, bond holders receive fixed regular coupon payments while equity holders receive dividends which are uncertain and variable (depending on corporate performance and corporate policy). Some other features of bonds are discussed below.

Par Value

Bonds are usually issued at par. The par value is also known as principal, face, redemption or maturity value. The par value is typically \$1,000 per bond. Over the life of the bond, its price changes because of changes in interest rates.

Valuation

The basic valuation formula for a security with at least one year maturity is as follows:

$$(P_0) = \sum_{t=1}^n \frac{\text{Cash flows}}{(1+r)^t} = \text{PV of future cash flows}$$

$$(P_0) = \frac{CF_1}{1+r} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n}$$

Where,

P_0 = Intrinsic value of the security today (the true value of the stock)

CF_i = Cash flow at the end of Year i

r = Discount rate reflecting the risk of the security. A more risky security would have a higher discount rate. Usually, the discount rate is the required return.

n = Tenure of the security, in terms of compounding periods

The value of a bond is described by the following equation:

$$(P_0) = \sum_{t=1}^n \frac{\text{Coupons}}{(1 + YTM)^t} + \frac{\text{Par value}}{(1 + YTM)^t}$$

Where,

P = Price of bond

YTM = Yield to maturity

t = End of period t

The YTM is inversely related to price of the bond P_0 . If YTM rises, bond price falls and vice versa.

Bond Price Quotes

The bond price is quoted as a percentage of par values. Hence, a bond with a price quote of 102.35 will be priced at \$1,023.50. This is calculated as:

$$\frac{102.35}{100} \times \$1,000 = \$1,023.50$$

Unlike most other bonds which are priced as per above, US Treasury prices are quoted in 32nds. Hence, a price quote of 91:29 translates into $91 \frac{29}{32} = 91.90625\%$ of par value. See Exhibit 3.1.

Exhibit 3.1 Bond Price Quote

	Par value	Price quote	Price as % of par	Price
US Treasury Bonds	US\$1,000	91:29	91.90625	US\$919.0625
SGS Bonds	S\$100	102:23	102.23	S\$102.23

Note that the minimum denomination for trading in SGS bonds is S\$1,000.

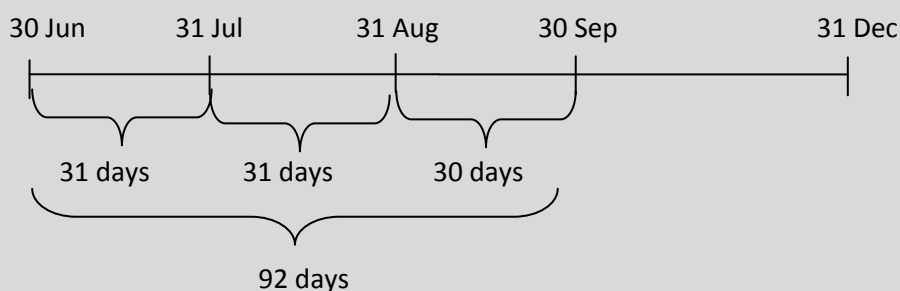
The quoted price is called the *clean price*. Besides the bond price, the bond buyer usually has to pay the accrued interest if he buys the bond between coupon dates. Hence, total price paid by investor is known as the *dirty price*.

$$\text{Dirty Price} = \text{Clean Price} + \text{Accrued Interest}$$

Timeline of Accrued Interest

Consider a bond with a coupon rate of 6% which pays coupons semi-annually on 30 June and 31 December. If the investor buys the bond on 30 Sep, the bond buyer must pay the seller about 3 months of accrued interest* as the interest earned from 30 June to 30 September should accrue to the seller.

This is like paying the seller the coupon in advance as the buyer will only receive the next coupon on 31 December.



Accrued interest = $92/184 \times (3\% \times \text{S\$}1,000) = \$15.00$

This calculation is based on an actual/actual basis, which is used in Singapore.²

If the calculation is done based on a 30/360 day count convention (US companies and municipal bonds use this), then:

Accrued interest = $(90/360) \times (6\% \times \text{S\$}1,000) = \$15$

SGS bonds pay a fixed semi-annual coupon on the 1st or 15th of the coupon paying month each year. If the payment date falls on a public holiday, the coupon payment will be made on the next business day. Coupon interest accrues from the previous coupon date (inclusive) to the settlement date (exclusive). SGS trades ex-coupon three working days prior to the coupon date.

Bonds priced above par are called **premium bonds**, while those priced below par are called discount bonds. Bonds selling at par are called **par bonds**.

Exhibit 3.2 Singapore Government Securities - Benchmarks

Closing Levels												
	Treasury Bills		Bonds									
	3-Mth	1-Year	2-Year	5-Year	10-Year	15-Year	20-Year					
Issue Code	BQ11116W	BY111100X	N508100V	N511100W	NY05100N	NY09100H	NZ10100F					
Coupon Rate			1.625%	1.125%	3.250%	3.000%	2.875%					
Maturity Date	28-Jul-11	2-May-12	1-Apr-13	1-Apr-16	1-Sep-20	1-Sep-24	1-Sep-30					
	Yield	Yield	Price	Yield	Price	Yield	Price	Yield	Price	Yield	Price	Yield
20-Apr-11	0.24	0.33	102.23	0.47	99.96	1.13	106.41	2.48	101.65	2.85	97.44	3.05
21-Apr-11	0.25	0.33	102.19	0.49	99.94	1.14	106.33	2.49	101.62	2.85	97.38	3.06
25-Apr-11	0.25	0.33	102.25	0.46	100.02	1.12	106.56	2.46	101.87	2.83	97.75	3.03
26-Apr-11	0.26	0.34	102.24	0.46	100	1.12	106.6	2.46	101.93	2.83	97.87	3.02
27-Apr-11	0.27	0.44	102.23	0.46	100.01	1.12	106.71	2.44	102.02	2.82	97.96	3.01
28-Apr-11	0.27	0.44	102.22	0.47	100.05	1.11	106.79	2.43	102.12	2.81	98.00	3.01

Notes: Data reflect bid rates quoted by SGS primary dealers. Yield is quoted as % p.a. Price is quoted in S\$ per S\$100 of principal amount.

² In Singapore, as in the US, UK, France and Germany, the day count convention for bond computations is on an actual/actual basis. Coupon is paid on the first or 15th of the particular month, or the next business day if it falls on a public holiday.

Exhibit 3.3 Example of US Corporate Bond Quotes

	Coupon	Maturity Date	Bid \$	Yield %
AGT Lt	8.800	Sep 22/25	100.46	8.75
Air Ca	6.750	Feb 02/04	94.00	9.09
AssCap	5.400	Sep 04/01	100.01	5.38
Avco	5.750	Jun 02/03	100.25	5.63
Bell	6.250	Dec 01/03	101.59	5.63
Bell	6.500	May 09/05	102.01	5.95
BMO	7.000	Jan 28/10	106.55	6.04
BNS	5.400	Apr 01/03	100.31	5.24
BNS	6.250	Jul 16/07	101.56	5.95
CardTr	5.510	Jun 21/03	100.52	5.27
Cdn Pa	5.850	Mar 30/09	93.93	6.83
Clearn	0.000	May 15/08	88.50	8.61
CnCrTr	5.625	Mar 24/05	99.75	5.68
Coke	5.650	Mar 17/04	99.59	5.80

|
Column 1

|
Column 2

|
Column 3

|
Column 4

|
Column 5

Column 1: Bond Issuer

Column 2: Coupon rate

Column 3: Maturity Date: For year, 25 means 2025, 04 is 2004, etc.

Column 4: Bid Price - A bond with a bid of 93 is trading at 93% of its par value

Column 5: Yield to maturity

Coupon Rate (also called nominal yield)

Exhibit 3.3 illustrates the main features of a Bond published in a typical financial newspaper. Most bonds carry a fixed semi-annual or annual coupon. The coupon is the income component of a bond. The yield takes into account the bid prices of the bonds. In general, riskier bonds will bear higher coupon rates to compensate investors for the higher risk.

Bonds may have coupons which are variable. *Floating rate bonds* have coupons which are pegged to some published rates, e.g. inflation rate or LIBOR (London Interbank Offered Rate).

Time to Maturity

The maturity date is the time when the principal or par value of the bond will be repaid. The greater the time to maturity, all other things being equal, the greater should be the coupon rate on the bond. Correspondingly, a longer term bond will display greater volatility than a short term bond. Bonds may be characterised by their maturity. Hence, they may be classified as:

- Medium-term notes: 1-10 year maturity
- Long term bonds: Maturity of 10 years or more
- Perpetual bonds are bonds that do not have any maturity date

Perpetual bonds pay coupons forever and their cash flows are therefore like those of perpetuity. Hence, they are valued as follows:

$$\text{Price of a Perpetual Bond} = \frac{\text{Annual Coupon}}{\text{Interest Rate}}$$

Most perpetual bonds are callable (redeemable at the issuer's discretion), with the first call date typically at least five years from the date of issue.

Yield-to-Maturity (YTM)

The yield to maturity considers the current income generated by the bond as well as any change in its value when it is held to maturity. The YTM considers the timing of cash flows and interest-on-interest and assumes that bond proceeds such as coupons can be reinvested at the YTM at the time they are paid. The YTM on a bond can be broken down into two components, the capital gain and the income, just like for stocks.

$$\text{YTM} = \text{Capital Gains Yield} + \text{Current Yield}$$

Where,

Capital gains yield = Price appreciation on the bond

Current yield = $\frac{\text{Annual Coupon}}{\text{Bond Price}}$

Example

If the coupon rate on an annual-pay bond is 6%, and the bond is selling in the market at \$1,020, then:

$$\text{Current Yield} = \frac{\$60}{\$1,020} = 5.9\%$$

The relationship between coupon rate, current yield and YTM are summarised below:

Discount bond	Coupon Rate < Current Yield < YTM
Premium bond	Coupon Rate > Current Yield > YTM
Par bond	Coupon Rate = Current Yield = YTM

In the example above, the bond is a premium bond (priced above par value of \$1,000) as the coupon rate (6%) > current yield (5.9%).

Currency of Denomination

Some companies, banks, governments or other entities may issue bonds in foreign currencies for various reasons. Some of these reasons include the ability to access cheaper investment capital in foreign markets or to hedge against foreign exchange risk if the country has foreign operations.

Companies which issue bonds in a foreign country are subject to economic conditions prevailing in that country. Hence, a US company could issue an SGD corporate bond in Singapore to take advantage of the low interest rate environment here. This bond issue will be governed by the law of the market of issuance, in this case Singapore. In addition, it will be subjected to the SGD yield curve, and not the USD yield curve.



Tips / Key Points

Bonds have the following key features:

- They are usually issued at par, which is also the principal value repayable at maturity.
- The price of a bond is the present value of the coupons and the par value to be received at maturity.
- Price quote is in percentage of par value, and most bonds have a par value of \$1,000. Hence, a price quote of 102 refers to price of \$1,020. US Treasury bonds are traded in 32nds, hence a price quote of 102:25 means 102 25/32 or \$1,020.78125.
- Bond investors pay the *dirty price* which is the *clean price + accrued interest* if the purchase is made in between coupon payment periods.
- Premium bonds: Priced above par value.
- Discount bonds: Priced below par value.
- Par bonds: Priced at par value.
- The coupon rate or nominal yield is the income component of a bond and is fixed at issue for fixed rate bonds.
- Time to maturity describes the time remaining until principal

repayment. The longer the time to maturity, the greater the risk and hence, longer term bonds have higher coupon rates, other things being equal.

- The yield-to-maturity (YTM) represents the expected total return on the bond and is made up of two components, the capital gains yield and the current yield.

3.3.3 Types of bonds

Straight Bonds

Plain vanilla bonds with no additional features. In contrast, there are various types of bonds which come with embedded options.

Callable Bonds

Bonds which give the issuer the right to redeem the bond issue before maturity at a pre-specified call price. Callable bonds usually come with a period of call protection during the initial years when they cannot be called.

Hence, a callable bond is a bond with an embedded call feature giving the issuer the option to redeem the bond when conditions are favourable to the issuer. The embedded call option enables the issuer to replace existing higher interest debt with lower interest debt when interest rates fall. This is called refunding. Some callable bonds come with non-refunding provisions which prohibit issuers from retiring a bond for the purpose of reissuing a new bond at lower interest rates to reduce its interest payments.

The call feature is a disadvantage to the investor and hence, callable bonds come with higher coupons and a call price which is usually set at or above the par value. The investor stands to lose if the bond is called when interest rates decline, forcing him to reinvest at lower interest rates. The call feature also limits the price appreciation of the bond as it is more likely to be called when interest rates have fallen to a low level, at which time bond prices would be high and rising.

Puttable Bonds

Bonds which give the investor the option to sell the bonds to the issuer before the maturity date. A puttable bond comes with a lower coupon rate than a straight bond because the put option embedded in the bond is favourable to the investor.

If interest rates rise, the investor can put (sell back) the bond to the issuer at par value. This also limits his capital loss as the bond price will fall when interest rates rise. The bondholder also benefits because after putting the bond to the issuer, he can then reinvest his bond proceeds, principal plus coupon, at the prevailing higher interest rate.

Floating Rate Bonds

Bonds whose coupons are reset at predetermined intervals and tied to some floating rate e.g. LIBOR, inflation, Federal funds rate.

Deferred Coupon Bonds

Deferred coupon bonds usually do not pay interest in the initial years, in an attempt by the issuer to conserve cash flows early in their business life. Their coupon rates are usually higher to entice investors.

Zero Coupon Bonds (or Zeros)

Zero coupon bonds do not issue coupons and the investor earns the difference between the price paid and the par value. Hence, they are usually sold at a deep discount to their par value.

Step-up Notes

Step-up notes have coupons which increase according to a predetermined schedule.

Convertible Bonds

Convertible bonds are straight bonds (bonds without any embedded features), with an embedded call option, that allow the investor to convert the bond into a fixed number of the common shares of the issuing company at a pre-specified price, usually above the current market price. Because this conversion option is attractive to the investor if the stock market goes up, the coupon rate on such bonds is usually lower than straight bonds.

A convertible bond offers the investor the potential upside of a stock and the downside protection of a bond. If stock prices rise, the investor can convert his bonds into shares and participate in the rising stock price. If stock prices are down, the investor will not convert and can still continue to receive the coupons from the bonds.

Income Bonds

A bond whereby the issuer is only required to pay interest when its earnings are adequate to pay the coupon; and the only obligation is to pay the par value at maturity.

Mortgage-Backed Securities (MBS)

Cash flows from loans are sometimes packaged and sold to investors as mortgage-backed securities (MBS), in the case of mortgage loans, or as asset-backed securities (ABS), in the case of car loan payments or credit card payments. This process is

called securitization and it enables the original lender to have immediate access to the cash that it has lent out to homeowners, car owners, etc.

The cash flows to investors have bond-like characteristics but differ in terms of frequency and content. For example, in the case of MBS, investors receive both principal repayment and interest on a monthly basis, instead of the typical coupons on a semi-annual or annual basis.

The cash flow yield, which is similar in concept to the YTM of ordinary bonds, is subject to significant reinvestment risk because borrowers (the entities which take up the mortgage loan) may prepay ahead of schedule.



Tips / Key Points

Bonds usually provide a higher return than money market instruments but entail more risk as they are longer in tenure. Tenures can range from one year to infinite, as in the case of perpetual bonds.

Bonds provide two forms of returns to investors:

- Regular coupon income; and
- Potential capital gain if interest rates fall

A bond investor can avail himself to the multitude of bonds both domestically and internationally. Investment can be made directly in individual bonds or indirectly through bond funds, which offer benefits of greater diversification and professional management.

3.3.4 Key Risks of Bonds

Credit or Default Risk

An indication of the relative creditworthiness of a bond is provided by the ratings of the major credit rating agencies such as Moody's and S&P's. The long-term obligation ratings of major rating agencies such as Moody's and S&P reflect their opinion of the relative credit risk of bonds with an original maturity of one year or more and the likelihood of default. These ratings are displayed below.

Exhibit 3.4 Credit Ratings

<u>Standard & Poor's</u>	<u>Moody's</u>	<u>Description</u>	
AAA	Aaa	The highest possible rating. Extremely strong ability to repay interest and principal.	Investment Grade
A	Aa	Very strong capacity to repay interest and principal.	
A	A	Strong capacity to repay interest and principal, but somewhat susceptible to circumstances.	
BBB	Baa	Adequate capacity to repay; more susceptible to changes in economic conditions.	
BB	Ba	Speculative ability to repay obligations.	Speculative Grade
B	B	Highly speculative; ability to repay over time is questionable.	
CCC	Caa	CCC/Caa and CC/Ca rated bonds may already be in default or facing imminent default.	
CC	Ca		
C	C	In default on interest repayments with little prospects for recovery.	

Bonds rated BBB (S&P) or Baa (Moody's) and above are considered investment grade or having low risk of default. Bonds below Baa (Moody's) or BBB (S&P) are considered high risk and are variously called non-investment grade bonds/ speculative grade bonds/ junk bonds/ high yield bonds. Bonds such as these can be riskier than stocks.

Higher risk bonds will command higher coupons to compensate for the risk of holding them. The difference between two bonds based on their credit risk is called the credit spread. This spread varies over time, depending on market sentiment, the general economic condition and pricing of risk premiums.

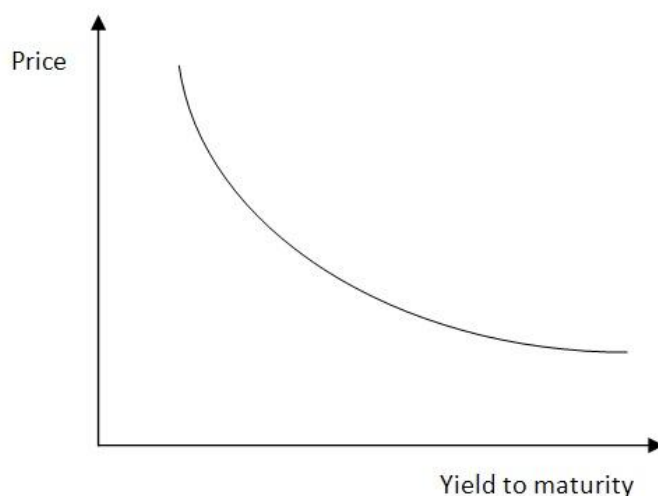
Credit spreads tend to increase in recessionary times as investors avoid junk bonds and go for the relatively safer government bonds. This flight to quality causes prices of junk bonds to go down (and their yields to go up) while prices of government bonds are bid up (and their yields go down).

Interest Rate Risk

Price of a bond and interest rates are inversely related. When interest rates rise, bond prices fall as investors gravitate towards bonds bearing higher interest rates; and when interest rates fall, bond prices rise. Previously issued bonds with lower yields will now have to provide a higher yield to attract investors. To achieve this, bond prices have to fall (see equation below).

$$P = \sum_{t=1}^n \frac{\text{Coupons}}{(1 + \text{YTM})^t} + \frac{\text{Par value}}{(1 + \text{YTM})^t}$$

Exhibit 3.5 Yield vs. Price



- (1) Bond price and yield to maturity are inversely related.
 - (2) As yields rise, bond prices fall at a decreasing rate. However, as yields fall, bond prices rise at an increasing rate. This bond property is called *convexity*.
-

Interest rate risk refers to the risk of bond prices falling if interest rates rise. If a bond is sold before maturity, the investor gets either a capital gain or loss as prices of bonds change over time. The sensitivity of a bond to changes in interest rates is called duration, and it depends on the following factors:

Exhibit 3.6 Interest Rate Sensitivity

Factor	Interest Rate Sensitivity	
	High	Low
Maturity	Long-term bonds	Short-term bonds
Coupon	Low coupon bonds	High coupon bonds
Yield	Low initial yield	High initial yield

Modified Duration

Duration measures the weighted average time to maturity of a bond in present value terms. It considers the size and timing of cash flows over the bond's life. The earlier the investor recoups his investment in the bond, the shorter the duration of the bond. The duration for a zero coupon bond = term to maturity. For all other bonds, duration is less than time to maturity.

A related measure, modified duration, measures the sensitivity of a bond's price to a change in interest rates. If a bond has a modified duration of 2, and if its yield now changes by 100 basis points, then the price of the bond will change by about 2%.

$$\Delta P/P = -D_{\text{modified}} \times \Delta YTM$$

$$\Delta P/P = -2 \times 1\% = -2\%$$

Where,

- D_{modified} = Modified duration

Note that bond prices and yields are inversely related; hence a 1% *rise* in yields as seen in the equation above leads to a 2% *fall* in price if modified duration is 2.

Modified duration increases with:

- Increase in time to maturity
- Fall in coupon rates
- Fall in bond yields



Tips / Key Points

Basis Point (bp):

A bp is one-hundredth of a percentage point. E.g. the yield difference between bond A with a yield of 3.21% and bond B with a yield of 3.50% is 29bp. Yield differences between bonds are quoted in bp, especially relative to Treasuries or to bonds with similar characteristics.

Price Value of a Basis Point (PVBP):

The PVBP is the price change that will occur if a bond's yield changes by 1 bp. This measure is useful in determining bond volatility as it enables the investor to convert a yield change to price change. E.g. suppose a bond is priced at \$980 and its PVBP is \$1. A yield increase of 5 bp will cause the bond price to fall by \$5 to \$975 (Price and yields move inversely).

Reinvestment Risk

Reinvestment risk refers to the risk of having to reinvest your bond proceeds – coupons and/or principal – at an interest rate below your YTM. If interest rates fall, you can now only reinvest your bond proceeds at the prevailing *lower* interest rates. Hence, the return you earn on your bond will be lower than the YTM.

Reinvestment risk is greatest for bonds with large coupons, higher coupon payment frequency, long maturities, selling at a premium. Zero coupon bonds have no reinvestment risk, while callable bonds have high reinvestment risk.

Inflation Risk or Purchasing Power Risk

Coupons, which are the income component of bond returns, are denominated in fixed dollar amounts. Their purchasing power will be eroded if inflation rises, reducing the real return to investors. [Real returns are the returns after accounting for inflation]. Floating rate bonds such as inflation-protected bonds (TIPS or Treasury inflation-protected securities in the US), or LIBOR-indexed bonds, offer some protection against inflation because of the regular resetting of the coupon rates.

Liquidity Risk

This is the risk that the investor is unable to sell the bonds quickly and at a fair price as and when he needs the funds. Not all bonds suffer from this risk; it depends on whether there is a liquid secondary market for those bonds. Even for relatively liquid bonds, in times of panic or financial market turmoil, liquidity could dry up, leaving few buyers.

Call Risk

This risk applies to callable bonds and it refers to the risk that the issuer will call (redeem) the bonds before the maturity date. This is disadvantageous to the investor as the bond is likely to be called when interest rates have fallen to a low level. If the bond is called, the investor can now reinvest the proceeds, which comprise both coupons and principal, at the prevailing low interest rates.

The callable feature also limits the price appreciation potential of the bond at a time when bond prices are going up. Finally, the call feature creates uncertainty in the cash flow pattern of a callable bond. Usually, the call price is set higher than the par value of the bond. The amount by which the call price exceeds the bond's par value is called the call premium.

Sector Risk

This is the risk that economic or other operating conditions move adversely against a sector, affecting all the companies within that sector.

Volatility Risk

This applies to bonds with embedded options, such as callable bonds and puttable bonds. Their prices are affected not just by interest rates but also by changes in volatility. For example, the value of a call option increases when interest rate

volatility increases. This will reduce the value of the callable bond because the bond holders will have sold a call option to the issuer.

Event Risk

This is the risk of an unexpected event causing a change in the ability of a company to repay its principal and/or interest on its bonds. These events include leveraged buyout, corporate restructuring, regulatory change, natural disaster, or industrial accidents.

Political Risk

This is the risk associated with investing in countries where policy changes may adversely affect the value of the bond. These changes include an imposition of withholding taxes where none previously existed, or of capital controls to curb hot money flows into the country.

Exchange Rate Risk

This risk is relevant if the investor invests in a foreign-currency denominated bond when his reference currency is domestic, say in Singapore Dollars. If the foreign currency depreciates against the Singapore Dollar, his investment in the foreign bond will decline in value even if the price of the bond has not changed.



Tips / Key Points

Key risks of investing in bonds include the following:

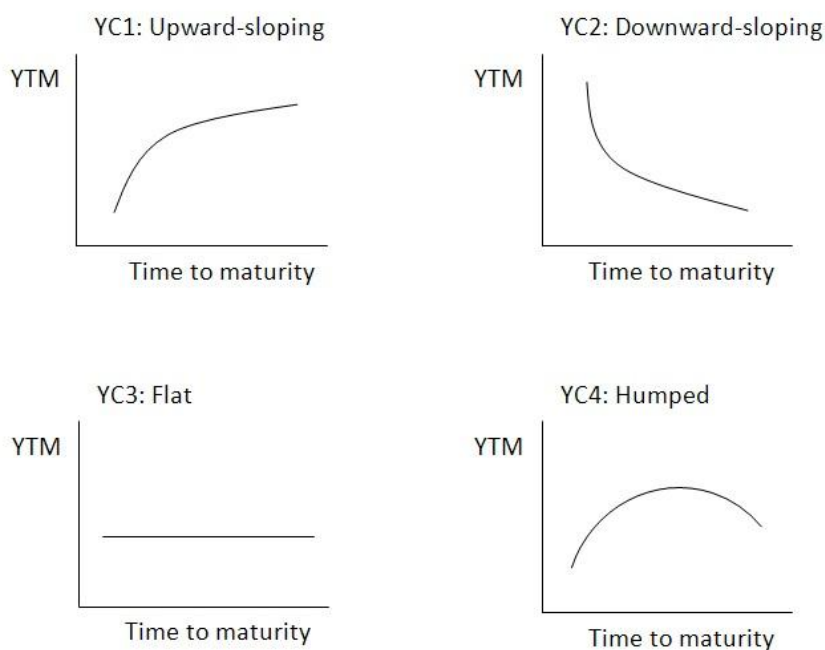
- Credit or default risk: Risk that the issuer is unable to repay the principal and/or coupons when these come due.
- Interest rate risk: The risk that interest rates rise, causing bond prices to fall. Sensitivity to interest rate risk can be measured by calculating modified duration.
- Reinvestment risk: The risk that interest rates fall, causing reinvestment of bond proceeds to be made below the YTM.
- Inflation or purchasing power risk: Risk that inflation increases erode the purchasing power of coupons.
- Liquidity risk: Risk that the investor is unable to sell the bond quickly and at a fair price.
- Call risk: Applies only to callable bonds, this is the risk that the bond will be called when interest rates fall to very low levels, thus truncating the returns of the investor. When a callable bond is redeemed, the investor receives both the coupons and the principal back and has to reinvest these amounts at the prevailing *lower* interest rates.

- Sector risk: Adverse economic or other conditions affecting a specific sector.
- Volatility risk: Applies to bonds with embedded options.
- Event risk: Refers to unexpected events affecting the ability of the issuer to repay.
- Political risk: Risks relating to investing in particular countries, such as changes in policy and government.
- Exchange rate risk: Risk due to exposure to foreign-denominated securities.

3.3.5 Theory of Yield Curves

The relationship between yield-to-maturity (YTM) and time to maturity can be captured graphically in a yield curve. This curve is useful in developing a bond portfolio strategy taking into account market expectations regarding future interest rates. There are various shapes that the yield curve can take:

Exhibit 3.7 Types of Yield Curves



Theories regarding the shape of the yield curve are called "Theories of the Term Structure" and they centre along the following arguments:

Pure (Unbiased) Expectations Theory

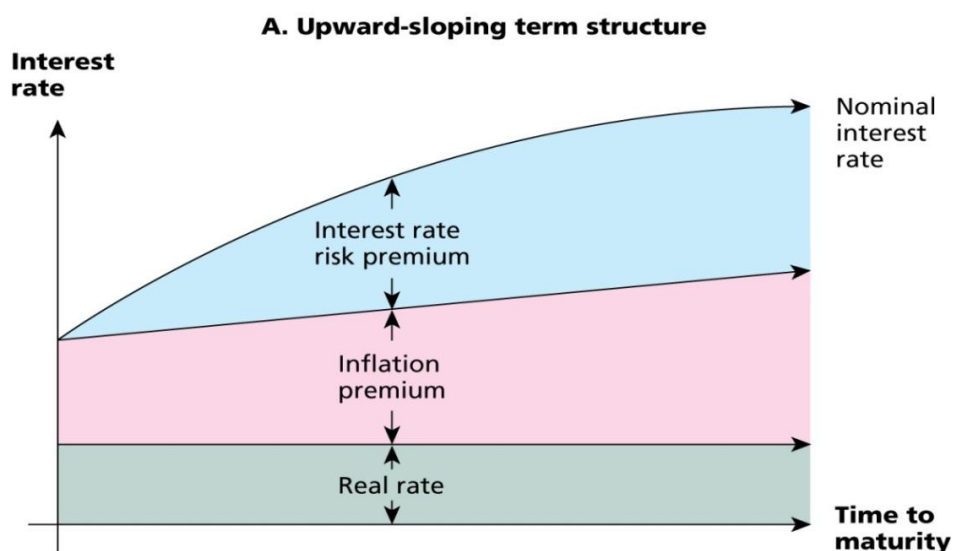
This theory states that the shape of the yield curve reflects the market's expectations of future interest rates. If the market expects future interest rates to rise, then the yield curve will be upward-sloping (YC1) and longer term bonds will have higher yields than short term bonds.

If the market expects future interest rates to fall, then the yield curve will be downward-sloping (YC2), and longer term bonds will have lower yields than short term bonds. If the market expects future interest rates to be unchanged, then the yield curve will be flat (YC3), and long and short term bonds will have similar yields. The pure expectations theory is consistent with any shape of the yield curve.

Liquidity Preference Theory

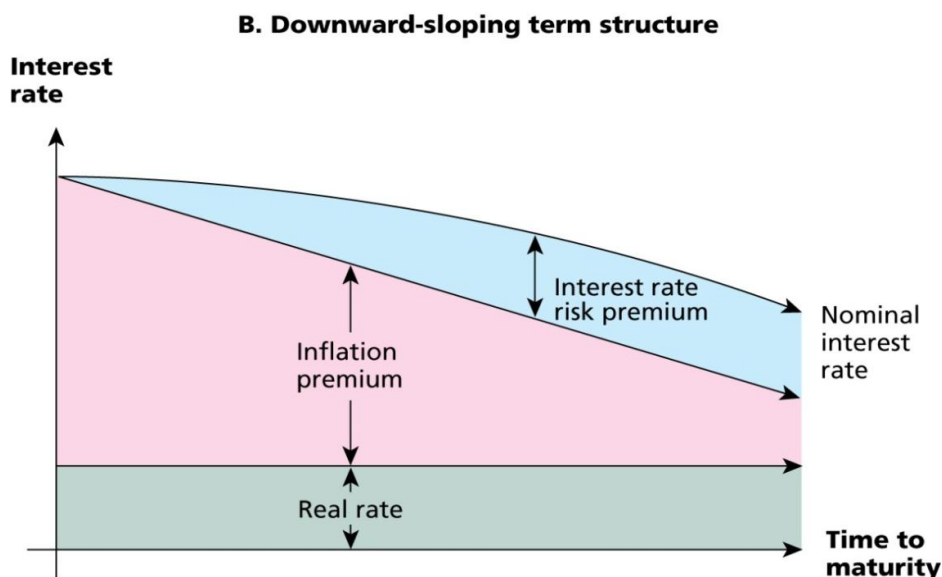
This theory posits that investors prefer liquidity and must be compensated with additional return to induce them to give up liquidity for longer term bonds. This higher yield, called a term premium (or interest rate risk premium), rises with increases in time to maturity. If this theory holds, the yield curve can only be upward-sloping. However, in reality, the yield curve is not always upward-sloping as there are other factors at work.

Exhibit 3.8 Upward-sloping Term Structure



Source: *Fundamentals of Corporate Finance* by Ross, Westerfield and Jordan

Exhibit 3.9 Downward-sloping Term Structure



Source: Ross, Westerfield and Jordan, *Fundamentals of Corporate Finance*. McGraw-Hill

Market Segmentation Theory

The market segmentation theory is consistent with any shape of the yield curve. This theory states that investors and issuers have preferred maturity segments located at specific regions of the yield curve. Pension funds and insurance companies, for example, are restricted to investing only in long bonds to offset their long term liabilities, while money market funds are restricted to short term assets.

Hence, demand and supply along the various maturity segments dictate the shape of the yield curve. If there is high demand for short term bonds and large supply of long bonds, prices of short term bonds will rise and their yields will fall. Long term bond prices will fall due to excess supply, and their yields will rise. Together, this will cause the yield curve to be upward-sloping.

3.3.6 Bond Investment Strategies

An examination of the shape of the yield curve can render some insights into the expected direction in which future interest rates are headed. For example, since interest rates and bond prices are inversely related, an investor should start to build a position in bonds when market interest rates have peaked and are starting to fall.

This situation may be reflected in a downward-sloping yield curve, also called an inverted yield curve. By implication, this is a forecast that bond prices may rise. Other investment strategies include:

Trading Strategy

If the investor is taking a trading stance, as opposed to a buy-and-hold strategy, the best time to buy into bonds is when yields are expected to fall - which is when bond prices are expected to rise. In anticipation of such a scenario, the bond investor can extend the duration of his portfolio.

For example, if the bond has duration of 3 years, a 1% fall in interest rates will lead to a 3% rise in the bond's price. For a 1% fall in yield, the rise in bond prices will be larger if the duration is longer. If the investor is expecting an increase in interest rates, a suitable strategy would be to reduce the duration so as to minimize possible losses from rising rates.

Buy and Hold Strategy

In a buy-and-hold strategy, the investor will receive the quoted YTM at the time of purchase if he holds the bond till maturity, if the bond does not default and if interest rates do not change.

If interest rates change, then his YTM will differ from the original quoted YTM because the coupon proceeds that he receives periodically will be reinvested at a different interest rate from the YTM. If the investor intends for his bond portfolio to provide a source of regular income using a buy-and-hold strategy, credit quality is an important consideration.

3.3.7 Best Case / Worst Case Scenario

The best case scenario for a bond investor would occur where interest rates are trending downwards throughout the life of the bond. For a buy-and-hold investor, he would have locked in a high interest rate which he consequently earns over the life of the bond. For an active investor with a trading stance, he can capitalise on the fall in interest rates and realise the capital gains from the falling interest rates. A falling interest rate environment is favourable for bond investments as it means that bond prices would be rising.

The worst case scenario for a bond investor is default by the issuer. This refers to the failure or inability to repay the principal and/or interest on the debt. The bond investor may lose some or all of his capital if the issuer defaults. It is not only the speculative grade bonds which can default. It is possible for an investment grade bond to default if the issuer's fundamentals suddenly change, although the likelihood of such a default is lower than that of speculative grade bonds.

3.3.8 Vehicles for Bond Investment

Individual investors can gain an exposure to the bond market via direct investment in the bond market, via actively-managed fixed income unit trusts or via bond index

unit trusts if a passive exposure is desired. Direct investment may be the desired route if the minimum investment amount is small and the bond market is liquid. This varies depending on the country and the instrument (corporate/ government, investment grade/high yield, long maturity/short maturity, etc.) being considered.



Tips / Key Points

When taking a trading stance, an investor should extend duration of his portfolio if he expects interest rates to fall as this will magnify the price increase in his bond portfolio. He can do this by investing in bonds with longer maturity, lower or zero coupons and lower yields.

In a recession, investors should go for credit quality and avoid distressed securities and speculative grade bonds unless they are mispriced. The key risk of investing in these bonds at this time is the risk of default.

3.4 Equities

Equities refer to common shares or common stock. A share or stock represents ownership in a corporation. Unlike bonds which are mainly bought for their income-producing property, common shares are usually bought for their capital appreciation potential, and to a lesser extent, income in the form of dividends.

3.4.1 Key Features of Common Equity Securities

Common shares have the following characteristics:

- **Voting Rights** – Common shareholders may vote on important corporate matters, usually one vote for one share
- **Limited Liability** – The maximum the common shareholders stand to lose if the company goes bankrupt is the amount they have invested in the company and nothing more.
- **Pre-Emptive Right** – Common shareholders have the pre-emptive right to subscribe to new shares of the company in proportion to their ownership in the company. Rights issues enable the existing shareholders to subscribe to new shares of the company usually at a price below the prevailing market price. Existing shareholders can choose to exercise these rights, sell them or allow them to lapse.
- **Participation in Profits** – If the company does well, it may distribute the excess profits to shareholders. Even if no distribution is made, shareholders benefit from an increase in the share price as investors bid up the price in anticipation of high profits.

- **Residual Claim** – If the company goes bankrupt, common shareholders will have residual claim on the assets and income of the company. If assets are liquidated, common shareholders are entitled to receive the proceeds after all other claimants - tax authorities, employees, suppliers, bondholders, other creditors, preferred shareholders - have received theirs.
- **Dividend Payment** – This is not guaranteed and it is up to the discretion of the company whether to pay a dividend in any given year.

3.4.2 Types of Equities

Investors with a slightly higher risk tolerance may consider an investment in equities. There are three major categories within the equities universe, although the last category is not strictly equity until they are converted to equities:

- Common shares
- Preference shares
- Convertible shares

Common shares

Within common stock as an asset class, there are companies which have different characteristics, such as those which provide income and others which are known as “Blue Chips”. Income stocks have a history of consistently paying above-average and relatively stable dividend yields e.g. utilities companies and Real Estate Investment Trusts (REITS).

Such stocks are attractive to pension funds, insurance companies, endowment funds, charitable and educational foundations as well as to conservative investors seeking some exposure to profit growth while at the same time enjoying high dividend pay-out.

Blue chip stocks would cater to *conservative* investors who seek safety and stability of returns as they belong to large well-established financially sound companies with a long track record and consistent earnings. Asset stocks are valued by investors when they trade below their net asset values i.e. (Price per share/Book value per share is less than 1).

Preference Shares

Preference shareholders receive fixed dividends. If the company goes bankrupt, preferred shareholders will be paid ahead of common shareholders but behind bond holders. However, preferred shareholders do not have voting rights.

The fixed dividends may not be paid every year depending on the type of preferred shares. For **cumulative preference shares**, any unpaid dividends are accrued and eventually paid. In contrast, for **non-cumulative preference shares**, there is no

obligation on the part of the issuer to pay the dividends. Hence, in any one year, the company may choose not to pay the preferred dividend.

Preferred shares possess characteristics of both equities and bonds. They have both equity and bond characteristics in that they have an infinite life and pay a fixed dividend. They may also be likened to perpetual bonds which pay a fixed coupon forever. However, if preferred shares are callable, then their life could be finite. Also, unlike bonds, failure to pay dividends on preferred shares in any given year does not constitute a default.



Tips / Key Points

Common Shares vs Preferred Shares

The appeal of common shares as compared to preferred shares lies in the former's ability to provide participation in the profits of a company. Preferred shareholders receive a fixed return in the form of preferred dividends, while common shareholders receive variable dividends depending on how the company is performing.

It is more attractive to invest in a company's common shares than its preferred shares when the company is returning strong profits. For such a company, common shareholders can also expect substantial capital gains, unlike preferred shares which are less volatile.

Where there is concern over its profitability and survival, an investment in preferred shares is safer than an investment in the common stock of the same company. This is because preferred shareholders will be paid ahead of the common shareholders in the event of liquidation. Preferred shareholders also rank ahead of common shareholders in the payment of dividends and in recessionary times, common shareholders may not receive any dividends.

Convertible Securities

A convertible security is a security that can be converted at the option of the investor. For example, a *convertible bond* gives the bond holder a right to convert his holdings into common stock at a certain conversion ratio or price. The holder receives a regular stream of income in the form of coupons, but has the opportunity to take advantage of an increase in the underlying stock's price through conversion.

Convertible preferred shares allow the investor to exchange his preferred shares for a fixed number of common shares. Because the conversion feature allows the investor to capture any upside potential should the underlying common shares rise

in price, such preferred shares usually carry lower dividends than non-convertible preferred shares.

A *warrant* is like a call option. It is issued by a company as a means to raise capital. An investor has the option to convert his warrants to the underlying common shares at a price which is usually higher than the current market price at the time of issue. The advantage of holding a warrant is that it enables the investor to hold a position in a stock through a smaller outlay than the underlying stock's price.

A warrant does not convey ownership rights to the underlying stock and hence the warrant holder will not receive any company pay-outs such as dividends, bonus issues or share buybacks. It does not have voting rights.

Corporate Actions

A corporate action refers to an event that causes material changes to a company and impacts its stakeholders. These stakeholders include bondholders, common and preferred shareholders. These events are approved by a company's board of directors and are usually voted on by common shareholders as well. Corporate actions can influence a company's share price and its returns. Examples of corporate actions are stock splits, dividends, rights issues, private placements and share buybacks.

▪ Stock Split

This divides each of the company's outstanding shares into more shares. For example, a 2-for-1 stock split means one share splits into two. Hence, an investor owning 1,000 shares previously will own 2,000 shares after the stock split. In addition, if the price before the split was \$20, it would be halved to \$10 after the split. The number of outstanding shares will be doubled and the price halved, leaving its market capitalisation unchanged.

The purpose of a stock split is usually to increase *liquidity* in the shares, especially if the share price has climbed up beyond the normal trading range on the exchange. With an increase in the number of outstanding shares, the stock will be made more available to interested buyers. This has the potential to drive the share price up, increasing the company's market capitalisation as well.

▪ Reverse Stock Split

This consolidates shares of a company into small amounts. For example, in a 1-for-10 reverse stock split, an investor owning 1,000 shares will own 100 shares after the reverse split. In addition, if the price before the reverse stock split was \$0.20, it would now be \$2 after the split. Its market capitalisation remains unchanged.

A reverse stock split may be used as a means to shed a company's image as a "penny stock", to drive out small investors, or to return share price to a

normal trading range. This was the case for Citigroup Inc. in 2011 when it entered into a 1-for-10 reverse stock split.

▪ **Stock Dividends**

A company may choose to issue dividends either in the form of cash or stock. For example, if a cash dividend of \$0.35 is declared, an investor owning 1,000 shares will receive \$35 in cash dividends. If a stock dividend of 10% is declared, a shareholder having 1,000 existing shares will receive an additional 100 shares. This has the effect of increasing the company's outstanding shares by 10%, which dilutes the earnings per share and hence, the stock price will decline.

▪ **Rights Issue**

A company may offer its existing shareholders the right to purchase additional shares in proportion to their current shareholdings. The additional shares usually will be priced below the prevailing market price within a specified time frame. The existing shareholders can choose to exercise this right, sell the rights in the stock market, or let the rights lapse.



Tips / Key Points

As compared to bonds, equities typically have higher risks and returns. There are three types of equities: Common shares, preferred shares and convertible securities. While common and preferred shares give an investor ownership rights in a company, convertible securities do not.

Common shares possess higher risk compared to preferred shares but the choice between the two depends on the economic environment, the objectives and constraints of the investor.

The return of equities, like bonds, is derived from two sources:

- Capital gains
- Dividend yield (income)

The motivation for investing in equities is usually to participate in a company's growth (price appreciation), which is the first of the above two components.

Return Calculations

The total return on a stock can be broken down into two components: capital gain and income, as follows:

Total stock return = Capital gains + Dividend yield

Example 1:

You bought ABC stock at \$1.20 per share. During the year, the stock paid a dividend of \$0.18 per share. You sold the stock one year later for \$1.26. What is your total return on the stock?

$$\begin{aligned}
 \text{Total return} &= \text{Capital gains yield} + \text{Dividend yield} \\
 &= \frac{P_1 - P_0}{P_0} + \frac{D_1}{P_0} \\
 &= \frac{\$1.26 - \$1.20}{\$1.20} + \frac{\$0.18}{\$1.20} \\
 &= 0.05 + 0.15 \\
 &= 0.20 \text{ or } 20\%
 \end{aligned}$$

Example 2:

Suppose an investor has held a stock for three years and sells it now at \$8.00 a share. He wishes to calculate his annual holding period return on this stock. He bought the stock for \$7.80 a share and owned 2,000 shares. He has received the following dividends:

Year 1: \$145

Year 2: \$145

Year 3: \$315

His annual holding period return is calculated based on the following time line:

Year 0: $-\$7.80 \times 2,000 = -\$15,600$

Year 1: \$145

Year 2: \$145

Year 3: Dividends = \$315

Return of capital = $\$8.00 \times 2,000 = \$16,000$

Total receipts = $\$315 + \$16,000 = \$16,315$

$$15,600 = \frac{145}{(1 + \text{IRR})} + \frac{145}{(1 + \text{IRR})^2} + \frac{16,315}{(1 + \text{IRR})^3}$$

His annual holding period return is the internal rate of return (IRR). This is solved from the above equation through iteration. A faster way to solve for the IRR is to use a financial calculator. In this example, the investor earned a return of 2.12% p.a. on this investment.

In general, a riskier stock will have a higher required return. Hence, its cash flows would be discounted at a higher discount rate or required return. In the above equation, if we put in required return in place of IRR, we would have an idea of how much the stock should be priced at.

3.4.3 Trading Platforms and Markets

Countries typically have one relatively large exchange, which dominates the market. Examples are the London Stock Exchange, the Frankfurt Stock Exchange, the Tokyo Stock Exchange and the Singapore Exchange. In some cases, there may be regional exchanges, which typically have less stringent listing requirements, as in the case of the US regional exchanges.

The Singapore Exchange Ltd (SGX) operates a number of markets. For the trading of securities, it has the Main Board and the Catalist. The Catalist is an Exchange-regulated but Sponsor-supervised market for fast-growing companies with limited track records. Because of this, Catalist stocks are considered to be riskier investments than their Main Board counterparts.

Investors wanting an exposure to stocks can consider investing either domestically or internationally. Investing internationally enables the investor to diversify into a wider range of assets with different risk return profiles and achieve a net reduction in risk with the potential to achieve higher returns. An exposure to international equities can be achieved through the following routes:

Direct Investment in Foreign Equities

Direct investments in individual stocks may present opportunities to diversify away from a domestically-oriented portfolio. However, investors who invest on their own may incur withholding taxes and significant transaction costs while some countries limit holdings by foreigners hindering direct investments in foreign equities.

American Depositary Receipts (ADRs)

ADRs represent shares of Asian companies. SGX offers ADR trading via its new quotation board, GlobalQuote. ADRs are denominated in US\$ and trade in board lot sizes of 10 ADRs. ADRs also enable investors to buy and hold foreign securities without going to the foreign exchange directly. ADRs also provide greater liquidity as they enable the investor to easily access foreign markets with high investment barriers, such as foreign ownership restrictions and capital controls.

Exchange-Traded Funds (ETFs)

ETFs are setup to mimic a broad-based index like the Straits Times Index (STI) or the Standard and Poor (S&P) 500. Investors who buy an ETF of a foreign market are acquiring an investment in the shares of the underlying foreign index.

Foreign Stocks with Dual Listings

Some stocks have a double listing, in Singapore and their home countries. Dual listing refers to the practice of listing a security on more than one exchange. To the investor, such listings enhance liquidity. Dual listings allow a stock to continue trading in another exchange after one exchange has closed.



Tips / Key Points

Investors wishing to have foreign equities in their portfolio have a number of options:

- Direct investment in foreign equities;
- ADRs representing Asian shares but denominated in US\$ trading on SGX;
- ETFs that mimic broad market indices of foreign markets; and
- Foreign stocks with dual listings in their home countries and on SGX.

Dual listing enables companies to trade their shares in multiple markets in multiple time zones and currencies, improving liquidity and the ability to raise capital.

3.4.4 Fundamental Analysis

Fundamental analysis looks at qualitative and quantitative factors affecting a company. It is grounded in the belief that markets are inefficient and may misprice a security in the short run. Profits can be made when markets eventually price the security correctly. Fundamental analysts attempt to arrive at a "fair price" or "intrinsic value" for a common share.

Before an investment in shares is made, it is important to analyse the broader fundamentals. The traditional approach is the top down approach; the analysis would proceed as follows:

- **Economy** – Expectations are formed regarding economic growth, monetary policy, fiscal policy, supply/demand fundamentals, etc.
- **Industry** – Having identified the economic prospects, this stage involves identifying industries which are poised to benefit from the expected economic scenario and industries which will be worst affected
- **Company** – This involves analysing the qualitative aspects of the company – such as suppliers, buyers, competitive environment – as well as the financial statements of the company.

Another approach that is not as popular is called the bottom up approach. This would be a reverse of the above process. The best companies to invest in are identified, followed by analysis of the industry in which it is in, and lastly analysis of the overall economy in which it operates.

One drawback of this approach is that it can result in a concentrated portfolio, as opposed to a diversified portfolio. Should this approach be adopted, limits are often set in terms of maximum weightage to a given sector or country, in order to achieve sufficient diversification.

Fundamental analysts believe there is a basic **intrinsic value** for the aggregate stock market, the industries and the individual securities. They attempt to determine this intrinsic value based on an assessment of variables comprising a company's financial fundamentals, industry and macroeconomic environment.

If the prevailing market price is below (above) the intrinsic value after transaction costs, the investment recommendation is to buy (sell or not to buy) the undervalued security. There are two major approaches to valuing common shares – Absolute valuation and Relative valuation:

Absolute Valuation

In this approach, a discounted cash flow (DCF) model is used to arrive at a fair value (called intrinsic value) of the stock. This fair value is then compared to the market price of the stock.

Fair value > Market price: Undervalued

Fair value < Market price: Overvalued

Fair value = Market price: Fairly valued

There are three major DCF methods:

- Dividend discount model;
- Free cash flow model comprising two versions, free cash flow to the firm and free cash flow to equity; and
- Residual income model.

The DCF approach is based on a company's fundamentals such as earnings, expected dividends, free cash flows, etc. Hence, the valuation that we arrive at is an absolute measure of fair value. The challenges of DCF models include the assignment of the *correct* discount rate to apply to each company and the forecast of future cash flows.

Relative Valuation

Unlike the method of forecasted fundamentals where value is derived based on the company's fundamentals, relative valuation uses the *method of comparables* using the market price of the industry or peers to calculate the price multiples and apply these to the subject company.

- **Price Multiples with Price in the Numerator** – This approach adopts price multiples such as P/E, PEG, P/B, P/S, P/CF, EV/EBITDA where:

P/E	=	Price Earnings Ratio
PEG	=	P/E to Growth Ratio
P/B	=	Price to Book Value Ratio
P/S	=	Price to Sales Ratio
P/CF	=	Price to Cash Flow Ratio
EV/EBITDA	=	Enterprise Value to Earnings before Interest, Taxes, Depreciation and Amortization

- **P/E Ratio** – Measures how much an investor is willing to pay per dollar of a company's earnings. Earnings may be historically computed or forecasted.

Historical P/E	=	Market Price / Historical Earnings in the past year
Leading P/E	=	Market Price / Forecasted Earnings in the coming year

Where,

Earnings Per Share = Net income / Number of common shares outstanding

And, a company is *overvalued* if:

$$\frac{\text{Market price of company}}{\text{Earnings per share}} > \frac{\text{Market price of peers}}{\text{Earnings per share}}$$

An analyst would prefer a leading P/E measure over the historical P/E. The reason is that the market price should reflect the future earnings of the company. In reality, most of the published P/E data available in the newspapers are on a historical basis.

- **PEG Measure** – This measure relates the P/E ratio of a company to its growth rate. This is calculated as:

$$\text{PEG} = \frac{\text{P/E}}{\text{Growth}}$$

A company may have a higher P/E than another company simply because it is growing at a faster rate. The PEG measure attempts to correct for this. The PEG measure assumes a linear relationship exists between P/E and growth. It ignores the length of the growth period. It does not take account other risks.

- **Price-to-Book Value (P/B)** – This measure relates the price of a share to its book value. This is calculated as:

$$P/B = \frac{\text{Market price per share}}{\text{Book value per share}}$$

Book value is the total value of the company's assets that shareholders will theoretically receive if a company was liquidated. It is the accounting value of the company's assets less all liabilities. Strong companies command large premiums over their book values, indicating that managers of these companies are adding value to the company through profitable investments.

In times of stock market dislocations, turmoil or uncertainty, these companies might sell for below their book values and hence provide good opportunities for investment. Book value does not consider future prospects of the firm because it is only a snapshot taken off the balance sheet. A going concern company should trade at a P/B above 1 if the market properly reflects the future prospects of the company and the upside potential of its share price.

An investor using this metric would look for a company with low P/B compared to its peers or industry average. It is not appropriate to use this measure in comparing companies with significantly different business models.

- **Price-to-Sales Measure** – This relates the market price to sales or revenue for the year. This measure may be used when the company has negative or cyclical earnings. It is not a popular measure as it ignores the cost side of the equation including debt financing. An investor using this measure would look for a company with a low P/S compared to its peers, all other things being equal.
- **Price-to-Cash Flow Measure** – This measure relates market price to the underlying cash flows of the company. Cash flow may be measured by the Earnings before Interest, Tax, Depreciation and Amortisation (EBITDA). Then, this measure will be defined as:

$$P/EBITDA = \frac{\text{Market price per share}}{EBITDA}$$

An investor using this measure would look for a company having low values of these multiples compared to its peers, all other things being equal.

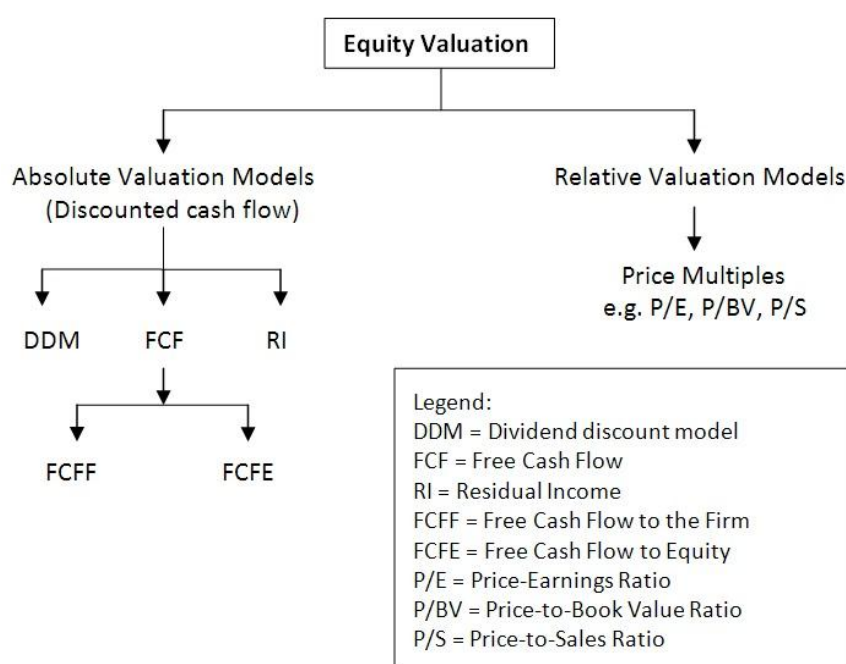
- **EV-to-EBITDA Measure** – This relates the total value of the firm, the enterprise value, to the EBITDA. It is useful for comparing companies with very different capital structures, accounting and depreciation choices, and is rarely negative.

However, because it ignores interest expense, depreciation and amortization, it is inadequate in taking into account the necessary expenditures in net working capital and fixed capital. An investor would

look for a company with a low EV/EBITDA compared to its peers, all other things being equal.

Usually, a combination of the above fundamental approaches is adopted in arriving at a decision on whether to invest in a stock. For example, an analyst can come up with a fair value of the stock using any of the DCF approaches and then supplement this with a combination of price multiples. Exhibit 3.2 summarizes the various methods of equity valuation.

Exhibit 3.10 Methods of Equity Valuation



Tips / Key Points

Two major approaches exist on the valuation of stocks:

- Absolute Valuation – Arrives at "fair value" for a stock. Uses discounted cash flow models.
- Relative Valuation – Arrives at "fair multiple" for a stock relative to peers. Uses price multiples such as P/E, PEG, P/B, P/S and EV/EBITDA.

Usually, a combination of these two approaches are used, and even within a single approach such as the relative valuation approach, it is important to use a variety of measures to determine value.

3.4.5 Technical Analysis

Technical analysis is a method of evaluating securities by studying various forms of pricing and volume information generated by market activity. The most common form of technical analysis is the use of graphical charts to identify patterns that can suggest future price and volume activity. Technical analysts do not seek to calculate the intrinsic value of securities as fundamental analysts do. The art of technical analysis is to try to identify trends or potential changes in trends.

In contrast to fundamental analysts, technical analysts do not require detailed knowledge of the security being analyzed, can apply their analysis across any time frame, and believe that the current price reflects the true value of a security as all information, including investor sentiment, is priced in.

Basic Assumptions of Technical Analysis

The assumptions behind technical analysis include the following:

- Market value or price is determined by supply and demand for a security;
- Supply and demand in turn are governed by rational factors such as the financial statements of the company and irrational factors such as the market sentiments;
- Prices tend to move in trends which can persist over a reasonable length of time;
- Changes in trend are caused by a shift in supply and demand for the security.

Technical Analysis often uses charts or graphs. There are four principal types of charts: Line, Bar, Candlesticks and Point & Figure charts (refer to appendix 3.3 for more details). In charting transacted prices and volumes, technical analysts would try to look for observed patterns. The patterns they look for include:

- **Support and Resistance Levels** – Support levels are levels at which actual or potential buying is sufficient in volume to halt further decline in prices for an appreciable period. Resistance levels are levels at which actual or potential selling is sufficient in volume to halt further increase in prices for an appreciable period.
- **Price Patterns** – Rising and falling trends are usually separated by periods of consolidation which can often be identified with a pattern or formation. In some cases, the consolidation is followed by a reversal in trend. The consolidation pattern may be in the form of rectangular, triangular or other shapes.

Contrarian and Price Momentum Strategies

Investors who adopt a *contrarian* strategy believe the best time to buy is when everyone is selling. Hence, he will attempt to sell when the stock price is near its peak and buy when it is near its trough. This strategy is grounded in the belief that

stock prices are mean reverting and they tend to overshoot their fair values as investors overreact to company news.

At the other extreme, a *price momentum* strategy assumes that stocks that go up in price will continue to do so, while those which are out of favour will remain so. This is based on the *under-reaction* hypothesis, that investors under-react to news that reaches them

Application

Technical Analysis may be applied in conjunction with fundamental analysis or on its own where charts are used to time trade execution and determine the possible entry and exits price points. For example, an investor which wishes to purchase a stock may wait for technical indicators like moving averages to time the purchase. Charts may also be used to determine stop loss and profit-taking price levels.

Technical analysis may also be used to supplement fundamental analysis in deciding on what to buy. For instance, the technical analyst would study the long term price pattern of several assets and pick the one trading near to its historical low to buy.



Tips / Key Points

Technical strategies are based on the assumption of an inefficient market. In technical analysis, investors predict future prices based on the pattern of past price movements. A *contrarian* strategy assumes mean reversion, while a *price momentum* strategy assumes continuation of the same trend in prices.

3.4.6 Stock Market and the Business Cycle

Industries may be classified according to their response to the business cycle:

- **Growth Industries** – E.g. biotechnology. Growth companies experience above average growth in sales and profits, regardless of the business cycle. Their performance is more influenced by secular trends.
- **Defensive Industries** – E.g. household necessities. These companies demonstrate relatively stable performance over a business cycle, as demand for their products is relatively independent of the business cycle
- **Cyclical Industries** – E.g. automotive, items involving discretionary spending. Companies in these industries are sensitive to business cycle, making high profits when the economy is doing well and doing badly in a recession as consumers cut back on their spending.

- **Counter-Cyclical Industries** – E.g. unbranded consumer staples. These comprise stocks whose financial performance is negatively correlated with the state of the economy. Hence, the stock's price will move in the opposite direction to the economy, falling when the economy is expanding and rising when the economy is contracting. Hence, they provide a good hedge in recessionary periods.

Companies may be classified by corporate life cycle stages:

- **Pioneering** – A new start-up;
- **Growth** – A company which is fairly well-established and experiencing growth in sales;
- **Matured** – A company which is well-established; and
- **Declining** – An older company which has slow or no growth and is experiencing declining market share for its products.



Tips / Key Points

Stocks may be variously classified under different categories, depending on the objective of the investor.

Classification by Business Cycle Reaction:

- **Growth Industry** – Stocks within this industry experience growing sales independent of the business cycle.
- **Defensive Industry** – Contain stocks with relatively stable performance throughout the business cycle.
- **Cyclical Industry** – Stocks are highly sensitive to the business cycle, and follow closely the performance of the economy.

Classification by Life Cycle Stage:

- **Pioneer Industry** – Start-up industry with high risks.
- **Growth Industry** – Sales are rapidly increasing, and industry is expanding.
- **Mature Industry** – Stock selection crucial for companies in this stage.
- **Decline** – Generally avoided, as competition is keen and market size is shrinking.

3.4.7 Styles of Investing

Investors may invest in equities with different approaches or “styles”. Common contrasting stock selection styles include:

- Growth vs value stocks;

- Small capitalization vs large capitalization stocks; and
- Sector Rotation.

Growth vs Value Investing

Growth and value investors focus on different aspects of the Price-Earnings Ratio (PER) in deciding whether to add a stock to their portfolio. *Growth-oriented investors* focus on stocks which will have rapid earnings growth in the future. They look for high price-earnings ratios. They expect the stock price to rise in line with the rapid earnings growth.

In contrast, *value-oriented investors* focus on the numerator, that is the price portion of the PER, and look for low ratios. They implicitly assume the PER is below its norm.



Tips / Key Points

Growth investors focus on the future economic story of a company with little regard to its price. Value investors focus on undervaluation of share price and potentially improving company fundamentals.

Small Cap vs Large Cap Investing

The definition of small capitalisation (small cap) stocks differs depending on the market. The MSCI World Small Cap Index currently includes companies whose market value ranges from USD200million to USD1.5billion. One of the attractions of a small cap investment is its huge growth potential which could result in a possible stock price gains as sales and profits rise.

Many large companies started as small cap companies and became large cap due to growth over the years. Small cap companies tend to be high growth companies, which entail more risk in terms of business potential, less access to borrowing facilities and greater vulnerability to economic downturns. Small caps tend to be less researched compared to big caps as institutional investors may be restricted from purchasing them. This also implies greater potential for mispricing.

Another risk relating to small cap companies is key man risk. These companies are often dependent on one or several key leaders who are the original founders who have built the business largely on the basis of personal relationships with customers, suppliers, banks and other stakeholders. Should anything happen to them, the company would be left rudderless.

In contrast, *big cap* companies are generally the well-established companies which have been in the business for some time and are considered less risky.



Tips / Key Points

Small cap stocks are bought for their growth and capital appreciation potential and tend to be more volatile and more risky because of economic disadvantages and vulnerability related to their size.

Large cap stocks are usually less volatile and provide lower returns than small caps but they comprise larger more established companies and are valued for their relative safety compared to small cap stocks.

Sector Rotation

A sector rotation strategy is an active strategy which attempts to correctly identify the sectors which are expected to outperform in the near future.

Sector rotation involves allocating from one sector to another based on a presumption that certain sectors will do better in a given stage of the business cycle than others. Sector rotation will be successful in timing the market only if the next stage of the business cycle can be accurately predicted.

Hence, if an economic upturn is expected, cyclical stocks should be over-weighted in a portfolio, while if a recession is predicted, then defensive stocks might be preferred. If interest rates are expected to fall, an overweighting in interest-sensitive stocks would be appropriate.



Tips / Key Points

Sector rotation is an active investment strategy which involves rotating out of sectors which are expected to underperform and into sectors which are expected to outperform based on a forecast of the stage of the business cycle.

3.4.8 Stock Market Indices

Stock market indices have a wide variety of applications. Besides being used as a benchmark for performance measurement as well as to gauge how stocks in a particular market are performing overall, it is also used in a passively managed portfolio to track the overall market and eliminate unsystematic risk. There are a number of ways to construct a stock market index. These include:

Market Value-Weighted or Capitalisation-Weighted Indices

This is the most popular method. In this category are the Standard & Poor (S&P) 500, Straits Times (STI) and Hang Seng (HSI) Indices. Stocks in these indices are weighted according to their market capitalisation, where market capitalisation is defined as “the price of a stock multiplied by the number of outstanding shares of that stock”.

Stocks with higher market capitalisations will have a larger impact on the stock market index. Hence, a stock with a market cap of \$10 billion will have two times greater weight in such an index than a stock with a market cap of \$5 billion.

Price-Weighted Indices

These indices include the Dow Jones Industrial Average (DJIA), Nikkei 225 (NSA 225). For example the DJIA is computed by summing up all the prices of the 30 companies in the index and dividing by 30 (with adjustments for stock splits). In this sense, stocks in these indices are arithmetic averages of share prices. Higher priced shares will have a larger effect on the overall index.

Equal-Weighted Indices

These indices include the Value Line Index which computes the geometric average of prices giving equal weight to each stock included in the index.

Understanding how stock indices are constructed is important as they are often used as benchmarks for performance measurement. One should be aware that in smaller markets, especially emerging markets, market indices may be dominated by just a few stocks.



Tips / Key Points

Stock market indices are used as benchmarks for performance measurement as well as in the construction of passively-managed index funds. Understanding how these indices are constructed is important in devising an investment strategy.

3.4.9 Investment Strategies

The investor can decide whether to adopt an active investment management strategy or passive strategy.

Passive Investing

If an investor believes that it is difficult to outperform the market, he may adopt a passive approach to investing. Passive investing is carried out by investing in an indexed fund or through Exchange Traded Funds (ETFs) which track a market index.

Active Investing

In contrast, active management is carried out by strategies which aim to capture outperformance (or “alpha”) through superior stock picking skills or market timing abilities. Alpha is the extra return above the benchmark achieved through active investing.

Hybrid Investing Using both Passive and Active Investing

Investors may also have a hybrid strategy using both passive and active strategies. One such approach is a core/satellite strategy. In this strategy, the manager employs a core portfolio, consisting of a diversified set of securities (such as ETFs) and also using another satellite portfolio which consists of opportunistic securities, which can provide outperformance returns or “alpha”.

3.4.10 Vehicles for Passive and Active Equities Investments

Passive Investment

One simple passive approach is having a buy-and-hold strategy using an indexed fund. Another approach is to purchase an exchange-traded fund. These vehicles are elaborated in the section on unit trusts and managed funds

Active Investment

Active investing uses a variety of strategies. These include:

- Direct investment in equities, through careful stock-picking and market-timing using sector rotation, style investing and technical analysis;
- Investing with actively managed unit trusts and hedge fund, investing in domestic equities and/or foreign equities; and
- Employing Contracts for Difference (CFDs).

A CFD is a leveraged instrument which is traded on margin and is available not just for indices, but also for equities and currencies. The investor can either buy or sell short the CFD based on his expectation of the price movement of the underlying asset.

He deposits a margin with the broker, usually 10%-20% for equity, and makes the difference between the opening and closing price of the underlying when he cashes out. The “small” margin implies that the full price of the security is not paid upfront and it is a leveraged transaction.

Leverage Effect of CFD Transactions

John bought \$100,000 worth of shares which then rise by \$3,000. This gives him a return of 3%. If he had bought a CFD on the same shares, with a margin of 10%, his outlay would have been \$10,000 ($\$100,000 \times 10\% = \$10,000$), and he would have made a return of $\$3,000 / \$10,000$ or 30%.



Tips / Key Points

The investor can choose to adopt either an active investment strategy or a passive investment strategy. Investors who adopt an active investment strategy attempt to outperform the overall market by either market timing or astute stock selection.

Alternatively, investors who adopt a passive investment strategy assume the CFD is efficient and it is impossible to 'beat' the market. For these investors, passive investments include ETFs (which provide advantages of liquidity, transparency and diversification in a single transaction), index unit trusts and CFD equity indices.

3.4.11 Best Case / Worst Case Scenario

The maximum loss that may be sustained by a common equity investor is the capital he has invested in the stock and this could occur if the company goes bankrupt. This is unlike derivatives where the investor can lose more than the amount invested.

Further, the equity investor has residual claim to the assets of the company and will be the last to be paid in the event of bankruptcy. Such an event could occur if the economy tips into recession and demand for the company's goods and services dip considerably; or it could be company-specific whereby the company is unable to remain competitive and profitable.

The best case scenario for a common equity investor would be if the company continues to be profitable, year by year. A profitable company may or may not give out dividends as it may need funds to finance their on-going activities. Nevertheless, common shareholders will benefit through the increase in the share price, which is a reflection of the company's future cash flows and profitability.

Generally, in an economic boom, companies tend to do well although company success could also be unrelated to the economic environment but opportunistic situations.

3.5 Chapter Summary

This chapter has covered many aspects of Fixed Income and Equity Securities. The salient points are:

- Money market securities offer investors desiring liquidity an avenue to achieve returns above bank deposit rates at a slightly higher risk. They have maturities of up to one year. Quotes are given in terms of yields rather than price.
- Investors with a slightly longer time horizon who want a regular income may wish to invest in bonds.
- Investment grade bonds, which are bonds rated BBB (S&P rating agency) and above, generally face less default risk than speculative grade bonds, which are those rated below BBB.
- Equities cater to investors who seek capital appreciation and growth and who have a longer time horizon. They possess higher risk than money market securities and bonds, and are correspondingly more volatile.
- There are three major categories within the equities universe, although the last category is not strictly equity until they are converted to equities:
 - Common shares
 - Preference shares
 - Convertible shares
- Within the equity asset class, risk varies widely depending on whether these are emerging market or developed market equities, domestic or international equities.
- Fundamental analysis looks at qualitative and quantitative factors affecting a company. It is grounded in the belief that markets are inefficient and may misprice a security in the short run.
- Technical analysis is a method of evaluating securities by studying various forms of pricing and volume information generated by market activity.
- Investors who believe the stock market is efficient can choose a passive approach and invest in an index fund.
- Investors who believe that they can outperform the market may to invest in an actively managed fund or select stocks themselves, with the intention to outperform the market index.

Appendices

Appendix 3.1: Bond Calculations

Straight Bond Mathematics

A 7% semi-annual coupon bond is selling for \$1,050 in the market. The bond has 2 years remaining to maturity. You decide to buy the bond.

- a) What is the coupon rate?
(Unless otherwise specified, the par value of a bond is assumed to be \$1,000, since the coupon is paid semi-annually)

$$\text{Coupon Rate} = 0.5 \times 7\% \times \$1,000 = \$35$$

- b) What is the yield-to-maturity of the bond?
(The yield-to-maturity is the rate of return you expect to earn on this bond)

You need to solve for YTM in the following equation:

$$\$1,050 = \frac{\$35}{1+YTM} + \frac{\$35}{(1+YTM)^2} + \frac{\$35}{(1+YTM)^3} + \frac{\$35}{(1+YTM)^4} + \frac{\$1,000}{(1+YTM)^4}$$

YTM can be solved by trial and error. If you have a financial calculator, this can be easily solved by entering the following inputs:

PMT = 35 (coupon per period)

FV = 1,000 (par value)

PV = -1,050

N = 4 (four semi-annual compounding periods)

Solve for YTM $\Rightarrow$ 2.18% per semi-annual period $\Rightarrow$ 4.4% p.a.

- c) What is the current yield?

$$\text{Current Yield} = \frac{\text{Annual coupon}}{\text{Price}} = \frac{\$70}{\$1,050} = 6.7\%$$

- d) What is the relationship between coupon rate, current yield and yield-to-maturity? *(Notice that our bond is a premium bond since it is priced above par. Price (\$1,050) > Par value (\$1,000))*

And the relationship between the coupon rate, current yield and yield-to-maturity is as follows:

Premium Bond: Coupon rate (7%) > Current yield (6.7%) > YTM (4.4%)

- e) One year later, interest rates have fallen and you decide to sell the bond. The YTM on your bond has fallen by 1.2%. At what price will your bond sell for?

The YTM has fallen by 1.2%, so it should now be $4.4\% - 1.2\% = 3.2\%$. This figure should be entered into the following formula in its original form without rounding. The bond now has only one year left to maturity, so:

$$P_0 = \frac{\$35}{1+0.016} + \frac{\$35}{(1+0.016)^2} + \frac{\$1,000}{(1+0.016)^2}$$

$$P_0 = \$1,037.11$$

- f) What is your holding period return?

You paid \$1,050 and held the bond for one year, during which period you received two coupons of \$35 each. Then you sold the bond for \$1,037.49. Your holding period return (HPR) should be solved as follows:

$$\$1,050 = \frac{\$35}{1+\text{HPR}} + \frac{\$35}{(1+\text{HPR})^2} + \frac{\$1,037.49}{(1+\text{HPR})^2}$$

HPR can be solved by trial and error, or alternatively by the use of a financial calculator:

PV = - 1,050 (a cash outflow as you paid money to buy the bond)

PMT = 35

FV = 1037.11

N = 2 (two semi-annual periods)

Solve for HPR $\Rightarrow$ 2.728% per semi-annual period $\Rightarrow$ 5.5% p.a.

- g) Why is the holding period return different from the yield-to-maturity when the bond was bought?

HPR $\neq$ YTM because interest rates have fallen. This caused the bond price to increase, enabling the bond holder to reap a capital gain if he sells the bond now. This illustrates the inverse relationship between bond prices and yields.

- h) In the example above, what would be the yield-to-maturity if the bond pays coupons annually?

You need to solve for YTM in the following equation:

$$\$1,050 = \frac{\$70}{1+\text{YTM}} + \frac{\$70}{(1+\text{YTM})^2} + \frac{\$1,000}{(1+\text{YTM})^2}$$

YTM can be solved by trial and error. If you have a financial calculator, this can be easily solved by entering the following inputs:

PMT = 70 (coupon per period)

FV = 1,000 (par value)

P = -1,050

$N = 2$ (two compounding periods)

Solve for YTM $\Rightarrow$ 4.3% p.a.

Convertible Bond Mathematics

Bond XYZ is a convertible bond which pays semi-annual coupons with the following terms:

Par value = \$1,000

Maturity = 8 years

Coupon rate = 3%

Yield to maturity = 4%

Conversion ratio = 60

Market price of XYZ convertible bond = \$1,050

Annual dividend = \$0.20 per share

- a) Calculate the straight value (or investment value) of the bond without the conversion feature.

$$\begin{aligned}\text{Bond Value} &= \$15/(1+0.02) + \$15/(1+0.02)^2 + \dots + \$15/(1+0.02)^{16} + \\ &\quad \$1,000/(1+0.02)^{16} \\ &= \$932.11\end{aligned}$$

- b) Calculate the conversion value assuming XYZ shares are currently trading in the market at \$12 per share.

$$\begin{aligned}\text{Conversion Value} &= \text{Conversion Ratio} \times \text{Share Price} \\ &= 60 \times \$12 = \$720\end{aligned}$$

The minimum price of the convertible bond is the higher of:

- its conversion value (\$720) OR
- its investment value (\$932.11)

XYZ bond's price must sell for at least at \$932.11. If it sold at \$720, the yield would be higher than comparable straight bonds and investors would bid up the price till its yield is 4%.

Conversely, if the conversion value > investment value, and the bond trades below the conversion value, you can buy the bond, convert it to stocks and sell it in the market for \$12, netting an arbitrage profit.

c) Calculate the conversion price.

$$\text{Since Conversion Ratio} = \frac{\text{Par value of XYZ Bond}}{\text{Conversion price}}$$

It follows that:

$$\text{Conversion Price} = \frac{\text{Par value of XYZ Bond}}{\text{Conversion ratio}} = \frac{\$1,000}{60} = \$16.67$$

d) Calculate the conversion parity price.

$$\text{Conversion Parity Price} = \frac{\text{Market price of XYZ Bond}}{\text{Conversion ratio}} = \frac{\$1,050}{60} = \$17.50$$

If the investor converts the bond, he is effectively paying \$17.50 for the company's common shares. This is the breakeven price.

Appendix 3.2: Valuation of Stocks

Dividend Discount Model

The company's future dividends are discounted by the cost of equity to find the fair value per share of the stock;

$$P_0 = \frac{D_1}{1+r_{CE}} + \frac{D_2}{(1+r_{CE})^2} + \frac{D_3}{(1+r_{CE})^3} + \dots + \frac{D_{\infty}}{(1+r_{CE})^{\infty}}$$

Where,

P_0 = Intrinsic value (or fair value) per share (what the stock is worth now)

D_1 = Dividend per share at the end of Year 1

D_2 = Dividend per share at the end of Year 2

r_{CE} = Cost of equity (or required return to equity)

Although this looks insoluble, generally a company which is growing rapidly will slow down to a final sustainable growth phase which continues forever. In a mature company, growth is assumed to continue at the same rate g as the economy's nominal growth rate forever, and the model can be simplified as follows:

$$P_0 = \frac{D_1}{r_{CE} - g}$$

Hence, if the company is currently growing faster than the economy's nominal growth rate, and is expected to slow down to a sustainable growth rate g in year n , then the stock would be valued as follows:

$$P_0 = \frac{D_1}{1+r_{CE}} + \frac{D_2}{(1+r_{CE})^2} + \frac{D_3}{(1+r_{CE})^3} + \dots + \frac{D_n}{(1+r_{CE})^n} + \frac{D_{n+1}}{(r_{CE}-g)} \frac{1}{(1+r_{CE})^n}$$

High growth phase Constant growth phase

This is a two-stage model where the initial high growth phase is followed by a mature constant growth phase which lasts forever.

For preferred stock, since the dividend is fixed and does not grow ($g = 0$), we can condense the following equation to arrive at intrinsic value per share of preferred stock:

$$P_0 = \frac{\text{Dividend}}{r_{CF}}$$

Free Cash Flow Model

There are two forms of the Free Cash Flow model – the Free Cash Flow to Equity (FCFE) Model and the Free Cash Flow to the Firm (FCFF) Model. Under the FCFE approach, the company's future FCFEs is discounted by the cost of equity to obtain the present value of equity. This is then divided by the number of common shares outstanding, to come up with a fair value estimate of a share of common stock.

$$\text{Value of Equity}_0 = \frac{\text{FCFE}_1}{1+r_{\text{CF}}} + \frac{\text{FCFE}_2}{(1+r_{\text{CF}})^2} + \frac{\text{FCFE}_3}{(1+r_{\text{CF}})^3} + \dots + \frac{\text{FCFE}_\infty}{(1+r_{\text{CF}})^\infty}$$

If the FCFE grows at a constant rate g forever, we can summarise the equation as:

$$\text{Value of Equity}_0 = \frac{\text{FCFE}_1}{r_{CE} - g}$$

If the company is in the supernormal growth phase, and we expect growth to slow to a sustainable rate g in year n , then:

$$\text{Value of Equity}_0 = \frac{\text{FCFE}_1}{1+r_{CF}} + \frac{\text{FCFE}_2}{(1+r_{CF})^2} + \dots + \frac{\text{FCFE}_n}{(1+r_{CF})^3} + \frac{\text{FCFE}_{n+1}}{(r_{CF}-g)} + \frac{1}{(1+r_{CF})^n}$$

$$\text{Intrinsic Value per Share} = \frac{\text{Value of Equity}_0}{\text{Number of Common Shares outstanding}}$$

Under the FCFF approach, the analyst discounts the FCFF by the Weighted Average Cost of Capital (WACC), and then subtracts the market value of debt to get the value of equity. This is then divided by the number of common shares outstanding to get the fair value per share.

$$\text{Value of Firm}_0 = \frac{\text{FCFF}_1}{1+\text{WACC}} + \frac{\text{FCFF}_2}{(1+\text{WACC})^2} + \frac{\text{FCFF}_3}{(1+\text{WACC})^3} + \dots + \frac{\text{FCFF}_\infty}{(1+\text{WACC})^\infty}$$

If the free cash flow to the firm grows at a constant rate g forever, we can summarise this equation into:

$$\text{Value of Firm}_0 = \frac{\text{FCFF}_1}{\text{WACC}-g}$$

If the company is in the supernormal growth phase, and we expect growth to slow to a sustainable rate g in year n , then:

$$\text{Value of Firm}_0 = \frac{\text{FCFF}_1}{1+\text{WACC}} + \frac{\text{FCFF}_2}{(1+\text{WACC})^2} + \dots + \frac{\text{FCFF}_n}{(1+\text{WACC})^n} + \frac{\text{FCFF}_{n+1}}{(\text{WACC}-g)} \frac{1}{(1+\text{WACC})^n}$$

$$\text{Value of Equity}_0 = \text{Value of Firm}_0 - \text{Market value of Equity}_{0V}$$

$$\text{Intrinsic Value per Share} = \frac{\text{Value of Equity}_0}{\text{Number of Common Shares outstanding}}$$

Residual Income Model

This method takes the beginning book value per share and adds the future value of residual income, where residual income is the amount of net income left over after subtracting the required return to equity.

In other words, residual income measures "abnormal" profit, or profit earned over and above the shareholders' required return. A company with a positive residual income will be adding value to the company and its Price/Book Value ratio will be above 1.

Intrinsic Value per Share = $B_0 + \text{PV of Residual Income}$

Where,

B_0 = Beginning book value of equity per share

Residual income = Net Income – Equity charge

Equity charge = Beginning book value of equity per share x cost of equity

Appendix 3.3: Technical Analysis

Types of Charts

- **Line Chart** – Displays a line by joining the closing prices of the security for each time frame. The time frame displayed in a chart may be as short as one minute or a fraction of it, to a time period of one month or even one year. The most common time frame used is for one day.



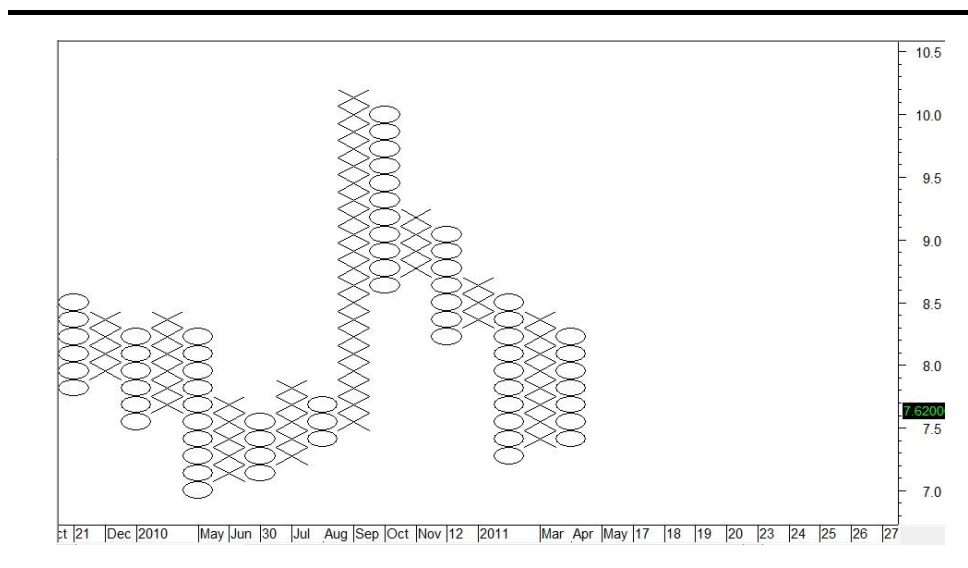
- **Bar Chart** – Each bar shows the opening, closing, high and low prices of the security for each time frame. The vertical bar show the range of prices traded during the time frame. The extremities of this bar represent the high and low price. The opening price is represented by the short horizontal line on the left of the vertical bar. The closing price is represented by the short horizontal line on the right of the vertical bar.



- **Candlestick Chart** – each candle displays the opening, closing, high and low prices of the security for each time frame. A white or transparent candle would mean that opening price is lower than the closing price. A black or solid candle would mean that the opening price is higher than the closing price. The upper and lower extremities of the candle, called the shadow, represent the high and low price of the security.



- **Point and Figure Chart** – This method focuses on the closing price of each time frame and the establishment of a unit of price in order for the movement to be plotted on the chart. This is intended to filter out insignificant price movements. Point and figure charts do not have a time axis. It only has a price axis and the “X”s and “O”s are always plotted in separate columns. Rising stock prices are shown with “X”s and falling prices are shown with “O”s. The chartist must also determine how much the price must move in the opposite direction for the chart to begin a new column.



Bar charts and candlestick charts are more commonly used than other types of charts. The figures below show the four types of charts described above for the same security based on a daily time frame.

Moving Averages

The objective of a moving average is to smooth price data so as to reduce whipsaws, and to display the underlying trend. Moving averages can be computed using equally-weighted price data (plainly known as moving average), or front-weighted price data (exponential moving average).

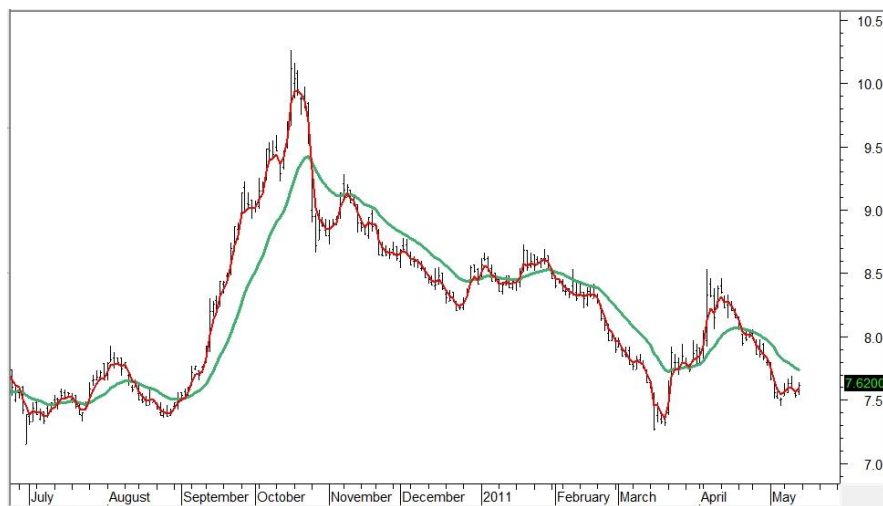
Crossovers between price and a short term moving average with a medium to longer term moving average are generally used as buy or sell signals. Moving averages could be used as dynamic support or resistance levels. The moving average line is plotted as an overlay on price.

Moving averages may be applied in a variety of ways:

- **Single Moving Average** – Mostly to represent the underlying trend.



- **Pair of Moving Averages** – Shorter term moving average to represent price and the medium or longer term moving average to represent the underlying trend.



Momentum Indicators

Momentum indicators are supplementary techniques that help support the reversal signals from price patterns. They are usually plotted below the price chart. They measure the velocity of price movements. These indicators can potentially give

advance warning when a reversal in price is about to take place. Some or all of the following principles apply to the momentum indicators discussed in this section.

Examples of momentum indicators include relative strength index, rate of change, moving average convergence divergence and stochastic. The five momentum principles discussed in this section are:

- **Overbought / Oversold Condition** – All momentum indicators have a line or pairs of lines that will move from one extreme to another. These extremes are known as the overbought zone and oversold zone, with an equilibrium line in between the two zones.

The momentum indicator may stay in the overbought or oversold zone for some time, so it will be prudent for the chartist to wait for other signals instead of jumping to a conclusion to sell (buy) the security the moment it hits the overbought (oversold) zone.

In the diagram below, the Relative Strength Index (RSI) shows an overbought position at point A in September and remains within the overbought zone until October when the indicator crosses point B. The security should only be sold at or after B. The RSI shows an oversold position at point C in February and remains within the oversold zone until March when the indicator crosses point D. The security should only be bought at or after D.



- **Divergences** – The price and momentum oscillator may not always move in tandem. It is possible that while price is trending up, the momentum oscillator is trending down. This is known as a negative divergence. The weakening momentum is an early warning signal of underlying weakness in the price trend.

Likewise, when the price is trending down and the momentum oscillator is strengthening, there is a positive divergence. No matter how significant the divergence may appear, it must be confirmed by price breaking its trend.

- **Trendline Violations** – Momentum moves in trends and some of the techniques of analyzing prices may be applied to momentum oscillators. For instance, if we see a reversal in trend for momentum, a similar reversal in price is not far away.
- **Price Pattern Completion** – Momentum oscillators can exhibit price patterns and they usually provide a leading indicator to potential pattern completion in price.
- **Moving Average Signals** – Momentum oscillators can be volatile and jagged and hence using one or two moving average computations to smooth them may produce a more reliable signal.

Momentum indicators are plotted using charting software and most practitioners would use the default settings provided. The indicator is plotted using the same time period as the price chart and is usually placed below the price chart.

Types of Momentum Indicators

- **Relative Strength Indicator (RSI)** – RSI measures the price velocity of a security relative to its past performance and not the market. The indicator is calibrated from 0 to 100. It uses a 14-period default and the overbought level is set at 70 while the oversold level is at 30. The first 4 momentum principles discussed in the earlier section applies to RSI.
- **Rate of Change (ROC)** – ROC compares the price today (or the most recent price) with the price n periods ago. Similar to the RSI, it is a relative measure of the price of the security to itself. Instead of having a standardized overbought and oversold level, ROC has an equilibrium level of 0 (similar to where the 50 level is for RSI).

The overbought and oversold levels are constructed for the ROC on a discretionary basis by the chartist. Both of these levels are equidistant from the 0 level. It should be drawn such that it encompasses the major rallies and sell-offs but exclude minor price fluctuations.

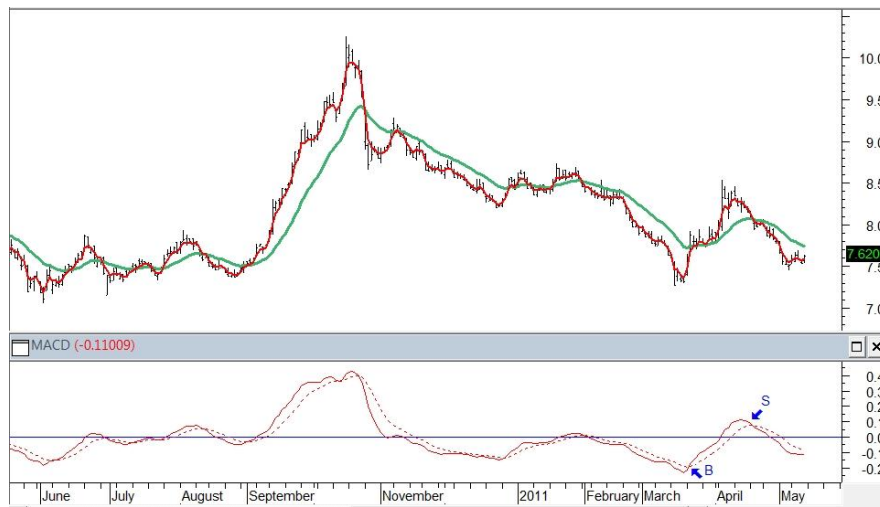
The other momentum principles such as divergences, trendline violations and price pattern completion apply to ROC.



- **Moving Average Convergence Divergence (MACD)** – The MACD indicator line is calculated by subtracting the value of a 26-period EMA from a 12-period EMA. A positive value is shown when the 12-period EMA exceeds the 26-period EMA – signalling that the price is increasing at a faster pace the higher the MACD line goes.

Most charting software will use this 26 and 12 period default. A 9-period EMA of the indicator line (the dotted "signal line") is displayed on top of the MACD indicator line. The settings for the signal line can be changed. A buy signal is triggered when the MACD indicator line goes above the signal line (point B in the diagram below) and a sell signal is triggered when the MACD indicator line goes below the signal line (point S in the diagram below).

The signal derived from the pair of EMA lines gives the MACD indicator the advantage of emphasizing peaks and troughs in the price pattern. However it is difficult to identify divergences, price pattern formations and trendline violations discussed above. Hence the MACD indicator should be used together with other momentum oscillators such as the RSI or ROC.



- **Stochastic** – The stochastic concept assumes that prices tend to close near the upper part of the trading range for a time period during an uptrend. The converse is true for a downtrend. The Stochastic Oscillator shown by the solid line in the bottom panel below, known as the %K line, always ranges between 0% and 100%.

A reading of 0% shows that the security's close was the lowest price that the security has traded during the preceding x-time periods. A reading of 100% shows that the security's close was the highest price that the security has traded during the preceding x-time periods. The dotted line in the bottom panel below, known as the %D line, is a moving average of the %K line.

Stochastic signals include divergences with the corresponding price pattern and crossovers of the %K and %D lines.

Chapter 4:

Options, Futures, Foreign Exchange and Swap Contracts

Objective

The candidate should be able to:

- ✓ Explain the key features of options, warrants, futures, foreign exchange and swap contracts
- ✓ Discuss the payoff and profit structure of options
- ✓ Describe the gain, loss and breakeven position of options, futures and foreign exchange contracts
- ✓ Distinguish the difference between equity, bonds and currency options
- ✓ Discuss the benefits of using options, futures and foreign exchange contracts
- ✓ Explain the risks in trading options, futures, foreign exchange and swap contracts
- ✓ Explain interest rate caps and floors
- ✓ Distinguish the difference between exchange-traded and OTC-traded options
- ✓ Explain an option, futures and foreign exchange term sheet
- ✓ Discuss the structure of the futures market
- ✓ Describe the mechanics of futures trading
- ✓ Describe the key drivers of currency movement
- ✓ Discuss foreign exchange rate quotations
- ✓ Discuss forward foreign exchange contracts
- ✓ Distinguish the difference between interest rate and cross-currency interest rate swaps

4.1 Introduction

Instead of buying fixed income securities or equities directly, investors can consider derivatives of equities and fixed income. Equity derivatives include securities such as stock options or stock index options while fixed income derivatives include securities like interest rate options.

In this chapter we will discuss the different types of derivatives available to investors, their key features, benefits, risks, and the differences between exchange traded and over-the-counter traded derivatives.

4.2 Equity Options

4.2.1 Key Features of Equity Options

Equity options are the most common type of equity derivatives. They are typically used by investors to hedge their equity positions and to take advantage of the volatility of the underlying equity asset.

An **option** is a financial security that gives the buyer of the option the right but not the obligation to buy or sell an asset at a pre-agreed price within a specific period of time. Options can be either American or European. American options may be exercised any time up to and including the expiration date. Options that may be exercised only on the expiration date itself are referred to as European options.

The person who sells the option is called the option writer. The price of the option is called the option premium. It represents the compensation the purchaser of the option must pay for the ability to exercise the option if it becomes profitable. The price at which the underlying asset may be bought or sold is known as the exercise price or strike price. The date after which an option is useless is the expiration or maturity date.

Rights of the Option Buyer

From the option buyer's perspective, options have value and are assets as they represent a right to buy or sell an asset at a set price within a specified period of time. The option buyer may, at his discretion, choose not to exercise this right. This usually happens if the option is out of the money or if the buyer is unable or unwilling to take physical delivery of the underlying asset.

Obligations of the Option Writer

From the option writer's perspective, options are contingent liabilities as they represent an obligation to sell or buy an asset at a set price within a specified period of time should the option buyer exercise his right. Since option writers are obligated to honour the liabilities, they will have to be prepared to deliver the underlying asset

(regardless if it's a physical or financial asset) or close off their position by buying a separate option which has terms that match the writer's liabilities exactly.

4.2.2 Call Options and Put Options

There are two basic types of options: the call option and the put option. Both put and calls are created by sellers who write a particular contract.

Call Options

A call option gives the holder the right to buy an asset for a specified price on or before a specific date. The holder of a call has the right to exercise the option if the market value of the security to be purchased exceeds the exercise price.

The option holder may either sell the option, or "call away" the security at the exercise price and reap a profit from immediately selling the security. Otherwise, the option may be left unexercised. If it is not exercised on or before the expiration date of the contract, a call option simply expires and no longer has value. In general, investors buy call options if they hold a bullish view that the price of the asset will go up.

The call option writer receives the premium income as payment against the possibility that he will be forced at some later date to deliver the security in return for an exercise price lower than the market value of security. If the option is left to expire because the exercise price remains above the market price of the security, then the call option writer makes a profit equal to the premium income derived from the initial sale of the option minus transaction costs.

Put Options

A put option gives the holder the right to sell an asset for a specified price on or before a specific date. While profits on call options increase when the security increases in value, profits on put options increase when the security value falls. The put will only be exercised if the market value of the security to be sold is less than the exercise price.

Similar to a call option, a put option has no value if it is not exercised on or before the expiration date. In general, investors buy put options if they hold a bearish view that the price of the asset will go down.

4.2.3 In the Money and Out of the Money

An option is said to be in the money when its exercise would produce profits for the holder. An option is out of the money when exercise would be unprofitable.

A call option is in the money when the exercise price is below the asset's value because a purchase at the exercise price would be profitable. It is out of the money when the exercise price exceeds the asset value as no one would want to buy an asset at a price higher than the market price. Conversely, put options are in the money when the exercise price exceeds the asset value because delivery of the lower-valued asset in exchange for the exercise price would be profitable.

4.2.4 Payoffs and Profits from Option Positions

The potential payoffs and profits of option positions depend on a few factors:

- The intrinsic and time value of the options;
- Whether it's a long or short position; and
- Whether it's a call or put position.

Intrinsic Value of Call Options

The intrinsic value of a call option is the price of the underlying asset less the exercise price of the option and benefit, if the asset price is above the exercise price. If the asset price is at or below the exercise price, the intrinsic value of the option is zero.

$$\text{Call Intrinsic Value} = \begin{cases} S_T - X & \text{if } S_T > X \text{ or} \\ 0 & \text{if } S_T \leq X \end{cases}$$

Where,

S_T = Value of the underlying asset

X = Exercise price of the asset

Example

If the exercise price of a call option is \$50 and the current stock price is \$54, the intrinsic value is \$4. On the other hand, if the share price declines to \$48, the intrinsic value of the call option is zero.

Intrinsic Value of Put Options

The intrinsic value of a put option is the exercise price of the option less the price of the underlying asset, if the asset price is below the exercise price. If the asset price is at or above the exercise price, the intrinsic value of the option is zero.

$$\begin{aligned} \text{Put Intrinsic Value} &= X - S_T \text{ if } S_T < X \text{ or} \\ &0 \quad \text{if } S_T \geq X \end{aligned}$$

Where,

S_T = Value of the underlying asset

X = Exercise price of the asset

Example

Information on the call and put option for a particular stock is given below. If the stock is currently priced at \$40, what is the intrinsic value and time value for both options?

	Expiration	Exercise Price	Market Price
Call Option	6-months	\$37.50	\$3.50
Put Option	6-months	\$40.00	\$2.80

Intrinsic value of Call option = \$40 - \$37.50 = \$2.50

Time value of Call option = \$3.50 - \$2.50 = \$1.00

Intrinsic value of Put option = \$40 - \$40 = \$0

Time value of Put option = \$2.80 - \$0 = \$2.80

Time Value of Options

In general, the price of an option will be greater than its intrinsic value before the expiration date. This difference between the actual option price and its intrinsic value is called the time value of the option.

$$\text{Time Value of Option} = \text{Actual Option Price} - \text{Intrinsic Value}$$

The option buyer hopes that some time before expiration of the option, there is a chance for capital gains. The longer the time to expiration of the option, the greater the chance it has to appreciate. Except for some very unusual market conditions, the time value of an option is generally positive. Only at expiration will the time value be zero.

Since time value of the option is zero at expiration, the value of the option at expiration is exactly its intrinsic value. The intrinsic value of the option is the minimum value of the option. However, is there an upper price boundary?

Since the value of the call option is based on the price of the underlying asset, the option can never sell for more than the underlying asset itself. The maximum price for a call option is thus the price of the underlying asset. In the case of a put option, the exercise price is its maximum intrinsic value.

An option almost never trades below its intrinsic value. If it does, arbitrageurs could realise immediate trading profits by purchasing the options and exercising them.

Determinant of Stock Option Price

There are various Option Pricing Models. Fischer Black and Myron Scholes developed a theoretical option pricing formula to determine value of European options in a classic article published in 1973. This is explained in Appendix 4.1.

Essentially, there are six factors influencing the option price:

- **Price of Underlying Stock** - The call option should increase in value as the price of the underlying stock increases (all things being equal). The reverse is true for a put option. As the price of the underlying stock increases, the price of the put option decreases.
- **Exercise Price** – All other factors being equal, the lower the exercise price, the higher will be the price of a call option. For a put option, the higher the exercise price, the higher will be the put option value.
- **Time to Expiration** – The longer the time to expiration, the greater will be the price of the option. The reason is that as the time to expiration decreases, the lesser the remaining time for the underlying stock price to rise or fall. As the time to expiration decreases, the option value approaches the intrinsic value.
- **Price Volatility of the Underlying Stock** – The greater the volatility of the underlying stock price, the higher will be the price of the option because of the increased probability for the price of the underlying stock to move in favour of the option buyer.
- **Risk-Free Interest Rate** – Call option values are affected by the level of interest rate because of the opportunity cost involved. Buying the underlying stock ties up one's money which can otherwise be available for investment at the risk-free interest rate. Consequently, holding all other factors constant, the higher the interest rate, the greater the opportunity cost of direct purchase of the underlying stock. Hence there is a positive relationship between interest rate and the call option value.
- **Expected Dividends During the Life of the Option** – Dividends have the effect of reducing the price of the underlying stock when it goes ex-dividend. A drop in the stock price will cause the call option value to decline. For put options, the reverse is true.

The following table summarises the relationship of these factors to the stock option price:

Price Sensitivity of Options		
<u>Option Price Sensitivity</u>	<u>Call Option</u>	<u>Put Option</u>
Price of underlying stock ↑	↑	↓
Exercise Price ↑	↓	↑
Time to expiration ↑	↑	↑
Volatility of underlying stock ↑	↑	↑
Risk-free interest rate ↑	↑	↓
Expected dividends ↑	↓	↑

Option Volatility

As long as there is value in an option, there is volatility. Managing this risk involves monitoring various “sensitivities” of the options, also known as “Greeks”. Each type of risk is represented by a Greek letter with the exception of Vega.

The sensitivities are expressed as a change in a parameter as function of a change in a second parameter:

- **Delta** – Change in option price vs change in the price of the underlying asset
- **Gamma** – Change in delta vs change in the price of the underlying asset
- **Theta** – Change in option price vs change in time to maturity
- **Vega** – Change in option price vs change in volatility
- **Rho** – Change in option price vs change in interest rates

4.2.5 Maximum Gain, Maximum Loss and Breakeven at Expiration

There are four basic option positions:

- Buying a call option (long a call option)
- Selling a call option (short a call option)
- Buying a put option (long a put option)
- Selling a put option (short a put option)

Buying Call Options

At expiration, the payoff to the call holder is the *intrinsic value* of the call option:

$$\text{Payoff to call holder} = \begin{cases} S_T - X & \text{if } S_T > X \text{ or} \\ 0 & \text{if } S_T \leq X \end{cases}$$

Where,

S_T = Value of the underlying asset

X = Exercise price of the asset

Example

Suppose an investor buys a call option on stock A that expires in one month and has an exercise price of \$100. If the stock price is \$110 at expiration date, the buyer can exercise his option to buy the stock at \$100 and simultaneously sell the shares at the market price of \$110 deriving a payoff of \$10.

On the other hand, if the shares sell below the \$100, the buyer can allow the option to expire.

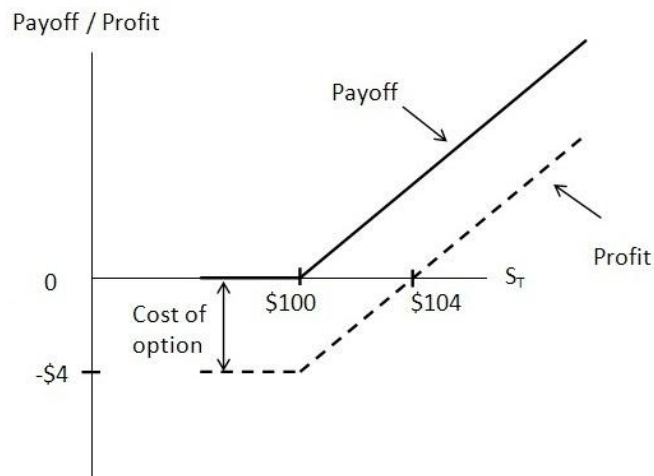
Assuming the price of the call option is \$4 and the option's exercise price is \$100, if the underlying is at \$110, the payoff to the option buyer is \$10 but the net profit after deducting the option price is \$6. Therefore the **maximum gain** to a call option holder is the payoff of the option less the price paid for it. This can, theoretically, be unlimited.

The payoff cannot be negative; meaning that the option is exercised only if S_T exceeds X . If S_T is less than X , exercise does not occur and the option expires with zero value. However the option buyer would have already paid for the option. Therefore the **maximum loss** is the price for the option.

The call option holder only breaks even, i.e. net profit equals zero if the price of the underlying rises above the exercise price by an amount equal to the price of the option. For example, if the price of the call option is \$4 and the option's exercise price is \$100, the **breakeven** point for the call option buyer is when the price of the underlying is exactly \$104.

The payoff and profit of the above example is depicted in the figure below. The solid line depicts the value of the call at maturity while the broken line depicts the net profit to the option holder. Exhibit 4.1 shows the payoff and profit diagram for the above example.

Exhibit 4.1 Option Payoff – Buying a Call Option


Selling Call Options

The writer of the call option incurs losses if the stock price increases. The writer will receive a call and will be obliged to deliver a stock worth S_T for only X dollars. The payoff to the call writer is exactly the mirror image of the call holder:

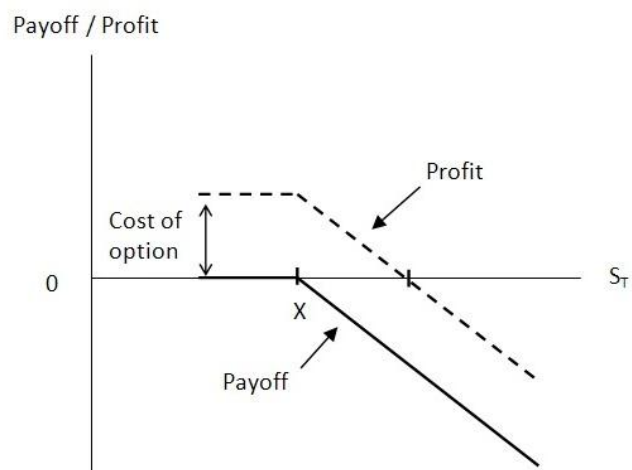
$$\begin{aligned} \text{Payoff to call writer} &= -(S_T - X) && \text{if } S_T > X \text{ or} \\ &= 0 && \text{if } S_T \leq X \end{aligned}$$

Where,

S_T = Value of the underlying asset

X = Exercise price of the asset

Exhibit 4.2 Option Payoff – Selling a Call Option



The **maximum gain** for a call writer is simply the price of the option that he had sold. This only happens in the situation when the price of the underlying is equal to or less than the option's exercise price.

The **maximum loss** for a call option writer is, theoretically, unlimited. This happens when the price of the underlying rises above the exercise price plus the price of the option; the higher the price of the underlying, the greater the loss to the option writer.

Why then would investors write call options if they had to bear unlimited liability when the price of the underlying asset were to increase? That is because in their view, they expect the price of the underlying asset to fall or to remain flat. If this happens, call holders will have no choice but to allow the option to expire, thereby allowing call writers to keep the call premium as profit.

However, if the price of the underlying asset were to increase, the call writer can offset his position and limit his liability by buying an identical contract in the market. But since the market has moved, the call writer will have to purchase the call option at a higher price than the option premium that he received.

The call option writer breaks even, i.e. net profit equals zero if the price of the underlying rises above the exercise price by an amount equal to the price of the option. Note that the **breakeven** point for both the call option buyer and call option writer is exactly the same.

Buying Put Options

At expiration, the payoff to the put holder is the *intrinsic value* of the put option:

$$\text{Put Intrinsic Value} = \begin{cases} X - S_T & \text{if } S_T < X \text{ or} \\ 0 & \text{if } S_T \geq X \end{cases}$$

Where,

S_T = Value of the underlying asset

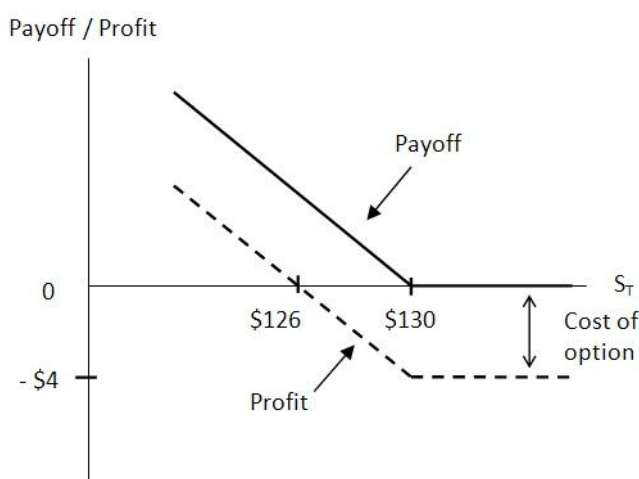
X = Exercise price of the asset

A put option conveys the right to sell an asset at the exercise price. In this case, the holder will not exercise the option unless the asset sells for less than the exercise price.

Example

If stock B were to fall to \$120, a put option with an exercise price of \$130 could be exercised to give a \$10 payoff to its holder. The holder would purchase stock B for \$120 and simultaneously deliver it to the put option writer for the exercise price of \$130. If the put premium is \$4, the profit to the holder would then be \$6.

Exhibit 4.3 shows the payoff and profit diagram for the above example:

Exhibit 4.3 Option Payoff – Buying a Put Option

The solid line illustrates the payoff at maturity to the put option holder. If the stock price at expiration is above the exercise price, the put has no value. If the stock price falls below the exercise price, the put value at expiration increases. The broken line represents the put option holder's profit at expiration, net of the initial cost of the put.

Therefore, similar to buying call options, the **maximum gain** to a put option holder is the payoff of the option less the price paid for it. The further the price of the underlying falls below the exercise price, the greater the put option's payoff.

The payoff cannot be negative; meaning that the option is exercised only if S_T falls below X . If S_T is above X , then exercise does not occur and the option expires with zero value. However the option buyer would have already paid for the option. Therefore the **maximum loss** is the price for the option.

The put option holder only breaks even i.e. net profit equals zero if the price of the underlying falls below the exercise price by an amount equal to the price of the option. Using example 4 above, if the price of the call option is \$4 and the option's

exercise price is \$130, the **breakeven** point for the put option buyer is when the price of the underlying is exactly \$126.

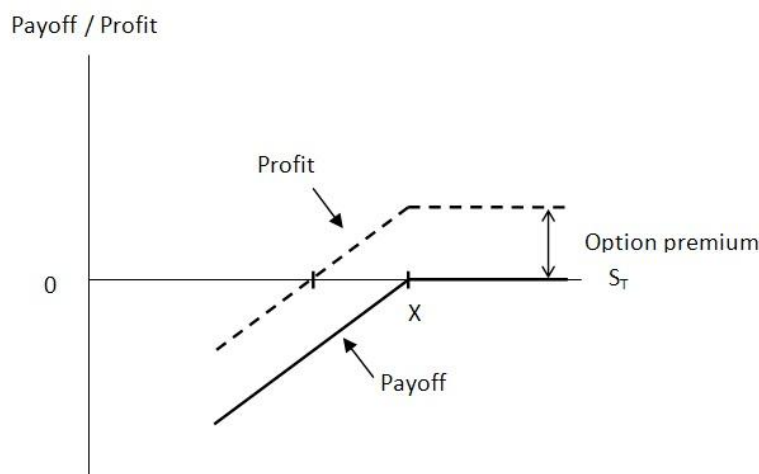
Selling Put Options

The put writer retains the option premium if the stock price rises above the exercise price, and loses if the stock price is less than the exercise price. The payoff to a put writer at expiration is:

$$\text{Payoff to put writer} = \begin{cases} -(X - S_T) & \text{if } S_T < X \text{ or} \\ 0 & \text{if } S_T \geq X \end{cases}$$

The maximum profit to be realised by the put writer is the option premium. The loss can be substantial should the share price drop significantly below the exercise price. Similar to the call writer, the put writer can offset his position by buying an identical contract in the market.

Exhibit 4.4 Option Payoff – Selling a Put Option



The **maximum gain** for a put option writer is equal to the price of the option that he sold. This only happens in the situation when the price of the underlying is equal to or greater than the option's exercise price. The **maximum loss** for a put option writer can be significant; the lower the price of the underlying, the greater the loss to the option writer.

Put option writers are willing to bear the risk of significant losses because they expect the price of the underlying asset to rise or to remain flat. If this happens, put option buyers will then let the option expire and the put writers can keep the put premium as profit. Put option writers can limit their losses by taking an offsetting position by buying an identical put option in the market.

The put option writer breaks even, i.e. net profit equals zero if the price of the underlying falls below the exercise price by an amount that is equal to the price of the option. Note that the **breakeven** point for both the put option buyer and put option writer is exactly the same.

4.2.6 Benefits of Using Options

Options allow risk-averse market participants to shift risk to those who willingly accept it. The first group of risk-averse investors are called hedgers. The second group of investors, who are prepared to take on the risk by speculating on the price movement of the underlying security, are called “speculators”.

Hedging

Hedging is similar to taking out an insurance policy as the gains and losses on the contract will eliminate some existing risk in the portfolio. Depending on his investment objective, an investor can use a variety of hedging strategies including protective puts. Every hedge has a cost (i.e. the option premium) and the savvy investor will need to consider if the benefit of downside risk protection justifies the expense.

Hedging is similar to buying insurance but it is not as precise. Insurance pay-outs are clearly defined when certain pre-agreed conditions are met while hedging is an inexact science. A perfect hedge, where an investor's risks are entirely eliminated, is hard to achieve as the hedger needs to take on positions which will fully cover all his potential losses.

Speculation

Speculators take positions against hedgers and bear the risks that the hedgers try to avoid. They may buy or sell puts and calls. Speculators would only gain if they are successful in predicting the price movements of the underlying asset. If their predictions are incorrect, they may incur severe losses.

The margin requirement for options is usually a small fraction of the underlying asset value, which means that there is a leverage effect, i.e. there is a large market exposure with only a small amount of initial cash invested. Leverage may amplify potential gains, but more importantly, it may also lead to large potential losses for the speculator.

Example

An investor owns shares of Company C which operates in the banking industry. He believes in the long term potential of the company and wants to hold the stock for the long run. However he is worried about short term losses in the banking industry. To protect himself from downside risk, he buys a put option.

In the event that the stock price of company C drops below the exercise price, the losses in his long term portfolio will be offset by the gain in his put option.

4.2.7 Risks in Trading Options

Market participants assume many different types of risks when trading options. It is important to understand the risks involved and attempt to manage or diversify them away.

Risks Assumed By Option Holders

Option holders run the risk of losing the entire premium paid for the option in a relatively short period of time. This is because options constantly lose time value and become completely worthless at expiration. An option holder who doesn't sell his option or who doesn't exercise his option prior to expiration will lose the option premium paid. This risk increases if the option is out of the money as expiration draws near.

European style options can only realise their value upon expiration. This lack of flexibility compared to American options increases the difficulty of making profits. This is because not only does the option holder need to correctly predict the price movement of the underlying asset; he also needs to correctly predict this at the exact time of expiration.

Specific exercise provisions of a specific option contract may create additional risks. This needs to be carefully monitored especially in the case of over-the-counter traded options.

Risks Assumed By Option Writers

Writers of naked call or put options are exposed to unlimited losses if the underlying asset rises or drops respectively.

As American options may be exercised by the option holder at any time on or before expiration, option writers effectively lose control over the timing and magnitude of their liabilities. Even if a trading market is unavailable or they are unable to execute a closing transaction, option writers are still obligated under the options that they sold.

Risks Assumed By Both Option Holders and Option Writers

Option trading can be volatile and relatively small price movements on the underlying asset may generate significant profits or losses. Therefore investors may find themselves very quickly out of the money if they make incorrect predictions on the direction of the market.

As options are very versatile and can be combined to create different strategies, the complexity of some option strategies may be a source of significant risk. Similar to an exchange for stocks and equity, an option exchange has the right to halt trading of any option. When this happens, investors are prevented from realising value.

Investors who trade internationally traded options assume additional risks due to the differences in timing and foreign currency exposure.

4.2.8 Stock Index Options

Stock index options were first introduced on the Chicago Board Options Exchange (CBOE) in 1983 and have gained wide popularity since then. Stock index options are options in which the underlying asset is a stock index. The underlying stock index may be a broad-based index like the S&P 500 Index or a narrow-based index like an index that is limited to a particular industry, for example the technology industry or the energy industry.

A stock index option gives the holder the right to buy or sell a specific stock index for a specified price on or before a specific date. Stock index options enable investors to trade on general stock market movements in the same way that they can trade on individual stocks. Thus, an investor who is bullish on the market can buy a call on a market index, and an investor who is bearish on the overall market can buy a put.

Unlike stock options which require the actual delivery of the stock upon exercise, buyers of the index options receive cash (i.e. cash settled) from the sellers upon exercise of the contracts. The amount of cash settlement is equal to the difference between the closing price of the index and the exercise price of the option multiplied by a specific dollar amount. The multiplier for the S&P 500 is US\$100 for example.

Benefit of Using Stock Index Options

Stock index options offer a cost effective way to hedge or speculate on the stock index. This is because the price of the stock index option is significantly cheaper than the value of the stock index so even clients with small trading accounts are able to take positions on the stock index.

Globally, there is a large variety of stock index options available to an investor and he would likely be able to find a particular stock index option which suits his trading or investing purposes. Moreover, compared to individual stocks, the price

movement of a stock index may not be as volatile. Consequently, the likelihood of significant losses in option positions within a short period of time is less.

Risks of Trading Stock Index Options

Apart from the popularly known stock indices like the S&P 500, STI Index, Nikkei 225, the variety of stock indices is very diverse. Some stock indices may track only a very small, select segment of the market or may track illiquid markets. Consequently, trading options in those stock indices may be very volatile and may expose the investor to significant counterparty default risk.

4.2.9 Difference between Exchange-Traded and OTC-Traded Options

Options can be traded both on an exchange and over-the-counter (OTC). Both trading structures have their pros and cons. Moving from an exchange to an OTC trading structure essentially involves a trade-off between greater guarantee of contract performance and greater customization of the option contract.

Exhibit 4.5 Difference between Exchange-Traded and OTC-Traded Options

Exchange Traded	OTC Traded
▪ Originated and traded on an organised and regulated exchange. Overseen by extensive government and industry regulation. ▪ Standardized terms (e.g. type, notional size, expiration date) in the form of a public transaction. ▪ Settled through a clearing house associated with the exchange. This guarantees performance of the option contract. ▪ Daily mark-to-market price available because contracts are standardized.	▪ Does not trade on an exchange or through a formalized trading system. There is less regulation from government and industry bodies. ▪ Contracts are not standardized. Option terms are tailor-made and can be fully customized to suit the needs of the parties involved. Transactions are private. ▪ There is no clearing house, leading to weaker performance guarantees. Counterparty risk needs to be managed and the selection of a reputable or financially sound counter-party or broker becomes an important factor. ▪ Daily mark-to-market procedures may not be readily available.

4.2.10 How to Read an Option Term Sheet

Below is an example of a typical term sheet for an option contract. Although the key features of all option contracts are generally the same, specific features may be different depending on the agreement between the counterparties, whether the contract is over-the-counter or the exchange rules.

Over-the-counter traded options tend to be more customised to each counterparty's needs. For example, the strike price and contract months can be set as desired and there may not be any price or position limits.

Exhibit 4.6 Key Features of an Option Term Sheet

Example of an Option Term Sheet		Explanation
Ticker Symbol	CS – Calls PS – Puts	This is the contract's ticker symbol when it's being traded on the exchange.
Contract Size	One MSCI Singapore Index Futures Contract	This states the underlying asset. The investor then knows that this is an option on the Singapore stock index futures.
Contract Months	2 nearest serial months and March Quarterly Cycle (Mar, Jun, Sep, Dec)	This particular option has 6 expiration dates. Mar, Jun, Sep and Dec plus the 2 nearest serial months. E.g. in Jun 2011 an investor is looking to buy options, he will see options listed for Jun, Jul, Aug, Sep, Dec '11 and Mar '12. Jul and Aug '11 represent the nearest 2 serial months outside of the quarterly cycle.
Trading Hours	<u>T Session</u> Order Cancellation: 8:15am – 8:30am / Opening: 8:30am – 5:15pm <u>T+1 Session</u> Order Cancellation: 6:00pm – 6:15pm / Opening: 6:15pm – 2:00am	T Session is the equivalent of the regular trading session while T+1 is equivalent to extended trading session. This denotes the market trading hours for this particular contract. Some exchanges may have a cancellation or matching period which in this case, is 8:15am to 8:30am and 6:00pm to 6:15pm.
Trading Hours on Last Trading Day	8:30am – 5:15pm	Trading hours for expiring contracts may be set slightly differently from non-expiring contracts.
Last Trading Day	The second last business day of the expiring contract month	The last trading day for expiring contracts is usually set differently from contracts that are not yet expiring. In this example, the last trading day is the second last business day rather than the usual last business day for non-expiring contracts.
Tick Size	0.1 index point = SGD 20	This shows the monetary value associated with each tick movement.
Strike Price Intervals	5 index points interval	Exchange trade options have pre-defined strike prices. OTC options will have strike prices set in agreement with both counterparties.
Exercise Procedure	European Style	Options can either be exercised European or American style. European options can only be exercised on expiration date while American options can be exercised up to and including expiration day.
Settlement at Expiration	Cash settled. In the absence of contrary instructions, in-the-money options will be automatically exercised at expiration	This states whether the contract is cash-settled or delivered.
Daily Price Limits	5% up or down	Exchange rules may halt or trading may be suspended when price limits for instruments are breached, perhaps in extremely volatile markets. These "circuit breakers" are meant to slow down price movements from cascading into a sharp market plunge or increase.
Position Limits	10,000 contracts net on the same side of the market, and in all contract months combined	Exchange rules may specify the position limit that each investor can take. This is meant to limit the risk that the exchange has to bear for each participating investor.

4.3 Bond Options

4.3.1 Key Features of Bond Options

A bond option gives the option buyer the right but not the obligation to buy or sell a bond at a pre-agreed price on or before the option's expiration date. Unlike equity options, bonds are typically traded over-the-counter (OTC). Therefore bond prices are not available on an Exchange and there is a lack of observed market prices. Hence pricing and valuing bond options is more complex than pricing and valuing equity options.

As the price of a bond approaches its par value as it nears maturity, the uncertainty of the bond's price disappears. Therefore, the expiration of a bond option is usually set significantly before the bond's maturity date. An investor typically buys a bond call option if he expects interest rates to fall, thereby causing the bond's price to increase. Conversely, an investor buys a bond put option if he expects interest rates to increase and bond prices to drop.

Example

Bank A buys a call option on US Treasury bonds from Bank B. The bond option's strike price is \$102 while bond's current spot price is \$101. Bank A pays an option premium to Bank B. At the option's maturity, Bank A either exercises the option if the spot price of the bond is above the option's strike price or lets the option expire.

4.3.2 Benefit of Using Bond Options

Holding long bond option positions allow the investor to participate in the bond market at his discretion instead of being compelled to hold long bond positions. Also investors with large bond portfolios can use bond options to hedge against future interest rate movements.

4.3.3 Risks of Trading Bond Options

A risk in trading bond options is invariably linked to the risks inherent in bonds. Usually the volatility in bond markets is not as significant as the commodities, futures and FX markets. However the risk related to specific bonds can be significant if it's lower than investment grade.

Rating downgrades by rating agencies such as Standard and Poors, Moody's etc. can very quickly increase the yields investors demand to hold that particular debt issue.

Macroeconomic factors also play a role in determining bond yields and prices. A country with poor economic fundamentals may eventually come under the scrutiny of “bond vigilantes” as demonstrated by how global investors pushed down prices of bonds from Greece, Spain and Portugal during the Eurozone debt crisis in 2010.

4.3.4 Interest Rate Options

Interest rates options are related to bond options. However these types of options focus on interest rates alone. Interest rate options are options in which the underlying asset is an interest rate. The exercise price of an interest rate option is also an interest rate.

An interest rate option gives the option buyer the right but not the obligation to make (call) or receive (put) a known interest rate payment. For example, an investor buys an interest rate put option on SIBOR with an exercise price of 2.0%.

This means that at expiration, if the prevailing SIBOR rate is lower than 2.0%, the investor will exercise his option and will receive 2.0% in lieu of the prevailing SIBOR rate. If SIBOR is above 2.0%, the investor will let the option expire.

Interest rate options are cash settled. Upon exercise, the underlying securities do not have to be delivered, but the differences between the interest rates are settled using a scale of 100. Interest rate options are exercised in the European style, i.e. option can only be exercised on a specified date and not before that.

Benefits of Using Interest Rate Options

Purchasing options as a way of taking a view on interest rates is attractive because the trader’s risk is limited to the premium of the option but the upside potential is unlimited. Therefore, a high level of gearing can be achieved. For interest rate options, only the differences between the interest rates are cash settled so there is no risk of the principal.

The premium received from selling a call option against an existing investment or selling a put option against a debt issuance can be used to improve the investment yield or lower the borrowing cost, respectively. But of course, this need to be weighed against the possibility of unlimited liability should the market turn against the option writer.

Risks of Trading Interest Rate Options

As with all types of options, the seller of interest rate options is, theoretically, exposed to unlimited downside risk. The risk can be significant in volatile interest rate markets such as when the central bank drastically lowers the discount rate³ which, in turn, affects the prevailing interest rate in the financial market.

³ The discount rate is the rate that a Central Bank charges banking institutions for borrowing funds.

4.3.5 Interest Rate Caps and Floors

An interest rate cap is a series of interest rate call options with the same strike rate where the expiration dates of the options correspond to the maturity schedule of the underlying interest rate. Each individual call option in the series can be exercised independently and are referred to as “caplets”.

The buyer of the cap receives payments at the end of each period in which the interest rate exceeds the agreed strike price. An example of a cap would be an agreement to receive a payment for each month the LIBOR rate exceeds 2.5%. The gain from the payments is used to offset the increases in interest rates.

An interest rate floor is a series of interest rate put options with the same strike rate where the expiration dates of the options correspond to the maturity schedule of the underlying interest rate. Each individual put option in the series can be exercised independently and are referred to as “floorlets”.

The buyer of the floor receives payments at the end of each period in which the interest rate is below the agreed strike price. The gain from the payments is then used to offset the drop in interest rates.

4.4 Currency Options

4.4.1 Key Features of Currency Options

A foreign exchange or currency option is a contract which grants the option buyer the right but not the obligation to buy or sell a predetermined amount of foreign currency at a fixed exchange rate on or before a specified date. Currency options can be traded OTC or on exchanges like the Philadelphia Exchange or the Chicago Mercantile Exchange.

Example

An investor who lives in New York and holds Canadian dollar-denominated government bonds in his portfolio. It is January and when the bonds come due in two months he will need to convert the proceeds back into US dollars which expose him to a possible weakening in the Canadian currency.

Accordingly, he buys the March put on the Canadian dollar with an exercise price of USD 0.77 / CAD⁴ for a total price of US\$1,965 (= 50,000 x 0.0393). This option would allow the holder to sell C\$50,000 in March for a total price of US\$38,500 (= 50,000 x 0.77).

⁴ Note that CAD/USD is 0.77. However, the market convention is to quote USA/CAD at 1.2987.

The investor will only exercise the contract if the spot CAD/USA price prevailing in March is less than 0.77 (that is, if the Canadian dollar weakens relative to the US currency from 77 cents to 76 cents).

This contract is a put on the Canadian dollar but is also a call on the US dollar as it allows the exchange of Canadian dollars for US dollars.

4.4.2 Benefits of Trading Currency Options

Currency options can be used to hedge against an identified currency exposure. For example corporations with foreign subsidiaries are likely to face currency exposure when repatriating profits back to their home country. Currency options can be used to hedge against such risks.

4.4.3 Risks of Trading Currency Options

The foreign exchange market is the largest and most liquid financial market in the world. At any given point in time, the foreign exchange market is a barometer of macroeconomic trends, investor perceptions and risk appetites and geopolitics.

Trading in currency options is risky as the notional value involved is extremely large. And because the FX market is open literally 24 hours a day and operates every day except weekends, traders need to be able to monitor their positions or risk being caught out by fast moving events.

Managing counterparty risk is especially important if it's an over-the-counter traded currency option as option contracts are fulfilled by actual delivery of the currencies.

4.5 Warrants

4.5.1 Key Features of Warrants

A warrant is a derivative that gives the investor an option to buy or sell a stated number of shares of an underlying instrument at a specified price (exercise or strike price) within a specified time period. Unlike options traded on the derivatives exchange, warrants are already paid in full upfront and therefore, not subject to margin calls.

Warrants are traded on the SGX-Securities Trading Ltd, and warrant holders do not receive or pay any dividend income or cash distributions even if the underlying instrument pays a dividend or cash distribution. Relatedly, investors may purchase structured products which are derivatives issued by third parties, that are not the same entity as the issuer of its underlying security.

Some structured products, including structured warrants are listed and traded on the SGX-Securities Trading Ltd. There is a special type of warrant which is called the zero strike warrant. A zero strike warrant is essentially a synthetic stock.

By definition, a zero strike warrant has an exercise price of zero, meaning that it's always in-the-money. The cash settlement and price of the warrant is equal to the closing price of the underlying stock.

Conversion Ratio

Conversion ratio is the number of warrants needed to be exercised to buy or sell one unit of the underlying security.

Exercise or Strike Price

The Exercise Price is the price at which the holder is able to buy or sell the underlying instrument.

Expiry Date

The expiration date of the structured warrant is the last day on or before which the warrant has to be exercised.

Issue Price

The issue price or the warrant price is the price at which the structured warrant is sold by the issuer to the holders.

Physical Delivery or Cash Settlement

Warrants may be settled by physical delivery of shares or by cash. Upon exercise of the warrant, depending on whether the warrant is a call or put, the holder can expect to deliver to, or receive from, the issuer the underlying share (or cash). The settlement style is made known when the warrant is issued. Structured warrants listed on the Singapore Exchange are settled in cash.

Gearing

Structured warrants are usually priced at a fraction of the share price. This allows you to trade more warrants than the underlying asset for the same investment outlay. Trading warrants therefore, offer benefits of gearing.

For instance, a small percentage gain in the underlying share price may lead to a larger percentage gain in the value of the call warrants. Conversely, a fall in the price of the underlying share may lead to a larger percentage loss in the value of the warrants.

Right and Obligation of Warrants Issuer

Generally, the issuer will have the obligation to deliver the underlying security if the holder exercises the warrant which gives him the right to buy at the exercise price⁵.

Right and Obligation of Warrants Holder

The warrant holder has the right but not the obligation to buy or sell a stated number of shares of an underlying security at a specified price (exercise or strike price) within a specified time period.

4.5.2 Company Warrants

Listed companies sometimes issue warrants to existing shareholders as a “sweetener” attached to an offering of bonds or a rights issues for shares. In the case of a bond issue, the “sweetener” may allow the issuer to offer a lower interest rate to investors. The warrants are detachable from the host instrument and may be listed and traded separately.

A company warrant is usually a long-dated call option as its maturity is typically three to five years, whereas options and structured warrants have expiration dates of less than one year. Company-issued warrants are “American-style” options and they may be exercised any time during the lifetime of the warrant. When these warrants are exercised, the company will issue the new shares in exchange for the warrants. This will result in a slight dilution of the company’s earnings per share.

4.5.3 Structured Warrants

Formerly known as covered warrants, structured warrants are issued by a third party financial institutions based on various underlying instruments such as single stocks, stock indices, investment funds and commodities. Different types of structured warrants are designed for different investment objectives and risk profiles of investors.

Similar to options, a “structured” call warrant gives the holder a right to buy the underlying asset while a put warrant gives the holder a right to sell the underlying asset at a predetermined price, on or before the expiry date, depending on the exercise style of the warrant. Structured warrants in Singapore are “European-style” options, meaning that they can only be exercised on the expiry date.

Structured warrants on stocks are subject to adjustments to take into account any corporate action arising from the underlying stock. Corporate actions such as rights issue, bonus issue, stock splits, special dividends or consolidation may lead to a diluting or concentrative effect on the theoretical value of the underlying stock. This

⁵ Warrants are usually call options. There are put warrants which grant the holder the right to sell the underlying security at a specified price to the issuer.

may require adjustments to the conversion entitlement, the exercise price or other variables of the structured warrants which would be effective on the ex-date⁶ of the corporate action.

Market-Making of Structured Warrants

To create liquidity, issuers of structured warrants may commit to create a “market” for the structured warrants they issued. By committing to make a market, the designated market-maker (DMM) appointed by the warrants issuer is obligated to provide competitive bid and offer prices for the structured warrants. The listing document will state whether the warrant issue has committed to make a market in the structured warrants.

Settlement Issues of Structured Warrants

Structured warrants trade and settle like common stocks and company warrants listed on the local exchange, that is through physical or cash settlements.

- **Physical Settlement** – To exercise a call warrant, the holder will need to submit an exercise notice with the cash payment (based on the exercise price) to the warrant agent. The warrant agent will then inform CDP and the issuer on this, and CDP will credit the account of the holder with the underlying securities and debit the account of the issuer with the same.

In the case of a put warrant, CDP will debit the account of the holder for underlying securities and credit the account of the issuer for the same, and the warrant agent will make a cash payment on behalf of the issuer to the holder. Regardless of the type of warrant, exercise expenses are payable by the holder to the warrant agent.

- **Cash Settlement** – Most structured warrants are automatically exercised and settled in cash on expiration date. The proceeds are calculated based on the difference between the market price of the underlying and the exercise price.

All structured warrants in Singapore have the Asian style of expiry settlement, where the last trading day of a structured warrant is different from its expiry day. In the case of a structured warrant with automatic exercise feature, its last trading day in the “Ready” and “Unit-Share” markets is at least 3 business days before its expiry date (such that the settlement or due date for the trade is at least a day prior to the expiry date).

Investors can only trade the structured warrant on or before the last trading day. The settlement price is based on the arithmetic average of the official closing price of the underlying for 5 market days prior to expiration date.

⁶ The Ex-Date is the date on or after which the benefits of a corporate action, such as dividends or rights issue, would not accrue to the underlying security. An example is the Ex-dividend Date.

The gains from trading in structured call and put warrants are shown in Exhibit 4.7.

Exhibit 4.7 Gains from Call and Put Warrants

	Call Warrant	Put Warrant
Scenario	When underlying share price rises above the warrant exercise price.	When underlying share price falls below the warrant exercise price.
Gain from (a) Exercising the warrant	Difference between the current share price and the warrant exercise price	Difference between the warrant exercise price and the current share price.
(b) Selling the warrant	Difference between the prevailing warrant price and the warrant price paid by the investor. Warrant price would have increased with the rise in the share price	Difference between the prevailing warrant price and the warrant price paid by the investor. Warrant price would have increased with the fall in the share price

4.6 Futures Contracts

4.6.1 Key Features of Futures Contracts

A futures contract is an agreement to buy or sell a stated commodity (such as silver) or financial claim (such as stock index) at a specified price at some future specified time. The price at which the parties agree to transact in the future is called the futures price. The designated date at which the parties must transact is called the settlement date or delivery date.

There are two broad categories of futures contracts:

- **Commodity Futures** – Agricultural, metals and energy-related.
- **Financial Futures** – Foreign currencies, interest rates, stock indexes.

When an investor takes a position in the market by buying a futures contract, he is considered to be in a long position. If he decides to sell a futures contract, he is deemed to be in a short position. The buyer of a futures contract will realise a profit if the futures price increases while the seller of a futures contract will realise a profit if the futures price decreases.

Futures vs Forward Contracts

The futures markets basically evolve from the forward markets. A forward contract is simply a deferred delivery sale of some asset with the sales price agreed on

currently. The parties of a forward contract have to negotiate the terms of the contract such as quality of the asset, date of delivery and the forward price.

Futures markets replace forward contracts with highly standardised contracts, establishing such features as contract size, delivery dates, place of delivery and condition of the items that can be delivered. Only the price and the number of contracts are left for futures traders to negotiate.

Let us look at the Chicago Board of Trade (CBOT) oats contract. This contract calls for the delivery of 5,000 bushels of No. 2 heavy or No. 1 grade oats to Chicago or Minneapolis - St Paul at a 7 cents per bushel premium. In addition, these contracts are written to become due in March, May, July, September, and December. It is through the standardisation of contracts that a highly liquid secondary market is developed.

To prevent large price movements from occurring because of speculative excesses, daily price movement limits are set by the exchange. For example, the CBOT imposes 10 cents per bushel (\$500 per contract) price movement limit above and below the previous day's settlement price of oats contract. If the price moves down by an amount equal to the daily price limit, the contract is said to be limit down. If it moves up by the limit, it is said to be limit up.

Right and Obligation of the Futures Buyer

Unlike an option buyer, a futures buyer has the obligation to fulfil the contract on delivery date. If the buyer bought a commodity futures contract, he must take physical delivery of the underlying asset. However, physical delivery only happens in a minority of futures contracts. Most buyers of commodity futures will cover the long position by selling an identical contract to liquidate his position.

In the case of financial futures which are cash-settled, the futures buyer has the right to receive cash from the futures seller if he had made a profit. However if the futures buyer has made a loss, he is obligated to pay cash to the futures seller.

Right and Obligation of the Futures Seller

A futures seller has the obligation to fulfil the contract on delivery date. If the seller sold a commodity futures contract, he must physically deliver the underlying asset to the buyer. As previously discussed, physical delivery only happens in a minority of futures contracts. Most sellers of commodity futures will cover the short position by buying an identical contract to liquidate his position.

For cash-settled financial futures, the futures seller has the right to receive cash from the futures buyer if he had made a profit. However if the futures seller has made a loss, he is obligated to pay cash to the futures buyer.

4.6.2 Structure of Futures Markets

There are many futures exchanges which trade in futures contracts. These include the following:

- Chicago Board of Trade (**CBOT**);
- Chicago Mercantile Exchange (**CME**);
- London International Financial Futures Exchange (**LIFFE**);
- Tokyo International Financial Futures Exchange (**TIFFE**); and
- Singapore Exchange Derivatives Trading (**SGXDT**).

A futures exchange provides an organised central trading place for transactions. It also encourages confidence in the futures market by allowing for effective regulation of trading. The various exchanges set and enforce rules, as well as collect and disseminate information on the commodities being traded.

Associated with every futures exchange is a clearinghouse. The main purpose of the futures clearinghouse is to guarantee that all trades will be honoured. The clearinghouse interposes itself as the buyer to every seller and the seller to every buyer.

This allows the parties of the contract to reverse their positions before maturity without having to find the original buyer (or seller). Because no trades occur directly between individuals, but between individuals and the clearinghouse, buyers and sellers realising gains in the market are assured of payment.

4.6.3 Mechanics of Futures Trading

Futures trading is a zero-sum game. For every futures contract, there is a party who sold it short and another party who holds it long.

Liquidating a Position

Futures contracts can be settled either by delivery or by offset. Delivery or settlement of the contract occurs in months that are designated by the various exchanges. The party purchasing a futures contract accepts delivery of the underlying asset and the seller liquidates the position by delivering the asset at the agreed price.

Deliveries normally occur in a very small proportion of futures transactions. Some financial futures, such as the stock index futures, are settled in cash because it is inconvenient or impossible to deliver the underlying asset. The typical method of settling a futures contract is by offset.

This means that the buyer or seller reverse their positions by selling or buying the same number of identical futures contracts before the settlement date. If a futures contract is not offset, it must be closed out by delivery. The open interest indicates the number of contracts that are not offset by opposite transactions.

It is important to note that the futures price on the delivery date will equal the cash market price or spot price on that date. This is because a futures contract with no time left until delivery is equivalent to a cash market transaction. Thus, as the delivery date approaches, the futures price will converge to the cash market price. This is called the convergence property.

Margin Requirement

At the initial execution of a trade, the investor is required to deposit a minimum dollar amount per contract as specified by the exchange. This amount, known as initial margin or margin requirement serves as “good faith” money made by the investor to ensure that the investor will satisfy the obligations of the futures contract.

The initial margin normally consists of cash and/or near cash securities as is referred to as the investor’s equity. The clearinghouse determines the initial margin requirement. Individual brokers may however require higher margins from their clients than those specified by the exchange.

All futures contracts are “marked to market” on a daily basis; this means that profits and losses on a contract are credited or debited to each investor’s account every trading day. If the market price of the futures contract declines and the margin account drops below a critical level called the maintenance margin, he will receive a margin call from his broker.

The investor is required to deposit additional margin known as variation margin to bring the equity in the account back to its initial margin level or close out his position. This process of marking to market helps to safeguard the position of the clearinghouse. Should there be any excess in the margin account; the investor may withdraw the excess.

The exchange uses a “settlement price” for the purpose of settling an investor’s position. It is a value determined by the exchange at the close of each trading day. The settlement price may be different from the closing price.

Example

Consider an investor who buys a stock index futures contract on the MSCI Singapore Free Index at 370 on day 1. The size of each futures contract will be \$200 multiplied by the MSCI Singapore Free Index. The value of the contract is \$74,000 ($\370×250).

Assume the initial margin and maintenance margin for the contract are 10% or \$7,400 and \$5,920 respectively. The table below shows the balance in the margin account at the end of each trading day:

Day	Settlement Price	Profit / (Loss)	Margin Balance	Margin Call
1	370		\$7,400	
2	372	\$400	\$7,800	
3	366	(\$1,200)	\$6,600	
4	360	(\$1,200)	\$5,400	\$2,000

As long as the balance in the margin account is above the maintenance margin level of \$5,920, there will be no margin calls. In the example, at the end of Day 4, when the investor's account balance falls to \$5,400, he is required to put up additional margin of \$2,000 to bring the equity up to the initial margin of \$7,400. If he is unable to put up the variation margin, the clearinghouse will close his position.

Leverage

Futures are traded on margins. The initial margin is normally a percentage of the contract value. In the previous example, the investor needs only 10% or \$7,400 to trade on a futures contract valued at \$74,000. The losses sustained by him at the end of four days amounted to \$2,000 or 27% (based on a capital of \$7,400). During this time, the decline in the index was 2.7%.

This illustrates the leverage effect of futures trading. The gains or losses on a futures contract are magnified even though the price movement on the underlying security may be small.

4.6.4 Gain, Loss and Breakeven

For cash settled futures contracts, the gain, loss and breakeven position is based on an underlying reference rate such as a short term interest rate index such as Libor, or the closing value of a stock market index.

For example, an investor buys one S&P 500 stock index futures contract at 1,200 points. The reference rate of this contract is its closing value. He decides to hold the contract till expiration at which the actual S&P 500 stock index is at 1,250 points.

The futures buyer makes a profit equal to how much the reference rate has changed multiplied by the dollar value of each index point. For a S&P 500 futures contract, each index point is \$250. Therefore, the futures buyer's gain is \$12,500 (i.e. 50 points X \$250). Correspondingly, the futures seller's loss for this contract will be \$12,500.

A breakeven with no gain or loss to either party only occurs if the reference rate is exactly the same when the buyer opened and liquidated his position. Therefore, the

gain, loss and breakeven depends on how much the reference rate has changed and the dollar value of each tick or point.

Different types of futures contracts will have different tick sizes and dollar value for each tick. For example, each index point of a S&P 500 contract is \$250 while one tick of a Dow Jones futures contract is \$10.

4.6.5 Benefit of Using Futures Contracts

There are two groups of participants in the futures market - the hedgers and the speculators. A speculator uses futures contracts to profit from movements in futures prices while a hedger uses futures contracts to protect against price movements.

Speculating

Unlike hedgers, speculators typically do not deal in the physical commodity or financial instrument underlying the futures contract. If they believe that prices will increase, they will take a long position in the futures contract and vice versa.

Speculators are essential to the proper functioning of the futures market, absorbing the excess demand or supply generated by hedgers and assuming the risk of price fluctuations that hedgers want to avoid. They also contribute to the liquidity of the market and reduce the variability in prices over time.

Why does a speculator buy a futures contract rather than buy the underlying security directly? First, the transaction costs are significantly lower in the futures market than in the cash market. Second, when dealing in futures for a physical commodity, there is no storage costs involved. Third, the speed at which orders can be executed also gives the advantage to the futures market. The last reason is the leverage that futures trading provide.

Hedging

Hedgers protect themselves from price fluctuations (price risk) by buying or selling futures contracts taking positions opposite to that, which they are holding. Price risk is eliminated because the profits/losses on their underlying positions will be offset by losses/profits on their futures contracts.

Consider an investor who is holding a quantity of silver as inventory for his business. He is concerned that the price of silver is falling. To protect the value of his inventory, he sells silver futures (i.e. takes a short position in silver futures). Consequently, the investor has locked-in the price of silver for his inventory. Protecting the value of an asset with short futures position is called a short hedge.

A position that is short in the cash market and long in the futures market is referred to as a long hedge. A long hedge is appropriate when a company knows it will have to purchase a certain asset in the near future and wants to lock in a price now. An

investor, who currently does not have any silver inventory but intends to purchase silver later, is in effect, shorting the cash market. To protect against rising silver prices, he can take a long position in the futures market.

4.6.6 Risks in Trading Futures

Basis Risks

The basis of a futures contract is the difference between the futures price and the cash price. That is,

$$\text{Basis} = \text{Cash Price} - \text{Futures Price}$$

If the asset to be hedged and the underlying security in the futures contract are the same, the basis will be zero on the maturity date of the contract. Before maturity, however, the futures price for later delivery may differ substantially from the current spot price.

If the futures contract and asset are liquidated before maturity, the hedger is confronted with basis risk as the futures price and cash price do not move in perfect lockstep at all times before delivery date. In this case, gains and losses on the contract and the asset need not exactly offset each other.

The effective price that is obtained for the asset with hedging is therefore:

$$\begin{aligned} P &= S_2 + F_1 - F_2 \\ &= F_1 + S_2 - F_2 \\ &= F_1 + b_2 \end{aligned}$$

The value of F_1 is known at the time when the hedge is initiated and the term b_2 represents basis risk. Note that basis risk can lead to an improvement or a worsening of a hedger's position.

Example

Consider an investor who initiates a short hedge. The spot (S_1) and futures (F_1) prices at the time the hedge is initiated are \$4.20 and \$4.80 respectively. The basis (b_1) is therefore \$0.60.

The next day, the spot price (S_2) increases to \$4.25, while the futures price (F_2) increases to \$4.81. The basis (b_2) has narrowed to \$0.56. The investor enjoys a capital gain of \$0.05 on the spot price and a loss of \$0.01 on the futures contract. The net gain from the hedge is \$0.04, which is due to a decrease in the basis.

Basis risks arise because of a number of factors, including:

- The asset, whose price is to be hedged, may not be exactly the same as the asset underlying the futures contract;
- The hedger may be uncertain as to the exact date when the asset will be bought or sold;
- The hedge may require the futures contract to be closed out well before its expiration date; and
- The size and unit of measurement of futures contract may not correspond with the size of the underlying hedged position. As a result, a position may be under-hedged or over-hedged.

Liquidity Risks

Although futures contracts are exchange traded, there is no assurance that liquidity will be available at all times. This is important to keep in mind because unless an investor is willing to take delivery of the underlying asset (if the contract cannot be cash settled), the only way for an investor to close his futures position is to offset it with another futures contract.

Liquidity could dry up in a rapidly rising or falling market. For example, if the price of a particular futures contract has risen or fallen by the maximum allowable daily limit and there are no counterparties willing to transact, there will be no trades. In such a “lock limit” market, an investor will not be able to offset his existing position.

Useful indicators to gauge the level of liquidity for a particular futures contract and for particular delivery months are **open interest** (the number of open futures contracts not yet closed off by an offsetting trade or not yet delivered) and trading volume.

Leverage Risks

As explained earlier, futures contracts are traded using margins. A relatively small price movement may generate significant profits or losses through the leverage effect. Therefore investors may find themselves very quickly out of the money if they make incorrect predictions on the direction of the market.

4.6.7 How to Read a Futures Term Sheet

Below is an example of a typical term sheet for a futures contract. Futures contracts are generally traded on an exchange and the key features of a contract are standardised as per the rules for that particular exchange.

Exhibit 4.8 Key Features of a Futures Term Sheet

Example of an Futures Term Sheet		Explanation
Ticker Symbol	SP	This is the contract's ticker symbol when it's being traded on the exchange.
Contract Size	\$250 x S&P 500 futures price	This states the underlying asset as well as the dollar multiplier to derive at the notional value of the contract.
Tick Size	0.10 index points = \$25	This shows the monetary value associated with each tick movement.
Contract Months	One month in the March Quarterly Cycle (Mar, Jun, Sep, Dec)	This particular futures contract has only 4 expiration dates. March, June, Sept and Dec. For example, in April 2011, an investor is only able to trade in contracts that expire in June 2011, Sept 2011, Dec 2011 and Mar 2012.
Trading Hours	Regular trading hours: 8:30am – 3:15pm / Extended trading hours: 3:30pm – 8:15am	These are the trading hours where this contract can be traded on the exchange.
Last Trade Date and Time	3:15pm on Thursday prior to the 3 rd Friday of the contract month.	The last trading day and trading hours for expiring contracts is usually set differently from contracts not yet expiring. In this example, the last trading day is the Thursday before the 3 rd Friday of the month. And trading for this expiring contract must conclude by 3:15pm.
Final Settlement Procedure	Cash settled. All open positions at close of last day of trading are settled in cash.	This states whether the contract is cash-settled or delivered.
Daily Price Limits	Regular trading hours: 10% downside limit / Extended trading hours (overnight): 5% up or down	Exchange rules may halt or suspend trading when price limits for instruments are breached, perhaps in extremely volatile markets. These "circuit breakers" are meant to slow down price movements from cascading into a sharp market plunge or increase.
Position Limits	20,000 net long or short in all contract months combined.	Exchange rules may specify the position limit that each investor can take. This is meant to limit the risk that the exchange has to bear for each participating investor.

4.7 The Foreign Exchange Market

4.7.1 Key Features of the Foreign Exchange Market

The foreign exchange market is a market where financial institutions, business firms, central banks, and individuals buy and sell foreign currencies. The participants include investors and speculators.

Decentralised Over-The-Counter Market

Unlike the stock market or futures market, the foreign exchange market does not refer to any specific geographical location, such as a stock exchange. Instead, the foreign exchange market is a global and decentralized over-the-counter market consisting of a number of interconnected marketplaces such as London, Tokyo and Singapore where different currencies are continuously traded 24 hours a day except for weekends.

Transactions are done through a complex communication network of computer terminals, telephones, faxes, telexes, cables and communication satellites. As such there is no single or centrally cleared market for the majority of foreign exchange transactions and there isn't a single exchange rate but instead there are a number of different rates depending on which bank or market maker is trading and where.

In reality, rates quoted in different financial centres are often very close as any significant discrepancies would otherwise be exploited by arbitrageurs.

Gold Standard to Floating Exchange Rates

The modern foreign exchange market was formed during the 1970s when countries gradually switched to floating exchange rates from the previous fixed exchange rate regime which was created by the Bretton Woods agreement in July 1944.

By the early 1970s, the old structure of the Bretton Woods agreement which pegged the US dollar at US\$ 35 per ounce of gold had become untenable as the US struggled with budget and trade deficits brought about by government expenditure on the Vietnam War and social programmes.

In 1971, US President Richard Nixon dramatically closed the gold window and took the US off the gold standard. From then onwards, the US was no longer pegged to a specific exchange rate as denominated by the amount of gold that could be exchanged for it. By 1976, all the major currencies were, like-wise, floating.

Most Liquid Financial Market

The foreign exchange market is the most liquid financial market compared with other asset classes. The huge volume of foreign exchange trades is largely due to the increased trading activity of high-frequency traders, banks trading for their own profit (i.e. proprietary trading desks) and on behalf of their clients as well as online trading by retail investors.

The introduction of online, electronic trading especially algorithmic trading has been instrumental to this growth. Because of its high trading volume, liquidity and a continuous trading window, the foreign exchange market is regarded as closest to the ideal of perfect competition (notwithstanding the occasional central bank intervention).

Right and Obligation of the Counterparties

When two parties enter into a foreign exchange contract, they are obligated to deliver the respective currencies at an agreed exchange rate and at an agreed time period (delivery date). Foreign exchange contracts are not cash settled and actual delivery must occur as both parties must fulfil the contract.

4.7.2 Key Drivers of Currency Movement

Many macroeconomic factors affect the movement of currency and the determination of exchange rates. These include:

- Economic factors;
- Political factors; and
- Market sentiments.

Economic Factors

One of the most important economic drivers of currency movement is the economic policy of the country. Economic policy refers to both fiscal policy (the government's taxation and budgeting / expenditure policies) and monetary policy (the central bank's approach towards the growth of money supply and the setting of interest rates).

In general, the market usually reacts positively to narrowing government budget deficits and negatively to widening budget deficits. Governments with surplus budgets or narrowing budget deficits are less likely to "monetize their debt" by printing money (and thereby increasing the money supply substantially) to pay for goods and services.

With less of a country's currency circulating in the financial markets, the supply of currency will be in step or less than the demand of currency thereby maintaining that currency's value. Markets usually react positively to an increase in a country's interest rate as this encourages both international and local investors to buy and hold that country's currency as they would be earning a higher yield relative to other currencies.

Inflation erodes the purchasing power of a country's currency. Countries with high inflation will usually have weak currencies while countries with low inflation will usually have strong currencies. However, a currency may sometimes strengthen when inflation rises because of expectations that the central bank will raise short-term interest rates to combat rising inflation.

Political Factors

Exchange rates are influenced by the level of political stability of the country in question. In general, countries with strong, stable governments and consistent

policies have stable exchange rates. Conversely, political instability, inconsistent policy-making and social strife may have negative effects on the country's economy and thus its exchange rate.

The political stability of a country and its effects on currency movement should not be viewed in a vacuum as events in one country may have positive or negative effects in a neighbouring country. This will also then influence that country's exchange rate.

Market Sentiment

Market sentiment and the perceptions of market participants influence how the currency markets behave. In times of political uncertainty, war or fear, investors typically will move their assets to markets or countries perceived to be relatively safe.

In the foreign exchange market, this flight to quality is reflected by investors existing positions in minor currencies for the US dollar, Euro, Swiss franc or gold. The knee-jerk reaction to negative events is usually very obvious to observers of intra-day currency charts as within a span of minutes, exchange rates will jump a few pips. Often, these reactions are reflected in technical charts and analytics.

Market sentiment is also driven by how the market views government policy. For example, any quantitative easing measures and low interest rate regime implemented by the US Federal Reserve will lead to bearish views of the value of the US dollar. Likewise, severe government budget deficits would cause added negative views.

4.7.3 Foreign Exchange Rate Quotations

A foreign exchange transaction involves the purchase of one currency against the sale of another currency for settlement or delivery on a specified date. The rate of exchange is the price per unit of one of the currencies expressed in units of the other currency. Foreign exchange rates can be quoted on a direct, indirect or cross basis.

Indirect Quotes / European Convention

Most currencies are quoted in European terms. It is also referred to as an indirect quote as it expresses the value of one US dollar in the foreign currency it is quoting. For example:

- USD / CNY 6.4921 Means that 1 USD = 6.4921 Chinese yuan
- USD / JPY 81.30 Means that 1 USD = 81.30 Japanese yen
- USD / SGD 1.2243 Means that 1 USD = 1.2243 Singapore dollars

In the USD / SGD quotation, the US dollar is referred to as the base currency, while the SGD is referred to as the counter-currency or term currency. As the US dollar is

the most widely accepted currency, most quotations are the in the indirect convention.

Direct Quotes / American Convention

An exception to the indirect quote is the old 'sterling bloc' such as the Australian dollar, the New Zealand dollar and the UK pound sterling. The quote expresses the value of one unit of foreign currency in US dollar terms. A quote expressed in this way is known as a direct quote. For example:

- GBP / USD 1.6706 Means 1 Pound sterling = 1.6706 USD
- AUD / USD 1.0971 Means 1 Australian dollar = 1.0971 USD

Cross Quotes

Although most foreign exchange transactions involve the US dollar as one of the currencies, the increasing importance of other major currencies in international transactions has resulted in direct quotation of two non-US dollar currencies. These quotes are referred to as cross rates. For example:

- EUR / GBP 0.8841 Means 1 Eurodollar = 0.8841 Pound sterling

Cross rates can be calculated easily from the US dollar rates for each currency.

Example

Given the following exchange rates, calculate the GBP / CNY cross rate:

GBP / USD = 1.6706

USD / CNY = 6.4921

The calculation here is different because of the different method of quoting the sterling.

GBP 1 = USD 1.6706, therefore

USD 1 = GBP 1 / 1.6706 = 0.5967

GBP 0.5967 = CNY 6.4921

GBP 1 = CNY 6.4921 / 0.5967 = 10.8800

Therefore GBP / CNY = 10.8800

Reciprocity

An interesting point to note in the exchange rate quotation is its reciprocity. For example SGD / USD 0.8168 can be expressed as USD / SGD 1.2243. The calculation is as follows:

- SGD 1 = USD 0.8168
- USD 1 = SGD 1 / 0.8168 = 1.2243

Bid-Offer Spread

A bank or an authorised foreign exchange dealer will always provide two prices when asked for a quote. The first price is the one at which the bank will buy a currency (known as the bid) and the second is the price at which the bank will sell a currency (the offer).

For example USD/SGD 1.2243-53 means that the bank dealer will pay SGD 1.2243 to buy 1 USD. He will sell 1 USD for SGD 1.2253. The spread of 10 basis points is the bank's profit margin. The spread varies according to market conditions and the value of the transaction involved. The greater the market volatility the wider will be the spread.

Settlement Dates

The date when both parties of a foreign exchange transaction exchange their currencies is known as the value date or settlement date.

Spot Transactions

A spot transaction is a contract in which one currency is exchanged for another for settlement usually in two working days. For example, a foreign exchange deal transacted on Tuesday, 2nd May 2011 will be settled on Thursday, 4th May 2011.

The two-day period gives ample time for both parties to exchange the currencies. As Saturdays and Sundays are not considered working days, a spot deal done on Thursday and Friday will be settled on the following Monday and Tuesday respectively.

Forward Transactions

Forward transactions are contracts to exchange one currency for another at some future date, later than spot, at a rate of exchange which is determined on the trade date. The purpose of forward deals is to establish and fix the cost of exchange for a known foreign exchange commitment at some future date.

4.7.4 Forward Foreign Exchange Contracts

There are two reasons why an investor will enter into a forward exchange contract: first, to hedge a currency exposure, and second, to speculate by taking a position in the currency.

An investor who has bought or sold foreign securities may want to cover his foreign currency exposure by fixing a forward exchange rate. By covering his exposure, through locking into a known forward foreign exchange rate, the investor also forfeits potential gains from favourable movements in market rates. These transactions, where there is only a single forward transaction, are normally referred to as 'outright'.

There are those who do not have an initial currency exposure but hope to profit by taking a view on the direction of the foreign currency. They would buy the foreign currency if it is expected to appreciate and vice versa.

Forward Rates

Many factors interact to determine the future value of a currency in foreign exchange markets. These include interest-rate differentials, inflation rates and market expectations, as well as forecasts of trade figures, capital inflow, and the level of foreign reserves, economic policy action and market sentiment.

The major factor determining the forward rate of one currency against another is the difference in interest rates in the two countries over the specific forward period. The interest differential is converted into forward points which are either added to or subtracted from the current spot rate.

The forward rate thus has two components - the spot rate and the forward points (also referred to as forward margin or swap points). That is:

$$\text{Forward exchange rate} = \text{Spot rate} \pm \text{Forward points}$$

It is important to note that the forward points are only a reflection of the relative interest rates of the currencies. They are not an indication of what the future spot rates of the currencies will be. Generally, the currency of the country with the higher interest rates will trade at a forward discount to that of the other; the currency with the lower interest rates will trade at a forward premium.

4.7.5 Gain, Loss and Breakeven

The gain, loss and breakeven for foreign exchange traders depend on how much the exchange rate for that particular contract has changed when the trader opened and liquidated his position.

For example, a trader sells USD 100,000 for SGD 122,430 at the rate of USD / SGD 1.2243 in anticipation that the USD will depreciate against the SGD. After a few days, the exchange rate is USD / SGD 1.2203; the trader then sells SGD 122,430 for USD 100,328, making a profit of USD 328.

Correspondingly, this trader would have made a loss if the USD had appreciated instead. A breakeven with no gain or loss to the trader occurs if he opened and liquidated his position at the same exchange rate. Foreign exchange contracts are typically traded with large notional values (e.g. in millions of dollars) so a small change in the exchange rate may result in large gains or losses.

4.7.6 Benefits of Using Foreign Exchange Contracts

Facilitation of International Trade

Governments and corporations use foreign exchange contracts to facilitate international trade. Commercial agreements to buy and sell goods and services usually denote the local currency or the US dollar (or some other major currency like the Euro or Yen) as the settlement currency.

For example, when BMW and Mercedes Benz export its cars to the Singapore, it expects to be paid either in US dollars or in Euros as revenue generated from sales needs to be repatriated back to Germany where most of their factors of production (e.g. employees, factories, and suppliers) are located.

Hedging Foreign Currency Exposure

Foreign exchange contracts are also used to hedge foreign currency exposure. Hedges are not just used by financial professionals and traders but also corporations and governments with long term and/or large exposure to foreign currency.

However, the volume of foreign exchange transactions with non-financial customers has been on a slow decline. It is estimated this represents about 10 to 15% % of all global foreign exchange transactions.

Speculation

The most popular use of foreign exchange contracts is for speculation and profit making in the financial market. This is largely driven by the increase in the number of hedge funds, traders and traditional funds that view foreign exchange as an asset class that can produce high yields, and the rise in online, electronic trading and algorithmic trading.

4.7.7 Risks of Foreign Exchange Trading

Leverage

Similar to how options and futures contracts are traded, foreign exchange trading is usually leveraged. Low-cost, online trading platforms offer up to 100 times leverage for the purposes of trading in the major currencies.

The use of leverage can significantly improve the return on investment (ROI) of profitable currency trades as contracts with large notional values can be traded with only a small fraction of cash invested. However the use of leverage will also amplify losses in unprofitable trades.

Volatility

The foreign exchange market can be very volatile especially in times of market uncertainty or political instability. Even though long term currency trends can be predicted with some degree of certainty, short term fluctuations can be unexpected and hard to predict.

Global Round-the-Clock Exposure

Because the foreign exchange market has an almost continuous trading window, events that transpire in a different time zone or continent may influence an investor's overnight currency position. This can be mitigated by not carrying positions overnight (i.e. close out all existing positions before the investor goes to sleep) or by implementing strict automated stop loss orders.

As foreign exchange is the "pulse" of the global economy, new currency traders may be overwhelmed by the need to monitor so many diverse factors. The risk lies in the inability to process the heavy volume and different types of information.

Counterparty Credit Risk

Foreign exchange is an over-the-counter traded product. As such, there is a counterparty credit risk that is not present in exchange-traded products.

4.7.8 How to Read a Foreign Exchange Term Sheet

Spot foreign exchange contracts typically have the following key terms in the terms sheet:

- **Trade Date** – The date on which the transaction was agreed upon by the counterparties.

- **Settlement Date** – The date on which the foreign currencies are exchanged between the counterparties. Typically, the settlement date is a few days after the trade date to allow the counterparties time to arrange the transfer of funds.
- **Deal Size** – This would be the notional amount involved for both currencies being traded.
- **Currency Pair** – This term clearly states the currencies involved in the transaction.

4.8 Swaps

4.8.1 Key Features of Swaps

A swap is an exchange of one entitlement for another. A common example is Borrower A (who has a fixed interest rate loan) exchanging his obligations with Borrower B (who has a floating interest rates loan).

Swaps differ from Futures in that swaps are typically traded over-the-counter (OTC) and are “tailor-made” to each parties’ specific needs though some standardized swaps are traded on Exchanges such as the Chicago Mercantile Exchange.

Swaps are typically marked to market and mature at dates specific to that particular transaction. Another difference between swaps and Futures is that Futures contracts are usually traded on margin (use of leverage) and have standard maturity dates as determined by the Exchange on which the Futures contract is traded.

Swaps have developed rapidly since the 1980s. They have become one of the most important and flexible instruments available to bank and corporate treasurers for asset and liability management.

4.8.2 Interest Rate Swaps

Institutions sometimes find that their assets and liabilities are unavoidably mismatched. For instance, a bank will generally borrow short and lend long. A rise in interest rates would result in losses for the bank. Its cost of funds and therefore liability payments would increase, as depositors demand higher rates, while its cash flow from existing assets (long-term loans) would remain the same.

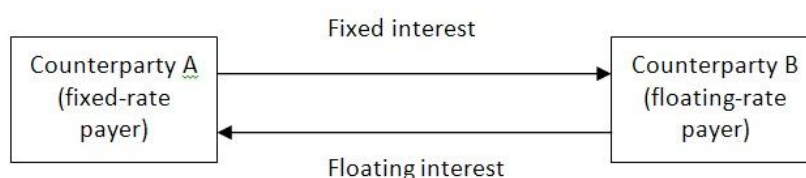
In contrast, a non-financial institution may borrow long and invest short by issuing long-term bonds and investing in floating-rate securities. If interest rates fall, the institution loses. Clearly, these two institutions could reduce their interest rate risk exposure by exchanging the interest rate elements of their positions.

In an interest rate swap, there is an exchange between two parties (referred to as counterparties) of interest obligations (payments of interest) or receipts (investment income) in the same currency on an agreed amount of notional principal for an agreed period of time. The agreed amount is called 'notional principal' because the principal amount is not exchanged initially or repaid at maturity.

The most basic form of interest rate swap is the plain vanilla agreement or "fixed-for-floating" transaction. The transaction involves the switch of interest flows from a fixed to a floating basis, or the reverse, in the same currency. The party who agrees to pay fixed interest payments is referred to as the fixed-rate payer while the other party who agrees to make interest payments that float with some reference rate, is referred to as the floating-rate payer.

This arrangement is illustrated in Exhibit 4.9.

Exhibit 4.9 Interest Rate Swap



Plain vanilla swaps are generally used to restructure the cash flows of an interest-sensitive asset or liability. For example, an institutional investor, who expects interest rates to rise, may want to convert his fixed-rate bond to a floating-rate note so that his coupons would rise as future money market rates increase. With the swap agreement, the investor is effectively paying the fixed-rate coupons he receives from his bond in exchange for receiving floating-rate coupons.

The reference rates used for the floating rate in an interest rate swap are those on various money market instruments such as Treasury bills or the London interbank offered rate (LIBOR). LIBOR is the interest rate at which major banks offer to lend money to other major banks in the London interbank market. The rate at which they bid to borrow funds from other banks is known as the London interbank bid rate or LIBID. Both LIBOR and LIBID are reference rates against which floating-rate payments of interest rates are calculated.

4.8.3 Cross-Currency Interest Rate Swap

A currency swap is an agreement between two or more parties to exchange interest obligations (payments of interest on borrowings) or interest receipts (interest income received from a loan) between two different currencies. In the swap, two currencies are involved. Because the associated cash flows are in different

currencies, the principal amounts usually are exchanged at the origination and maturity dates of the contract.

The interest rates defining the transaction can be expressed on either a fixed or floating-rate basis in either or both currencies. The predominant currency swap exchanges a fixed rate in the foreign currency for a floating interest rate in the US dollar. This structure can be illustrated in Exhibit 4.10.

Exhibit 4.10 shows two distinct types of exchanges:

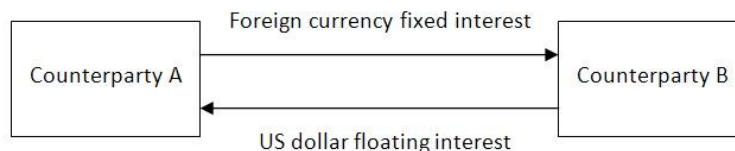
- Principal at both origination and maturity
- Interest on all of the settlement dates

Exhibit 4.10 Cross-Currency Interest Rate Swap

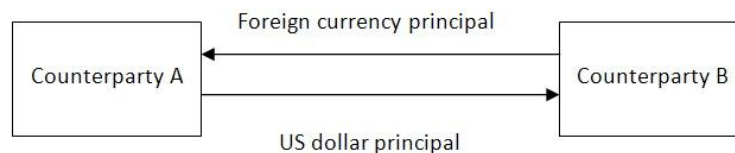
- (1) At origination, the counterparties exchange the principal amounts of the swap at an agreed spot rate. This will produce principal amounts in the two currencies upon which the payments of interest will be made.



- (2) On each settlement date (including maturity)



- (3) On maturity date, the counterparties re-exchange the principal amounts along with the final periodic payment of interest.



Currency swaps are used extensively in conjunction with bond issues. Borrowers have been able to achieve lower funding costs by executing a currency swap to create funding in the currency desired. For example, the World Bank issues bonds in many currencies but these are often swapped to obtain “cheaper” floating-rate US dollar funding.

4.8.4 Risks of Using Swaps

Foreign exchange swaps, interest rate swaps and currency swaps all have two common inherent risks.

Interest Rate Risk

The value of the amount being swapped between two parties is dependent on interest rate used as a reference in that particular contract. Volatility in interest rates can lead to significant uncertainty over the amount that each counterparty receives and pays out. However, this type of risk can be hedged by taking offsetting positions using related derivative instruments, like interest rate futures, currency futures, etc.

Credit Risk

As swaps are traded over-the-counter, there is always a risk of default by the counterparties involved.

4.9 Chapter Summary

This chapter has covered many aspects of Options, Futures, Foreign Exchange and Swap Contracts. The salient points are:

- An option is a financial security that gives the buyer of the option the right but not the obligation to buy or sell an asset at a pre-agreed price within a specific period of time.
- When an investor is bullish about the asset price, he will buy a call option or write a put option. If he is bearish, he will write a call or buy a put.
- The option price comprises of the option's intrinsic value and its time value. The intrinsic value of an option is the minimum value of the option and is the difference between the share price and the exercise price. It cannot be negative.
- Stock index options allow investors to buy puts and calls on broad market indexes. Settlement is in cash.
- Interest rate options can be used for interest rate risk management while currency options can be used to hedge against a currency exposure.
- A futures contract is an agreement to buy or sell a stated commodity or financial claim at a specified price at some future specified time.
- Most futures contracts are settled by offset. The open interest reflects the number of contracts that are not offset by opposite transactions.

- The margin requirement refers to the “good faith” money made by investors to ensure that the obligations of the futures contracts will be satisfied.
- All futures contracts are marked to the market daily, which means that profits and losses are credited or debited to each investor’s account every trading day.
- If the margin account balance drops below a critical level, called the maintenance margin, the investor will be required to deposit additional margin, known as variation margin, to bring the account back to its initial margin.
- Currency movements are influenced by many factors. Both fundamental and technical factors need to be considered.
- A foreign exchange transaction involves the purchase of one currency against the sale of another currency for settlement or delivery on a specified date. Foreign exchange rates can be quoted on a direct, indirect or cross basis.
- A spot transaction is a contract in which one currency is exchanged for another for settlement usually in two working days. A forward transaction is a contract to exchange one currency for another at some future date, later than spot, at a rate of exchange which is determined on the trade date.
- The major factor determining the forward rate of one currency against another is the difference in interest rates in the two countries over the specific forward period. The interest differential is converted into forward points or swap points which are either added to or subtracted from the current spot rate.
- Generally, the currency of the country with the higher interest rates will trade at a forward discount to that of the other; the currency with the lower interest rates will trade at a forward premium.
- An interest rate swap is an agreement between two parties to exchange interest payments in the same currency on an agreed amount of notional principal at designated times.
- A cross-currency interest rate swap is an agreement between two or more parties to exchange interest obligations or interest receipts between two different currencies.

Appendices

Appendix 4.1: Black Scholes Option Price Model

A stock option's price is based on the price of the underlying stock, the option's exercise price, the time to expiration of the option, the price volatility of the underlying stock, the risk-free interest rate and the dividends expected.

Fischer Black and Myron Scholes derived a model for pricing call options. The Black-Scholes option pricing formula computes the theoretical price of a European call option on a non-dividend paying stock. The formula is:

$$C_0 = S_0 N(d_1) - \frac{X}{e^{rt}} N(d_2)$$

$$d_1 = \frac{\ln(S_0/X) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

Where,

C_0 = Call option price

X = Exercise price

r = Continuously compounded riskless rate of interest on an annual basis

t = Time to maturity in years

σ = Standard deviation of the annual rate of return on the underlying stock

$\ln$ = Natural logarithm function

e = 2.71828, the base of the natural log function

$N(d_1)$ = Cumulative density function of d_1

Four of the inputs in the above formula, namely, exercise price, stock price, time to maturity and risk-free interest rate are easily observed. The fifth factor, the stock's volatility, as measured by the standard deviation, has to be estimated.

Because the standard deviation must be estimated, it is likely that discrepancies exist between the observed option price and its theoretical value as computed by the Black-Scholes model.

Appendix 4.2: Warrants

Example

Share price	S	= \$5.90
Exercise Price	X	= \$6.00
Warrant price	WP	= \$0.305 (Call Warrant with expiry in 150days)
Conversion ratio	n	= 2 (i.e.2 warrants for 1 share)

$$\begin{aligned}
 \text{Gearing ratio} &= \text{Share price} / (\text{Warrant price} \times n) \\
 &= \$5.90 / (\$0.305 \times 2) \\
 &= 9.67
 \end{aligned}$$

Prices after 50 days	Decrease in Share price	No change in Share price	Increase in Share price
ABC Ltd	\$5.30	\$5.90	\$6.50
% change in ABC's price	- 10%	0%	+ 10%
Warrant Price	\$0.115	\$0.245	\$0.425
% change in Warrant Price	- 63%	- 20%	40%

From the table above, we can see that a certain percentage change in the price of the underlying results in a larger percentage change in the warrant. When ABC company's share price remains unchanged, the warrant price declines nonetheless, due to time value decay of the warrant.

Example

Suppose we have a Put Warrant with the following information:

DEF Ltd

Share price S = \$22.20
 Exercise Price X = \$22.00
 Warrant price WP = \$0.295 (Put Warrant with expiry in 172days)
 Conversion ratio n = 5

Prices after 57 days	Decrease in Share price	No change in Share price	Increase in Share price
DEF Ltd	\$20.00	\$22.20	\$24.40
% change in DEF's price	- 10%	0%	+ 10%
Warrant Price	\$0.485	\$0.24	\$0.10
% change in Warrant Price	+ 65%	- 19%	- 65%

$$\begin{aligned}
 \text{Gearing ratio} &= \text{Share price} / (\text{Warrant price} \times n) \\
 &= \$22.20 / (\$0.295 \times 5) \\
 &= 15.05
 \end{aligned}$$

Delta

The delta of a warrant is the rate at which its price changes with respect to changes in the price of the underlying asset. Call deltas are always positive and put deltas are always negative. For example, at-the-money warrants have a delta of around 0.5. This means that a one cent change in the value of the underlying asset will result in a 0.5 cent change in the value of the warrant.

Options which are deeply in-the-money have a delta of close to 1. In other words, a one cent change in the value of the underlying asset will result in a one cent change in the value of the warrant.

$$\text{Delta} = n \times \delta\text{WP} / \delta\text{S}$$

Where,

Delta = Rate at which warrant price changes with change in share price

δWP = Change in warrant price

δS = Change in share price

n = Conversion Ratio

For investors, the concept of delta is important when deciding on the strike price of a warrant. Investors expecting a small move in the price of the underlying asset should buy warrants which are at-the-money or slightly in-the-money while investors expecting a large move can buy out-of-the-money warrants.

Effective Gearing

Effective gearing of a warrant is calculated simply by multiplying the delta of a warrant by its gearing.

$$\text{Effective Gearing} = \text{Delta} \times \text{Simple Gearing}$$

Example

Given the same information as above, and if Call Warrant's Delta = 0.543 (or 54.3%)

Simple Gearing = 9.67 times

Effective Gearing = $9.67 \times 0.543 = 5.25$ times

This means that a 1% increase in the underlying price results in a 5.25% increase in the warrant price illustrating its effective (or actual) gearing of 5.25 times.

In the case of the Put Warrant example, if the Delta = -0.422 (or -42.2%)

Simple gearing = 15.05 times

Effective gearing = $15.05 \times -0.422 = -6.35$ times

Call warrants with high strike prices provide more gearing but are not appropriate if only a small price move is expected. In general, the strike price chosen should be between the current share price and the expected share price. The maturity of the warrants should be carefully chosen to avoid excessive time decay loss.

Implied Volatility

Implied volatility is the volatility implicit in the market price of the warrant. Implied volatility can also be looked at as a measure of risk. The higher the volatility of the underlying shares, the more expensive a warrant is.

Implied Volatility = f (share price, warrant price, strike price, maturity, interest rate and dividend yield)

Interpreting the Trading Name of a Structured Warrant

The trading name of a structured warrant indicates some of its basic features.

Example

Example: ABC XYZ eCW080228

- Underlying instrument ("ABC") --> ABC Ltd Shares. In the case where the underlying is an index, the exercise price of the warrant is indicated immediately after the underlying's name.
- Issuer ("XYZ") --> XYZ financial institution.
- Exercise style ("e") --> European Style. If it is an American-style warrant, there is no prefix to the Type of warrant.
- Type of warrant ("CW") --> Call Warrant. The notation for a Put Warrant will be "PW" and for a Discount Certificate it will be "DC".
- Expiration date ("080228") --> 28 February 2008.

Warrant Valuation

- **Intrinsic Value (IV)** – The difference between the exercise price of a warrant and the market price of the underlying asset; only in-the-money warrants have intrinsic value.

Call Warrant

$$IV = \text{MAX}\{0, (S - X)/n\}$$

Based on the example above, $IV = \text{MAX}\{0, (\$5.90 - \$6.00)/2\} = 0$

Put Warrant

$$IV = \text{MAX}\{0, (X - S)/n\}$$

Based on the example above, $IV = \text{MAX}\{0, (\$22.00 - \$22.20)/5\} = 0$

- **Conversion Price** – This is the total price which the investor pays by converting the warrants into the underlying security, is also known as the breakeven price.

Call Warrant

$$CP = X + nWP$$

For example, $CP = 2 \times \$0.305 + \$6.00 = \$6.61$.

Put Warrant

$$CP = X - nWP$$

For example, $CP = \$22 - (5 \times \$0.295) = \$20.525$

- **Premium** – The difference between the warrant price and intrinsic value is the premium. This term is different from the one used in options, where the premium is actually the price of the option.

This premium is largely the time value of the warrant (refer to the factors affecting time value in the previous chapter on Options). Exhibit 4.11 shows the profile of a call warrant's price, its minimum/intrinsic value and its maximum/underlying value.

$$\text{Call Warrant Premium (\$)} = nWP + X - S$$

$$\text{Put Warrant Premium (\$)} = nWP - X + S$$

Premium is usually expressed as a percentage of the underlying share price. By definition, premium is the percentage by which the underlying share price needs to have moved at maturity for the investor to break even.

The premium for a Call Warrant is calculated as follows:

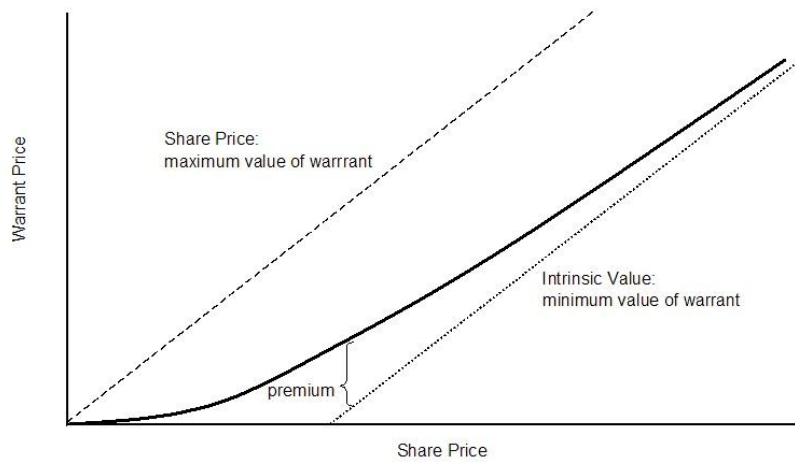
$$\text{Premium (\%)} = [(nWP + X - S) / S] \times 100$$

For example, $\text{Premium (\%)} = [(2 \times \$0.305) + \$6.00 - \$5.90] / \$5.90 \times 100$
 $= 12.03\%$

In the case of the Put Warrant example,

$$\text{Premium (\%)} = [(nWP - X + S) / S] \times 100$$

For example, $\text{Premium (\%)} = [(5 \times \$0.295) + 22.20 - \$22.00] / \22.20×100
 $= 7.55\%$

Exhibit 4.11 Call Warrant Premium**Appendix 4.3: Foreign Exchange Contracts****Calculating Forward Points**

The general formula for calculating forward points is as follows:

$$\text{Forward points} = \frac{S \times (R_c - R_b) \times n / 360}{1 + R_b \times n / 360}$$

Where,

S = Spot rate

R_c = Annualised interest rate of counter-currency

R_b = Annualised interest rate of base currency

n = Number of days from spot date

Example

Suppose the prevailing market rates are:

Spot USD / JPY = 81.30

$R_b = R_{US} = 0.25\%$ p.a.

$R_c = R_{JPY} = 0.00\%$ p.a.

n = 30 days

$$\text{USD / JPY forward points} = \frac{81.30 \times (0.00 - 0.0025) \times 30/360}{1 + 0.0025 \times 30/360} = -0.0169$$

The 30-day USD / JPY forward rate will be quoted at a discount which is USD / JPY 81.2831.

Quotation of Spot Rates and Forward Points

It is a common practice for banks to quote only the last two figures of an exchange rate. If the exchange rate at a particular point in time is GBP/USD 1.5850/55, it means GBP/USD is 1.5850 to 1.5855. A spot trader is likely to quote 50-55.

Forwards are quoted as “forward points” to be deducted from the spot rate. Thus,

$$\text{Forward Exchange Rate} = \text{Spot Rate} \pm \text{Forward Points}$$

$$\text{Forward points} = \text{Forward Exchange Rate} - \text{Spot Rate}$$

or

$$\text{Forward points} = \text{Spot Rate} - \text{Forward Exchange Rate}$$

Example

Assume the spot AUD / USD is quoted as 1.0970 / 80 and the 3-month outright forward rate is 1.1150 / 70. What are the forward points?

Spot rate: AUD/USD	1.0970	1.0980
Forward points	<u>180</u>	<u>190</u>
Forward Outright	<u>1.1150</u>	<u>1.1170</u>

That is,

The bid forward points = 1.1150 - 1.0970 = 0.018 or 180 basis points

Similarly, the offer forward points = 1.1170 - 1.0980 = 0.019 or 190 basis points.

Answer: Forward points are 180/190.

Interest Rate Parity

The value of the premium or discount can be theoretically computed from the interest rate parity theorem. This theory states that (except for the effects of small transaction costs) the forward premium or discount should be equal to the difference in the national interest rates for securities of the same maturity.

Should the interest rate parity relationship be violated, arbitrageurs will be able to make risk-free profits in foreign exchange markets with zero net investment. Their actions will force futures and spot exchange rates back into alignment.

According to the interest rate parity theorem, the relationship between the spot rate and forward rate is:

$$F = S \times \left(\frac{1 + R_c (n/360)}{1 + R_b (n/360)} \right)$$

Where,

F = Forward rate

S = Spot rate

R_c = Annualised interest rate of counter-currency

R_b = Annualised interest rate of base currency

n = Number of days

We can illustrate the interest rate parity theorem by using two currencies, the US dollar (counter-currency) and the British pound (base currency). For example, if $R_{US} = 0.25\%$ and $R_{UK} = 0.50\%$ annually, while pound spot is US\$1.6706, then the forward price for a 1-year contract would be:

Theoretical Forward Price, $F = \text{US\$1.6706} (1.0025/1.0005) = \text{US\$1.6664443}$ per pound

This is the rate which will provide no arbitrage gains if an investor borrows pounds to invest in dollars for one year. That is, this is the zero profit or breakeven rate, as illustrated in the 4 steps below:

- **Step 1** – Borrows £1,000,000 for 1 year at 0.5% per annum. After one year, the arbitrageur will need to repay £1,005,000.
- **Step 2** – Exchanges the borrowed pounds into dollars at the spot exchange rate of US\$1.6706. The arbitrageur receives US\$1,670,600 (e.g. £1,000,000 x 1.6706).
- **Step 3** – Invests the dollars for one year at 0.25% per annum. At the end of one year, he will receive US\$1,674,777 (e.g. US\$1,670,600 x 1.0025).
- **Step 4** – Enters into a one-year forward contract to buy £1,005,000 at forward price of US\$1.6664443. At the end of one year, the arbitrageur needs to pay US\$1,674,777 (e.g. £1,005,000 x 1.6664443).

After one year, the arbitrage position is closed out as follows:

- Receive US\$1,674,777 from the US dollar deposit (See Step 3).
- Use US\$1,674,777 received to obtain £1,005,000 at an exchange rate of \$1.6624 (See Step 4).
- Use £1,005,000 received to pay the principal and interest on £1,000,000 loan (as in Step 1).
- At the end of one year, the arbitrageur's net position has no liability and no asset. He makes zero gains, illustrating the interest rate parity theorem.

However, if the 1-year forward is 1.6625, there may be opportunities for an arbitrageur to make some gains:

- He contracts at the forward rate of 1.6625 for £1,005,000 and needs to pay US\$1,670,813 (e.g. $1,005,000 \times 1.6625$) at the end of the year.
- He receives US\$1,674,777 at the end of the year from the pound investment (See Step 3).
- He will make a gain of US\$3,964 (e.g. $US\$1,674,777 - US\$1,670,813$).

This arbitrage gain was accomplished by simultaneously borrowing in one currency, investing in another currency, and covering the exchange position in the forward exchange market. The entire process is known as Covered Interest Arbitrage.

Chapter 5:

Structured Products

Objective

The candidate should be able to:

- ✓ Understand the key features of the various structured products
- ✓ Understand the risks, obligation and rights of buyers and sellers of structured products
- ✓ Understand the generic risks of structured products
- ✓ Understand the uses and suitability of structured products
- ✓ Understand the common strategies and types of structured product
- ✓ Understand the key terms of each structured product
- ✓ Perform the calculation of the maximum gain, loss, and breakeven under various scenarios

5.1 Introduction

A structured product is a complex financial product that is designed to provide investors with highly targeted investments that are tied to a specific risk/reward profile and market expectations. It is usually part of a pre-packaged investment strategy consisting of two or more financial instruments with one or more embedded derivative components.

The building blocks of the structured product include bond, equity, equity index, foreign exchange, commodities, asset-backed securities (ABS) and collateral debt obligations (CDO) and derivatives products such as options, and credit default swaps (CDS), etc. The variety of products illustrates the fact that there is no single, uniform definition of a structured product. Therefore, the scope of structured products is very wide.

Growth of Structured Products

As compared to the traditional investment asset classes such as bond, equity and commodities, the amount invested in structured products (as an asset class) is still

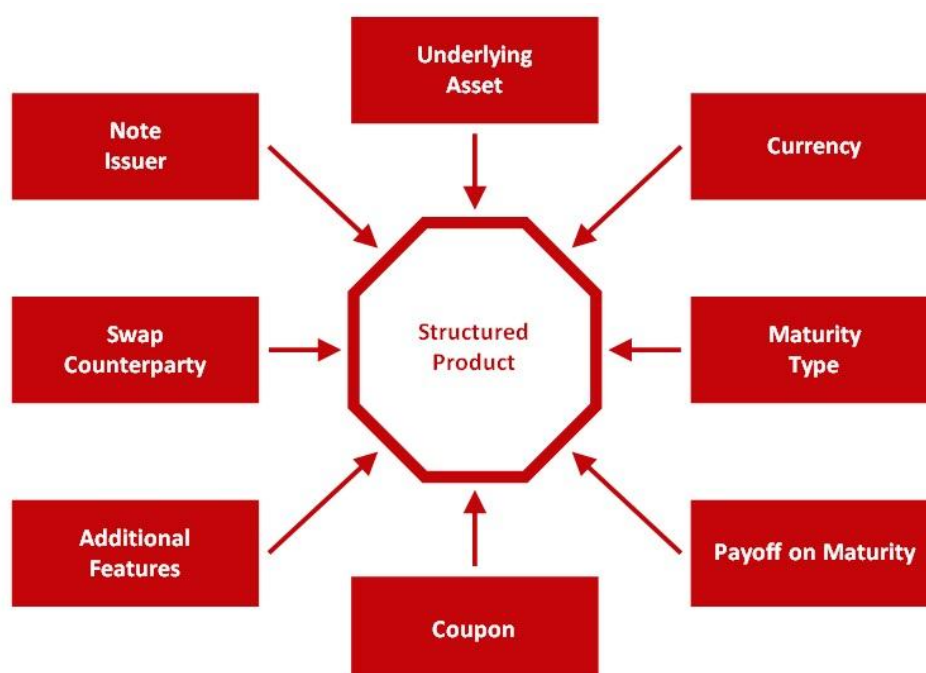
relatively small. However, demand for structured products has grown over the years. The reasons for the growth are as follows:

- Low interest rates environment, which motivates investor to look for extra yield;
- The extreme volatility in the equity market, which encourages investors to invest in structured products which offer upside participation, yet preserve the principal capital against the negative returns associated with a volatile or bear market; and
- Marketing effort by banks to offer structured products.

5.2 Key Components and Features of Structured Products

To understand structured products, it is important to understand its key features. Exhibit 5.1 summarises the key features and components of structured products.

Exhibit 5.1 Components and Key Features of Structured Products



A good understanding of the components and the key features would allow both the buyer and seller of the structured product to know the key risks, the drivers of performance and what to monitor during the life of the investment.

In addition, it would also allow them to know the rights and obligations of the investors and issuers, and perform scenario analysis. It will also assist the investor to compute the maximum gain, maximum loss and breakeven level under each scenario. This would help both Covered Persons and clients to carefully assess the uses and suitability of a structured product.

5.2.1 Selling of Structured Products

Structured products help investors to capitalise on certain market views and achieve return (income and capital growth) objectives within their risk parameters. The structured product chosen should match the client's market views, achieve his/her return objective if his/her view is correct and confine the loss in case his/her view is wrong.

Therefore it is important for the Covered Person to help clients understand the product features and make full disclosure of the potential risks that may prevent the investor from achieving his/her objectives.

5.2.2 Underlying Asset

Structured products are investments which typically offer defined rates of return linked to the performance of an underlying asset or index over a fixed term. The return performance will track that of the underlying assets / financial instruments / indices that can be chosen from the equity, fixed income, funds, commodities asset classes, and currency, interest rates and derivatives markets. These instruments include:

- Bond and other fixed income derivative instruments ;
- Equity, equity index and exchange traded funds (ETF);
- Fund (mutual funds and hedge fund) and fund index ;
- Commodities, and commodity index and futures;
- Foreign exchange (FX) and non-deliverable forwards (NDF);
- Interest rates and interest rate futures such as LIBOR, SOR, T-bond yield; and
- Credits such as Credit Default Swaps (CDS) and credit-linked notes (CLN).

There are many ways for the issuer to choose and combine these instruments. The variations include:

- A single underlying instrument / asset
- Two or more underlying instruments from the same asset class
- Two or more underlying instruments from different asset classes

Moreover, the pay-off can be derived from the price performance of a single underlying asset or two or more assets on a best-performing, worst-performing, average price or even a spread basis.

5.2.3 Maturity Type, Payoff and Coupon Structure

There are various types of maturity. Most structures have a normal maturity with cash repayment. Some may have features such as callable, putable, early redemption trigger, knock out redemption, first-to-default redemption, and some even have features like maturity repayment in an underlying asset (physical settlement) such as a stock, a bond, a fund or cash in an alternate currency.

The maturity payout may have a minimum return of principal on maturity. Such structures usually employ a zero coupon bond with a long-call strategy. The performance of the structure would depend on the performance of the underlying financial instrument, strike price and the participation rate.

A minimum return of principal product may also employ CPPI (constant proportion portfolio insurance) strategy where no options are involved (see below examples in section 5.6.11). A maturity payout which does not have a minimum return of principal usually employs short options strategies.

The coupon in a structured product may be fixed, floating, inverse floating, daily accrual, step-up, step-down, zero, an “all-or-nothing” pay-off depending on certain conditions being met, or participating where the coupon is depending on the performance of the chosen underlying financial instrument. Other possible variations of coupon include:

- Same coupon or coupon formula throughout the product’s life;
- Different coupon or coupon formula for different coupon periods, e.g. fixed for first coupon period, formula-based for subsequent coupon periods; and
- Conditional coupon formula for each coupon period dependent on certain pre-specified condition, e.g. x% if condition A is met during coupon period T1 and y% if condition B is met during coupon period T2.

Structured products use all forms of derivatives including plain vanilla put, call, exotic digital and barrier options. To achieve the desired coupon and pay-out profile, structured products employ either long or short or a combination of long and short strategies, and with various strike and tenors. Structured product may also employ CDS.

The information on the underlying asset, maturity tenor as well as the coupon and pay-off formula and profile can be found in the term-sheet of each structured product.

5.3 Types of Structured Products

The term “wrapper” refers to the legal form in which a structured product is offered to the end investor. A structured product is usually wrapped as a:

- Structured Deposit;
- Structured Notes; or
- Structured Fund

5.3.1 Structured Deposits

A structured deposit is essentially a combination of a deposit and an underlying financial instrument, where the return is dependent on the performance of the underlying financial instrument. Structured deposits are a type of deposits and differentiated from structured notes. The distinguishing feature is that for a deposit, the principal sum, with or without interest, has to be repaid in full at maturity. In the case of structured notes, investors may potentially lose part or the whole principal sum.

Structured deposits can only be issued by banks and must meet the definition of “deposit” as defined under the Banking Act. However, they are not covered by the Deposit Insurance Scheme under the Deposit Insurance Act 2005 (MAS guidelines on structured deposit). If an investor withdraws his/her deposit before the maturity date, he may lose part of the return and/or principal. The amount payable depends on the market value of the underlying financial instrument that the structured deposit is linked to.

From one perspective, structured deposits are generally less risky than direct investments in the underlying asset / financial instruments such as stocks and bonds because the bank is obliged to repay the principal in full at maturity or when it redeems the deposit before the maturity date (applicable to structured deposit with early redemption features).

However, from another perspective, structured deposits are riskier than traditional deposits because their returns are dependent on the performance of the other financial instruments that they are linked to. In some scenarios, the investor may only get back the principal with no returns.

5.3.2 Structured Notes

A Structured note is a debt instrument / debenture, which is governed by the Securities and Futures Act (SFA). The performance of the note, such as coupon payment or the market value of the note is linked to the performance of other underlying instruments such as a stock, indices, interest rates, credit markets (CDO and CDS) or even a basket of these instruments. These structures usually have one or more embedded options, or may employ other strategies using derivatives.

A structured note is subject to SFA prospectus requirements, except if the note is offered to only “accredited investors”. A structured note can be directly issued by a financial institution or a Special Purpose Vehicle (“SPV”) set up by the bank.

- Under direct issuance, the bank offers the notes to note-holders (as investors) under its MTN (medium-term note) programme. It is not uncommon that the notes issuer may be a different entity from the Covered Entity, which distributes the notes. In addition, the issuer of the structured note has a swap transaction with another institution to derive the cash flow so as to pay investors, for coupons (if any) and redemption on maturity or early redemption date. The investor bears the credit risk of the note issuer as well as the swap counter-party.
- Under the SPV model, the SPV issues the notes to the note holders. It is common for the SPV to use the proceeds from the notes issued to buy fixed income securities such as zero coupon bonds, debts securities, CDOs, or other asset-backed securities either to preserve the principal capital (capital preservation product) or use them as collaterals against the options or credit insurance which they sell to the market (as yield enhancement products).

5.3.3 Structured Funds

A structured fund is a product that is offered as fund combining a security such as equities or fixed income with derivatives to help investors achieve a specific risk / reward profile. Fixed income products are usually used to preserve principal investment with the added gain of interest payments. Options, futures, and other derivatives are typically used to provide exposure to capital appreciation.

Structured funds are also governed by the Code for Collectives Investment Scheme (CIS) under the Securities and Futures Act. Structured funds are packaged and can only be managed by an asset management company which needs to have a Capital Market Service license.

Unlike deposit and note where the fees associated with the structured products are usually embedded into the pricing of the products (e.g. through a theoretical lower yield or lower participation rate in the performance of the underlying asset), a structured fund usually has separate fees such as services and management fees.

The fund manager owes the investor a fiduciary duty to look after their interest. They therefore have a responsibility to ensure that they get the best prices when they execute trades for the fund. In addition, structured funds have to provide regular Net Asset Values (NAV) as required under CIS, while structured notes and structured deposits are not required to do so.

5.3.4 Other Structured Products

In addition to being wrapped as a structured deposit, note or fund, a structured product can also take the form of an over-the-counter (OTC) product. An example of an OTC structured product is an unfunded Accumulator where client is only required to place a margin with the distributing Bank. An unfunded Accumulator will be covered below in section 5.6.10.

5.4 Generic Risks of Structured Products

Different structured products have different risk profile. It is therefore essential to ascertain the terms and conditions of each structured product and the particular risks involved from the term sheet and product descriptions.

5.4.1 Credit and Counter-Party Risks

An important risk factor is credit risk. An investor should check and know exactly the credit rating and standing (e.g. credit outlook and credit watch) of the issuer before investing in the product. For notes issued by a SPV and involving CDS, it is important to be sure about the credit quality of the pool of underlying debts securities in the SPV and reference entities linked to the SPV.

As mentioned earlier, in a structure where there is a swap agreement between the issuer and another institution, the investor bears the credit risk of both the issuer and the swap counterparty.

5.4.2 Liquidity Risks and Secondary Market Trading

Structured products are designed for investors who are willing to hold the investment to maturity. As they are customised product tied to a specific risk / reward profile, there is typically a limited secondary market making it hard for investors to sell it before maturity date.

Structured products are illiquid, because the underlying derivatives components such as options (especially exotic options) and credit default swap may not be actively traded. Investors may also lose some part or even a substantial part of the principal sum if they were to liquidate the investment before maturity.

However, the issuer or its affiliates usually provide a secondary trading market for the structured product after issuance, even though they may not be obliged to do so. This is an important feature. Therefore, clients should be advised to assess the issuing bank with regard to its quality, commitment, as well as its track record of providing liquidity for secondary market trading.

In spite of the bank's commitment to provide secondary trading, investors should be made aware that the price of a structured product would ultimately depend on prevailing market conditions and levels / prices of the underlying instruments, such as yield, terms of maturity and marked-to-market valuation of any underlying derivative instruments.

5.5 Assessing Product Suitability for Investors

In assessing product suitability for the investors who would like to invest in structured notes, a Covered Person, must be clear about the investors' objectives and make full disclosure of the potential risks that may prevent investors from achieving their objectives.

At the same time, if the product does not suit the investor's profile in terms of his/her wealth and risk tolerance, the Covered Person must advise the investor accordingly. However, if the client insists on proceeding with acquiring this product, the Covered Person needs to ensure that there are proper documentation and acknowledgement by the client.

A Covered Person should help his/her clients understand that the potential higher payoff is only possible from assuming higher risk or forfeiting some of the principal preservation benefits. They should be aware of the implications of these structures. Before committing to invest in a structured product, clients should consider a number of factors, including:

- **Market View** – Apart from considering whether the structured product matches / expresses the same market view as his/her own, a client should also be advised to study whether in the scenario where his/her market views materialise, the return offered by the structured product will meet the return objectives as expected.

Some structured products impose limits, caps and barriers that affect their return potential. In addition, changes in market conditions and time remaining until maturity can expose investors to downside vulnerabilities such as loss of partial or all of the principal, and the possibility that at maturity, the investor will own the underlying at a depressed price.

- **Liquidity** – The amount invested usually will not be fully redeemable during the tenure of the structured product. Early withdrawal will result in loss of part or all of the principle invested. A client who invests in structured product should be prepared to hold them until maturity.
- **Risks** – Structured products have higher risks due to the different the layers of risks involved. Apart from implicit leverage within products like options, futures, swaps and other derivatives some clients employ further leverage to acquire the structured products.

Structured products that utilise leverage typically provide partial or no principal preservation but offer the potential of leveraged returns on the value of the underlying assets. The covered person should help the client understand the higher downside risks involved and whether the products fit within his/her risk appetite.

- **Returns** – Returns of the structured products are tied to various underlying instruments. Investors should be aware that their returns will vary based on

the performance of the underlying instruments and there are possibilities that the instruments' principal values will be affected.

In some structured products where the pay-out is formula-based, the value of the structured products prior to maturity can be substantially different from the value derived by the formula. This is because of demand / supply dynamics and the illiquidity of the embedded financial instruments.

- **Tax** – Due to the yearly capital or income distribution feature of some structured products, there might be tax implications for some investors. It is the investor's responsibility to consult his/her own tax adviser before making the investment decision.
- **Terms and Conditions** – Clients should review the term sheet as it will explain to the investors their rights and obligations involved and also certain risks that investors are subjected to.
- **Fees and Charges** – Before investing, clients should be familiar with the fees and charges before investing. Management fee is likely to be higher than standardised financial instruments due to the complexity of structured products. As for the underlying instruments utilised, it is likely that a transaction cost will be incurred for every transaction. Clients need to understand how these costs are eventually passed on to them.

The general principle is that clients should not purchase structured products if they do not understand how the products work. They should not buy any product until they have sought proper clarifications and advice.

Structured products are generally not suitable for investors who:

- Want the possibility of higher returns than traditional products such as bonds and deposits but are not prepared for variable returns which include the risk of losing all or a substantial part of their original principal sum;
- Do not understand how the pay-out in a structured product works. For example, the investor is unclear about how returns are computed or the factors and scenarios that can affect the returns;
- Do not understand the risks associated with the structured product. For example, the full extent of the credit risks associated with the issuer as well as any other reference entities;
- Are not prepared to leave their money invested until the maturity date as an early redemption of the structured note on the part of the investor may result in the investor receiving significantly less than his/her original investment amount; and
- Are not prepared for their investment to be redeemed earlier at the issuer's option or for other specified reasons.

Structured Deposit Linked to Equity Index

Product description: Structured deposit linked to S&P index

Reference index: S&P spot index

Guarantee: 100 % principal sum at maturity

Principal investment: SGD 1,000,000

Tenor: 1 years

Current spot price of S&P Index: 1000

Strike Price: 1000

Participation rate: 40% of reference asset performance

If the S&P Index closes at 1200 (20% increase) at maturity, the interest rate earned on the equity deposit = $40\% \times 20\% = 8\% = \text{SGD } 80,000$.

In the event that the S&P index ends below 1,000 at maturity, investors will earn a zero return. They will however get back 100% of the principal sum at maturity.

5.6 Strategies and Common Types of Structured Products

Structured products employ various strategies with features based on individual specific market view, risk and return objectives. Below are some examples of the various strategies / structured products.

5.6.1 Zero Coupon Fixed Income Plus Option Strategy or “Zero Plus” Option

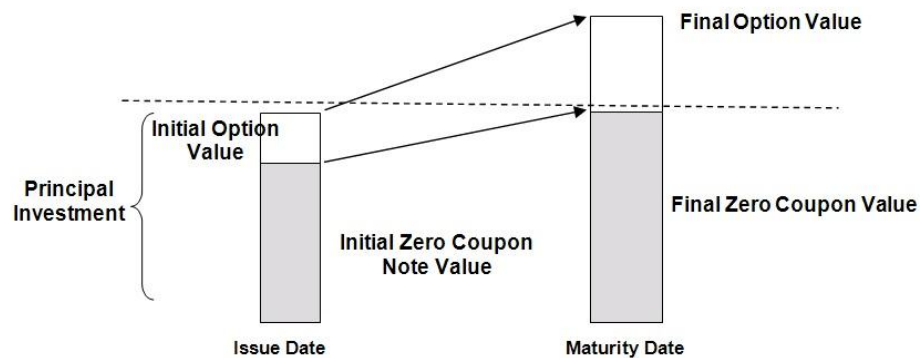
A zero plus option strategy employs a zero coupon fixed income instrument, usually a zero coupon note with a call option on an underlying financial instrument. The underlying financial instrument can be a stock or an equity index, a currency pair, a commodity or a commodity index etc. The returns of such strategy would depend on the performance of the underlying financial instruments.

The key terms of a zero plus option strategy structured product are:

- **Reference Asset (or Financial Instrument)** – The underlying financial instrument which the performance of the structured product is linked.
- **Strike Price** – The level of the underlying financial instrument / asset where the participation in performance of the underlying financial instrument kicks in.
- **Participation Rate** – The percentage increase in return of the structured product for every 1% upside performance of the underlying financial instrument.

- **Fixing Date** – The day where the closing level of the underlying financial instrument is used to determine the pay-out of the structured product at maturity.

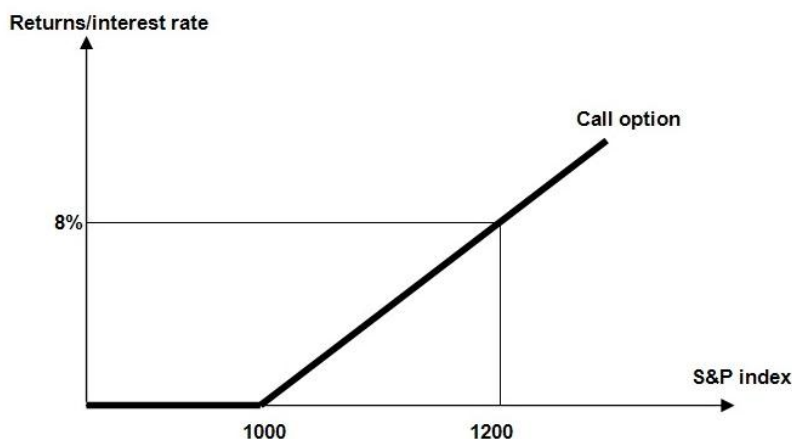
Exhibit 5.2 Zero Plus Option Strategy



As illustrated in Exhibit 5.2, as long as there is no credit event in the issuing bank, the investor will at least get back 100% of the principal sum. This strategy is also known as a capital preservation strategy. The return of such a structure product is a function of the performance of the **underlying asset** over and above the **strike price** and **the participation rate**.

If the structure uses at-the-money option, then the strike price is the prevailing market level of the underlying financial instruments at the start of the product's life. The fees on a zero plus option strategy structured product are usually embedded in the structure, and there is no separate fee that investor would have to pay.

Exhibit 5.3 Pay-Out of a Zero Plus Option Strategy



Uses and Suitability of the Zero plus Option Structured Deposit / Note

A zero plus option strategy structured product is usually more appealing to conservative investors. In the worst case scenario, the investor will still have his/her principal back at maturity subject to the creditworthiness of the issuing bank. An investor for such a product typically has a moderate bullish view on the underlying financial instrument / asset and so seek participation with principal preservation.

The investor must be made aware that in the event that the underlying asset performs exceedingly well, the return from this strategy is going to underperform the underlying asset. In the event that the underlying close at or below the strike price, an investor will not earn any returns on his/her investment.

The investor must be prepared to hold the investment for the entire tenor as the structured product is usually not liquid and investor may lose part of the principal if the investment is liquidated before maturity date.

5.6.2 Range Accrual Note

This is a structured note where the investor receives a target level of return if a reference index “falls” within an agreed range, failing which the client receives less / no interest, but his/her principal will not be affected. The reference index can be a stock index such as Dow Jones index or an interest rate benchmark such as LIBOR or Singapore SOR (Swap offer rate).

The interest is usually accrued and calculated on a daily basis for the days when the reference index falls within the range stipulated in the terms and conditions. In short, it is a structure where the final interest pay-out depends on observed price movements of the reference index.

A range accrual note is long digital options. As long as the reference index falls within the range, investor will get a fixed or floating coupon. In the event the reference index close outside of the predefined range, the investor will receive less / no interest as there is no pay-out on the digital options. A general expression for the payout of a range accrual is:

Reference index inside range: Payout = $P1 \times (n/N)$
 Reference index outside range: Payout = $P2 \times (N-n)/N$

Where,

N = Total number of observations within a period

n = Total number of observations when the index is inside range

P1 = Payout when the index is inside the range

P2 = Payout when the index is outside the range

The observation period can be daily, week or monthly, though a daily observation is very common. It is also common for the payout to be zero when the index is outside the agreed range. The key terms of a range accrual note are:

- **Reference Index** – The index observed for determining any payout.
- **Range** – The range of prices of the reference index within which coupon will be accrued.
- **Inside Range Payout** – The coupon earned by investor if the reference index falls inside the range.
- **Outside Range Payout** – The coupon earned by investor if the reference index falls outside the range.
- **Observation Period** – Time period over which returns are observed, e.g. daily or weekly.
- **Settlement Period** – Coupon payment interval.

Scenario Analysis: Range Accrual Note

Description: Notes in SGD with yield linked to 3-mth SGD SOR
 Nominal investment: SGD 100,000
 Term: 1 year
 Coupon: 5% p.a. daily for each day the 3mth SGD SOR closes within 1% to 1.5% range, 0% if 3-mth SGD SOR closes outside the range
 Maturity: 100% principal sum subject to the credit of issuer, ABC bank
 Coupon payment: Semi-annually
 Basis of coupon rate calculation: ACT/360
 Coupons fixing dates: 22 Jun 2010 and 22 Dec 2010

Given, from 23 Dec 2009 to 22 Jun 2010, the number of days 3-mth SOR closes within the range is 110 days out of 182 calendar days and from 23 Jun 2010 to 22 Dec 2010, the number of days the 3-mth SOR closes within range is 160 days out of 183 calendar days, calculate the coupon payment fixed on 22 Jun and 22 Dec 2010.

1st coupon paid on 22 Jun 2010 = $110/182 \times 5\%/2 = 1.51\%$.

2nd coupon paid on 22 Dec 2010 = $160/183 \times 5\%/2 = 2.16\%$.

Variations of Range Accrual Notes

As the market evolves, some Range Accrual structures are now combined with an equity-linked structure or DCI structure. In a Daily Range Accrual Equity Linked Structure, the coupon during the life of the structure will be subjected to the Daily

Range Accrual conditions, while the payment on maturity will be dependent on an equity-linked note structure. In such cases, the Range Accrual Note will be a principal-at-risk product, instead of a product with 100% minimum return of principal on maturity.

It is also possible to link the Range Accrual condition to multiple underlying assets, for example, a basket of 3 underlying shares. In such cases, the structure will only accrue coupons if all 3 underlying shares fall within the agreed range, thereby greatly increasing the risk of not accruing coupons on certain days.

5.6.3 Short Option Strategy

A short option strategy is used for the purpose of yield enhancement or for the accumulation of the underlying asset / financial instruments in the case of a structure with physical settlement. Because of the yield enhancement feature, such a strategy is popular in a low interest environment.

Since this strategy has a short-option, the upside is limited but the downside can be substantial and there is a negatively skewed risk-reward ratio. However, when an investor takes physical delivery of the financial instrument / asset at the strike price and then subsequently the price of the underlying financial instrument / asset appreciates in value, a gain is available.

Structure products which employ the short option strategy include:

- Equity linked note;
- Credit linked note;
- Bond linked note;
- Dual currency investment;
- Range accrual note; and
- Accumulator

5.6.4 Equity Linked Note (ELN)

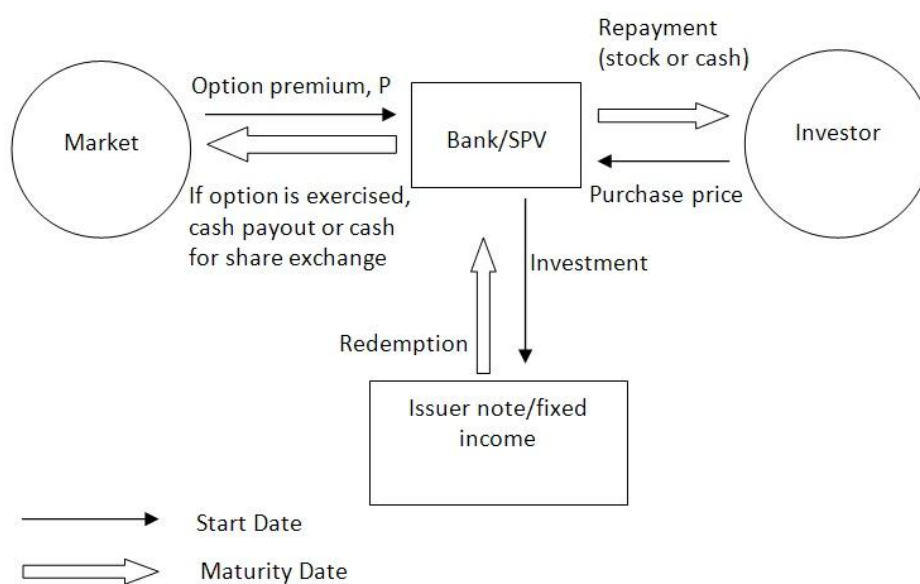
An equity linked note employs an issuer's note and a short-put, which is either linked to a stock index or a particular stock. An ELN may pay a higher yield than market rate of interest arising from the value of the short option position associated with the underlying stock or index.

An ELN can be structured to allow for physical or cash settlement at maturity. When the underlying is an index, physical settlement (i.e. where the investor take delivery of the underlying asset) is not possible and the ELN will be settled in cash.

The key terms of an ELN are:

- **Issuer** – The entity which issues the note. It could be a financial institution or a SPV.
- **Reference Financial Asset** – The underlying asset linked to the put option.
- **Strike Price** – The price level of the underlying financial instrument below which the option will be exercised.
- **Price of Note** – A discount to the face value of the note. The discount calculates the yield of the note if the put option is not exercised.
- **Mode of Settlement** – Cash or physical settlement.
- **Fixing Date** – The day when the closing price level of the underlying asset / financial instrument is used to determine the pay-out of the structured product at maturity. It is usually 2 business days from the maturity date.

Exhibit 5.4 Equity Linked Note



Scenario Analysis: Equity Linked Note (ELN)

Product description: Structured note linked to ABC Ltd stock price
 Reference underlying asset: ABC Ltd shares
 Notional Investment: S\$ 1,000,000
 Purchase price of ELN: S\$0.90
 Maturity: 100% notional sum or 100,000 ABC shares.
 Tenor: 1 year

Current ABC share price: S\$12.00

Strike Price: S\$10.00

At maturity, there could be 3 scenarios:

Best case scenario: Where the ABC Ltd share is fixed at or above the strike price

In this scenario, the put will not be exercised and the investor will receive S\$1,000,000 at maturity earning 11.1% yield on the S\$900,000 initial investment.

Moderate case scenario: ABC Ltd share fixed below the strike price (e.g. S\$9.50)

In this scenario, if the terms and conditions of the ELN specify a cash settlement, then the investor will receive $100,000 \times S\$9.50 = S\$950,000$ cash at maturity, and he still makes S\$50,000 out of his / her S\$900,000 initial investment. The break-even level of ABC share price is S\$9.00, where the investor will receive only the principal sum and zero return on his / her investment.

If the terms specify physical settlement as the mode of settlement, then the investor will receive 100,000 ABC Ltd shares. If the investor sells the shares in the market immediately, he will receive S\$950,000, a scenario which is no different from cash settlement.

However, if the investor decides to hold on to the shares, he has a direct exposure to the subsequent performance of the ABC Ltd shares. The ultimate return would depend on the performance of the stock price and the timing of the sale of the shares.

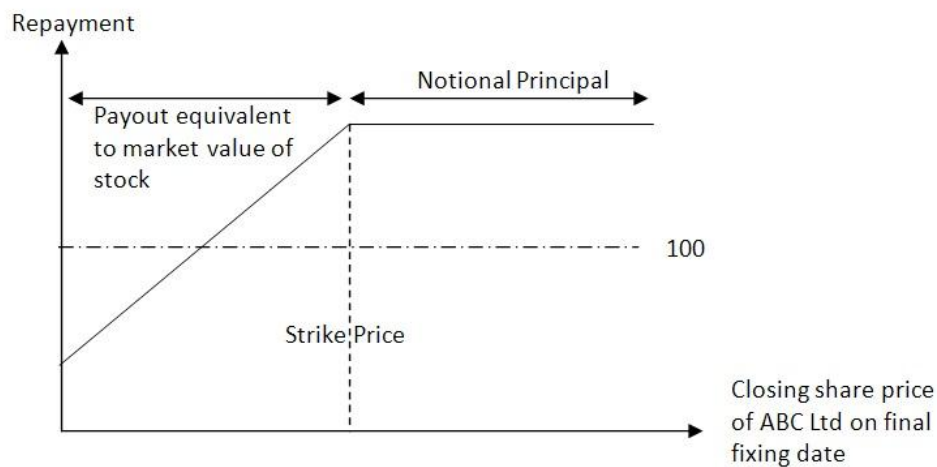
Worst case scenario: Price of ABC Ltd share dropped to zero value

Under the cash settlement mode of settlement, investor will lose the entire principal investment i.e. he gets nothing back on maturity date.

Under the physical settlement mode of settlement, investor will receive 900,000 of ABC Ltd shares. The investor will not be able to sell the shares immediately in the market as the value has dropped to zero. However, he can hold on to the shares and wait for any possibility of turnaround in the company and the recovery in share price of ABC Ltd.

There is also a possible scenario where ABC Ltd winds down the company business and liquidates all its assets. In this scenario, there may be a residual value to shareholders after ABC Ltd uses the asset sale proceed to pay off all the company's liabilities.

Exhibit 5.5 Payoff of an Equity Linked Notes


Uses and Suitability of Equity Linked Notes

Equity Linked Notes with physical settlement can be used by an investor who would like to own the shares of Company ABC Ltd, but feels that the current price of ABC stock is too high. Thus, a strategy for such an investor is to buy an Equity Linked Notes that has an out-of-the-money put on ABC Ltd shares.

At maturity, if the share price of ABC Ltd is traded below the strike price, the investor will be able to buy ABC Ltd shares below the current market price of ABC Ltd shares. In fact, because the ELN is sold at a discount, investor will buy even below the strike price.

An ELN is not a structured note which tries to preserve the principal investment. If the closing price of the underlying asset is below the strike price at the final fixing date, the investor is exposed to the downside movement in the price of the underlying asset. The underlying asset can trade considerably below the strike price, which may lead to a partial or even a total loss of the principal capital.

The investor also bears the credit risks of the issuer or the credits of the debt securities backing the SPV. A credit event may trigger an early termination of the ELN and the investor will be paid the liquidation value of the various components of the ELN, which could be substantially below the principal investment.

As shown in the payoff profile, an ELN's potential return is limited on the upside, which is usually pre-determined in the term sheet. If the underlying stock price rises considerably, the investor may have obtained a better return by a direct investment in the underlying stock.

An ELN may have little or no liquidity. Investor should be prepared to hold the ELN for the entire tenor. Furthermore, if a secondary market exists, the marked-to-

market value of the ELN will fluctuate depending on a number of factors such as the movement in the underlying share price or index, its volatility, the performance of the fixed income investment, and the creditworthiness of the issuer.

5.6.5 “Worst of” Equity Linked Note

A “worst of” ELN is like a normal ELN described above, except that it is linked to more than one underlying share or index and its return depends on the performance of the worst performing underlying in the basket. The worst performing underlying stock or index is determined by comparing its closing price on the final fixing date against its initial price in percentage term.

For a “Worst of” ELN, you are exposed to the downside risk of the worst performing underlying security, even if, for all other underlying securities, the closing prices are above their strike prices on the fixing date. Such an ELN pays a higher yield (i.e. deeper discount) as compared to a normal ELN because it is a much more risky investment, with exposure to the downside movement of not just a single stock or single stock index, but multiple stocks or indices.

5.6.6 Credit Linked Notes (CLN)

A credit linked note is another type of yield enhancement strategy structured product. As the name suggests, it has exposure to the credit markets. In a CLN, the manager offers credit insurance known as credit default swaps (CDS) linked to a particular company (referred to as “reference entity” in the term sheet). It has a fixed income investment, which is used as collateral against the credit insurance sold in the market.

A CDS is akin to providing credit insurance on a reference entity. It is a contract between two counterparties, whereby the “buyer” pays periodic payments (insurance premium) to the seller in exchange for the right to a pay-out if there is a credit default of the “reference entity”. It is a derivative that is traded over the counter.

A credit default is defined as the event where the reference entity:

- Fails to pay interest on its loan or bond; and/or
- Is unable to repay the bond or loan on maturity date.

In the event of a credit default, there are 2 ways to settle the CDS contract:

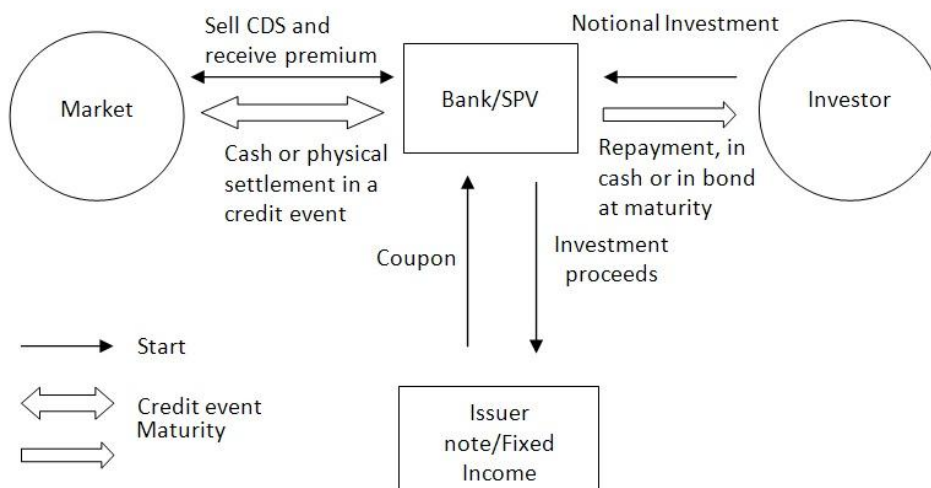
- The bank being both the seller of the CDS and the issuer of CLN pays the CDS buyer the principal amount of the CDS in cash and receive a debt obligation (i.e. a bond) of the reference entity that is now in default. The market term for such settlement is called physical settlement. In this scenario, the CLN investors end up owning the bond of the reference entity, which would have a value determined by the market.

- The bank pays the CDS buyer the difference between the par value and the market price of a specified debt obligation of the reference entity, typically determined in an auction. The market term for such settlement is called cash settlement. In this scenario, the CLN investors will bear a loss equivalent to that difference.

The key terms of a CLN are:

- **Issuer** – The entity which issues the note. It may be a financial institution or a SPV.
- **Reference Entity / Entities** – Company / companies which are linked to the CDS.
- **Mode of Settlement** – Cash or physical settlement of reference entity in a credit event.
- **Type of CDS** – Single reference entity default or “first to default” for multiple reference entities.
- **Early Termination** – Event that may trigger an early redemption of the note, e.g. a credit event of a reference entity.
- **Coupon** – Periodic interest rate payable to note holder.
- **Coupon Payment Period** – Time interval between coupon payments.

Exhibit 5.6 Credit Linked Notes



Credit Risks

Like an ELN, a CLN investor receives a higher than market rates reflecting the risk associated with the sale of the CDS linked to the reference entity. In a CLN, investor is exposed to two different credit risks.

The two credit risks are:

- Credit of the note Issuer or bond(s) held by the SPV as collateral for the CDS; and/or
- Credit of the “reference entity” linked to the credit default swap.

The first form of credit risk is investment grade. It is quite common that the first form of credit risks is the credit quality of the issuing bank. This must be good because it is used as collateral for any potential loss that may arise from the sale of the CDS.

Scenario Analysis: Credit Linked Note (CLN)

Description: Note linked to the credits of ABC Ltd

Reference entity: ABC Ltd

Type of CDS: Single credit of ABC Ltd

Mode of settlement: Physical settlement with ABC Ltd bond (maturing Jun 2015)

Issuer: XYZ Bank, rated “AA”

Early termination trigger: Credit default of ABC Ltd

Term: 5 years

Coupon: 10% p.a. payable annually, unless there is an early termination of the note.

Maturity: 100% of principal if there is no credit event on the “reference entity”

There are 2 scenarios:

Best case scenario: No credit default of the reference entity

In this scenario, the CLN holder will receive the 10% annual coupon annually and 100% of their principal investment at maturity.

Worst case scenario: Credit default of the reference entity

In this scenario, the issuing bank will use the cash to exchange for the reference bond of ABC Ltd at face value, as agreed in the CDS. The note will be redeemed and investor will receive the reference bond. The investor can choose to hold on to the bond or sell out in the market immediately upon receiving the bond. If the investor sell out immediately in the market, he is likely to suffer a substantial loss as it is not likely for a defaulted bond to trade close to the par value.

5.6.7 “First to default” Credit Linked Note

In some CLN structure, the issuer offers CDS insuring multiple credits, on a first to default basis. This structure is a much more risky as compared to a normal CLN. If any of the credits defaults (first one to default), the cash with the issuer will be used to compensate the CDS buyer by way of cash settlement or physical settlement as discussed above. In such an event, the investor is likely to lose a substantial part of his/her principal investment.

5.6.8 Bond-Linked Note (BLN)

A BLN is also a yield enhancement structured product. It embeds a short-put on a bond. Potentially, investor may end up owning a bond at maturity like a CLN described above. However there are differences between the two structured products. The CLN sells a CDS on a reference entity, while a BLN sells a put option on a bond which has a strike price.

The pay-out of a CLN is dependent on whether there is a credit event on the reference entity while the pay-out of a BLN is dependent on the price of a bond. Besides a credit event, the price of a bond can be affected by other factors such as a credit downgrades, widening spreads and volatile interest rates. In a BLN, investor may end up owning a bond even if there is no credit event in the reference credit / bond.

5.6.9 Dual Currency Investment (DCI)

A dual currency investment involves a base currency, an alternate currency and an option in the currency pair. It can also be considered as a type of yield enhancement structured product. The MAS has specific guidelines on DCIs.

Dual currency investments are considered riskier than traditional deposits and other structured deposits because they expose the investor to additional foreign exchange risks. DCI accepts principal amounts in one currency (called the base currency) at the beginning.

Depending on exchange rate movements, the bank may pay out the principal plus interest in another currency (called the alternative currency) based on a pre-agreed exchange rate at maturity. DCI generally offers higher interest rates than traditional foreign currency fixed deposits. Part or all of the interest earned represents the premium of the sale of the currency option.

The key terms of a DCI are:

- **Base Currency** – The currency which the investor place with the bank.
- **Alternate Currency** – The currency which the investor may receive at the end of the tenor.

- **Strike Price** – The level to convert the base currency to the alternate currency.
- **Fixing Date** – The date where the currency rate is observed. The prevailing exchange rate on the Fixing Date will be used to compare with the strike price to determine if the investor will receive the base currency or the alternate currency.

Scenario Analysis: Dual Currency Investment (DCI)

Investment sum: SGD 100,000

Base currency: SGD

Alternate currency: USD

Tenor: 1 year

Interest rate: 5%

Strike price of USD/SGD rate: 1.2500 (current USD/SGD rate: 1.3000)

USD sum to be received if USD/SGD hit below 1.2500: USD 84,000

At maturity, there could be 3 scenarios:

Best case scenario: Where the USD is traded at or above 1.2500 against the SGD

In this scenario, the put will not be exercised and the investor will receive SGD 105,000 at maturity earning 5% yield.

Moderate case scenario: USD trade below 1.2500 against SGD, but it is not below 1.1905 SGD to 1 USD

In this scenario, the investor will receive USD 84,000 at maturity. When investor convert the USD 84,000 back into SGD at that point of time, the investor will still receive more than SGD 100,000. The breakeven USD/SGD level is 1.1905.

Worst case scenario: If the USD trades below 1.1905 SGD to 1 USD

In this scenario, the investor will receive USD 84,000. But because the USD has fallen below the breakeven level of 1.1905 SGD to 1 USD, the investor will receive less than SGD 100,000 when he converts the USD back to SGD.

This not only results in the investor not earning a single cent on the DCI, the investor will, in addition, lose part of the principal sum in SGD term.

Risks, Suitability and Uses of DCI

Investors should be advised to consider the exchange rate risk of receiving his/her money at maturity in the alternate currency. If the maturity proceeds (principal plus interest) are paid in the alternate currency, the investor could experience a loss on his/her principal when he converts it back to the base currency. Foreign exchange movements of the alternative currency affect the investor's returns from DCI.

A DCI might provide opportunity to earn potentially higher interest rate than time deposits for an investor who has a natural demand for the alternate currency, for example, if the investor has a child who is studying in Australia and needs AUD for both living and education expenses. The investor will buy a DCI with AUD as the alternate currency. If the SGD is not converted to AUD at maturity, the investor will earn an enhanced yield on his/her SGD funds.

In the scenario when the investor receives the AUD at maturity, the investor uses AUD for the expenses of his/her child in Australia. The risk for that investor is that in a scenario where the alternate currency keeps on appreciating against the base currency, he/she may never be able to buy the alternate currency at a good rate because the option will never be exercised.

5.6.10 Accumulators

An accumulator allows an investor to purchase a pre-determined quantity of a reference stock at regular intervals. The product involves a long-call and a short-put on the stock with the same strike price. An accumulator can be structured as an OTC option where investor is only required to put up a margin (i.e. unfunded basis) or wrapped as a structured note (fully funded). An unfunded accumulator is more common.

Both types of accumulator may have a knock-out (KO) barrier (although this is not a standard feature), where if the daily closing price of the underlying shares is at or above the barrier, the derivative agreement will be terminated. With such a feature, the upside (quantities purchased at discounted price) for the investor is limited by the KO barrier. As long as the daily closing price of the underlying is below the KO barrier, investor will accumulate a predetermined quantity of the shares daily and take delivery periodically, e.g. monthly.

The purchase price (strike price) is fixed at the outset and usually represents a discount to the market price of the shares at the time the investor enters into the OTC accumulator agreement with the Covered Entity. The discounted strike price reflects the risk that the investor will bear as there is a requirement to purchase the reference stock at the strike price throughout the tenor of the agreement even if, on any purchase date, the market price of the shares has dropped below the strike.

In a "1X2 gear" scheme, with a KO Barrier Accumulator, an investor is required to buy 2X the predefined quantities of the reference stock when the price is below the strike price. However, if the share price is above the strike and below the KO barrier,

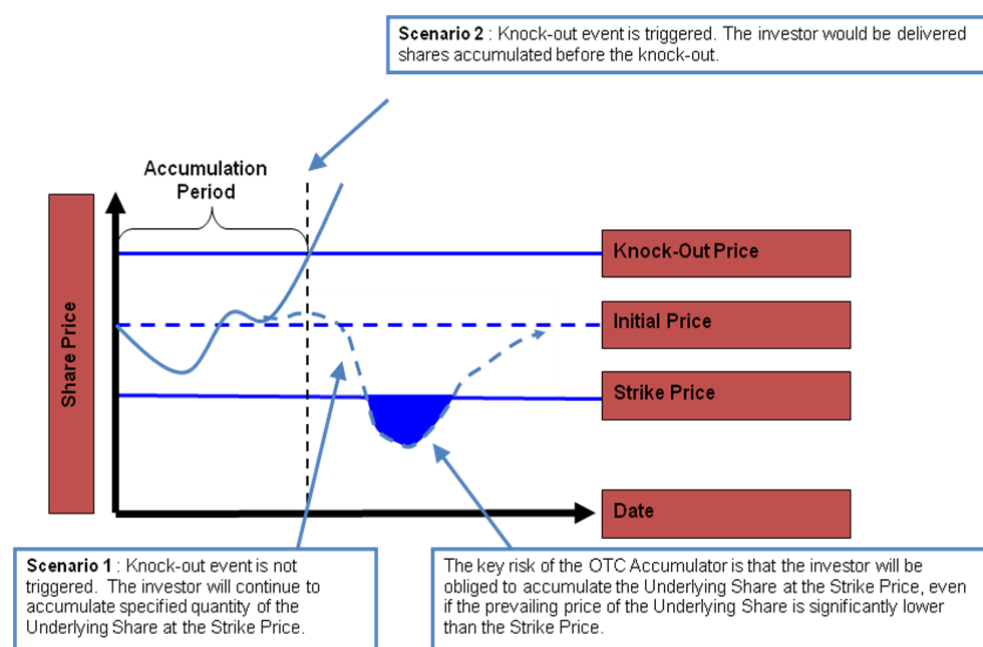
investor can only buy 1X of the predefined quantities of the shares of the “reference stock”.

If the closing price of the underlying shares is at or above the KO barrier, the agreement will be terminated immediately. Investor will have to pay for and take delivery of any shares accumulated at settlement date, which is usually 2 or 3 business days after termination of agreement.

The key terms to know:

- **Reference Stock** – The share that the investor is ‘accumulating’.
- **Strike Price** – The price at which the investor buys the reference stock.
- **Knockout Price** – The share price which will trigger the termination of the Accumulator contract.
- **Full Observation Period** – The tenor of the note.
- **Settlement Period** – Time interval for settlement, e.g. daily or month-end.
- **Quantity of Shares** – The quantity of reference stocks to be accumulated per observation.

Exhibit 5.7 Illustration of an Accumulator



Scenario Analysis: Accumulator

Reference share: ABC Ltd
 Tenor / observation period: 1 year (250 business days)
 Spot reference price: S\$1.10
 Strike price: S\$1.00
 KO barrier: S\$1.30
 Number of shares per daily observation: 10,000
 Settlement: Daily observation, monthly settlement

Best case scenario: The closing price of the shares ABC shares stays above the strike price and is below the KO barrier on each business day during the tenor of the agreement

In this scenario, investor will accumulate 10,000 shares at a price of S\$1.00 per share on each business day and will pay for and receive the accumulated shares on a monthly basis.

If the prevailing market price of the shares is above S\$ 1.00 when investor receives the shares, he can sell the accumulated shares immediately in the market and realise a gain. The investor may also choose to hold on to the shares if he is of the view that the share price of ABC Ltd will head higher.

Moderate case scenario: The closing price of the shares of ABC Ltd stays above the strike price but on one trading day during the tenor, the closing price is at or above the KO barrier

In this case, investor will accumulate 10,000 shares at a price SGD 1.00 per trading day up to but excluding the day on which the closing price is at or above the KO barrier. When that happens, the agreement will be terminated.

If the prevailing market price of the shares is above SGD 1.00 when the investor receives the ABC Ltd shares, he can sell the accumulated shares in the market and realise a gain. However, the gain is limited as investor will not be able to purchase the maximum number of underlying shares at the strike price, which is probably at a discount to the prevailing market price of the shares.

Worst case scenario: The closing price of ABC Ltd shares is zero for each business day during the tenor

In this case, investor have to purchase 10,000 shares per day for 250 business days, or 2,500,000 ABC Ltd shares at S\$1.00 per share. Investor will not be able to sell the shares as they are valueless, and incur a marked-to-market loss of S\$2,500,000, the maximum loss under this agreement.

If the accumulator operates on a non-funded basis (i.e. margin facilities), the bank would probably have asked for the client to top up margin to the full extent, i.e. to fork out S\$2,500,000, way before maturity of the agreement.

Risks of an Accumulator with a KO barrier

- There is no capital preservation feature in an accumulator. If the closing price of the reference stock is below the KO barrier at any time during the tenor of the Accumulator, investor will have to purchase the underlying shares at the strike price for the tenor even if the market price of the underlying shares remains substantially below the strike price for most or all of the tenor, which could result in a substantial loss.
- If the closing price is at or above the KO barrier on any day during the tenor of the accumulator, the accumulator will terminate immediately and investor will not be able to purchase the underlying shares at the strike price. In other words, the investor's potential gain is limited by the KO barrier.
- Investor is exposed to larger potential losses if it is a 1X2 gear accumulator where the investor is required to buy 2X the predefined quantities of reference stock when the price of the stock trades below the strike price.

Uses and Suitability of an Accumulator

Before investing in accumulators, an investor should be advised to consider whether the reference stock is a security that he would like to accumulate at the strike price, which may be higher than the prevailing market price when the accumulation of the share takes place.

The investor must be made aware of the maximum quantities of shares which he is required to acquire throughout the tenor of the accumulator, and in the worst case scenario, where the share price dropped to zero value, the maximum loss he will bear.

If, during the tenor of an Accumulator, the prevailing market price of the underlying shares remains above the strike price but below the KO barrier, the accumulator allows investor to realize a gain by acquiring the pre-defined number of underlying shares at the strike price and selling these shares at the market price upon receipt.

For unfunded accumulators that provide leveraging / margin facilities, investors may not be required to make an initial payment. The leverage works for and against investor, with potentially significant gains and losses. With a 1X2 geared accumulator, and a KO barrier, the risk-reward is skewed as the gain is limited by the KO barrier while the loss will be further magnified by the 2X predefined quantity that the investor is required to buy when the stock is below the strike.

The Covered Entity which offers unfunded accumulator will have a margin monitoring system to track the volatility of the reference shares price and the marked-to-market value of the accumulator. In the event of a significant fall in the shares price of the reference entity, margin call or liquidation of the underlying shares will be enforced if the investor is not able to top up the margin within a stipulated time.

In such a situation, the Covered Entity may also, at its sole discretion, decide to close out the accumulator and/or initiate any other steps as set out in the credit facility agreement. The investor will bear all the costs and expenses incurred in connection with the closing out of the accumulator.

The investor is not allowed to terminate the accumulator before maturity without the consent of the bank. If the Covered Entity agrees to terminate the accumulator at the investor's request, the investor will be required to pay the "break" costs which can be substantial. Therefore, an investor should be advised to consider his/her level of certainty that the reference share will trade within range between the strike and the KO price till its maturity.

In addition, investor is not entitled to receive any dividends or any other rights or interests attached to or derived from the underlying shares prior to delivery. The company that issues the underlying shares may be the subject of certain events and corporate actions such as stock split, rights and bonus issues, or one-off extraordinary dividend payment, which may trigger the bank to adjust the key terms such as the strike price, the KO barrier level (if any) and the quantities to be accumulated, according to pre-defined terms and conditions.

The company shares may also be suspended as a result of unforeseen disruption events which may give the bank the right to terminate the accumulator according to pre-defined terms and conditions.

5.6.11 Constant Proportion Portfolio Insurance (CPPI) Strategy

A structured product (fund or note) employing CPPI technique dynamically allocates funds between a "risk free" asset (a bond) and a risky underlying asset (an equity index or an equity fund or hedge fund, etc). Under CPPI, a floor value of the total portfolio is calculated at regular intervals, the objective of which is to preserve the principal.

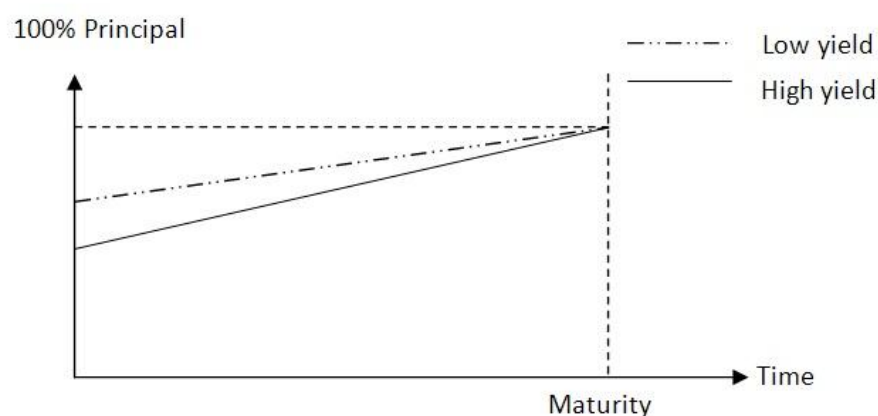
In other words, when the portfolio value drops to the floor, allocation shifts to the risk free asset, so that the investor will get back the principal sum when the structured product matures. To make the principal more secured, the manager may buy insurance from another financial institution (usually a bank) to guarantee the principal sum, at maturity for a yearly fee of 1 to 2%.

It is important for an investor to note that when the floor value is reached, all his/her funds are allocated to the risk free asset and there will no longer be any participation in the subsequent upside performance of the underlying asset. In that situation, the investor will only receive the principal sum at maturity, assuming the "risk free asset" does not default.

The Workings of CPPI

- **The “Floor Value”** – Changes during the life of the structured fund, depending on the remaining term to maturity and the yield curve at each point in time. As the investment moves towards maturity, the floor value approaches 100% of principal sum.

Exhibit 5.8 Floor Values of a CPPI

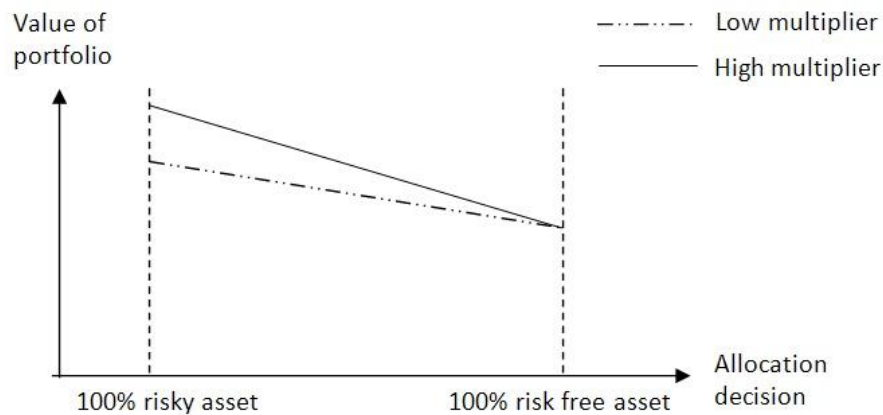


- **Crash Size** – The percentage drop in value of the underlying asset in any one period in a crash scenario. The **multiplier** is equal to 1 divide by the crash size percentage. For illustration purpose, if the manager is of the view that the largest percentage crash for the chosen risky asset is 20%, then multiplier = $1/20\% = 5$.
- **Asset Allocation** – Adjustment is done periodically, usually once a month. The difference between the value of the total portfolio and the floor value is known as the cushion. If the cushion is large, the manager can afford to take more risk, i.e. higher allocation to risky asset, and vice versa.

$$\text{Allocation to the Risky Asset} = \text{Multiplier} \times \text{Cushion Value}$$

If the cushion value is 15%, and the multiplier is 5, then the allocation to the risky asset = 75%.

Exhibit 5.9 Asset Allocation


CPPI Strategy

Investment amount: S\$100
 Crash Size: 20%
 Multiplier: $1/20\% = 5$
 Risky Asset: S&P Index fund
 Tenor: 5yrs

Asset allocation on Day 1

Floor value: 85% (or cushion 15%)
 Multiplier: 5
 Investment in Risky Asset = $5 \times 15\% = \$75$.
 Investment in Risk-free Asset = $\$100 - \$75 = \$25$

Asset allocation at the end of period 1

Let's assume that the new floor value increases from 85% to 86% of initial principal sum. At this point, assuming the price of the Risky Asset has remained unchanged, the manager will have to adjust the allocation to Risky Asset as per the new floor value as follows :

Floor value: 86% (or cushion of 14%)
 Multiplier: 5
 Investment in Risky Asset: $5 \times 14\% = \$70$
 Investment in Risk-free Asset: $\$100 - \$70 = \$30$
 Total value of CPPI Structure: \$100

Scenario 1: Value of portfolio falls to 86%

Say the price of the Risky Asset falls by the full crash size of 20%. The value of the risky asset would have fallen from \$70 to \$56. The total portfolio value at

this point would have been \$86 computed as follows:

Investment in Risky Asset = \$56
 Investment in Risk-free Asset = \$30
 Total value of CPPI Structure = \$86

Because the total value of the structure is now at \$86 which equals the floor value, the entire allocation to Risky Asset will be liquidated and re- allocated to the Risk free asset. When this happens, investor will no longer enjoy the subsequent price appreciation in the S&P index fund and can only look forward to receiving their principal sum at maturity.

Scenario 2: Value of the portfolio falls to 96%

The price of the Risky Asset falls by 5.71%, such that the value of the investment in risky asset declines from \$70 to \$66.

The allocation to Risky Asset equal $5 * (96\% - 86\%)$, which is 50%. The Manager will have to sell part of the \$66 investment in Risky Asset to adjust the asset allocation to \$50 Risky Asset and \$46 Risk-free Asset.

	Before Re-allocation	After Re-allocation	Change
Risky Asset	\$66	\$50	- \$16
Risk-free Asset	\$30	\$46	+ \$16
Total Value	\$96	\$96	

Scenario 3: Value of portfolio rises to \$110

The price of the Risky Asset rises by 14.29% such that its value increases from \$70 to \$80. Then, the allocation of Risky Asset equals $5 * (110\% - 86\%) = 120\%$.

In this case, all the fund will be allocated to the Risky Asset as follow:

	Before Re-allocation	After Re-allocation	Change
Risky Asset	\$80	\$110	+ \$30
Risk-free Asset	\$30	\$0	- \$30
Total Value	\$110	\$110	

In some CPPI Structure, the Manager may be permitted to take on leverage to further increase the allocation to the Risky Asset to meet the calculated allocation as follows:

	Before Re-allocation	After Re-allocation	Change
Risky Asset	\$110	\$120	+ \$10
Risk-free Asset	\$0	-\$10	- \$10
Total Value	\$110	\$110	

If after taking on the leverage, the Risky Asset suffers a price decline equals to the Crash Size of 20%, such that the value falls from \$120 to \$96, the re-allocation will take place as follows :

	Before Re-allocation	After Re-allocation	Change
Risky Asset	\$96	\$0	- \$96
Risk-free Asset	- \$10	\$86	+ \$96
Total Value	\$86	\$86	

Notice that the 20% fall in price of the Risky Asset has taken the CPPI Structure back to the Floor of 86%. The manager will thus have to liquidate all \$96 of the Risky Asset, pay off the leverage of \$10, and allocate the remaining \$86 into the Risk-free Asset.

The scenario analyses and asset reallocation will be carried out periodically. Take scenario 2 as an example. If at the end of period 2, the value of the portfolio (value of Risky Asset plus value of Risk-free Asset) rises to 105%, the Manager will then sell the Risk-free Asset and reallocate the sales proceeds to the Risky Asset, according to the formula described above.

Risks of CPPI Strategy Product

According to the CPPI asset allocation process discussed above, there is more allocation to the Risky Asset as its value appreciates, and vice-versa. This CPPI technique may result in a situation where the investor ends up buying high and selling low in a range bound market.

If there is a prolonged period of range bound market, there would also be a higher chance that the portfolio value may drop to the floor value, forcing the manager to allocate the entire fund into the Risk-free Asset. Once that happens, there shall no longer be any participation for any subsequent appreciation in the value of the underlying asset.

For the above scenario described above, it is important to highlight to the investor that such a strategy assumes the following characteristics of the underlying asset:

- Asset that appreciates in value over time;
- Low volatility; and
- Limited drawdown as measured by percentage decline from peak to trough.

Illustrative Details of a Structured Fund using CPPI

Product description: Structured fund linked to ABC's fund of hedge funds (FOHF)
 Base currency: USD
 Term: 10 years
 Return: Variable depending on the volatility and performance of ABC's FOHF
 Maturity: 100% principal guaranteed
 Guarantor: XYZ Bank
 Sale fees: 5% one-off
 Guarantee fee: 1.5% p.a.
 Management fee: 2% p.a.
 Style of Management: CPPI

Suitability of Structured Fund Employing CPPI

The structured fund described above does not have much liquidity in the secondary market and investors must be prepared to hold the product till maturity.

The advantage of this structured product compared to a direct investment in a hedge fund (or fund of hedge funds) is that there is a capital preservation feature where an investor will receive 100% of the principal sum at maturity unless a credit event occurs in the guarantor bank.

The disadvantage for this structured fund is that the overall fees are much higher than a traditional investment product. There are usually structuring, sales and guarantee fees as well as management fees. The return of the fund is likely to be affected by these fees.

In addition, in a volatile market or range-bound performance market, there is always the possibility that the floor value is reached. In this situation, the manager may be required to allocate all the funds into the Risk-free Asset. When that happens, there shall be no more participation in any subsequent upside.

5.7 Chapter Summary

This chapter has covered many aspects of Structured Products. The salient points are summarized below:

- A structured product is a complex financial product that is designed to provide investors with highly targeted investments that are tied to a specific risk / reward profile and market expectations. It is a pre-packaged investment strategy wrapped with 2 or more financial instruments with one or more embedded derivative components.

- The building blocks of the structured product include bonds, equities, equity indices, foreign exchange products, commodities, funds, asset-backed securities (ABS) and collateral debt obligations (CDO) and derivatives products such as options, and credit default swaps (CDS), etc.
- A Covered Person needs to analyze the key features of a structured product before recommending it to the client. The features include the appropriate:
 - underlying asset;
 - maturity type; and
 - payoff profile and coupon structure.
- A structured product can also take the form of an OTC investment product or wrapped as a structured deposit, structured note or part of a structured fund.
- In addition to market risks, investors of structured products are usually exposed to the credit risk of the issuer. Investors also are exposed to the counter-party risk if there is an embedded swap transaction. Structured products are generally not liquid.
- In assessing product suitability for the client, the Covered Person must understand clearly about the client's need and objectives and make full disclosure of the potential risks. Clients who do not understand the complexities behind the pay-out and the risks involved should not invest in structured products.
- A "zero plus" option strategy employs a zero coupon fixed income instrument, usually a zero coupon deposit or a zero coupon note, and a call option on an underlying financial instrument. The underlying financial instrument can be a stock or an equity index, a currency pair, a commodity or a commodity index etc. The returns of such strategy would depend on the performance of the underlying financial instruments.
- The short option strategy is used for the purpose of yield enhancement or for the accumulation of the underlying asset / financial instruments in the case of a structure with physical settlement. Since this strategy has a short-option, the upside is limited but the downside can be substantial.
- Equity linked note (ELN) employs a short option strategy and is made up of the issuer's note and a short-put, which is either linked to stock index or a particular stock. An ELN can be structured to allow for physical or cash settlement at maturity.
- A "worst of" ELN is like a normal ELN, except that it is linked to more than one underlying share or index and its return depends on the performance of the worst performing underlying in the basket.
- In a credit linked note (CLN), the structure sells credit insurance known as credit default swap (CDS) linked to a particular company (referred to as "reference entity" in the term sheet).

- In the event of a credit default, the CLN investor would either take delivery of the bond of the “reference entity” (physical settlement) or bear a loss equivalent to the difference between the par value and the market price of a specified debt obligation, typically determined in an auction (cash settlement).
- In a “first to default” CLN, the Issuer / SPV sells CDS insuring multiple credits, on a first to default basis.
- A bond-linked note embeds (BLN) a short-put on a bond. Unlike a CLN where the pay-out is dependent on whether there is a credit event on the reference entity, the pay-out of a BLN is dependent on the price of the bond, which is affected not only by a credit event, but also a credit downgrade, credit spread widening as well as interest rates movements.
- Dual currency investment (DCI) involves a base currency, an alternate currency and an option in the currency pair. DCI accepts principal amounts in one currency (called the base currency) at the beginning. Depending on exchange rate movements, the bank may pay out the principal plus interest in another currency (called the alternative currency) based on a pre-agreed exchange rate at maturity.
- A range accrual note is one where the client receives a return higher than market interest rates if the reference index falls within the certain range, failing which the client receives less / no interest, but his / her principal will not be affected.
- An accumulator allows an investor to purchase a pre-determined quantity of a company shares at regular intervals. The product involves a long-call and a short-put on a stock with the same strike price. An accumulator can be structured as an OTC option where investor is only required to put up a margin (i.e. unfunded basis) or wrapped as a structured note (fully funded).
- A common feature (although non-standard) in an accumulators is a knock out (KO) barrier, where if the daily closing price of the underlying shares is at or above the barrier, the derivative agreement will be terminated.
- A structured product (fund or note) employing CPPI technique dynamically allocates funds between a “Risk-free Asset” and a risky underlying asset (an equity index or an equity fund or hedge fund, etc). Under CPPI, the floor value is calculated at regular intervals, the objective of which is to preserve the principal.

The other factors which affect the allocation to the risky and “risk-free” assets are the cushion value (portfolio value less the floor value) and the crash size in an allocation period.

Chapter 6:

Unit Trusts and Fund Products

Objective

The candidate should be able to:

- ✓ Understand what a unit trust is
- ✓ Differentiate the role of the trustee and the manager
- ✓ Understand the advantages and disadvantages of unit trusts
- ✓ Understand the main features of different types of funds
- ✓ Explain the fees charged to the funds
- ✓ Calculate portfolio returns and risk
- ✓ Know what to look out for when evaluating a unit trust

6.1 Introduction

This chapter focuses on Collective Investment Schemes, which includes unit trusts, investment trusts and mutual funds. After reading this chapter, you should be familiar with the roles of parties that are involved in a typical unit trust and mutual fund. This chapter also highlights the advantages and disadvantages of investing in unit trusts and gives an overview of some of the criteria when evaluating unit trusts.

6.2 Key Features of Unit Trusts, Mutual Funds and Closed Ended Funds

6.2.1 Unit Trusts

A trust is defined as a legal arrangement in which a person or organization controls property and/or money for another person or organization. Following from there, a unit trust is a product whereby the organization (the trustee) controls the assets of the trust while fund manager manages the assets in line with a defined investment objective. The investor owns units of this trust.

In Singapore, we refer to unit trusts as collective investment schemes (“CIS”). The rules and regulations which collective investment schemes comply with are the Securities and Futures Act and Regulations. Responsible entities would also need to comply with the provisions within the Code of Collective Investment Schemes, although it is non-statutory.

As CIS are professionally managed, there are important roles and responsibilities which have to be undertaken by different parties involved. Typically, these parties are the trustee and the fund manager. Other parties like the registrar and the administrator may be engaged to perform certain service for the CIS.

The document which details the operation of the CIS is called a trust deed, while the offering document is called a prospectus.

Trustee

The main role of the trustee is to look after the interests of the unit holders. The trustee’s main responsibilities include:

- Ensuring that the fund manager manages the CIS in accordance with the investment objective and restrictions stated in the trust deed and prospectus. And if there are any breaches, the trustee will inform the Monetary Authority of Singapore within 3 business days after becoming aware of the breach;
- Ensuring that the accounting records are kept properly and the CIS is audited after which the unit holders receive the semi (within 2 months from the end of the period covered by the accounts or reports) and annual reports (within 3 months from the end of the period covered by the accounts or reports) in time;
- Take legal ownership of all assets in the CIS and ensures that these assets are held independently from the fund management company; and
- Ensuring that the CIS is managed in the interest of the unit holders. For example, if there are any changes proposed by the fund managers, the trustee will determine if these are beneficial to unit holders, if necessary an Extraordinary General Meeting may be required to obtain the approvals of the unit holders before the changes can be implemented.

Fund Manager

The roles of the fund manager include:

- Managing the assets of the CIS in line with the defined investment objective and within the investment restrictions as stated in the offering document;
- Preparing the semi-annual accounts, annual accounts, semi-annual report and annual report. These reports are to be furnished to the trustee so that

the trustee can have them audited before dissemination to the unit holders;
and

- Taking responsible for the creation and the redemption of units.

In summary, the fund manager manages and invests the fund's assets and is responsible for the fund's performance while the trustee holds the property on behalf of the unit holders and ensures the fund manager carry out his duties in accordance with the trust deed.

6.2.2 Mutual Funds

In other jurisdictions, there are references to mutual funds. Mutual funds are different from unit trusts as they are not set up as trusts. Some mutual funds are set up as fund within an investment company. Such structures are common in Luxembourg where there are tax benefits to set up Sociedade de Investimento de Capital Variável (SICAV) companies. Mutual funds do not have trustees as they are not trusts.

6.2.3 Closed-Ended Funds

A closed-ended fund is also a collective investment vehicle. It is usually available for subscriptions for a period of time. After that period, no more subscriptions will be allowed into the fund. Such a feature is common for a fund investing in structured products which will mature in a pre-determined number of years. Investors can still redeem their holdings from the fund but there are usually penalties or they may be realising their holdings at a loss.

Another form of common closed-end funds invests in closed markets like Vietnam or China "A" shares. The manager of such funds may not be able to liquidate the positions easily or remit the money of these countries to meet redemptions. Investors, who invest in such funds, should be aware that they may not be able to redeem their units as freely.

Some closed-ended funds are listed and have a secondary market where the shares are traded. However, as they are infrequently traded, investors should be aware of the liquidity risks.



Tips / Key Points

Investors may invest in various types of collective investment vehicles which include unit trusts, mutual funds and closed-end funds.

6.2.4 Advantages of Investing in Unit Trusts

Diversification

A typical unit trust is invested in many securities. Some unit trusts hold up to a hundred securities. Compared to what a typical investor has in his portfolio of just a few securities, the number of securities in a unit trust provides the investor with extensive diversification.

Economies of Scale

Fund management companies usually enjoy economies of scale when they purchase securities simply because of the number and the size of trades they put in. The brokerage fees may be lower than what an individual investor pays. These savings are usually passed back into the unit trust, benefitting the unit holders in the end.

Lower Capital Outlay

Equities and bond offerings are normally available only if purchased with a minimum amount. The amount required may not be appropriate for an individual investor as he may find it difficult to come up with a high capital outlay. In contrast, funds from many investors are pooled together in a unit trust. Through this larger pool of funds, investors can participate in more companies and their bonds. Some unit trusts also offer regular savings plans which are usually affordable for most investors.

Liquidity

Unit trusts are usually valued every business day (some are valued on a weekly basis) and this allows the investors to redeem their units and have access to their money easily.

Professional Management

Fund managers are competent professionals who spend time watching the markets and the economic environment. They also have access to systems and research materials which help them in their investment strategy. Investors can tap on the expertise of these professional fund managers to manage their funds as most investors would not have the time and the real-time access to market/economic data.

Flexibility When Risk Appetite Changes

Most fund managers offer a range of unit trusts. In the event that the investor's risk appetite changes, they can rebalance their portfolios by switching from one fund to another at a reduced cost. Some fund managers even offer a certain number of free switches within a calendar year.

Regulated

Unit trusts will need to be registered with the Monetary Authority of Singapore (MAS) before they can be offered to the public. They are also required to comply with the rules, regulations and guidelines issued by the MAS. In addition, the trustees and the fund managers operating and managing the unit trusts are properly licensed by the MAS.



Tips / Key Points

Understand the advantages of investing in unit trust and consider them before recommending unit trusts as an option to your client.

6.2.5 Disadvantages of Investing in Unit Trusts

Unit Trusts Incur Sales Charges

Fairly high sales charges are incurred when an investor invests in a unit trust. Thus, it is costly for the investor to buy and sell on a frequent basis. Investors in unit trusts should have a longer term investment horizon and should not treat unit trusts like stocks.

Management Fees

There are annual management fees associated with a unit trust. These fees are charged to the unit trust regardless of whether the fund is performing well or not. Thus, it is important for the investor to compare fees across different funds. In the long run, high management fees would affect the returns of a unit trust.

Performance of Fund Managers

Some unit trusts have spectacular performances which may be due to the exceptional skills of particular fund managers. If a manager with superior performance leaves the fund, the unit trusts may not continue to perform as well. Thus, an investor should try to monitor staff movements regularly to evaluate if there are any changes in the investment objective or style of the Unit Trust invested in.



Tips / Key Points

It is important to be able to understand and highlight to your client the potential disadvantages of investing into a unit trust.

6.3 Types of Funds

6.3.1 Equity Funds

Equity funds are unit trusts which have substantially all of their holdings in stocks. There are a myriad of equity funds in the market, for example, global equity funds, regional equity funds, single country funds and single sector funds.

As these funds contain mostly stocks, the risk and the return generally track the returns of equity markets. On a comparative basis, pure equity funds generally carry a higher level of investment risks and are more volatile than fixed income funds, balanced funds and money market funds.

Within this categorization, the investment risk associated with global equity funds are lower compared to sector funds. The reason is that the more focused the investment objective, the more concentrated the investment universe will be and this in turn increases the risks. For example, investing in a single-industry fund will cause the investor's returns to be mainly susceptible to the cyclical economic changes of that particular industry.

Conversely, investing in a global equity fund exposes the investor to a diverse set of countries and industries. This spreads out the investment risks so as the underperformance of any particular country or sector will not have a substantial impact on the overall fund.

As equity funds carry higher investment risks and are more volatile, these funds are generally recommended for longer-term investments. They are more appropriate for investors who have a longer investment horizon and are able to withstand significant volatility.

6.3.2 Fixed Income / Bond Funds

Fixed income or bond funds are unit trusts which have substantially all of their holdings in fixed income securities. Some bond funds also pay-out dividends regularly. Bond funds' performances are usually less volatile than equity funds as their performance usually depend on both fixed coupon payments and bond price appreciation.

Similar to equity funds, within the category, the investment risks associated with global bond funds are lower compared to a single country or sector bond fund. The risks associated with the bonds in the fund are dependent on credit quality of the issuers and "duration" (which has been discussed later).

Bond securities can either be corporate or sovereign issues. These bonds are usually rated by rating agencies like S&P, Moody's and Fitch. These rating are proxies for probability of default. Generally, an investment grade bond would mean having a rating of at least BBB/Baa or better by S&P, Fitch or Moody's.

Bonds with longer duration may also be considered more risky than those with shorter duration, as the prices are more sensitive to changes in the interest rates. Duration measures the sensitivity of the price of the bond security to a change in the interest rate.

Bond funds are usually less volatile than equity funds and provide a source of regular income. Generally, Bond funds are suitable for investors who are averse to the high volatility of the equity funds.

6.3.3 Money Market Funds

Money market funds are unit trusts which have substantially all their assets in short-term deposits with financial institutions and money market instruments or debt securities such as government and corporate bonds, Treasury bills, bank certificates of deposits, banker's acceptance, floating rate notes, commercial papers, trade bills, asset-backed securities and repurchase agreements.

The Code of CIS indicated that for such funds, 90% of the securities in the funds should have a term to maturity of not more than 366 calendar days. The remaining 10% may be invested in permissible securities with a term of maturity of not more than 732 calendar days.

As money market funds typically invest most of its assets into short-term instruments, such funds are suitable for investors who are looking for higher returns than fixed deposit but with less volatility compared to bond and equity funds.

However, it does not mean that there are no risks involved in money market funds. As there could be short-term bond securities in the fund, default risk is still present. Money market funds are typically less risky than bond funds. These unit trusts are most suitable for investors seeking a low-risk alternative to cash rates.

6.3.4 Balanced Funds

Traditionally, balanced funds typically consist of investment in two asset classes, equity and fixed income securities. Currently, there are various funds in the market with varying weightings in both these asset classes.

A typical balanced fund may contain 50% allocation to both equity and fixed income securities whilst allowing for a variance of 10% for strategic and/or tactical asset allocation to be made by the fund manager in response to market conditions.

The aim of a balanced fund is usually to provide a long-term capital growth while generating recurrent income for the investor. As these funds contain both equity and fixed income securities, it carries the risk of both asset classes. The volatility typically falls between pure equity and fixed income funds.

6.3.5 Multi-Asset Funds

Multi-asset funds generally consist of more than 2 asset classes. In addition to the equities and fixed income securities, multi-asset funds include other assets like private equity and commodities.

The premise of multi-asset funds is that by diversifying into various differing asset classes, the funds will be able to withstand different investment conditions as each asset class contributes to the returns at different points of the market cycle. In addition, the fund manager will provide his professional service of making tactical calls.

Consequently, the investor would not need to rebalance his portfolio when market conditions change. Multi-asset funds are suitable for investors who wish to be well diversified and are willing to rely on a fund manager to make the investment calls.

6.3.6 Multi-Manager Funds

Multi-manager funds are unit trusts which invest in two or more external funds managers or other unit trusts. The main selling point of a Multi-Manager Fund is availability of a larger set of expertise. For example, a global equity multi-manager fund may be investing into several regional mutual funds. Each of these regional unit trusts is managed by a professional who specializes in that specific region.

Multi-manager funds are easily confused with Multi-asset funds as multi-asset funds may also engaged several underlying fund managers to manage the different asset classes. Both use a set of subsidiary fund managers. It should be noted that more fees are incurred when more sub-fund managers are involved.

6.3.7 Fund of Funds

A Fund of funds is defined in the Code of Collective Scheme as one which substantially invests all of assets with at least five different underlying fund managers on a dedicated basis or invested in pooled investment schemes. A Multi-Manager fund is a Fund of funds.

6.3.8 Index Funds

Index funds are unit trusts which track a particular stock market index. In other words, the fund manager of the index fund will typically purchase most of the stocks which are in a particular index. This is termed “passive investing” as compared to “active investing”.

6.3.9 Exchange Traded Funds (ETFs)

Exchange traded funds (“ETF”) are funds which track an index, a basket of securities or commodities. ETFs are like listed stocks and they are traded on stock exchanges around the world. ETFs market has been experiencing a rapid growth in the Asian region in recent years. The ETFs in the market include Blackrock’s iShares, State Street’s SPDR (pronounced as Spiders) and Vanguard’s range of Vanguard ETFs®.

Features and Uses of ETFs

ETFs have lower management fees compared to mutual fund/unit trusts as there are no sales/upfront charges and other transaction costs associated with mutual funds. Investors use ETF for many reasons, including strategic reasons to gain exposure to particular industry sectors or countries or for hedging against other investment positions.

Portfolio managers using the core-satellite approach also uses ETFs, often as the core portfolio. ETFs can also be used for tactical reasons, such as gaining exposure to a specific market rapidly while searching for specific securities to invest in. They are used to provide for thematic exposures (for example, selected dividend-paying stocks).

There are several reasons why investors buy ETFs. ETFs also offer liquidity and flexibility as they are listed on exchanges and can be traded any time the market is open. In addition, they are priced continuously throughout the day. ETFs can be a good vehicle for diversification as an investor in an ETF gains exposure to a basket or a large number of securities.

Moreover, there is a broad range of asset classes available, for example, equities, fixed income, cash, alternatives, currency and commodities. ETFs generally incur relatively lower transaction costs and are also a cost-effective way for investors to have a diversified market exposure.

ETFs, like stocks, can be shorted. However, there may be rules imposed by the respective security market regulators. There are risks involved in shorting of ETFs and if executed correctly, can lead to disastrous results. Thus, typically, only sophisticated investors or traders and hedge fund managers engage in shorting ETFs.

Types of ETFs

ETFs can be broadly classified into equity, fixed income, cash, alternatives, currency and commodities ETFs. ETFs can adopt leveraged strategies as well as inverse strategies (allowing investors to achieve inverse performance of the basket of currencies).

- **Equity** – Categories of equity ETFs include:
 - Global Markets
 - Capitalisation Size (large, mid or small);

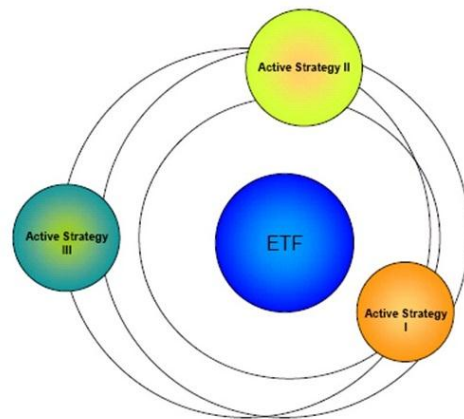
- Industry Sectors;
 - Country-based;
 - Emerging markets; and
 - Investment Styles (e.g. active investing, dividend-paying and Shariah-compliant).
- **Fixed Income** – There are also ETFs which solely focus in fixed income securities. Some categories of these ETFs include:
 - Government Securities;
 - Corporate Securities
 - Credits;
 - High yield Securities;
 - Mortgaged- back Securities; and
 - Emerging markets Bonds.
 - **Cash** – Cash ETFs have their holdings in short term money market instruments. These ETFs hold money market securities, such as Treasury Bills, Municipal and Corporate Bonds that mature in less than a year.
 - **Alternatives** – ETFs that focus on alternatives are mainly invested in listed hedge funds. These ETFs provide a secondary market for the less liquid hedge funds. There are also ETFs that are tracking the prices of “carbon emissions allowance futures”.
 - **Currency** – Currency ETFs track the performance of a basket of currencies from developed or emerging markets.
 - **Commodities** – Commodity ETFs track broad commodity indices, for example S&P GSCI (Goldman Sachs Commodity Index) and RICI (Rogers International Commodity Index). They also track other commodity sub-indices, such as energy, livestock, precious metals, industrial metals and agriculture. For example, a commonly known ETF is the SPDR Gold. Alternatively, such ETFs can be based on futures and forwards of various commodities.
 - **Index** – Index ETFs are usually passively managed funds where the manager will replicate most of the securities within a particular index. Some Index ETFs are leveraged. There are also others which replicate the inverse performance of the index by utilising derivatives. Such ETFs are known as inverse or short ETFs.

Typically, investors have the impression that if they buy an index ETF, they would have full exposure to all the stocks within the index. This may not be accurate. Some ETF Managers do not own every security in the index but replicate index as close as possible by minimizing the tracking error. However, the tracking error is not always totally eliminated.

ETF in the Core

In the illustration below, an ETF linked to a broad market index, such as the DJ Euro Stoxx 50 or MSCI World Indices, is used as the core of the portfolio. The ETF is passively managed and will deliver returns in line with the index. With the core in place, investors can have satellite portfolios using active strategies to enhance returns according to their investment objectives and risk tolerance.

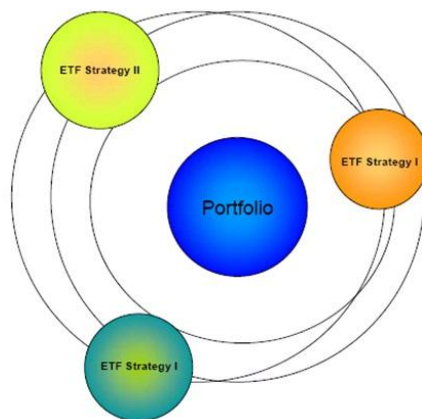
Exhibit 5.1 Using ETFs in the Core in Core-Satellite Investing



ETFs as the Satellites

In the illustration below, the ETFs are used as satellites around the portfolio. These ETFs are primarily used to generate incremental returns or “alpha” for the portfolio. They are typically very focused in terms of industry sector or style. They can also be linked to individual countries or regions.

Exhibit 5.1 Using ETFs in the Core in Core-Satellite Investing





Tips / Key Points

Understand that broad classifications of funds. There are hybrids and many innovative schemes in the financial market.

6.4 Charges and Fees

As CIS are professionally run by various parties, there are costs and expenses involved. The respective parties in a fund are entitled to fees which are charged either at the onset of the investment or on an on-going basis. List below are the typical fees which are incurred in running a unit trust.

6.4.1 Sales Charge / Initial Fee / Upfront Charge / Front End Load

This charge is usually retained by the distributor as their remuneration for selling the unit trust to the investor. This fee charge range from as low as 0% (for money market funds) to 5% on the initial investment amount. For funds included under the CPF Investment Scheme, the maximum sales charge allowed to be charged is 3%.

6.4.2 Switching Fee

As mentioned above, if an investor is already invested in one of the funds managed by the fund manager and would like to switch to another fund which is under the same umbrella, he only needs to pay a switching fee which is usually 1% as compared to the full sales charge. Some funds allow investors one free switch per calendar year. As with sales charges, switching fees are usually retained by the distributors.

6.4.3 Redemption Fee / Exit Fee / Back End Load

Some unit trusts do not charge upfront fees or sales charge but they may charge the investor a redemption fee (if the investor redeems within a pre-defined period) or termed a back-end load. For example, a unit trust may not charge a front end load of 5% on condition that the investor stays invested for at least 5 years. If the investor pulls out after the 2nd year, he may need to pay $[5\% - (5\%/5 \times 2)] = 3\%$ on his initial investment amount.

6.4.4 Management Fee

The fund manager is entitled to a management fee for providing the fund management expertise. Management fees vary for different types of funds. They

can be as low as 0.2% for money market funds to 1.5% for equity funds. Some other unit trusts, like certain hedge funds, may have management fees of up to 3%, depending on the extensiveness of the fund management involved.

However, management fees are usually dependent on market forces. If a fund manager prices too high, he may not be able to garner sufficient investor participation. Management fees are quoted per annum and are usually accrued on a daily basis. Typically, the prices of unit trusts have already taken into consideration the management fees charged to the fund.

6.4.5 Trailer Fees

Trailer fees are given to the distributors by fund managers as long as the assets they have gathered stay with the fund manager. Trailer fee usually comes out of the management fee which is already charged to the fund. Depending on the amount of assets the distributor has helped to gather, the trailer fee can range from 25% to 50% of the annual management fee.

6.4.6 Trustee Fee

The trustee, in performing its duties to the investors of the unit trust, charges a trustee fee. Trustee fees typically range from around 0.10% to 0.15%. This fee, like the management fee, is accrued daily and is usually accounted for in the prices of the unit trust.

6.4.7 Custody Charges

Custody charges are incurred by a fund for a custodian to hold the invested securities safely on behalf of the unit trust owners. The charge is typically transaction-based and country dependent. For example, the custodian charge can be higher for transactions in a relatively difficult market to access like India.

6.4.8 Performance Fees

Certain types of unit trust charge performance fees. Such fees are usually charged when the fund manager outperforms a specific benchmark. The prospectus will typically highlight how the fees is computed and charged.

6.4.9 Expense Ratio

The Expense Ratio is an expression of the total operating expenses charged to the fund expressed as a percentage of the fund's average asset for the year. The

Investment Management Association of Singapore (IMAS) has recommended including the following items:

- Management fee;
- Trustee fee;
- Administration fee;
- Accounting and valuation fee;
- Custodian/sub-custodian fee;
- Registrar fee;
- Legal and professional fee;
- Printing and distribution fee;
- Audit fee;
- Amortised expenses;
- Performance fees ; and
- Goods and Services Tax (GST) on expenses

Fees which can be excluded include interest expense, brokerage and other transaction costs associated with the sales and purchases of investments, foreign exchange gains and losses, tax deducted at source or arising on income received, including withholding tax, front end loads, back end loads, and dividends and other distribution paid to unit holders.



Tips / Key Points

It is important to highlight fees to your client as fees can affect the performance of the funds and in turn affect the expectations of your client.

The Expense Ratio consists of the total operating expenses charged to the fund expressed as a percentage of the fund's average asset value for the year. It has many components including custody, legal, audit, valuation, management and performance fees.

6.5 Bid, Offer Prices and Net Asset Value (NAV)

Some unit trusts quote their prices with a bid-offer spread. The bid price is the price which the investor gets when selling their units. The offer is the price which the investor pays when buying units. The difference between them is called the spread and this is in fact the sale charge which is retained by the distributor.

For example, the bid-offer price quoted for a particular unit trust is \$0.9250 (bid) – S\$0.9713 (offer). The price which the investor buys units will be S\$0.9713. And if the difference expressed over a percentage (over the bid price) is 5%, the sales charge paid by the investor. If the investor sells, the price will be S\$0.9250.

More unit trusts are quoted on a single price basis. Instead of looking at two prices, there will only be one price, called the net asset value (NAV). For such unit trusts, the sales charge is first deducted from the investment amount before the remaining amount is used to purchase units at the prevailing NAV.

Example

Prevailing NAV of unit trust = S\$1.2025
 Investment amount = S\$10,000
 Sales charge = 3%

Remaining amount after deduction of sales charge = $\$10,000 - (3\% \times \$10,000)$
 = S\$9,700

Amount of units purchased = $\text{S\$}9,700 / \text{S\$}1.2025 = 8066.53$



Tips / Key Points

Ensure you understand what bid and offer mean and what NAV refers to.

Forward Pricing vs Historical Pricing

When investors buy or sell units of a unit trust, they will only be getting an indicative price. The actual price will not be available until the next day or the day after. This arrangement is called forward pricing. The reason for this is due to a need for the valuation of the unit trust by the Administrator.

In order to arrive at a price for the unit trust, the administrator will need to value each of the underlying securities, taking into consideration the new subscriptions, redemptions and the expenses of the unit trust. Some administrators may use the closing prices of the quoted securities for that particular day. The pre-defined time is known as the valuation point.

As a result, the price will only be available the next day or even the day after. This also depends on the markets which the securities are listed. Global unit trusts typically hold securities listed on exchanges all over the world and the time differences and different closing times affect the valuation point. Investors will usually receive an order confirmation a few days after their transaction which will show the actual price at which the units were transacted.

Historical pricing is not as common as forward pricing. As the name implies, the investor will know exactly the price which they are transacting on a historical basis. This practice is more suitable for unit trusts which hold less volatile securities.



Tips / Key Points

Most funds in the markets use forward pricing for the subscription and redemption of units in a fund.

6.6 Reading and Understanding Factsheets

6.6.1 Return Calculations on Fund's Factsheet

An investor of a fund is interested about how well his fund will perform. Although the historical performance of a fund is no assurance on how the fund will perform in future, it gives the investor some clues as to the fund manager's ability in managing his money.

Computing a Fund's NAV to NAV

These returns measure how the fund's Net Asset Value (NAV) has appreciated or depreciated over a specified period, e.g. 1 month, 3 months, 1 year, 3 years and since inception of the fund. They do not take into account the actual returns to an investor because the investor will incur an initial sales charge when they buy into the fund. A Fund's NAV to NAV return is also called **Bid-to-Bid** returns.

Computing a Fund's NAV to NAV (net of charges)

These returns measure what the returns would have been if the initial sales charge (also called front end load) is taken into account. The amount of the sales charge can be as up to 5% and is usually stated in the prospectus and term sheet as well.

Example

If an investor puts \$100,000 into a fund and the initial sales charge is 5%, the amount of money that is put into the fund for investment is \$95,000 while the \$5,000 is used to pay the distributors of the fund.

If the NAV of the fund increases by 10% over the period, his \$95,000 would have grown to (\$95,000 x 1.10) or \$104,500. Since he came out with \$100,000, his return on investment using NAV-to-NAV (with charges applied) is calculated as follows:

$$\begin{aligned} \text{Return on Investment} &= \frac{(\$100,000 - \$5,000) \times 1.10}{\$100,000} \\ &= 4.5\% \end{aligned}$$

While the NAV-to-NAV is a more realistic return from the standpoint of the investor, several things have to be taken into consideration:

- The investor may not be paying as high a sales charge as used by the fund in the calculation. For example, while the sales charge is up to 5%, he may be able to buy the fund at a lower sales charge (after getting a promotional discount); and
- The sales charge has nothing to do with how well the fund manager performs and taking this into consideration is not appropriate if you want to determine the ability of the fund manager to generate good returns.

The NAV to NAV (with charges applied) is also called the **Offer-to-Bid** returns. This is because the investor buys into the fund at its offer price, which includes the initial sales charge.

Annualized Returns

A fund's returns may also be stated on an annualized basis to enable comparison with other funds. The annualized returns allow the investor to compare the fund's returns with the returns of other funds with different inception periods.

Example

A fund has appreciated by 33.1% since its inception three years ago. What is its annualized return?

$$\begin{aligned}
 \text{If } x \text{ is the annualized return, } (1 + x)^3 - 1 &= 33.1\% \\
 (1 + x)^3 &= 1.331 \\
 x &= (1.331)^{(1/3)} - 1 \\
 x &= 0.1 \text{ or } 10\%
 \end{aligned}$$



Tips / Key Points

Study a fund's return carefully to gauge if the fund manager has managed the fund well.

6.6.2 Risk Adjusted Returns on Fund's Factsheet

Sharpe Ratio

This is a measure of a fund's annualized return that is in excess of the risk-free rate of interest and adjusted for the fund's **total risk**, as defined by its standard deviation. The risk-free rate may differ depending on the case. If the investor must use only cash to buy a fund, the relevant risk-free rate is the bank deposit rate. If the investor is allowed to use his CPF savings to buy the fund, the relevant risk-free is the CPF interest rate.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where,

R_p = Return on the portfolio

R_f = Risk-free rate of return

σ_p = Standard deviation of the portfolio.

The higher a fund's Sharpe Ratio, the better its return relative to the amount of total risk taken.

Example

A fund has an annual volatility of 15% for the past one year. The risk-free rate is 2.5% p.a. and the 1-year return on the portfolio is 10%. What is the fund's Sharpe Ratio?

$$\begin{aligned}\text{Sharpe Ratio} &= \frac{10\% - 2.5\%}{15\%} \\ &= 0.5\end{aligned}$$

Treynor Ratio

The Sharpe Ratio relates the excess return of a fund to the fund's absolute volatility. However, if the investor already owns a diversified portfolio comprising unit trusts and individual securities, it is not the absolute volatility of the fund that adds on to the risk of his investments but rather the fund's **systematic risk**, as defined by its beta.

The Treynor Ratio relates the excess return of a fund to the fund's beta and is calculated by the formula:

$$\text{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

Where,

R_p = Return on the portfolio

R_f = Risk-free rate of return

β_p = Systematic risk or “beta” of the portfolio.

The higher a fund’s Treynor Ratio, the better its return relative to the amount of systematic risk taken.

Example

A fund has a beta of 1.5 for the past one year. The risk-free rate is 2.5% p.a. and the 1-year return on the portfolio is 10%.

$$\begin{aligned}\text{Treynor Ratio} &= \frac{10\% - 2.5\%}{1.5} \\ &= 0.05\end{aligned}$$

Jensen Alpha

The Jensen measure of performance is based on the CAPM and measures the risk-adjusted return as an “*alpha*”. It is a good measure only if two portfolios under comparison have the same beta. As such, it is not a good measure of performance in its own right.

According to the CAPM, a portfolio’s expected return, $E(R_e)$ is given by:

$$E(R_e) = R_f + \beta_p [E(R_m) - R_f]$$

Where,

$E(R_e)$ = Expected return on the security, e

$E(R_m)$ = Expected return of the market, m

R_f = Risk-free rate

β_p = Beta of portfolio

$$\begin{aligned}\text{In a regression, the ex post equation is} \quad Y &= A + B(X) \\ (R_e - R_f) &= A + B(R_m - R_f)\end{aligned}$$

$$\text{Thus, Jensen’s Alpha, A, is given by} \quad A = (R_e - R_f) - B(R_m - R_f)$$

The *alpha* is the difference between the portfolio's actual excess return during the period and the excess return measured by the CAPM. A positive *alpha* reflects superior performance while a negative *alpha* indicates inferior performance of the manager.

Sortino Ratio

The Sortino Ratio is similar to the Sharpe ratio in that it is a risk-adjusted measure of return based on the volatility of returns in the portfolio. However, it is a better measure of risk than the Sharpe Ratio because while the Sharpe Ratio penalizes both upside and downside volatility equally while the Sortino Ratio penalizes only downside volatility.

$$\text{Sortino Ratio} = \frac{R_p - R_f}{\sigma_d}$$

Where,

R_p = Return on the portfolio

R_f = Risk-free rate of return

σ_d = Standard deviation of negative asset returns

The higher a fund's Sortino Ratio, the better its return relative to the amount of total risk taken.



Tips / Key Points

Do not rely on return measures alone to select a fund. Make sure you know how the fund has performed on a risk-adjusted basis.

6.6.3 Risk Factors and Ratios for Equity Funds

Standard Deviation (or Variance)

Variance and standard deviation are measures of a fund's *absolute* risk or risk on a standalone basis. They measure the dispersion of a fund's returns from its average return. Variance is defined as the average of squared differences from the mean.

The differences from the mean are squared to make all differences into positive numbers thereby avoiding negative numbers from reducing the variance. The higher the variance, the higher the fund's total risks

Example

Suppose there are two funds with the following monthly returns.

Fund A: 5%, 4%, 5%, 6%

Fund B: 5%, -10%, 20%, 5%

Although both funds have the same average monthly return of 5%, Fund B is riskier because the returns fluctuate greatly versus its average return. The variance in this example is calculated as follows:

$$\begin{aligned}\text{Fund A's variance } (\sigma^2) &= \frac{(5-5)^2 + (4-5)^2 + (5-5)^2 + (6-5)^2}{4} \\ &= 0.5\end{aligned}$$

$$\begin{aligned}\text{Fund B's variance } (\sigma^2) &= \frac{(5-5)^2 + (-10-5)^2 + (20-5)^2 + (5-5)^2}{4} \\ &= 112.5\end{aligned}$$

The variance is an “exaggerated” number because all the differences are squared in the calculation. Therefore, to arrive at a true number to describe the fluctuation, you take the square root of the variance to arrive at the standard deviation.

$$\text{Fund A's standard deviation } (\sigma) = \sqrt{0.5} = 0.7$$

$$\text{Fund B's standard deviation } (\sigma) = \sqrt{112.5} = 10.6$$

As Fund A has a standard deviation of 0.7, it means that while the fund has a mean monthly return of 5%, the returns tend to fluctuate by 0.7% around this mean.

As Fund B has a standard deviation of 10.6, it means that while the fund has a mean monthly return of 5%, the returns tend to fluctuate by 10.6% around this mean.

Beta

This measures the volatility of a fund's return *relative* to the benchmark index. The higher the beta, the more volatile a fund's return is compared to the benchmark. If a fund has a beta of 1.5, if the benchmark goes up by 10% during a period, it will on average go up by 15% and if the benchmark goes down by 10%, it will on average go down by 15%.

R-Squared

This measures the percentage of the fund's return that is attributable to movements in its benchmark index. An R-squared of 1.0 means that 100% of the fund's variance is attributable to the index and therefore the fund is tracking the index exactly. A lower R-squared implies that the fund's returns are not fully explained by the variance of the benchmark because the fund has other specific risk factors not explained by the benchmark.

A person who invests in a fund typically does so because he wants exposure to the investment universe represented by the benchmark. For example, if he is optimistic about the growth of equity markets in the Asian Pacific region, he might go for a fund with a benchmark that is the MSCI AC Asia Pacific Index. If the R-squared of the fund is low, this implies that the fund has a variance not reflecting the market benchmark.

Tracking Error

Another method to measure how well the portfolio is tracking a benchmark is the tracking error. Unlike the standard deviation of the fund, which measures the volatility of the fund's absolute returns, the tracking error is the standard deviation of the *difference* between the fund's return and the benchmark index returns.

Funds that attempt to track a particular benchmark closely will have a high R-squared and almost no difference between the fund's return and the benchmark's return. In other words, its tracking error would be close to zero.

Liquidity Risk

For open-ended funds, the investor will need to redeem his shares if he wants his money back. That means that the fund manager may have to sell shares in the fund to facilitate redemption. Typically, the fund will contain a percentage of its holdings in cash so as to be able to handle redemptions.

However, if the size of the redemption on a given day is large, the manager may not be able to sell all the shares. In such cases, the redemptions will be delayed and the investor has to remain invested longer than he intended.

A fund's investment in illiquid securities may also affect the liquidity of the fund because the fund manager will have more difficulty selling securities when he wants to. Typically, funds that invest in small-cap stocks of emerging markets face more liquidity problems in the sale of the underlying shares.

It is important to check the dealing frequency of the fund as some funds do not allow redemptions every day of the week and specifies which days redemptions are allowed. This presents liquidity risks for the investor because he may not be able to get his money when he needs it.

As mentioned earlier, for close-ended funds, if an investor wants to buy into a fund, he has to buy it from someone who wants to sell it in the secondary market, just like one would trade in the stocks of an individual company.

Therefore, it is important to know the average daily trading volume of the fund units. If the volume is very thin and the investor has a relatively large investment in the fund, he may not be able to liquidate all of his holdings in one day. There will also be market impact cost i.e. he may depress the price of the fund if he tries to sell his position aggressively.

Currency Risk

A Singapore investor will usually be interested in how much returns an investment will come back to him in his domestic currency, say in Singapore Dollars (SGD). However, the fund may be denominated in USD, thereby exposing him to USD/SGD fluctuations.

Furthermore, the fund may be invested in Japanese stocks, causing USD/JPY fluctuations to affect his return as well. The client relationship manager has the responsibility to ensure that his client understands that he is taking direct and indirect currency risks

Example

A Singapore investor buys a USD-denominated fund that is invested in US shares with Singapore Dollars. In a one-year period, the fund appreciated by 10% in US\$ terms. In the same period, the S\$ depreciated by 5% against the US\$. What is the return of the fund to this investor in S\$ terms?

$$\begin{aligned}\text{Investor's return in S\$ terms} &= \frac{1.10}{1.05} - 1 \\ &= 4.8\%\end{aligned}$$

Use of Derivatives

Derivatives are financial products which are derived from other financial products. Some funds are allowed to use derivatives and this may increase the risk exposure for an investor. For example, a bond fund that buys total return swaps has exposure to the returns of a bond and the risk of default of the bonds without having to pay full price for the bond.

An equity manager that buys into equity futures also gets exposure to equity markets without having to pay the full price of the underlying shares. What these mean is that the use of derivatives may give the fund leveraged exposure to the markets.

Using derivatives also result in counterparty risks as the fulfilment of the contracts depends on the performance of other parties. If these parties are unable to fulfil their part of the transaction, the outcome may turn out to be different from expected.

Transparency

It is important for a fund to be transparent and release all pertinent information to the investor so that he can make an informed decision. Information required by the investor includes information about the fund's return, risks and costs.



Tips / Key Points

Always make sure that the fund you choose has risks that are acceptable to the investor.

6.6.4 Interpreting Bond Fund Factsheets

Equity funds factsheets has information which may include betas, standard deviations and Sharpe Ratios. Bond funds have other statistics that are specific to bonds which include the following:

Average Coupon

The coupon of a bond represents the periodic interest payments made to the holder of the bond over the life of the bond. The coupon rate when multiplied by the par value of the bond provides the dollar value of the coupon payment.

The average annual coupon rate of a bond fund is the average coupon rate of all the bonds in the fund weighted by their market value within the fund. It is stated as a percentage of total par values of the bonds.

Information on average coupon rate of a bond fund is important to the investor because if he expects a regular income payout from the fund. For example, a retiree may depend on the coupons for his living expenses. He would choose funds which have high coupons.

Yield to Maturity (also called Yield to Redemption)

The Yield to Maturity is a measure of the average rate of return that will be earned on the bond fund if it is bought now and all the bonds are held to maturity. It is the interest rate or discount rate that will make the present value of the cash inflows from the bonds equal to the current market value of the bond fund.

This is an important yield indicator for investors in bonds because it does not just take into account the coupon rate of the bonds but also the current market price of the bonds and the length of maturity.

Credit Rating

There are several well-known credit rating agencies that provide credit rating for bonds such as Moody's and Standard & Poor's. Bond factsheets often give a composition of what percentage of the fund is invested in bonds of various credit ratings. Funds may also report the average credit quality of bonds in the fund, weighted by their market value within the fund.

This is an important piece of information for the investor because the risk of bond fund he is investing in has to match his own risk profile. If the investor has a low risk profile, he would want to invest only in bond funds that hold high quality bonds.

Duration

Duration is a weighted average of the maturity of all the income streams from a bond. There is an inverse relationship between bond prices and interest rates. When interest rates rise, bond prices fall. The duration of a bond is also an indication of the sensitivity of the bond to changes in interest rates. A duration of 5 years means that the bond will decrease in value by about 5% if interest rates rise by 1% and vice versa.

Bond funds usually provide the duration of the fund, which is the average duration of all the bonds in the fund weighted by their market value. If the bond fund has duration of 5 years, it also means that if interest rates go up by 1%, the fund will drop in value by about 5%. An investor should note if the duration of the fund matches his risk profile. For a low-risk investor, he would not want to be exposed to high interest rate risks and see the value of his bond fund decline when interest rates rise.

6.7 Evaluation of unit trusts

When evaluating whether a unit trust is a suitable product for a prospective investor, there are a number of points which need to be considered which are discussed below.

6.7.1 Diversification

A common motivation for investing in a unit trust is to diversify through the various securities that is in the fund. However, depending on the investment objective of the unit trust, some of these funds are very focused and they will only invest in a very limited universe of stocks. For example, a unit trust whose investment objective is to achieve growth by investing in China-related large cap companies will

have less diversification than another unit trust which aims to achieve growth by investing in global large-cap companies.

In addition, some unit trust may only hold up to 30 stocks in their portfolios (as the fund managers have high conviction on only these 30 companies). Comparatively, such unit trusts are less diversified than other unit trusts which have hundreds of stocks in their portfolios. Depending on the circumstance of the investor, they may want to have such focused unit trusts in their portfolio as they are also holding other types of investment or other unit trusts.

From an overall portfolio perspective, these unit trusts may offer the diversification they need. In addition to the annual and semi-annual report where an investor receives, he can refer to the factsheets which would contain more recent information (rather than waiting for the financial year end of the fund). These typically contain the top 10 holdings of the funds and other useful information.

A quick assessment of the fund's diversification can be obtained by looking at the percentages of the top 10 holdings. For example, if the top 10 holdings accounts for more than 50% of the fund, it is quite likely that the fund is not well-diversified. This should be compared with the investment objectives of the fund. Some funds have very specific or focused investment objectives which would account for the low diversification.

6.7.2 Total Expense Ratio

Expenses can affect the performance of the unit trust in the long run. However, it should not be the only criteria when evaluating a unit trust. For example, when comparing between two funds with similar objectives but differing expense ratios of 1.75% for fund A and 1.70% for B, one should not just drop off Fund A without looking into the performance and other factors like stability of returns and consistency of investment approach.

The reason is that the fund with a lower expense ratio could constantly underperform its peers. In addition, if there is a high turnover of fund managers, there is a possibility that the investment approach may not be implemented consistently, resulting in erratic returns. It is a regulatory requirement for fund managers to calculate and disclose the Total Expense Ratio in the annual or semi-annual reports of their respective funds. This data is actually available in all the annual or semi-annual reports of the fund managers.

6.7.3 Investment Objectives and Time Horizons

Investors have different investment objectives and time horizons. Depending on their needs, certain unit trusts may be more suitable for them. More often than not, investors themselves are not very sure what they are after. A client relationship manager may need to help his client to ascertain the appropriate investment objective and time horizon.

Having the correct information would facilitate the appropriate choice of funds. For example, if the investor is expecting very high returns within a short period of time and is not ready to take a loss in his investments, a mismatch of expectation will be created if he is recommended to go into a risky equity unit trust with illiquid stocks. Though he may have a better chance at achieving the high returns objective in a short time, he may also run the risk of bearing a loss on his investment. Generally, riskier unit trusts are more suitable for investors with a longer-term investment horizon.

It is also important for the investor to understand how the fund manager is going to achieve the investment objective. For example, does the fund manager intend to utilise derivatives liberally or engage in leverage. Such information could affect the decision of the investor on whether to proceed with investing into the fund. Ultimately, the investor must know what they are getting into and they should not be losing sleep over their investments.

6.7.4 Performance

A fund is usually managed relative to a benchmark so that the investor is able to compare the fund's return versus the benchmark and see if how the fund manager has performed. If the fund's return is higher than the benchmark return, the fund manager has done well and vice versa. However, one cannot look at this number in isolation.

The fund may have done well because the fund manager has taken excessive risks. Thus, it is necessary to look at both risks and return using risk-adjusted measures like the Sharpe, Treynor and Sortino Ratios. When comparing a fund's return against its benchmark, it is important to make the comparison across various time periods, 3 months, 1 year, 3 years, etc.

This is because a fund may outperform its benchmark not because of the fund manager's overall superiority but due to isolated investment decisions. For example, if he chooses to invest in lower beta stocks or chooses to have a high percentage of the fund in cash, the fund may outperform the benchmark when the benchmark is going down. However, if the trend reverses, the fund may quickly underperform the benchmark.

Likewise for a bond fund, it is good to compare the performance of the fund versus its benchmark for periods when interest rates are rising as well as declining. Past performance is not indicative of future performance. This sentence is usually disclosed in most of the marketing materials for unit trusts as it is a regulatory requirement.

This is especially true if the performance of a unit trust is heavily dependent on a star fund manager and should the fund manager decide to leave the Fund Management Company, the performance of the unit trust could potentially be affected. Yet, there are funds which may have been performing especially well due to market rallies for that particular sector or country which the fund objective is to invest in.

6.7.5 Fund Sizes and Capacity

A small fund size can affect the performance of the unit trust as the fixed expenses in running the fund offsets the performance. For example, a S\$10,000 legal expense which is billed to a S\$8 million fund is insignificant (0.00125%) and does not impact too much on the fund.

However, when billed to a S\$1 million fund, it becomes more apparent as it is 0.01% of the fund. Such expenses will in turn contribute to the total expense ratio of the fund. On the flip side, when a unit trust becomes too huge, it may run into capacity issues. What this means is that the fund manager will have difficulty adding value to the fund.

For example, if the unit trust objective is to invest in Singapore equities and due to its popularity, the fund size may grow to S\$10 billion, which will be huge relative to the market. Consequently, the fund manager will not be as nimble and may face difficulty in adding value. Thus, when a Unit Trust reaches its optimal capacity, it will typically stop accepting any fresh subscriptions.



Tips / Key Points

Ensure you know where all these information can be accessed in order to make a good assessment.

6.8 Chapter Summary

This chapter has discussed Unit Trusts and related fund products. The salient points are summarized below:

- Unit trusts in Singapore have to comply with the Securities and Futures Act and other regulations. Covered Entities would also need to comply with provisions in the Code of Collective Investment Schemes, although it is non-statutory.
- The main role of the trustee is to look after the interests of all unit holders.
- The key role of a fund manager is to manage the assets of a Unit Trust in line with the defined investment objectives of the fund and within the investment restrictions as stated in the offering document.
- Mutual funds are different from unit trusts as they are not set up as trusts.
- Closed-ended funds are typically unit trusts which are available for subscriptions for a period of time. After that period, no more subscriptions will be allowed into the fund.

- Some advantages of investing in unit trust include diversification, economies of scale, lower capital outlay, liquidity, professional management, flexibility when risk appetite changes and they are regulated.
- Unit trusts are not suitable for investors with a short-term horizon.
- There are various types of funds available in Singapore for example, equity fund, fixed income/bond funds, money market funds, balanced funds, multi-asset funds, multi-manager funds, fund of funds, index funds and exchange traded funds.
- When investing in a unit trust, an investor should be aware of the fees that are charged to the fund. Some examples of these fees include sales charge, switching, custody, redemption management, trailer, trustee and performance fee.
- The Expense Ratio is a summation of the total operating expenses charged to the fund expressed as a percentage of the fund's average asset for the year.
- The bid price is the price which the investor gets from selling his units. The offer is the price which the investor pays when buying units. The difference between them is called the spread.
- There are different methods of measuring fund returns. Returns can be measured on a bid-to-bid, offer-to-bid or annualised basis.
- Sharpe Ratio, Treynor Ratio and Information Ratio are measures of risk-adjusted returns for funds.
- The standard deviation (or variance) measures the absolute risk. Beta measures the volatility of the fund's returns relative to a benchmark index. The statistical measure R-squared, measures the percentage of the fund's returns that is attributable to movements in its benchmark index. Tracking error measures how well the portfolio is tracking the benchmark.
- Some factors to consider when evaluating unit trusts are diversification, the total expense ratio, the investment objectives, time horizons, performance, fund sizes and capacity.

Chapter 7:

Alternative Investment Products and Services

Objective

The candidate should be able to:

- ✓ Understand the key features and key risk elements
- ✓ Measure performance of investment
- ✓ Understand disclosure obligations
- ✓ Understand the key features and key risk elements of Hedge Funds, Private Equity Investments and Commodity Investments
- ✓ Measure performance of investment
- ✓ Understand disclosure obligations

7.1 Introduction

In this chapter, the more commonly used financial alternative investment assets will be discussed in details. They are Real Estate, Hedge Funds, Private Equity, Commodities and Passion investments including antiques, coins and stamps that are well sought after by Private Banking clients.

7.2 Real Estate Investments

Real Estate investments can be carried out in many ways. Given its tangible form, many investors prefer physical real estate over securities with a claim on real estate. Nevertheless, with the many forms of real estate investment available, investing in this asset class can be complex and challenging.

7.2.1 Key Features of Physical Real Estate

Physical real estate can be broadly classified into 5 main types. Each type of physical real estate has its own unique characteristics.

- **Residential** – This includes apartments, condominiums and houses.
- **Industrial and Commercial Buildings** – This includes warehouses and office spaces.
- **Retail** – This includes shops and shopping malls.
- **Hotels / Service Apartments** – This category includes hotels, motels and serviced apartments.
- **Land** – This includes commercial and industrial land, virgin forest and agricultural land.

Returns from Physical Real Estate

Investing in Physical real estate may provide investor three types of returns:

- **Capital Gains** – This refers to the potential increase in the capital value of a property over time. Due to unique factors such as location and design, no two properties are exactly similar. Hence, this may result in a significant capital appreciation in value over time when there is demand for that property due to inflation, lack of supply and/or its unique features.
- **Income** – This refers to the rental from leasing out the property.
- **Leverage** – This refers to opportunity to borrow with the property as collateral. Consequently, the cash outlay is often only a small percentage of the full value of the property. Hence, the capital gains and income as a percentage of the outlay are magnified providing the investor financial leverage.

Significant Transaction Cost

As real estate transactions are carried by private treaties or through auctions, the transaction costs are relatively higher compared to transactions in financial securities. Transaction cost is also less standardized. The main components of transaction costs include:

- **Agent's / Broker's Commission** – The commission charged by real estate agents range from 0% to 2% for Singapore residential properties and 3% - 6% for overseas residential properties.
- **Stamp Duty** – Stamp duty is a fee payable to the government computed on completed property transactions.

- **Legal Fees** – Legal fees are payable by both the seller and buyer to transact a property. For simple transactions, a standard scale of fees is chargeable. For complex transactions, additional charges may be applicable.
- **Valuation Fees** – There may a need to value a property before the sale or purchase. This requires a professional valuation for which a fee is payable.

Various Applicable Taxes and Other Costs

- **Property Tax** – Real Estate owners are required to pay a property tax annually. The tax levied is a percentage of the annual value of the property as assessed by the Inland Revenue Authority of Singapore (IRAS).
- **Tax on Rental Income** – The income generated by rental is subjected to income tax or corporate tax depending on the entity owning the property. The tax rate applicable is thus the personal income tax rate or corporate tax rate.
- **Capital Gains Tax** – Depending on the jurisdiction that governs the real estate investment, a tax may need to be paid if there are capital gains. There is no capital gains tax in Singapore. However, as an anti-speculation measure, properties sold within 2 years after purchase will incur a capital gains tax.
- **Property Management Cost** – There is a cost in maintaining a property. This includes conservancy and other related charges.

7.2.2 Risks Associated with Real Estate Investments

The key risks for investing in Real Estate Investments include:

Market Risks

Real Estate may be subjected to severe price fluctuations due to market conditions. Property prices are highly correlated to the health of the economy. When the economy is experiencing positive growth and the stock market is on the uptrend, it is likely that Real Estate will appreciate in value.

Income Risks

One of the main sources of real estate returns is the rental income generated from leasing of properties to tenants. Real estate investment is thus subjected to income risks resulting from the lack of tenants or the inability of a tenant to pay the rent.

Leverage Risks

Real estate investments often use leverage by borrowing against the value of the underlying properties. In bearish markets, property prices and rentals may be depressed. Investors may, then, experience loan servicing difficulties which may result in foreclosures by lenders.

Interest Rate Risks

If funds are borrowed to invest in real estate, the investment is exposed to interest rate risks.

Regulatory Risks

Real Estate investments are exposed to regulatory risk as changes in regulations may be introduced from time to time. Demand for real estate may be managed by the authorities to curb speculation. Supply can be managed by controlling the sale of land for new developments.

Physical Risks

Properties may be damaged by inclement weather or fire resulting in losses. To mitigate this, property insurance are often purchased.

7.2.3 Performance Measures and Monitoring

To evaluate the performance of real estate investments, the common measures include:

Property Valuation

Professional valuation of a property at different time intervals is one good way to assess how well the investment has performed over the investment period. However, the actual transacted price may deviate from the valuation due to market sentiments.

Rental Yield

The formula for calculating Gross Rental Yield is as follows:

Gross Rental Yield	=	$\frac{\text{Gross Annual Rental Income}}{\text{Gross Property Transacted Price}}$
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The Gross Rental Yield is useful for comparison between different properties. It is widely used in the industry. To derive the actual rental yield, investors need to take into account transaction cost and rental expenses.

Cap Rate

Similar to Gross Rental Yield, Cap Rate is a common measure used to track the performance of the recurring income through Rental. The difference is that Cap Rate uses Net Operating Income in the numerator instead of Gross Rental income (which takes into account the property expenses). Cap Rate is a more accurate gauge for the income received by an investor. The formula is as follows:

Cap Rate	=	$\frac{\text{Annual Net Operating Income}}{\text{Property Transacted Price}}$
Where,		
Net Operating Income	=	Rental Income – Property Level Expenses

Equity Yield

Equity Yield looks at the actual cash outflow during investment and the actual cash collected during the investment period. The formula is as follows:

Equity Yield	=	$\frac{\text{Annual Cash Inflow}}{\text{Total Amount Invested}}$
Where,		
Annual Cash Inflow	=	Rental Income – All Expenses – Interest Cost – Tax + Depreciation & Amortization
Total Amount Invested	=	Property Transacted Price – Loan Amount + Transaction Cost

Return on Investment

This is a more complete measure that takes into account all elements of revenue and expenses from real estate investment. The formula is as follows:

$$\text{Return on Investment} = \frac{\text{Total Amount Generated} - \text{Total Amount Invested}}{\text{Total Amount Invested}}$$

Where,

$$\text{Total Amount Generated} = \text{All Annual Cash Inflow} + \text{Property Valuation} - \text{Remaining Loan} - \text{Transaction Cost upon Sale}$$

This measure calculates the total return for the investment assuming that the value of the property is sold at the point of calculation. The total amount generated includes both capital appreciation and income from rentals. It is a useful measure for investor to compare real estate investments with other asset classes.

It is important for a Covered Person to have an understanding with his client at the onset regarding the objective of real estate investments. A key consideration for performance monitoring is the choice of an appropriate benchmark to compare against the computed measures. Some benchmarks commonly used include:

- **Real Estate Indices** – There are agencies that compile indices for different types of real estate investments. These may be used as benchmarks to assess the real estate investment returns of the investor.
- **Proforma Measures** – Before major investments are undertaken, it is a common practice to estimate proforma numbers for consideration. After the investment, the actual returns may be compared with these proforma numbers to assess the returns.
- **Neighbouring Properties** – The Rental Yield and Cap Rate may be compared with neighbouring projects to assess how well the real estate investment has performed.

7.2.4 Other Avenues of Real Estate Investment

Besides the purchase of physical real estate assets, there are other ways to invest in real estate. These include:

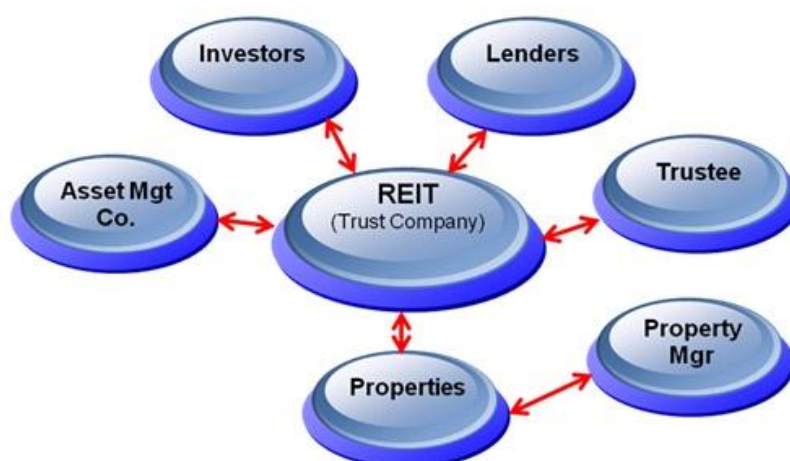
- REITs ;
- Investment in Real Estate companies; and
- Real Estate Funds.

7.2.5 Real Estate Investment Trusts (REITs)

A REIT is a vehicle structured for investors to pool their capital together to invest in large-scale income-producing properties. REITs have almost similar features, risks and performance measures as compared with physical real estate investments for

which their underlying assets are physical properties. However, REITs possess additional characteristics which investors should understand before making an investment.

Exhibit 7.1 REIT Structure



A REIT is a trust company which is used to purchase real estate assets. The trust company raises funds to purchase properties by issuing shares to investors. To qualify as a REIT, the structure will have a Trustee that holds the legal title of the properties, an Asset Management Company (REITs Manager) that manages all investment related work and a Property Manager that manages the physical properties. Lenders are involved in most REITs to enhance the funding for the property purchases.

In this structure, the main implications for the investors under this structure are as follows:

- **Investors are not the legal owners of the properties** – From the legal perspective the investors do not own the properties but only own shares of the trust company that holds the properties. Therefore, the investors do not have any rights to decide on the purchase or sale of the underlying properties.
- **Asset Management Company manages all investment related activities** – All investment-related activities, including asset enhancements, are carried out by Asset Management Company. The executive officers within the Company will seek approval from the Board of the Asset Management Company for all investment-related activities. The Board is in turn answerable to the investors.

The main benefit for having a REIT structure is that there is tax exemption at the Trust Company level. Income generated from the properties will not be taxed at the Trust Company level but only subjected to the personal income tax of investors. For a non-REIT structure, where the properties are held by a company, the income

from properties will be taxed at the corporate rate before they are available to the investors.

To enjoy the tax exemption, a REIT is subjected to mandatory distribution rules and debt ratio limits. In Singapore, at least 90% of the profits are required to be distributed while amount of debt is capped at 35% of the property value.

Portfolio of Properties

With REITs, investors will be investing in a portfolio of properties. With more properties in a portfolio, there will be risk diversification and returns will likely to be more stable.

Additional Cost

In addition to the usual expenses associated with properties, REITs have additional costs, which include:

- Asset Management Fees; and
- Trustee Fees

The cost to set up the structure could be significant as it involves legal work and application to the relevant authorities for the REIT to be approved.

REIT's Business Strategy

Unlike pure real estate play, the main investment objective for most REITs is for recurring rental income rather than capital appreciation. REITs investors should invest for the dividend rather than the potential for capital appreciation. As REITs focus on the income that properties generate rather than its capital appreciation, it is not a REIT's primary purpose to cash in any capital appreciation when its intrinsic valuation is high.

Dividend Yield

With rental income as the focus for generating returns, the main performance measure used to assess REITs will be Dividend Yield. The formula is as follows:

Dividend Yield	=	$\frac{\text{Annual Dividend Distributed}}{\text{REIT Price}}$
Where,		
Annual Dividend Distributed	=	Rental Income – All Expenses – Interest – Property Level Tax
REIT Price	=	Purchase price of REIT on IPO or when traded in Stock Exchange

Disclosure Obligations

As REITs are more regulated than plain physical real estate investments, there is more documentation. A REIT has a term sheet, prospectus and annual reports for investors to analyze. It is the investors' responsibility to review these documents.

7.3 Hedge Funds

The Hedge Funds industry has transformed over the years. Hedge Funds were initially created to bet against the market using derivative products and short-selling strategies. However, Hedge Funds today adopt different variations of investment strategies. For example, not all hedge funds engage in hedging activities. Many capitalize on arbitrage opportunities to derive profits.

7.3.1 Main Characteristics of Hedge Funds

Aside from differing investment strategies, all hedge funds possess some similar characteristics which include:

- **Absolute Returns Targets** – The performance of most Hedge Funds are not measured against a pre-assigned benchmark but rather evaluated based on the absolute returns it can generate. These Funds tend to make specific bets for the purpose of generating higher than average market return.
- **Usage of Leverage** – It is usual for Hedge Funds to use some form of leverage as it will increase their returns. Some Hedge Funds may be very highly leveraged which greatly increase the risk level of the funds for investors. Leveraging can be created through:
 - Borrowing of external funds;
 - Short-selling securities;
 - Borrowing through brokerage margin accounts; and
 - Using financial instruments or derivatives with margin mechanisms.
- **Unrestrictive Legal Structure** – Hedge Funds are usually set up in such a manner that they are relatively unrestricted in their strategies. Specifically, the managers take long or short position of any assets, use different kind of derivatives and conduct leveraging without the restrictions of many traditional funds. Hence, many of these funds prefer incorporation in places like British Virgin Islands, Cayman Islands and Bermuda as these territories allow flexible legal structures.
- **Higher Incentive Fees for Managers** – Hedge Funds have fee structures focusing on incentives when they outperform agreed return hurdles and high-water marks, in addition to receiving management fee. Incentive fees are usually very attractive to motivate the Hedge Fund Managers. The fees could go as high as 30% of the total profits generated.

7.3.2 Types of Hedge Funds

Even though it is difficult to categorize all Hedge Funds in a generic way, there are 4 main broad classifications used in the industry which should capture most of the Hedge Funds available:

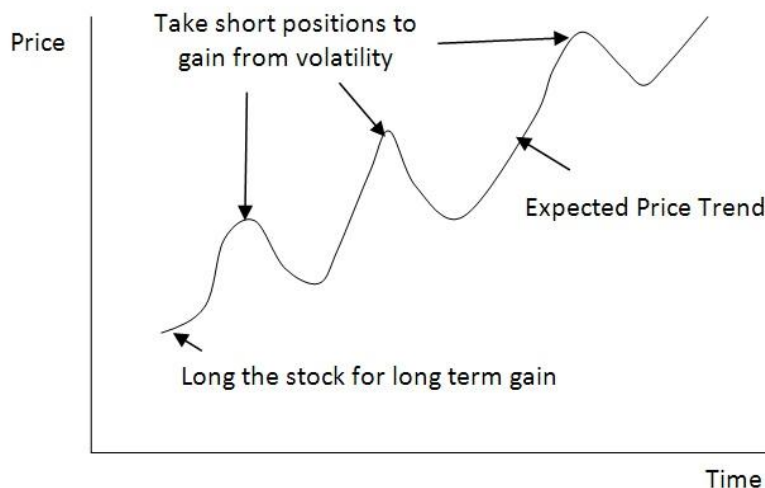
- Equity (e.g. Equity Long/Short, Equity Market Neutral, Short Sellers);
- Global Macro Funds;
- Event Driven Funds (e.g. Merger Arbitrage, Distressed Debt); and
- Arbitrage Funds.

Equity Long / Short Funds

Such funds basically allow both long and short position of common stocks in their portfolio. The concept of long-short funds is simply the buying of stocks with a potential for price increase while shorting of stocks with a potential for price decrease. Long short funds thus provide more avenues for returns as compared to long-only funds.

Various investment strategies could be implemented with the long-short Strategies. For example, if the price of a stock is believed to have a long term upward trend with short term volatility, it may make sense to long the stock and take several short positions, either directly or through the use of derivatives, over the investment period. Doing this will maximize the potential profit.

Exhibit 7.2 Example of Long / Short Strategy



Leveraging is often used to enhance the profits of a Hedge fund. For example, with an investment amount of \$100, the fund manager will long a stock with upside potential using the \$100. The manager could use the stock to support a short

margin position and sell another stock with downside potential. With a short position, the manager will receive incremental cash, for example of \$30.

He would use this amount to long more of the stock with upside potential. In this case, he would have \$130 long position and \$30 short position in its portfolio. Hence with \$100 the manager has created an investment worth \$160 through leveraging effect.

Market Neutral Funds

Such funds use long-short techniques to eliminate market risk of its investment, hence enabling the manager to focus on the excess return a stock is able to generate above the market return. For example, if there are 2 stocks in the automobile industry, Stock A and Stock B, and Stock A is a better performing stock than Stock B when the automobile market is on an uptrend and will decrease less than Stock B when the market is on a downtrend.

Assuming a Market Neutral Fund is uncertain about the market movement, the manager will long into Stock A and short into Stock B. In the event of a market uptrend, there will be loss from the short position which the long position will be able to cover over and above. In the event of a market downtrend, there will be loss from long position which the short position can well cover. A numerical example is provided in the Exhibit 7.3 below.

Exhibit 7.3 Numerical Example of Market Neutral Funds

Stock	Current Price	Price assumption during market uptrend	Price assumption during market downtrend
A	\$100	\$120	\$90
B	\$100	\$110	\$80

Market Neutral Strategy – Long Stock A and Short Stock B

a) During market uptrend:

Stock	Current Price	Price assumption during market uptrend	Profit / Loss
A	\$100	\$120	\$20
B	\$100	\$110	- \$10
Total Profit			\$10

b) During market downtrend:

Stock	Current Price	Price assumption during market downtrend	Profit / Loss
A	\$100	\$90	- \$10
B	\$100	\$80	\$20
Total Profit			\$10

The above strategy will enable the fund to profit regardless of the market movement if the assumptions are correct. Market Neutral Funds may utilize other asset classes like fixed income and derivatives instruments to implement its strategy. For example, a manager could long an undervalued bond and short bond futures or fixed income derivatives. Various other combinations are available.

Global Macro Funds

Global Macro Funds take bets on the direction of any macroeconomic variable like stock market, currency, interest rate or commodity. Such funds tend to be highly leveraged and utilize derivatives instrument extensively. Macro strategies are generally opportunistic and directional. Macro strategies can also be long/short in order to lower the fund's volatility or to exploit opportunities.

Event-Driven Funds

Funds in this classification aim to profit from specific events of a company or a security. Usually such events will result in price fluctuation which the fund will try to benefit from. For example, there are Distressed Securities Funds, which look for companies having financial difficulties or are in distress.

It is likely that these distressed companies will offer debt or equities in the market at huge discounts. The funds will purchase the debt or equities and stand to make a profit if the company is able to turn around through bankruptcy reorganization. Such funds could also take short positions when they believe that the condition will worsen instead.

Arbitrage Funds

Arbitrage Funds usually aim for stable returns with low risk by entering into paired positions that have offsetting risks. It often involves the simultaneous purchase of an instrument on one market and the sale of the same or related instrument on another market.

However, the resulting position may not be totally risk-free although an arbitrage is meant to be a risk-free transaction. Some of these paired trades are based on exploiting small market inefficiency or mis-pricing and require high leverage to make the gains attractive.

An example of arbitrage funds is one that uses a fixed income arbitrage strategy. Such strategy attempts to profit from the price difference in the swap-spread. This consists of taking opposing long and short positions in a swap and a Treasury bond. Such strategies provide relatively small returns but huge losses when not executed properly.

7.3.3 Key Risks of Hedge Funds

- **Managerial Risk** – Returns from Hedge Funds are highly dependent on the performance of individual managers in implementing the investment strategies. As such, there are managerial risks involved compared to other types of investments. Managerial Risk will arise from:
 - Ability of the managers in making the right investment decisions;
 - The level of risk that the managers are willing to adopt in implementing the investment strategies;
 - Manager focus – particularly as the fund size increases; and
 - Style or Strategy drift – some managers are very strong in particular asset categories or markets. If their area of specialization becomes temporarily less interesting there is a risk that the manager may explore other markets where he is less capable.
- **Liquidity Risk** – This risk arises when a fund loses its ability to raise liquidity efficiently. Some funds may invest in securities that are illiquid or utilize strategies that are dependent on the liquidity of specific markets. During extreme market conditions, the fund may suffer huge losses when they are not able to access the necessary liquidity to cover its positions.

For example, a Distressed Securities Fund would invest in the stock of a distressed company hoping that it will turn around. In the event that the condition of the company worsens instead, the stock could become highly illiquid and hence the Fund may not be able to sell off the stock or may need to lower the price substantially before it could obtain any liquidity.

Many funds have redemption restrictions which typically are one to three months but in rare cases may even be one year or more. Additionally, as per the example above, some funds that invest in particularly illiquid securities may return these actual securities to investors wishing to redeem from the fund rather than returning cash.

- **Pricing Risk** – Hedge Funds often utilize complex instruments which are only available over-the-counter. Pricing of such products will be more difficult as ready buyers or sellers may not be available unlike the stock market. Without a proper mechanism to determine prices, the transacted prices may not be a true reflection of the intrinsic value of the instruments. Hedge Funds may suffer loss when that happens.

For example a Hedge Fund may invest in a derivative instrument initially priced at \$10,000. The Fund is only required to put in a margin for the product and a margin call will only take place if the price falls below \$8,000. In a distressed condition, without a clear indication of pricing, it is likely for the broker of the product to adopt a conservative pricing policy and price

the instrument lower than the likely intrinsic value, triggering a margin call. The Hedge Fund will then be required to top up its margin account.

- **Counterparty Risk** – Also relating to over-the-counter instruments, the mechanism to ensure the fulfilment of obligations by counterparties may not be as robust. Either party may fail to deliver the specified securities or money to fulfil the transaction, creating counter-party risks.
- **Risk from Short Selling** – Hedge Funds may borrow stocks to carry out short selling strategies. In the event that owner of the borrowed stocks require their shares back, the fund will have to buy from the market and it may incur losses if prices spike up during the buyback exercise.
- **Risk from Taking Margin Positions** – In distressed conditions when margin calls are made, Hedge Funds with margin positions may suffer as they are required to top up their margin position with cash. If a fund does not have sufficient cash, it may be forced to sell its holdings at lower prices.

7.3.4 Performance Indicators

With the objective of searching absolute returns and the use of complex strategies, it is difficult to find benchmarks that could be used for comparison of Hedge Funds. Most Hedge Funds are evaluated based on the agreed target returns determined before investments commenced. Hurdle rates which affect the incentive fees for the managers may be put in place as benchmarks to gauge outperformance.

Nevertheless, there are Hedge Funds indices available that are published by data providers. Sub-indices for the various classifications of Hedge Funds are also available. Such indices may be used as references to evaluate the performance of a particular Hedge Fund, even though it may not be exactly suitable.

7.3.5 Disclosure Obligations

With the use of derivative instruments and complex investment strategies, Hedge Funds are generally classified as complex investment products that should only be accessible to accredited investors. Before investing in Hedge Funds, the investors should understand the instruments to be used and the leveraging involved. They should acknowledge that these are potentially high risk funds with the objective of seeking higher returns and at the same time may incur significant losses.

Like other forms of approved funds, documentations of the Hedge Funds should be made publicly available. Standard documentations include having prospectuses, term sheets, product summaries and annual reports. Investors should go through the documents to understand the mechanics of the fund before investing.

7.4 Private Equity Investments

Private Equity Investments are investments in privately held companies or companies that are not publicly listed. Such investments are more suited for accredited investors or corporates as the investment amount is usually large. The Private Equity Investment industry is actually quite active in terms of the total investment size and the number of private equity funds.

7.4.1 Key Features of Private Equity Investments

Different Investment Objectives of Private Equity Investments

The basic rationale of investing in privately held companies is that these companies are currently undervalued or their valuations have the potential for an increase in the near future. The main reasons for such re-valuations are as follows:

- **Initial Public Offering (IPO).** Companies that are slated for IPO or have the potential for IPO will be attractive for private equity investments. This is because the share price of these companies is likely to increase upon IPO. Private Equity funds are usually able to purchase the shares of these companies at a discount before IPO and hence make a profit after the listing of the companies.

Companies going for IPO may also be keen to have private equity investments before their IPO as this may boost their branding and marketability. Companies, with private equity investors as shareholders during IPO, are likely to draw more demand for their shares as the public will assume that these companies have undergone the due diligence process of the private equity funds.

Branding of the companies will also be enhanced if a well-known private equity funds has taken a stake in them. This is one of the main reasons why companies are willing to offer their shares at a discount to private equity firms prior to IPO.

- **Merger and Acquisition (M & A)** – Similar to the reasons above, private equity firms will be interested to invest in companies that are targets for M & A as there is a potential for the share price to increase when the M& A takes place.
- **Technology Advancement** – Private equity firms may also be attracted to companies working on technology advancement projects that have the potential of increasing firm value. Such projects are deemed to create products that will generate high revenue for the companies, enable the companies to be offered as an IPO or become targets of M&A.

Private equity firms may invest in these companies at the project research and development stage as valuation of such companies at that stage will

likely to be low. The private equity firms will then capture a large profit when the products are eventually launched.

- **Strategic Stakes** – Some private equity firms invest in companies because of strategic interests in the business that the companies are conducting. Such private equity firms seek returns from the dividends that these companies distribute and potential capital appreciation. Some private equity firms may also enhance the profits of the investee companies through strategic partnerships with other external parties.

For example, a private equity firm could partner with a US centric fashion chain to invest in an Asian fashion company. With the help of the US fashion company, the investment is likely to value add to the business of the Asian company and hence improve the profitability of the investee company.

Varied Forms of Private Equity Investments

There are numerous forms of Private Equity Investments. The most common forms include:

- **Leveraged Buyout** – This type of investments use significant amount of financial leverage in acquiring the shares of a private company. In this case, investors only need to provide a small portion of the capital for the acquisition.
- **Venture Capital** – These investments are typically made into less matured companies that are at their launch or early stage of development. It usually involves a new technology or product that is not available in the market.
- **Growth Capital** – Funds may be invested in companies that are seeking capital for expansion or implementation in new business initiatives.
- **Turnaround Capital** – These are funds invested in private companies that are in financial distress with plans to take over the company or seek high returns when the companies recover.
- **Mezzanine Finance** – A combination of debt and equity investment commonly used to finance the expansion of companies. Investors usually provide debt capital to such companies on the condition that the debt could be converted to equity interest of the company if the loan is not repaid.

Absence of Formalized Platform

There is a lack of formalized trading or information platform for private equity transactions. Private equity firms need to seek out the deals on their own usually with the help of arrangers or investment brokers.

Unlike the public market, there are no formalized platforms that share information on private equity deals transacted or in the pipeline. All transactions are directly concluded between investors and investee companies without a clearing market.

Transactions Backed by Legal Contracts

Private Equity Investments are unique deals between two parties and they are only bounded by the legal contracts that they have signed. Unlike the public market, there is no independent third party involved to ensure delivery of the obligations.

The contract between the 2 parties is the only document that governs the Private Equity Investment. With many of the private equity investment done across borders, legal matters have become more complex to ensure that the contracts are enforceable in different jurisdictions.

Difficulty in Valuing Investee Companies

As shares of the investee companies are not actively traded, it is difficult to determine their market values. The final transacted price is usually determined based on negotiation between the investors and investee companies. Various methodologies could be used to help them derive a reference fair value:

- **Cost Approach** – This approach looks at the value of the asset that makes up the company. This approach will be useful for companies with significant assets like real estate. Value of the company is the total value of the individual assets. However, this method will not work for companies where a significant part of the company resides in the human capital or intangible assets like intellectual property. It will be difficult to derive a price based on cost of these assets.
- **Comparable Approach** – This approach uses industry benchmark or similar companies that are actively traded to derive the value. A common indicator for this approach is the Price to Earnings (P/E) ratio. Information on P/E ratios for most industries or similar companies is available from the stock market. Value of the company can be derived by multiplying the P/E ratio with the earnings of the company. This method would be useful when the earnings of the company have reached a steady state.
- **Cash Flow Approach** – This approach derives value through discounting the expected future cash flow of the company. Realistically, this method may be difficult to implement as various estimation and assumptions are involved in deriving the future income and the appropriate discount rate to use.

Regardless of the approach used to derive the valuation, it is common to have a discount or premium factor added to the derived value. Whether it is discount or premium, the amount will depend on the negotiations between the investor and investee company.

7.4.2 Risks of Private Equity Investments

The main risks associated with Private Equity Investments include:

Liquidity Risk

Private equity investments are highly illiquid assets, especially during distressed conditions. As such, the exit strategy of private equity investment must be clear and due diligence must be properly done. If the exit strategy could not be implemented due to unforeseen circumstances, the investment is likely to suffer a significant loss.

For example, investing in pre-IPO companies requires an exit by selling the shares upon listing. In the event that the listing does not take place, it will be difficult for the Private Equity firm to sell the shares that they own to another private investor. Even if they are able to find another party that may be interested to purchase, it is difficult for the share price to be agreed upon.

Pricing Risk

As highlighted earlier, it is difficult to value investee companies as they are not actively traded. Private equity firms thus face the risk of inappropriate pricing which may result in losses to their investments. For example, if there is over-pricing of pre-IPO shares, losses will be incurred after the IPO with losses for the private equity firm.

Counterparty Risk

This risk stems from the lack of a formalized platform in clearing the deal. Private equity transactions are carried out directly between the investor and investee company and governed by the contract that they signed. Such transactions have counterparty risk when either of the party does not fulfill his obligations. This risk is especially real for cross-border trades with companies from different jurisdictions.

Risk Relating to Occurrence of Specific Events

Most Private Equity Investments are dependent on the occurrence of specific events to achieve the desired returns. For example, some investments will require an IPO or M & A to take place or a new product to be successfully launched. These events are in turn affected by various other risks that may prevent its occurrence. When that happens, returns of the Private Equity Investments will be adversely affected.

7.4.3 Performance Monitoring of Private Equity Investments

Similar to the situation for Hedge Funds, given the uniqueness of each Private Equity Investment, it is difficult to source a benchmark to provide a fair comparison. Private Equity firms should be evaluated based on its ability in delivering the returns

as they have set out. Nevertheless, there are data providers that publish prices of Private Equity funds or the expected returns from Private Equity Investments by vintage years (i.e. according to the calendar in which a fund is set up). Investors could use them as reference to gauge the performance of their investments.

7.5 Commodity Investments

A commodity is defined as a basic good which can be bought and sold. In the context of investments, commodities refer to goods that are tradable, e.g. Hard commodities like gold and soft commodities like sugar.

7.5.1 Key Features of Commodity Investments

Different Types of Commodities

Commodities for investment or trading purposes can be grouped into 3 major categories:

- **Agricultural Products** – Examples include wool, wheat, corn, soybeans, cocoa and coffee;
- **Energy** – Examples include crude oil and natural gas; and
- **Metals** – Examples include gold, silver, copper and aluminum.

Inherent Demand

The various types of commodities listed above have their own inherent demand for consumption or as raw materials for production. As a result, the pricing of commodities can fluctuate due to such inherent demand and supply availability.

Various financial instruments (e.g. Derivatives Instruments) are created to enable price confirmation of the commodities for the future inherent demand. The commodities market attracts different players including financial investors, commodities traders, hedgers and speculators.

Avenues to Invest in Commodities

Investment in commodities is seldom carried out directly as most investors would not want the hassle of warehousing the physical items. Instead, commodities are usually invested indirectly from the following avenue:

- **Futures Contract** – A standardized exchange traded agreement between two parties which dictates the type and amount of commodity that the

buyer would like to buy from the seller at an agreed price and a future delivery date.

- **Stocks of companies** – An investor may purchase shares of companies which produce the commodity, e.g. buying the shares of a gold mining company.
- **Bonds** – An investor may buy bonds which are indexed to the prices of a selected commodity.
- **Mutual Funds / Exchange Traded Funds** – There are many funds that invest either directly or indirectly in commodities.
- **Certificates of Ownership** – For example, it is possible to buy gold certificates.

Motives for Investing in Commodities

There are different motivations for investing in commodities:

- **Hedging** – Investors with businesses that use commodities as raw materials may want to invest in commodities futures to ensure that their production cost is kept in check. For example, a confectionary that requires lots of sugar for their daily operation may purchase sugar futures. In the event that the price of sugar indeed increases, the increase in cost would be offset by the profit from the futures.
- **Inflation Hedge** – Commodities are usually good inflation hedge as the prices of commodities usually increase during inflationary times. Hence, investing in commodities will be a good way to hedge against inflation.
- **Participating in Rapid Economic Growth** – For economies undergoing economic growth, it is likely that they will require huge quantities of commodities to fuel their growth. Prices of these commodities are likely to increase with a surge in demand. For example, with the rapid urbanization of China, steel is in great demand for the construction of buildings and railroads. Investing in aluminum may be a good avenue to participate in China's economic growth.
- **Store of Value** – Certain commodities, specifically precious metals like gold are excellent ways to “store value”. For example, gold has been one of the most accepted medium of exchange worldwide for centuries. When money and currencies devalue drastically in times of crises, gold maintains or increases in value. Investing in gold is a way to hedge against loss of value due to devaluation of currencies.
- **Speculative Motive** – Potential gains may be possible when there is limited supply of specific commodities. Oil is good examples of such commodities.

Inability to Generate Yield

Commodities do not have the ability to generate yield. Unlike investment in stocks that generate dividend and real estate that generate rentals, any returns from commodities can only be obtained from a realization of value. There will not be any returns generated during the holding period of the commodity (or instrument).

7.5.2 Key Risks for Commodities Investments

Risk Due to Physical Conditions

Supply of certain commodities may be dependent on physical conditions and seasonal factors. For example, the amount of agricultural products that can be produced will be affected by the weather. In extreme conditions like flooding, earthquake or dry spell, crops may be destroyed which will result in surging prices.

Risk Related to Exploration of Energy

For investors who invest in projects relating to exploration of energy, for example, investing in oil exploration, there is an additional risk of not achieving the expected yield.

Risk Associated with Investing in Securities and Derivatives

As mentioned earlier, investing in commodities is usually done indirectly through derivative contracts. Such instruments are subjected to its own set of risks. Investors should consider these risks before investing in commodities.

7.5.3 Performance Monitoring

Commodities index published by data providers are widely available. They will provide benchmarks to compare the actual returns from investments.

7.6 Passion Investments

Passion investment products have been slowly catching the attention of investors in the recent years. Underlying assets that fall into this category include art, antique, wine, collectibles and precious stone. Such investments are mainly for the truly affluent.

7.6.1 Key Features of Passion Investment

Limited Supply

The main desire to invest in passion investments stems from the uniqueness of the object. These objects are usually in limited supply or unique. Thus, the value of these objects is likely to increase over time. Hence passion investments can be highly rewarding.

Enjoyment for the Investor

Investors are able to derive pleasure from owning the passion investments. For example, wine connoisseurs invest in wines while art is beauty to the beholder. Investors are often willing to pay high prices for products or objects because of derived enjoyment.

Subjective Appeal

Many passion investments appeal to selective individuals. The values of these products are thus rather subjective depending on how much someone can appreciate the particular object or artifact.

7.6.2 Key Risks of Passion Investment

Liquidity Risk

Passion investment products bear high liquidity risk as the target market group can be rather small and individual preferences are subjective. It may be difficult to find someone that can appreciate the product just as much and willing to pay a high price for it.

Damage Risk

Passion investment products are susceptible to damages as they are required to be properly kept and maintained, e.g. in the case of Art or Wine. For example, the quality of wines may deteriorate if improperly stored.

7.7 Chapter Summary

This chapter has discussed Unit Trusts and related fund products. The salient points are summarized below:

- Real estate products are able to generate returns from these 3 ways; capital appreciation, rental and leverage.
- There are significant transaction costs involved in real estate investments.
- Taxes and other costs must also be considered when investing in real estate.
- The key risks associated with real estate are: market, regulatory, Interest rate, income, financial and physical property risks.
- The various performance measures linked to real estate investments are, property valuation, rental yield, cap rate, equity yield and return on investment.
- REITs are a vehicle structured for investors to pool their capital together for investment into large-scale income producing properties.
- The four major characteristics of hedge funds are:
 - Absolute Return Targets
 - Usage of Leverage
 - Unrestrictive Legal Structure
 - Higher Incentive Fees for Managers
- The majority of hedge funds use at least one of the following strategies:
 - Long / Short
 - Market Neutral
 - Global Macro
 - Event Driven
 - Arbitrage
- Investing in hedge funds exposes the investor to additional risks not typically faced by investors in Collective Investment Schemes.
- Private equity investments are investments in privately held companies or companies that are not publicly listed.
- The basic rationale of investing in privately held companies is that these companies are currently undervalued or their valuations have the potential for an increase in the near future.
- The four most common types of private equity investments are:
 - Leveraged Buyout
 - Venture Capital
 - Growth Capital
 - Turnaround Capital

- Mezzanine Finance

- There are various difficulties when it comes to valuing investee companies in private equity investments.
- The main risks associated with private equity investments are liquidity risk, pricing risk, counterparty risk and event risk.
- Commodities for investments include agricultural products, energy and metals.
- Investments in commodities are seldom done through the outright purchase of the underlying asset.
- The motivations for investing in commodities include hedging against inflation and price increases, and speculation.
- Investing in commodities exposes the investor to certain specific risks, such as, physical conditions and seasonal factors.
- The key features of Passion Investments are, they serve as a source of enjoyment for the investor, they are typically in limited supply and there exist a subjective appeal to selective individuals.
- Liquidity risk and the risk of damage are the 2 greatest risk associated with passion investments

Chapter 8:

Using Credit and Leverage

Objective

The candidate should be able to:

- ✓ Explain some best-practice approaches when assessing and addressing client needs for financing
- ✓ Articulate the benefits, risks and obligations of the client and the private banker when engaging in lending activities
- ✓ Discuss the approaches for evaluating the credit-worthiness of HNWI's
- ✓ Explain the use of collateralized loans and margin financing for various types of investments (including real estate)

8.1 Introduction

This chapter provides an overview of the bank lending environment and how lending activities in the private banking sector fit into the overall credit landscape. It also discusses the credit needs of private banking clients and how Covered Entities and Covered Persons can service these needs.

Further, this chapter explains how Covered Entities provide loans and margin financing for various investments including currency, equities, fixed income and derivatives products as well as for real estate. It highlights the risks when their clients are extended financing. Concepts like lending value (the amount of financing given to clients) and margin calls together with their related calculations are illustrated. Finally, this chapter also highlights some best practices and standards of care Covered Persons may adopt.

8.2 The Lending Environment

Lending is the traditional bread and butter business of commercial banks and remains a major determinant of their profitability. In full-service banks, lending to private clients typically make up only a small portion of the loan book which tends to be dominated by commercial and consumer loans. In contrast, boutique private banks tend to provide financing mainly for the investment activities of HNWI individuals and families and not for commercial business or retail clients.

To manage the potential losses arising from loan defaults, commercial banks charge higher interest rates are charged to those businesses and individuals assessed to be of higher credit risk. For private banks, there is a strong focus on having adequate collateral backing investments and they generally avoid providing loans on a clean basis regardless of the credit quality of individuals.

HNWIs often obtain lending or financing with different banks for different purposes. Many HNWIs would segregate their private investment from business activities. They would use credit facilities from private banks mainly for their personal investments and not for commercial interests. However, some clients do approach the Covered Entities for both personal investment and commercial loans. See Exhibit 8.1 for the Overview of Lending or Financing by different financial organizations.

Exhibit 8.1 Overview of Lending and Financing

<u>Retail Banking</u> Student Loans Credit Card Financing Housing/Mortgage Loans Bridging Loan Renovation Loans Hire Purchase Car Loans Personal Overdraft Facility	<u>Priority & Private Banking</u> Personal Overdraft Facility Housing/Mortgage Loans Bridging Loan Collateralized Loans for Investments Margin Financing
<u>Commercial Banking</u> Overdraft/Credit Facility Term Loans Trade-Finance & Guarantees Factoring Services Specialized Industry Specific Financing (aircraft, ship financing) Syndicated Loans	<u>Investment Banking</u> Margin Financing Lending to Hedge Funds Issuance of Stocks, Bonds, Derivatives & Structured Financing Asset Sales Syndicated Loans/Financing Structures
	<u>Brokerage Firms</u> Margin Financing

8.3 Determinants of Credit & Financing

8.3.1 Understanding the Needs for Credit of a HNWI

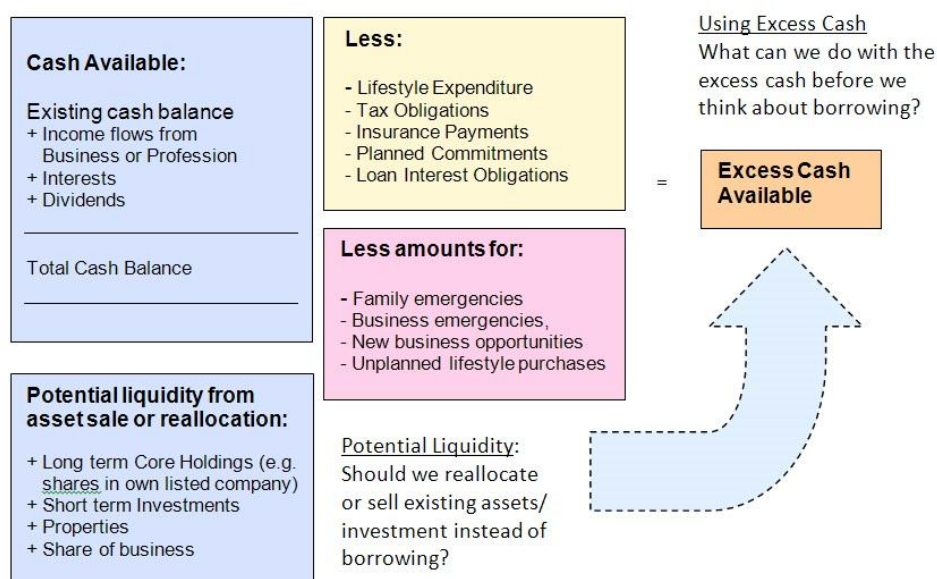
The lending or financing needs of a HNWI might vary. A few of the key ones are given below:

- **To Increase Liquidity** – Credit facilities provide additional liquidity to draw upon whenever required. Many entrepreneurs commit most of their assets and cash funds in their businesses or investments. Thus, they may use financing from private banks to meet some of the liquidity needs.
- **To Increase Investment Returns** – Clients often use financing for incremental funding to make a healthy spread over borrowing cost. HNWI's also use borrowed funds to invest in investment products which they are optimistic about the return even after factoring in the borrowing costs. Investors can augment returns with the use of leverage and margin financing when the investment is appreciating in value.
- **To Hedge Currency Risks** – Some HNWI's use financing as a natural currency hedge. This is done by borrowing in the same currencies that they are investing in. If an investment falls in value due to weaknesses in an underlying currency, the loan interest and principal would be repaid in that same currency, thereby mitigating the loss.

8.3.2 Assessing a Client's Financial Situation

Before offering any loans or credit facilities, a Covered Person should first review his/her client's financial situations to ascertain the needs for borrowing. Exhibit 8.2 illustrates this situation. By mapping out the financial resources and commitments of the client, a covered person could help the client to identify opportunity to re-allocate funds placed in underperforming investments before borrowing.

Exhibit 8.2 Assessing Client's Financial Situation



8.3.3 Understanding the Risk Profile and Risk Exposure of the Client

It is also important to assess the clients' risk profile. While some clients are averse to borrowing, others maximize the use of leverage to augment returns. Covered Entities need to be vigilant as some clients borrow from different sources and have multiple loan exposures.

Knowing the client's overall credit exposures and his/her needs allow a Covered Person to manage the risk of loan defaults for the Covered Entity that he is working for while also helping the client to manage his/her borrowing risk.

8.3.4 Credit Evaluation Criteria

To evaluate the credit-worthiness of a client, lenders consider among others the following criteria⁷:

- **Collateral** – The quality of the security;
- **Character** – The honesty and integrity of the borrower;
- **Capacity** – The borrower's ability to pay and handle the proposed new level of debt;
- **Capital** – The real net value of the borrower's assets which form back up liquidity for loan repayment; and
- **Condition** – Vulnerability to economic fluctuations.

Of the list above, Covered Entities particularly rely on collateral as the main protection for financing investments and they prefer collateral that is well-diversified, liquid, assessable, unencumbered, and of good credit quality. Stable or steady growth in market value is also preferred.

The amount of financing extended to a client will also depend on a lender's internal credit policy. The credit policy inherently reflects the business policy, strategies and risks propensity of the lender. The economy, market environment and regulations will also influence the amount of financing available to a borrower.

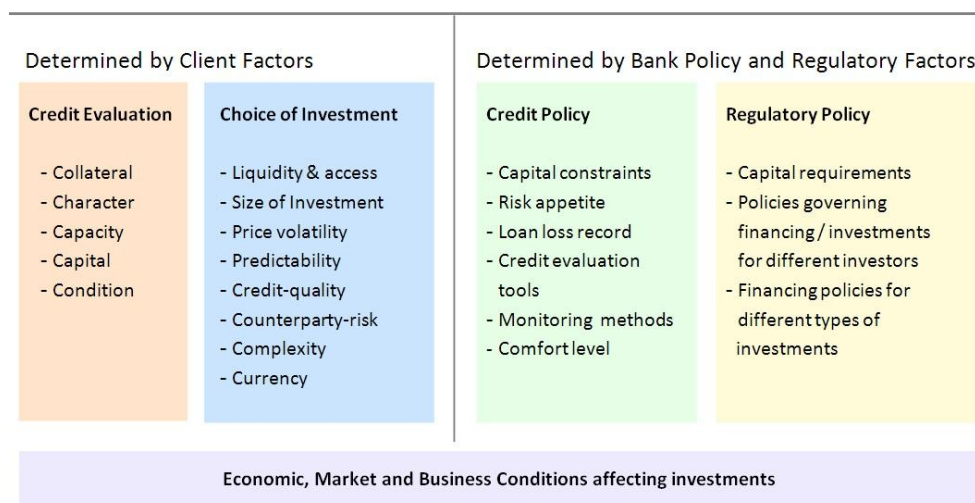
The choice of investment products also has an impact on the credit evaluation process. The risk of financing an investment is also dependant on the inherent risks of the underlying asset, its predictability, the hedging or protective structures put in place, trading liquidity, its complexity and the counterparties involved. Some Covered Entities may not be comfortable dealing in investment products that are too complex and less predictable.

The amount of financing extended to a client for any investment product will also depend on the Covered Entity's internal credit policy towards such products. The credit policy inherently reflects the business policy, strategies, financial health, and risks propensity of the lender. The economy, market environment and regulations will also influence the amount of financing available to a borrower. Regulations

⁷ See Joseph F. Sinkey, Jr., Commercial Bank Financial Management. Pearson Prentice Hall. 2009.

governing the financing of investments do vary between markets. These considerations are summarized in Exhibit 8.3.

Exhibit 8.3 Credit Evaluation of HNWIs



Generally, Covered Entities offer mainly collateralised loans and are relatively less dependent on the estimated cash flows of a client to determine the amount of credit facilities. However, not every asset can be used as collateral. Some Entities may be very stringent in qualifying assets as collaterals.

Typically, cash deposits, money market instruments, stocks, bonds and precious metals (e.g. gold) can be used as collaterals. These assets are liquid and relatively easy to determine the market values. Derivatives are not commonly used as collaterals. Covered Entities also look at asset concentration risks of the assets offered by clients as collaterals. If the asset composition is narrowly concentrated, the lending value will be capped to limit the exposure of the Covered Entity.⁸

8.3.5 Use of Leverage

The leverage ratio is measured as the ratio of borrowings to the market value of the invested assets. There are two ways of calculating this ratio. Both methods are acceptable as long as they are consistently used.

$$\text{Leverage Ratio} = \frac{\text{Loan}}{\text{Market Value of Assets} - \text{Loans}}$$

Or,

⁸ Lending Value is discussed in Section 8.5.1 in this Chapter.

$$\text{Leverage Ratio} = \frac{\text{Loan}}{\text{Market Value of Assets}}$$

Leverage can multiply profit or loss. The higher the leverage, the greater the potential for higher returns but higher the risk exposure. See the following examples.

Example

You have a client who has an existing USD10m investment portfolio of securities generating a return 4% annually. If the client uses this asset to raise USD5m to invest in similar type of securities providing similar return. He pays an interest cost of 0.75% above the interbank lending rate of 0.25% (USD LIBOR). What is the client's new Return on Asset (ROA)?

Scenario A

Client earn 4% on the USD10m	= USD 400,000
If the client earns (4% – 1%) on USD 5,000,000	= USD 150,000
New Return on Asset	= USD 550,000/USD 10,000,000
Return on Assets	= 5.5%

If client did not borrow he would only earn USD400,000 or 4%.

Scenario B

What would be the returns if the client was wrong? E.g. there was an unexpected shock to the financial markets and the invested assets generated a negative return on -4%? What happens to the client's ROA and how much are the losses?

Client suffers a loss of - 4% on the USD 10m	= - USD 400,000
Client suffers a loss plus interest cost (5% on USD5m)	= - USD 250,000
New Return on Asset	= - USD 650,000/USD 10,000,000
Return on Asset	= - 6.5%

If client did not borrow he would only lose –USD 400,000 or -4%.

If client borrows, he would only lose– USD 650,000 or -6.5%.

The leverage used here is rather low at 50% of Assets under Management (AuM). If the client borrows aggressively at say, 9 times the AuM, the profit and loss would be very dramatic.

Calculating the Amount of Loan Needed

Your client has SGD 3m in investments. He wants increase his investments using a loan but due to the bank's credit policy, he is allowed to borrow up to 30% of the total eventual assets held. What is the new loan amount, the new amount of investments and the equity position of your client?

Client has SGD 3m in investments. Loan should be no more than 30% of total invested assets:

$$\begin{aligned}
 \text{i.e. (Loan) / (Loan + Asset)} &= 30\% \\
 \text{Solving for L, } L / (L+3) &= 0.3 \\
 0.3 (L+3) &= L \\
 0.7L &= 0.9 \\
 \\
 \text{Loan amount, L} &= 1.285714\text{m} \\
 \text{Amount invested} &= 4.285714\text{m} \text{ [Check: } 1.285714 / (1.285714 + 3.0) = 30\%]
 \end{aligned}$$

8.4 Margin Financing

Financial institutions, including Covered Entities, offer margin financing to their clients for trading certain types of securities.

8.4.1 Margin Financing at Brokerage Firms

Share margin financing, offered by stockbrokerage firms, allow clients to increase their investment exposure to equities. Clients place cash and/or securities as collaterals for the facility.

In Singapore brokerage firms may offer margin financing up to 3.5 times the amount of cash deposits or 2.5 times the amount of marginable securities (with a minimum initial cash deposit of S\$5,000 or S\$10,000 worth of marginable securities) placed as collateral. An example of how margin financing works in local brokerage firm is illustrated below.

Calculating the Margin

A client has cash balance of SGD400,000 and securities in collateral of SGD500,000 in his account. The brokerage firm has a policy of lending 3 times against cash deposits. After reviewing the securities that the client owns, it is found that only SGD450,000 is marginable and the firm is willing to lend up to 1.5 times against these securities. What is dollar amount of margin credit line that the client is likely to be offered based on the criteria? What is the initial margin (%) extended by the firm to this client?

Cash = SGD400,000

Marginable Securities = SGD450,000

Margin line available = (SGD400,000 x 3) + (SGD 450,000 x 1.5) = SGD 1,875,000

If 100% of the margin line is used to purchase securities, the securities would be also worth SGD 1,875,000. This is the total amount of the margin credit line. Not counting the securities that cannot be used as collateral (non-marginable), what is the margin extended?

$$\text{Margin \%} = \frac{\text{Total Market Value of Securities}}{\text{Net Financing Provided}}$$

The "Total Market Value of Securities" includes the amount of securities financed as well as those purchased directly without financing. "Net Financing Provided" is the total amount of financing provided net of cash collateral.

Equity refers to:

The Marginable Securities	SGD 450,000
+ the allowable purchases	<u>SGD 1,875,000</u>
Total Market Value of Securities	<u>SGD 2,325,000</u>

Net Financing Provided

SGD1,875,000 – SGD400,000 = SGD1,475,000

The client maintains cash collateral of SGD 400,000 and marginable securities of SGD 450,000 to back the existing credit line. For the purposes of obtaining the initial margin position, we calculate the amount of the total value of client collateral, compared to the net credit provided. Initial Margin Position % = SGD 2,325,000/1,475,000 = 158%. The client has the option to use the entire line of SGD 1,875,000 or lower.

Suppose the client decides to use only SGD 1,500,000 to purchase securities instead of applying the full amount of SGD 1,875,000.

Net Financing provided:

Actual purchases	SGD 1,500,000
- Less cash collateral value used	<u>(SGD 400,000)</u>
Net Financing Provided	<u>SGD 1,100,000</u>

Actual Margin Position % = (SGD450,000+1,500,000)/1,100,000 = 177%

Maintenance Margin and Margin Calls in Brokerage Operations

Brokerage firms in Singapore are required to ensure that clients maintain a margin percentage⁹ higher than 140%. This is the minimum **maintenance margin** that a client has to maintain. At 140% or below, a margin call occurs. At 130% or below, forced selling may be implemented executed unless the client tops up immediately

⁹ Margin Percentage = Total Market Value of Securities / Net Financing Provided. This is also called the maintenance margin.

to restore the margin position. The following example illustrates the working of maintenance margins:

Margin Call and Closing Out

Based on our last example, the client purchased \$1,500,000 worth of securities. Suppose there is a downturn in the stock market occurs and equities plunge to 20% of the original purchased value on Day 1, and to 25% below the original value on Day 2, and to 30% below the original value by Day 3. When does the margin call occur?

Financing provided is the same:

Actual purchases	SGD 1,500,000
- Less cash collateral value used	<u>(SGD 400,000)</u>
Net Financing Provided	<u>SGD 1,100,000</u>

Event	Equity Value Calculation	Total Mkt Value of Securities	Margin %
Day 1, 20% decline	$(450,000 + 1,500,000) * (1-0.20)$	1,560,000	$(1,560,000/1,100,000) = 142\%$
Day 2, 25%	$(450,000 + 1,500,000) * (1-0.25)$	1,462,500	$(1,462,500/1,100,000) = 133\%$
Day 3, 30%	$(450,000 + 1,500,000) * (1-0.30)$	1,365,000	$(1,365,000/1,100,000) = 124\%$

Margin Percentage = Market Value of Securities / Net Financing Provided.

A margin call occurs on Day 2, when the client's margin percentage falls to 133%. The client must re-establish a margin percentage of at least 140%. No new purchases will be allowed. The client will usually be given 2 days to top-up the account by depositing cash or securities or by liquidating some securities. If he fails to do so by the second day, the brokerage firm will liquidate some counters to raise the margin percentage.

On Day 3, the margin percentage falls below 130%. Forced-selling may be implemented as the margin percentage has fallen below **130%**. Typically, the client is contacted immediately and has to top up margin positions on that same day. The client earns interest if the net balance in the share margin financing account is in credit. The interest rate will be based on an overnight deposit rate given by the bank and will be subject to change daily.

8.4.2 Margin Financing for Equities by Covered Entities

Unlike brokerage firms, Covered Entities tend to be more conservative in the size and lending value for margin financing. Covered Entities also have processes to warn clients to trim outstanding positions, top-up the insufficient collateral, cut losses and close out their loan positions. There are several specific trigger levels which will activate actions to be taken.

For example, when the margin position is eroded, a warning will be given to client. The client would be asked to manage any shortfall at this “**Warning Status**”. When there is a margin erosion of 25% or more, the client is at the “**Margin Call Status**”. Clients will be asked to commit more funds or assets (as collaterals) to cover their investment positions. When the margin erosion exceeds 50%, a Covered Entity may consider closing the entire investment position.

8.4.3 Margin Trading for Forex by Covered Entities

Unlike foreign exchange deposits, the use of foreign exchange margin trading facilities can be quite speculative. Gains and losses can be augmented significantly depending on reading of currency movements, timing, good trading discipline and the client’s holding power.

The rapid collapse of a currency accompanied by a margin financing position on that currency can have wipe out substantial amounts wealth. Similar to equities, foreign exchange margin trading will have "Warning Status", Margin Call Status and a final "Close-out Status". The following example illustrates how these are applied:

Margin Call and Closing Out

A client wants to use the margin financing for his foreign exchange trading for AUD/USD trade (currently trading at AUD/USD 1.0400-1.0410). He wants to put investing in AUD10m which he believes will appreciate significantly against the USD. The forex trading desk of the Covered Entity you work for has the following margin trading policy – the initial margin is set at 9%, at 7% there will be a margin call and at 5% they will close out the position.

For simplicity, the client posts a margin of USD1m for the USD10.4m trade into AUD10m which is above the initial margin requirement of USD 936,000 (0.09 x 1,040,000).

Warning Status: Now suppose, against the expectations of the client, the AUD weakens to AUD/USD =1.0336. At this point, the initial margin is reached and the client would have lost USD 64,000 (i.e. USD 10,400,000 – USD 10,336,000). If the AUD continues to weaken from this point, the Covered Person (or the FX team monitoring the client’s position) should warn the client that his account is now below the initial margin.

Margin Call Status: When the AUD weakens to AUD/USD = 1.0128, the client would have lost a cumulative amount of USD 272,000 (i.e. USD 10,400,000 – USD 10,128,000). The Covered Person (or the FX team monitoring the client's position) would call the client to add at least another USD 208,000 to restore the initial margin position of 9%. The client will have to be given some time to do this. Usually, this is a funds transfer or allocation from his accounts within the bank (or on occasion its remittance into the bank).

Close Out Status: However, currency movements can be quite rapid. Suppose the AUD weaken further to AUD/USD = 0.992 as commodities demand falls and is expected to continue to weaken. Suppose this happens before the client can restore the position by remitting funds. The Covered Entity will now have the right to close out the position immediately. The amount of time and flexibility accorded to the client to top-up on his position will depend on the degree of goodwill and credit that the client has with the Covered Entity.

AUD/USD	AUD	USD	Margin Status	USD Value of Margin	USD Added Loss	USD Total Loss
1.0400	10,000,000	10,400,000	Posted margin	1,000,000	0	0
1.0336	10,000,000	10,336,000	Initial margin 9%	936,000	-64,000	-64,000
1.0128	10,000,000	10,128,000	Margin call 7%	728,000	-208,000	-272,000
0.9920	10,000,000	9,920,000	Close out 5%	520,000	-208,000	-480,000

Besides foreign exchange margin trading, some of the more aggressive participants can take on overdraft facilities to finance forex margin trading or forex option trading. This makes it even more speculative since it is leverage on a leveraged position which can result in major losses and financing such trades is not usual.

8.5 Financing Foreign Exchange Carry-Trades

Some clients "invest" in currencies in countries with high interest rates and borrow or "short" currencies in countries that offer low interest rates in what is termed the "carry-trade". For example, they may borrow in **Yen** (e.g. JPY denominated loan at interest of 1-2%) and invest in higher yield currency investments (e.g. an AUD currency deposit with interest at 4-5% or in AUD-denominated bonds or equities). If the Yen borrowing appreciates and their investments in the AUD fall, then the investor loses both ways.

This strategy exposes the client to short-term currency risks as illustrated in Exhibit 8.4. In Exhibit 8.4, the volatility of the AUD/JPY currency pair moved from about 7-8% in March to 21-22% by April 2011. Clients who had invested in AUD/JPY currency in a dual currency investment will be exposed to severe risks.

Exhibit 8.4 AUD/JPY Exchange Rate and 1-Month Volatility



Source: UBS AG

8.6 Collateralised Loans, Lending Value, Shortfalls and Margin Erosion

Collateralized loans are offered by Covered Entities. A collateralized loan obligation is simply a cash loan granted to a client, backed by cash or pledged collateral in the form of marketable securities (stocks, bonds, ETFs). Clients can enhance their returns by:

- Borrowing on their existing investments; and
- Placing the borrowed funds in an investment with a higher yield than the interest cost of the collateralized loan.

8.6.1 Margin Erosion

Each Covered Entity can have its own internal way of calculating margin erosion and there will be variations in the approach and some will be more transparent with clients than others in the process of revealing how they calculate margin erosion. Three versions vary slightly with the more conservative formulae of (b) and (c) being more commonly applied in private banks. The key point is to warn the client before the margin call situation which is usually set at 25%.

Lending Value, Shortfall, and Margin Erosion

A client holds a portfolio of stocks in your private bank worth USD2m in value. He would like the bank to provide additional financing to purchase more stocks. Assume the bank has determined the Lending Value to be **60%**. After financing, the total portfolio will be $\$2m/0.4 = \text{USD}5m$. The client will be allowed to borrow up to \$3 million. The Lending Value is **\$3 million**.

		Margin Erosion Formulae				
Performance of Portfolio		0%	-10%	-15%	-25%	-40%
Market Value of Assets (not-financed)		2,000,000	1,800,000	1,700,000	1,500,000	1,200,000
Market Value of Assets financed		3,000,000	2,700,000	2,550,000	2,250,000	1,800,000
Total Market Value of Assets		5,000,000	4,500,000	4,250,000	3,750,000	3,000,000
Loan Exposure		3,000,000	3,000,000	3,000,000	3,000,000	3,000,000
(a)	Margin Erosion	0.00%	16.67%	25.00%	41.67%	66.67%
(b)	Margin Erosion	0.00%	16.67%	26.47%	50.00%	100.00%
(c)	Margin Erosion	0.00%	16.67%	26.47%	50.00%	100.00%
Formula (a) = (Loan Exposure - Market Value of Investments) / Lending Value						
Formula (b) = (Loan Exposure - Lending Value) / (Total Market Value - Lending Value)						
Formula (c) = {(Loan/Market Value) - (Lending Value/Market Value)} / (1 - Lending Value/Market Value)						
@where lending value is the value of assets being financed						
		Margin call would be activated at this point				

Suppose there is a 15% fall in the value of the stocks, the overall portfolio falls to USD 4.25m. There is a margin erosion of 25%. A margin call would be triggered by the Covered Entity.

Margin erosion may be defined as:

- (a) =
$$\frac{\text{Loan Exposure} - \text{Market Value of Investments}}{\text{Lending Value}}$$
- (b) =
$$\frac{\text{Exposure} - \text{Lending Value}}{\text{Total Market Value} - \text{Lending Value}}$$
- (c) =
$$\frac{(\text{Loan}/\text{Market Value}) - (\text{Lending Value}/\text{Market Value})}{(1 - \text{Lending Value}/\text{Market Value})}$$

8.6.2 Lending Value

When a portfolio of investments is used to finance additional investments, Covered Entities will assign a lending value for the investment and collateral used. The lending value represents the percentage of the total value of the investment that

the bank is willing to finance. The Lending value is determined by the type and quality of investments and collateral.

Typically the more volatile and risky the asset class is, the lower the lending value. The actual amount depends on the riskiness of the share counters and corporate policy. Generally, clients may borrow between 30-70% based on the value of their equity investments.

Currencies and money-market investments may receive a higher proportion of up to 90-95% while fixed income securities may have lending value of 75-85%. Since the formulae for margin erosion is affected by the lending value it is important to remind the client at the onset that the Covered Entity has the right to change the lending value of assets.

8.6.3 Mortgage Loans

Covered Entities may also provide real estate financing directly to their clients or in the case of full-service banks they can also be referred to their retail housing loan divisions. These schemes are akin to a series of short term advances or overdrafts which are rolled over annually. Clients have the flexibility to draw-down a larger or smaller quantity of the loan midway through the investment period.

Covered Entities usually offer customized financing arrangements of up to 60-70% of the market value of a property. Loans are usually only allowed for properties in selective countries where property laws are relatively clear and easy to enforce. Covered Entities are careful in extending large mortgage loans to their clients as it may lead to an outflow of funds away from financial securities.

Yet, the availability of mortgage loans allows clients to hold on to their financial investments with a Covered Entity without the need to realize the assets to purchase properties. In this context, mortgage loans may help Covered Entities to retain their clients and their assets under management.

A Covered Person needs to be aware that property investments may be used as a means to facilitate money-laundering. Unlike Covered Entities, property developers and real estate agents may not have stringent KYC procedures to ascertain the sources of funds for property purchases by foreign buyers. Some foreign investors may move funds into Singapore by first purchasing properties and subsequently, selling the properties for cash, and depositing the funds afterwards with a Covered Entity.

A covered person needs to look at the sources of funds of their clients (before the initial property purchase) to avoid facilitating “money-laundering schemes” and incurring reputation risks.

8.7 Chapter Summary

This chapter has discussed what a Covered Person should note in terms of the use of credit and leverage by the clients. The salient points are summarized below:

- A HNWI may need to borrow for many reasons, including the need for liquidity, to increase returns and to hedge currency risks.
- Before considering financing through loans, Covered Persons should help clients review their excess idle cash and the possibility of reallocating assets and the need for borrowing.
- The criteria of assessing credit worthiness of clients include the collateral used to back the investment, the character, capacity, capital of the borrower and the market conditions.
- Covered Entities look for collateral that is well-diversified, liquid, and assessable, unencumbered, of good credit quality, and reflect stable or growing market value.
- The amount of financing extended to a client will also dependent on a lender's internal credit policy, choice of investment products, as well as, regulations and the economic environment.
- The types of loans and financing provided by private banks include: margin financing, collateralized loans, mortgage loans.
- The lending value represents the percentage of the total value of the investment that the bank is willing to finance. Typically the more volatile and risky the asset class is, the lower the lending value.
- Clients must be made aware that Covered Entities reserve the right to change the lending value especially during market downturns and this can affect when margin call occurs.
- Margin erosion occurs when the market value of a client's portfolio declines clients and clients should be warned ahead of time to avoid margin call situations.
- When clients fail to cover margin calls, forced selling of investments may be activated.
- While clients can enhance their returns by borrowing on their existing investments, they should be made aware that leverage can multiply profit or loss. The higher the leverage, the greater the potential for higher returns but higher the risk exposure.

Summary of Formulas

1	$\text{Return on the Investment, } (R_i) = \frac{V_E - V_B + D}{V_B}$ Where, R_i = Return on the investment V_E = End value of the investment V_B = Beginning value of the investment D = Distributions (i.e. dividends or interest paid)
2	$1 + R = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)]$ $R = [(1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)] - 1$ Where, r_1, r_2, r_3, r_4 = Quarterly returns
3	$\text{Total Return} = (\text{NAV}_2 - \text{NAV}_1) / \text{NAV}_1 \times 100\%$ Where, NAV_1 = Original price NAV_2 = Current price n = Number of years
4	$\text{Annualised Return} = (1 + \text{Total Return})^{(1/n)} - 1$ Where, n = Number of years
5	$\text{Expected Return, } E(R_i) = \sum_{j=1}^N (\text{Probability of Return}) (\text{Possible Return}) = \sum_{j=1}^N (P_j) (R_j)$ Where, P_j is the probability of event j associated with return R_j for security, i
6	$\text{Required Return} = r + i + \sigma$ Where, r = Real interest rate i = Inflation rate σ = Risk premium
7	$\text{Capital Asset Pricing Model (CAPM), } E(R_i) = \text{RFR} + B_i (\text{RM} - \text{RFR})$ Where, RFR = Expected Return on government securities B_i = Beta or Relative risk of the Stock i to the Market RM = Expected Return on the Stock Market Index
8	$\text{Aggregate Expenditure, } (AE) = \text{Aggregate Income (wages, rent, interest, etc)}$ $AE = C + I + G + (X - M)$ Where, C = Consumption I = Investment G = Government Purchases X = Exports M = Imports $X - M$ = Net exports
9	$\text{Real GDP} = (\text{Nominal GDP}) \times 100 / (\text{Price Index})$

10	$\text{Return of a Portfolio, } (R_p) = \sum_{i=1}^n (w_i)(r_i)$ Where, R_p = Actual or expected return on the portfolio w_i = Portfolio weight for the i^{th} security r_i = Actual or expected return on the i^{th} security n = Number of different securities in the portfolio $\sum(w_i) = 1.0$
11	$\text{Portfolio Risk, } (\sigma_p) = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$ Where, w_1 and w_2 = Portfolio weights given to security 1 and security 2 respectively σ_1 and σ_2 = Standard deviation of security 1 and security 2 respectively $\rho_{1,2}$ = Correlation coefficient of security 1 and 2
12	$\text{Capital Market Line (CML), } E(R_p) = R_f + \frac{E(R_m) - R_f}{\sigma_m} \sigma_p$ Where, $E(R_p)$ = Expected return on any efficient portfolio on the CML R_f = Risk-free rate of return $E(R_m)$ = Expected return on market portfolio M σ_m = Standard deviation of the returns of market portfolio σ_p = Standard deviation of the efficient portfolio being considered
13	$\text{Security Market Line (SML), } E(R_i) = R_f + B_i[E(R_m) - R_f]$ Where, $E(R_p)$ = Expected return on security or portfolio R_f = Risk-free rate of return $E(R_m)$ = Expected return of overall market B_i = Beta of security or portfolio
14	$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$ Where, R_p = Return on the portfolio R_f = Risk-free rate of return σ_p = Standard deviation of the portfolio
15	$\text{Yield, } (Y) = \frac{\text{Face Value} - \text{Purchase Price}}{\text{Face value}} \times \frac{360}{\text{No of Days to Maturity}}$ $= \frac{D}{F} \times \frac{360}{t}$ Where, Y = Annualised yield on a bank discount basis (in decimal form) or discount basis yield D = Dollar discount F = Face value t = Number of days remaining to maturity

16	Valuation for a Security, $(P_0) = \sum_{t=1}^n \frac{\text{Cash flows}}{(1+r)^t} = \text{PV of future cash flows}$ $(P_0) = \frac{CF_1}{1+r} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n}$ Where, P_0 = Intrinsic value of the security today (the true value of the stock) CF_i = Cash flow at the end of Year i r = Discount rate reflecting the risk of the security. A more risky security would have a higher discount rate. Usually, the discount rate is the required return. n = Tenure of the security, in terms of compounding periods
17	Value of a Bond, $(P_0) = \sum_{t=1}^n \frac{\text{Coupons}}{(1+YTM)^t} + \frac{\text{Par Value}}{(1+YTM)^t}$ Where, P = Price of bond YTM = Yield to maturity t = End of period t
18	Price of a Perpetual Bond = $\frac{\text{Annual Coupon}}{\text{Interest Rate}}$
19	Modified Duration, $\Delta P/P = -D_{\text{modified}} \times \Delta YTM$ $\Delta P/P = -2 \times 1\% = -2\%$ Where, $-D_{\text{modified}}$ = Modified duration
20	Yield-to-Maturity, $(YTM) = \text{Capital Gains Yield} + \text{Current Yield}$ Where, $\text{Capital gains yield} = \text{Price appreciation on the bond}$ $\text{Current yield} = \frac{\text{Annual coupon}}{\text{Bond Price}}$
21	Historical P/E = $\frac{\text{Market Price}}{\text{Historical Earnings in the Past Year}}$ Leading P/E = $\frac{\text{Market Price}}{\text{Forecast Earnings in the Coming Year}}$ Where, $\text{Earnings per share} = \text{Net income} / \text{Number of common shares outstanding}$
22	PEG Measure, $PEG = \frac{P/E}{\text{Growth}}$
23	Price-to-Book Value, $P/B = \frac{\text{Market Price Per Share}}{\text{Book Value Per Share}}$
24	Price-to-Cash Flow, $P/EBITDA = \frac{\text{Market Price Per Share}}{\text{EBITDA}}$

25	Dividend Discount Model, $P_0 = \frac{D_1}{1+r_{CE}} + \frac{D_2}{(1+r_{CE})^2} + \frac{D_3}{(1+r_{CE})^3} + \dots + \frac{D_{\infty}}{(1+r_{CE})^{\infty}}$ Where, P_0 = Intrinsic value (or fair value) per share (what the stock is worth now) D_1 = Dividend per share at the end of Year 1 D_2 = Dividend per share at the end of Year 2 r_{CE} = Cost of equity (or required return to equity)
26	Intrinsic Value per Share = B_0 + PV of Residual Income Where, B_0 = Beginning book value of equity per share Residual income = Net Income – Equity charge Equity charge = Beginning book value of equity per share x cost of equity
27	Time Value of Option = Actual Option Price – Intrinsic Value
28	Call Intrinsic Value = $S_T - X$ if $S_T > X$ or 0 if $S_T \leq X$ Where, S_T = Value of the underlying asset X = Exercise price of the asset
29	Payoff to call holder = $S_T - X$ if $S_T > X$ or 0 if $S_T \leq X$ Where, S_T is value of the underlying asset X is the exercise price of the asset
30	Payoff to call writer = $-(S_T - X)$ if $S_T > X$ or 0 if $S_T \leq X$ where S_T is value of the underlying asset X is the exercise price of the asset
31	Put Intrinsic Value = $X - S_T$ if $S_T < X$ or 0 if $S_T \geq X$ Where, S_T = Value of the underlying asset X = Exercise price of the asset
32	Payoff to put writer = $-(X - S_T)$ if $S_T < X$ or 0 if $S_T \geq X$ Where, S_T is value of the underlying asset X is the exercise price of the asset

33	Black Scholes Option Price Model, $C_0 = S_0 N(d_1) - \frac{X}{e^{rt}} N(d_2)$ $d_1 = \frac{\ln(S_0/X) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$ $d_2 = d_1 - \sigma\sqrt{t}$ Where, C_0 = Call option price X = Exercise price r = Continuously compounded riskless rate of interest on an annual basis t = Time to maturity in years σ = Standard deviation of the annual rate of return on the underlying stock $\ln$ = Natural logarithm function e = 2.71828, the base of the natural log function $N(d_1)$ = Cumulative density function of d_1
34	Delta = $n \times \delta WP / \delta S$ Where, Delta = Rate at which warrant price changes with change in share price δWP = Change in warrant price δS = Change in share price n = Conversion ratio
35	Effective Gearing = Delta x Simple Gearing
36	Implied Volatility = f (share price, warrant price, strike price, maturity, interest rate and dividend yield)
37	Gearing Ratio = $\frac{\text{Share Price}}{\text{Warrant Price} \times n}$ Where, n = Conversion ratio
38	Intrinsic Value of Warrant, (Call Warrant) = $S - X / n$ (Put Warrant) = $X - S / n$ Where, S = Share price X = Exercise price n = Conversion ratio
39	Conversion Price of Warrant, (Call Warrant) = $X + nWP$ (Put Warrant) = $X - nWP$ Where, X = Exercise price WP = Warrant price n = Conversion ratio
40	Forward Exchange Rate = Spot rate $\pm$ Forward points
41	Forward Points = $\frac{S \times (R_c - R_b) \times n / 360}{1 + R_b \times n / 360}$ Where, S = Spot rate R_c = Annualised interest rate of counter-currency R_b = Annualised interest rate of base currency n = Number of days from spot date

42	Interest Rate Parity, $F = S \times \left(\frac{1 + R_c (n/360)}{1 + R_b (n/360)} \right)$ Where, F = Forward rate S = Spot rate R_c = Annualised interest rate of counter-currency R_b = Annualised interest rate of base currency n = Number of days
43	Range Accrual Note, (Reference index inside range) Payout = $P1 \times (n/N)$ (Reference index outside range) Payout = $P2 \times (N-n)/N$ Where, N = Total number of observations within a period n = Total number of observations when the index is inside range $P1$ = Payout when the index is inside the range $P2$ = Payout when the index is outside the range
44	Allocation to Risky Asset = Multiplier x Cushion value Where, Multiplier = $1/\text{crash size}$
45	Portfolio's Expected Return, $E(R_e) = R_f + \beta_p [E(R_m) - R_f]$ Where, $E(R_e)$ = Expected return on the security, e $E(R_m)$ = Expected return of the market, m R_f = Risk-free rate β_p = Beta of portfolio
46	Sortino Ratio = $\frac{R_p - R_f}{\sigma_d}$ Where, R_p = Return on the portfolio R_f = Risk-free rate of return σ_d = Standard deviation of negative asset returns
47	Gross Rental Yield = $\frac{\text{Gross Annual Rental Income}}{\text{Gross Property Transacted Price}}$
48	Cap Rate = $\frac{\text{Annual Net Operating Income}}{\text{Property Transacted Price}}$ Where, Net Operating Income = Rental Income – Property Level Expenses
49	Equity Yield = $\frac{\text{Annual Cash Flow}}{\text{Total Amount Invested}}$ Where, Annual Cash Flow = Rental Income – All Expenses – Interest Cost – Tax + Depreciation & Amortization Total Amount Invested = Property Transacted Price – Loan Amount + Transaction Cost

50	Return on Investment = $\frac{\text{Total Amount Generated} - \text{Total Amount Invested}}{\text{Total Amount Invested}}$ Where, Total Amount Generated = All Annual Cash Inflow + Property Valuation – Remaining Loan – Transaction Cost Upon Sale
51	Dividend Yield = $\frac{\text{Annual Dividend Distributed}}{\text{REIT Price}}$ Where, Annual Dividend Distributed = Rental Income – All Expenses – Interest – Property Level Tax REIT Price = Purchase Price of REIT on IPO or when traded in Stock Exchange
52	Leverage Ratio = $\frac{\text{Loan}}{\text{Market Value of Assets} - \text{Loans}}$ or Leverage Ratio = $\frac{\text{Loan}}{\text{Market Value of Assets}}$
53	Margin % = Client's Equity / Total Financing Provided
54	Margin Erosion = $\frac{\text{Loan Exposure} - \text{Market Value of Investments}}{\text{Lending Value}}$ or = $\frac{\text{Exposure} - \text{Lending Value}}{\text{Total Market Value} - \text{Lending Value}}$ or = $\frac{(\text{Loan}/\text{Market Value}) - (\text{Lending Value}/\text{Market Value})}{(1 - \text{Lending Value}/\text{Market Value})}$

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